

Coffeehouses and the Rise of Science*

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Abstract

Can social venues raise knowledge production by lowering the cost of learning from others? Historians have long credited coffeehouses, commercial venues where strangers met across social ranks, with fostering scientific activity during the Scientific Revolution. We test this claim in seventeenth-century London. Using geocoded records and a staggered event study across parishes, we find a persistent increase in mathematical practitioners and scientific instrument shops following a parish's first coffeehouse. Effects are concentrated among practitioners with weak ties to Oxford and Cambridge, connections between theorists and applied makers increase, and new shops turn toward experimental instruments. We argue that coffeehouses fostered this growth by lowering the cost of accessing scientific knowledge for those outside academic networks and by making experimental science interactive and public. These findings show that social venues can foster knowledge production outside formal academic networks.

Keywords: COFFEEHOUSES, SCIENCE, KNOWLEDGE DIFFUSION, SOCIAL NETWORKS

JEL Codes: N13, O31, O33, O43, Z13

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1. INTRODUCTION

Producing new knowledge is crucial for technological innovation and long-run growth (Romer, 1990), and how much researchers produce depends on how much of the existing stock of knowledge they can access (Mokyr, 2002). Though much of the existing knowledge is codified and accessible in print or online, frontier knowledge often contains tacit elements that are not yet articulated or difficult to articulate, making face-to-face interaction important for their transmission (Polanyi, 1966). In practice, access to tacit knowledge is costly, shaped by education, geography, social networks, family background, and wealth (Bourdieu, 1986; Glaeser et al., 1992; Jaffe et al., 1993). As quality is hard to assess in advance, universities, laboratories, and journals filter access to this knowledge, admitting, hiring, and refereeing to judge who and what is worth learning from. Such filtering protects quality but narrows the pool of participants. Institutions that broaden access to exchange could raise knowledge production (e.g., Stephan, 1996) or dilute it with noise when nobody is screened for quality (Akerlof, 1970). However, the long-run impact of such institutions (e.g., coworking spaces, open-source platforms) remains hard to quantify, as participation in them is rarely random and data are often scarce. This paper exploits a historical episode that helps overcome these challenges.

The new social institutions of the seventeenth and eighteenth centuries are credited as among the most transformative developments in early modern European history (e.g., Clark, 2000; Cowan, 2005; Mokyr, 2016). These were the French salons, German table societies, and English coffeehouses. Most famously, Habermas (1991) termed their introduction a “structural transformation of the public sphere,” and singled out the coffeehouse as a catalyst of Enlightenment culture, a public, interactive, middle-class space where reason, rather than rank, shaped debate. The coffeehouse “not merely made access to the relevant circles less formal and easier; it embraced the wider strata of the middle class, including craftsmen and shopkeepers” (Habermas, 1991, p. 33). On this view, coffeehouses turned geographic proximity into low-cost knowledge exchange, and Mokyr (2005) relatedly argues that they reduced access costs to “useful knowledge,” the type of knowledge oriented toward progress at the intersection of science and technology. Did these social venues foster intellectual progress by broadening access?

We answer this question by studying the coffeehouse in the context of science in seventeenth-century London, home to one of the first coffeehouses in Europe in 1652 and a center of the Scientific Revolution. For the price of a dish of coffee, a penny or about one-tenth of a day’s wage, strangers met and debated across social ranks, earning coffeehouses the nickname “penny universities” among contemporaries (Ellis, 1956). On 10 December 1660, Samuel Pepys (1633–1703) captured the appeal of the institution on what he called “the first time that ever I was there,” finding “much pleasure in it, through the diversity of company and discourse” (Pepys, 2026). Coffeehouses spread suddenly and unevenly, parish by parish, after 1652, and rich historical records track scientific activity over the subsequent decades. We focus on mathematical practitioners, the applied scientists of the period, recorded as teaching mathematics, writing on it, or making instruments, and on the shops that made, repaired, and sold those instruments. In a staggered event study design, the arrival of a parish’s first coffeehouse roughly doubled the number of mathematical practitioners and raised the number of instrument shops nearly fivefold. Both increases persisted for decades.

London was not closed to useful knowledge before coffeehouses. Gresham College offered public lectures, St. Paul's Churchyard supported an open book trade, workshops transmitted craft skills, and universities and households connected established scholars. The scholarly channels were the narrowest: of the 176 mathematical practitioners active between 1665 and 1699, only 12 are ever named in the *Philosophical Transactions*, the Royal Society's journal and the leading scientific publication of the period. The closest precedent was Robert Boyle's (1627–1691) “invisible college” of the 1640s, a network of natural philosophers that, despite having no building or charter, was closed to outsiders since joining required correspondence, introductions, and learned standing. Private patronage worked similarly, as seventeenth-century research depended on wealthy sponsors or personal fortunes secured through social standing rather than fees. In contrast, one penny bought news and a seat in a shared room without a credential, charter, or patron. As science shifted from Renaissance secrecy toward a public, competitive enterprise of discovery (Wootton, 2015), natural philosophers, instrument makers, artisans, merchants, and the middling sort met there repeatedly. Historical accounts place such institutions in the exchange of useful knowledge and Europe's transition to modern economic growth (Broman, 1998; Jacob, 1997; Mokyr, 2002, 2005, 2016; Stewart, 1992), where repeated contact built the reputation and trust that made exchange possible (Johns, 1999; Shapin, 1994; Sunderland, 2007).

The idea that London coffeehouses played a central role in the diffusion of knowledge can be traced to John Houghton (1645–1705), an English apothecary, writer on agriculture and trade, and Fellow of the Royal Society. In 1701, Houghton (1727, p. 132) observed:

Coffee-houses make all sorts of people sociable, the rich and the poor meet together, as also do the learned and unlearned: It improves arts, merchandize, and all other knowledge; for here an inquisitive man, that aims at good learning, may get more in an evening than he shall by Books in a month [. . .]. I have heard a worthy friend of mine [. . .] say, that he did think, that coffeehouses had improved useful knowledge, as much as [the universities] have [. . .].

Why might coffeehouses have mattered? They did not necessarily create skill. They connected the people who held it. Thomas Tompion (1639–1713) was already a successful clockmaker when he discussed dividing compasses with Robert Hooke (1635–1703) at Child's Coffee House and a spring-watch mechanism at Garraway's. Hooke brought scientific problems and designs, Tompion built, tested, and revised them, and their work continued in the workshop and through a presentation to the king (Iliffe, 1995; Robinson and Adams, 1935). The record does not show that the coffeehouse first brought the two men together. It shows how regular meetings could combine knowledge held by different people. The same exposure could draw in new entrants: craftsmen and scholars encountered scientific questions, potential collaborators, and demand for instruments, while theorists discovered what makers could build.

We take Houghton's hypothesis to the historical record. We hand-geocode every coffeehouse, mathematical practitioner workplace, and scientific instrument shop with a recorded address, including parishes, streets, alleys, and signboards, using digitized seventeenth-century maps and gazetteers. This places them within the 162 parishes of seventeenth-century London and yields a parish-by-decade panel from the 1600s through the 1690s. We also reconstruct the social connections among practitioners.

Treatment and the two main outcomes come from separate compilations assembled from different historical records by different authors, so the results are not merely an internal artifact of a single work or author.

As Figure 1 shows, coffeehouses appeared just as a sustained rise in the number of practitioners and shops began. These trends coincided with a broader cultural shift toward quantification. [Kelly and Ó Gráda \(2022\)](#) show that the publication of applied mathematics titles more than doubled in the 1650s, interpreting this as a sudden rise in demand as navigational and surveying technologies matured and came into widespread use. In the same decade, language oriented toward progress and associated with science and industry began to rise ([Almelhem et al., 2026](#)). While historians likewise note these trends and emphasize coffeehouses as a central driver of this shift (e.g., [Ellis, 2006](#); [Jacob, 1997](#)), no empirical evidence corroborating these claims exists.

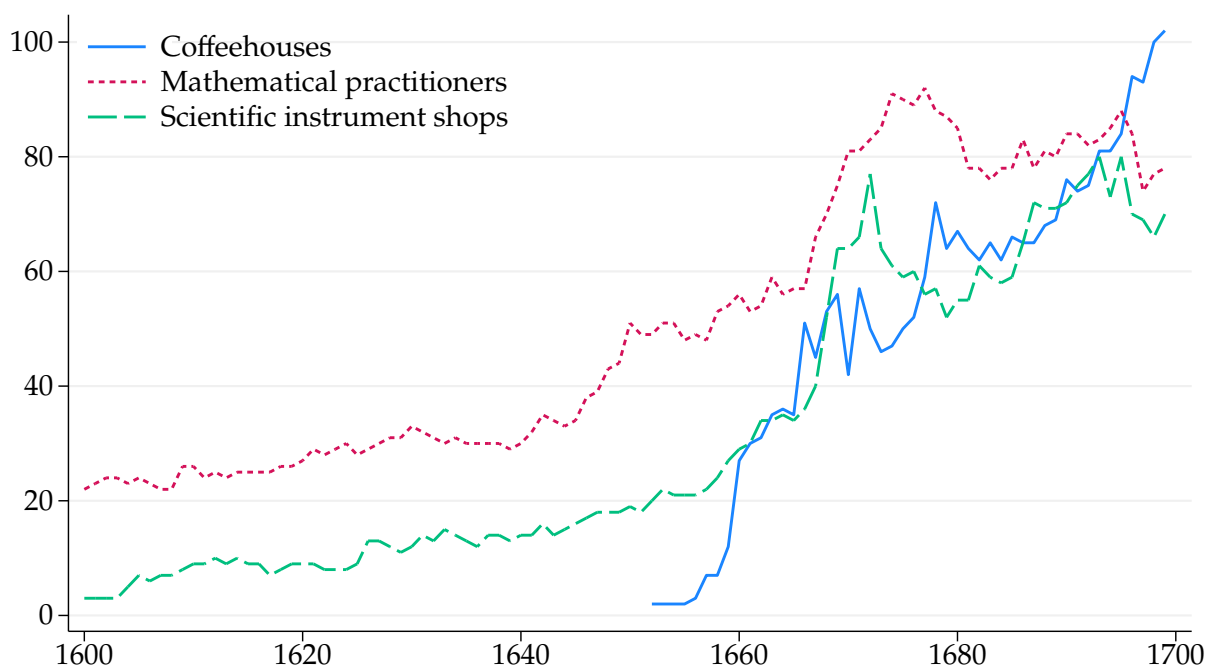


FIGURE 1. COFFEEHOUSES AND SCIENCE OVER TIME

Notes: Figure plots annual coffeehouses, mathematical practitioners, and scientific instrument shops in seventeenth-century London. Coffeehouse data come from [Lillywhite \(1963, 1972\)](#). Mathematical practitioner data come from [Taylor \(1954\)](#). Scientific instrument shop data come from [Clifton \(1995\)](#) and [Hay et al. \(2025\)](#). [Return to pages 3, 74]

However, the staggered and incomplete distribution of coffeehouses within London provides an ideal setting for hypothesis testing, as it helps separate the effect of these venues from rising aggregate demand. We identify the effect of coffeehouses from variation across London's parishes. These are small units, and 97 of the 162 lie within the City walls alone, so a parish that received a coffeehouse and one that did not were often only a few streets apart. Of the 162 parishes, 101 never received a coffeehouse before 1700. We compare activity before and after the first recorded coffeehouse with contemporaneous changes in these never-treated parishes. The baseline specification uses the staggered, heterogeneity-robust event study estimator of [Wooldridge \(2023\)](#).

We find large and persistent effects on both outcomes. Treated parishes averaged

0.62 mathematical practitioners in the decade before entry, and the number rises by 0.68 thereafter, roughly doubling. Scientific instrument shops rise from 0.33 to 1.57, a nearly fivefold increase. London gained 68 practitioners and 136 shops between the 1640s and the 1690s. Estimated effects in the 1690s account for 49 percent of the increase in practitioners and 63 percent of the increase in shops. On a second measure, entry accounts for 20 percent of practitioner-decades and 46 percent of instrument-shop-decades recorded between the 1650s and the 1690s. These shares are statistically attributable to entry under parallel trends. They sum effects within treated parishes and do not identify a London-wide equilibrium response.

The parish estimates do not by themselves show that London gained the same amount of activity. For shops, entry begins when a maker is first recorded, so moves among known London addresses cannot produce the result. Positive spillovers into neighboring parishes and scientific growth elsewhere in England also weigh against simple displacement within London. Moves from unrecorded locations or from outside London remain possible. Interpreting the aggregate shares as an enlargement of London's scientific sector requires assuming that such moves do not account for the response. The parish-level result does not require this additional assumption. A quantitative urban model, disciplined by these estimates and by the city-level differences in Europe, suggests that coffeehouses account for 31 percent of practitioner-decades between the 1650s and the 1690s once the roster of [Taylor \(1954\)](#) is completed with the instrument makers found by later research. Of the gain in parishes with a coffeehouse, 23 percent came from elsewhere in London and the rest was new to science (Section 7).

Coffeehouses did not open at random. Proprietors favored commercially active sites, and treated parishes began with more commerce and higher levels of both outcomes. The validity of the comparison is supported by flat pre-trends. Changes in both outcomes were similar before entry, and the pre-treatment coefficients are jointly indistinguishable from zero. The estimates also remain close after adjustment and reweighting for the observed predictors of entry. Population density, luxury shops, other commercial premises, and nonscientific correspondence do not rise with coffeehouse entry, which weighs against a coincident local boom. [Cowan \(2005\)](#) argues that the coffeehouse itself grew out of a national culture shaped by the virtuosi (natural philosophers and scholars). Such a change would affect London as a whole, while our design uses differences in when individual parishes received a coffeehouse. The local concern is that houses opened where science was already gathering. Coffeehouse founders and natural philosophers came from distinct social worlds. Scientists shaped what coffeehouses became rather than where they opened: they were early users, and they helped turn the rooms into venues for empirical discussion.

Survival in the historical record raises a separate concern. Coffeehouses could make existing practitioners easier to observe through advertisements, imprints, correspondence, or contact addresses. Using separate compilations for treatment and outcomes does not rule this out. The largest response, however, appears among instrument shops, which are often recorded through signed objects, guild records, and trade cards. The estimates remain similar when we restrict the sample to guild-linked makers and when we omit coffeehouses recorded in different types of sources. Separate trends for areas affected by the Great Fire address rebuilding and differential record loss. Conley errors, leave-one-parish-out estimates, alternative geographic units, and alternative estimators yield similar conclusions.

Customer demand is the leading alternative to knowledge exchange. Merchants and navigators may have visited shops and teachers as customers, not collaborators. Null results for ordinary retail alone do not rule out demand for scientific goods. Effects are strongest among practitioners whose surnames were least represented at Oxford and Cambridge. Connections among makers, and between makers and theorists, rise when both were active in the same decade. Shops producing experimental instruments such as microscopes and air pumps rise too, though their gap over practical shops stays imprecise. Makers take more apprentices, and Fellows associated with treated parishes rise in later decades. Earlier pub openings show no matching rise, arguing against ordinary sociability alone. The patterns do not identify a single channel, but are consistent with coffeehouses making scientists and skilled entrants easier to reach.

Contribution to the Literature. This paper contributes to several strands of literature. First, whether coffeehouses changed who did science has been argued since Houghton wrote in 1701, without evidence either way. We provide, to our knowledge, the first causal evidence that they increased applied scientific activity, resting on a newly assembled geocoded parish-by-decade panel of coffeehouses, practitioners, and instrument shops. Historians often read the coffeehouse as a product of London science. Cowan (2005, p. 260) calls it “the product of the cultural world forged by the virtuosi,” while Stewart (1999) describes particular houses as “other centres of calculation” alongside the Royal Society. We ask whether local coffeehouse entry also changed who took part in applied science. Activity rose where coffeehouses opened, when they opened, and not before. The evidence favors new activity over relocation.

Second, institutions that widened access to useful knowledge, and reduced the frictions in producing it, come in two kinds (e.g., Dowey, 2017; Galofré-Vilà, 2023). Some codify knowledge and carry it across distance (e.g., Mokyr, 2002): the printing press (Dittmar, 2019, 2011), the postal networks of the Republic of Letters (Cervellati et al., 2026; Melillo et al., 2026), the encyclopedias of arts and trades (Squicciarini and Voigtländer, 2015), the Uniform Penny Post of 1840 (Hanlon et al., 2022), and the state translation programs of Meiji Japan, whose comparative advantage then shifted toward the industries that stood to learn most from British patents (Juhász et al., 2026). Others organize exchange within a bounded membership: universities and academies (de la Croix et al., 2025; Koschnick, 2025b), and economic societies (Cinnirella et al., 2025). Coffeehouses did neither. They lowered the cost of repeated face-to-face contact across communities that otherwise met in more specialized settings. Modern evidence finds high returns to contact of this kind (e.g., Andrews, 2023; Andrews and Lensing, 2024; Atkin et al., 2022; Boudreau et al., 2017; Catalini, 2018), and network evidence documents spillovers beyond the people directly exposed (Fluegge and Bailey, 2024). The rise in connections we document gives one way that urban density could raise knowledge production, at the scale of streets and parishes rather than towns (Cox and Figueroa, 2025); pub openings produce no comparable increase. The same results offer large-sample support for the view of Fleck (1979) and Mokyr (2016) that scientific knowledge is produced in interaction among people, not by individuals alone.

Third, better information technology does not retire this margin. Access to codified knowledge has grown steadily cheaper, most recently through the internet (Goldfarb and Tucker, 2019) and frontier AI (Stanford Institute for Human-Centered Artificial Intelligence, 2026), but these advances do less for tacit knowledge, which travels with

people rather than with texts (Storper and Venables, 2004). Seventeenth-century London had the printed book, yet the coffeehouse still changed who did science. Tao (2026, p. 7) finds the same gap in mathematics today: a proof polished by a machine can be “easy to read and hard to learn from,” because the friction a human author leaves on the page carries part of the tacit knowledge of a field. That gap widens as frontier AI cheapens codified knowledge further, since the costly parts remain choosing which problem is worth solving and getting an answer absorbed by others. Venues for tacit exchange may matter more, not less.

Finally, what enlarged the number of people who translated scientific knowledge into practice is unsettled. Zilsel (1942) attributed the rise of the early modern mathematical practitioner to a weakening separation between university scholars and artisans, Taylor (1954) to demand from navigation, warfare, and surveying, and Kelly and Ó Gráda (2022) document how far practical mathematics spread. Related work traces tacit skill through apprenticeship (de la Croix et al., 2018), the skilled implementers behind industrial technological change (Meisenzahl and Mokyr, 2012), and the later emergence of professional engineers (Hanlon, 2025). Historical work already locates practical mathematics in the shops, lectures, and commercial communities of sixteenth-century London and treats the coffeehouse as another venue for a sociability that predated it (Cormack, 2017). We provide a causal test of this institutional margin: coffeehouse entry raised local entry and expansion. Coffeehouses connected established centers of knowledge to a wider body of skilled men, giving institutional content to accounts that link the associational world of Britain to the diffusion of useful knowledge (Clark, 2000; Habermas, 1991; Iliffe, 1995; Jacob, 1997; Mokyr, 2016; Squicciarini and Voigtländer, 2015). The practitioners and makers whose numbers rose are those that accounts of Britain’s later technological change place at its center. Their price of entry was a penny rather than a university place.

Roadmap. The paper proceeds as follows. Section 2 provides the historical background. Section 3 describes the data. Section 4 examines the effects of coffeehouse openings on mathematical practitioners and scientific instrument shops. Section 5 examines the mechanisms underlying these effects. Section 6 turns to European cities. Section 7 uses a quantitative urban model to gauge the effect on London as a whole, and what the city would have lost had Charles II’s proclamation of 1675 against the retail sale of coffee not been suspended. Section 8 concludes.

2. BACKGROUND

This section provides historical background for the emergence of coffeehouses in seventeenth-century London, from their origins to their role in scientific exchange.

2.1. LONDON

In the early 1600s, London was a booming commercial hub with dense trade networks, skilled artisans, and a vibrant community of merchants and instrument makers closely connected to practical science (Cormack, 2017; Harkness, 2007; Shaw-Taylor et al., 2024). The population surged from 200,000 in 1600 to 475,000 by 1670 (Wrigley, 2015, p. 61) while England shifted from centuries of stagnation to sustained productivity growth

(Bouscasse et al., 2025). Urbanization and the rise of wage labor increased disposable incomes, expanding demand for new consumer goods and exotic imports (Lake and Pincus, 2006). Coffee was one such import, reaching the city through contact with the Ottoman Empire, where a distinct coffee culture had flourished since the 1500s (Hattox, 2014). English travelers described the drink from 1600, and coffee consumption in England is first recorded in the late 1630s, more than a decade before the first coffeehouse opened (Cowan, 2005, pp. 5, 25). Merchants of overseas trading companies, such as the Levant Company founded in 1592, encountered coffee abroad, while natural philosophers interested in new forms of knowledge were among its early enthusiasts, reflecting a nascent culture of curiosity about the natural world associated with figures such as Francis Bacon (1561–1626) (Cowan, 2005, pp. 14, 55–56).

2.2. COFFEEHOUSES



FIGURE 2. A SEVENTEENTH-CENTURY LONDON COFFEEHOUSE

Notes: Drawing by an unknown artist, c. 1690–1700, showing the interior of a London coffeehouse. It depicts a maid behind a canopied bar and coffee boys taking clay pipes from a chest, tending pots at the hearth, and serving coffee, while customers in wigs and hats sit at communal tables, reading, conversing, and drinking beneath framed portraits and prints. [British Museum, 1931,0613.2](#). [Return to page 7]

Coffeehouses emerged in England in the 1650s as a novel kind of public house, similar to taverns, inns, and alehouses, where people could converse, exchange ideas, and build trust (Figure 2). Samuel Hartlib, a noted promoter of science, recorded in his *Ephemerides* (1654) the opening of “a cuphye-house or a Turkish, as it were ale-house” near the Old Exchange (quoted in Cowan, 2005, pp. 25–26). Bookshops, artisanal workshops, and literary circles already supported sociability of this kind, and the new houses took a familiar form. What set them apart at first was simply the drink they offered: coffee.

Like beer, wine, and ale, coffee encouraged conviviality, yet spared customers the fog of intoxication. Physicians, writers, and politicians soon praised it as a sober stimulant that sharpened the mind and increased productivity, while alcohol was increasingly condemned as a threat to social order. Although the medical benefits of coffee remained uncertain, promotional tracts had promised mental clarity and self-control since well before the 1650s, and coffeehouses prospered on that promise.¹

That appeal first took physical shape in 1652, when Pasqua Rosée, a Greek servant of the Levant Company merchant Daniel Edwards, opened the first coffeehouse in London.² To attract customers, he circulated a handbill advertising the virtues of coffee, claiming the beverage “quickens the Spirits, and makes the Heart lightsome” and “is excellent to prevent and cure the Dropsy, Gout, and Scurvy.” The idea spread quickly. By 1663, the City of London already counted 82 coffeehouses ([City of London Magistrates, 1663](#)). A penny, about £1.10 or \$1.45 in 2025 prices, bought a dish of coffee; two pence could cover a longer visit ([Peacham, 1667](#)).³ These low prices opened coffeehouses to men below the elite, turning them into public rooms for sober news, argument, and exchange ([Cowan, 2005](#), p. 52). Throughout the 1660s, coffeehouses became fixtures of urban life: Fellows of the Royal Society gathered at Garraway’s, merchants met near the Royal Exchange, and their growing influence so alarmed Charles II that in 1666 he even considered banning them. Their blend of low cost and lively discussion earned them the contemporary nickname “penny universities,” a phrase that underscored their reputation as informal centers of learning. In the London broadside *The Triumphs of London*, the poet Thomas Jordan (1675) vividly echoes this sentiment:

So great an university,
I think there ne’er was any;
In which you may a scholar be,
For spending of a penny.

[Habermas \(1991 \[1962\]\)](#) expands on this view by presenting English coffeehouses as the embodiment of the emerging “public sphere” in early modern Europe: bourgeois, urban spaces open to all, where rational, egalitarian debates on topics from literature to politics flourished, reason rather than social rank or political power settled matters, and philosophical discourse escaped rigid hierarchies. The coffeehouse, in this account, became a seedbed of Enlightenment ideals. [Mokyr \(2005, p. 312\)](#) strikes a similar note,

¹In *Theatrum Botanicum*, the herbalist John Parkinson (1640) offered the first full English description of coffee, praising it for “strengthening a weake stomacke, helping digestion, and the tumours and obstructions of the liver and spleene, being drunke fasting for some time together.”

²Pasqua Rosée’s Coffee House is often described as the first coffeehouse in London, and sometimes as the first coffeehouse in Christian Europe. The latter claim is contested. Some sources credit an Oxford establishment with priority, based on Anthony Wood’s diary entries from 1654–55, which report that a Jewish man named Jacob sold coffee near the Angel in Oxford in 1650. [Ellis \(2004, p. 38\)](#), a historian of the English coffeehouse, disputes this attribution. He argues that Wood’s entries show that coffee was sold in Oxford, but do not establish the existence of a true public coffeehouse. The European priority claim is also complicated by popular accounts that identify Venice as the site of Europe’s first coffeehouse in 1645. However, [Ellis \(2004, p. 85\)](#) dates the first Venetian coffeehouse to 1683.

³Price conversion uses the retail-price series from [Clark \(2026\)](#). The average-earnings series from the same source implies that, assuming 300 paid workdays per year, one penny was roughly 10 percent of a day’s earnings. This was a meaningful unit of everyday spending: [Peacham \(1667\)](#) notes that a penny could buy a dish of coffee, a small loaf, a candle, or seasonal fruit.

portraying the coffeehouse as a natural gathering point for polymaths and practical men alike, a locus where disciplines intertwined and a culture of improvement and empiricism took hold. “Much of the improved access to useful knowledge,” he writes, “took place through informal meetings [...], in coffeehouses and pubs, improvised lectures, and private salons.”

Understanding the social impact of coffeehouse culture requires comparison with earlier norms. As [Cowan \(2005\)](#) explains, coffeehouses replaced alcohol-centered public houses as key venues for elite exchange. In early seventeenth-century England, the virtuosi (natural philosophers and scholars) had gathered in aristocratic great houses, where interactions were formal and exclusive. Coffeehouses, by contrast, encouraged informal, cross-class dialogue. While they did not replace great houses as centers of virtuosi activity, they challenged traditional pathways to intellectual recognition. For less wealthy aspiring virtuosi, coffeehouses provided unprecedented access to information and social networks.

The promise mattered more than the drink. Coffee was consumed mainly in coffeehouses rather than at home, since London household inventories before 1700 rarely listed utensils for hot drinks, and Robert Hooke, a frequent visitor, seldom recorded drinking it, perhaps because it was too routine to note ([Withington, 2020](#), pp. 54, 59). Quantities were in any case small. At the height of the trade in the mid-1680s, English imports could have supplied at most about 50,000 people with half a liter a day, some 1.5 percent of the population aged fifteen or over ([Smith, 2001](#), pp. 252–3), too little for caffeine to explain a substantial change in productivity. What the houses supplied was the setting rather than the stimulant: accessible and predictable venues that lowered the cost of communication between theorists, craftsmen, and entrepreneurs from different social backgrounds, and that diffused new norms of civility and intellectual openness ([Klein, 1996](#)). Their contribution was institutional and cultural rather than physiological.

2.3. DIFFUSION, RECEPTION, AND CLIENTELE

The spatial diffusion of coffeehouses within London followed the city’s commercial and institutional geography ([Cowan, 2005](#), pp. 152–192). Early establishments clustered near the Royal Exchange, Fleet Street, and the Inns of Court, just beyond the western wall of the City, where merchants, lawyers, and printers intersected. From these nodes, coffeehouses expanded into nearby neighborhoods as proprietors captured foot traffic and repeat customers. By the early 1660s, coffeehouses were a stable feature of the metropolitan landscape, often differentiated by clientele, profession, or politics. Diffusion reflected integration into existing circuits of movement, credit, and information.

Public reception reflected both the coffeehouse’s role as a site of collective discussion and its positioning as a sober alternative to traditional public houses ([Cowan, 2005](#), pp. 45–46). Whether coffeehouses were uniformly sober in practice is secondary. What mattered was that they were marketed and widely understood as spaces of sobriety, in contrast to taverns, inns, and alehouses. This framing shaped patron expectations and official scrutiny. The concentration of news, opinion, and rumor drew attention from authorities during periods of political instability in the late seventeenth century. Regulatory efforts focused on publicity rather than consumption ([Cowan, 2005](#), pp. 193–224). Satire and polemic framed coffeehouses as idle or pretentious, but such criticism

did not slow their expansion.

Coffeehouse clientele was broad but bounded (Cowan, 2005, pp. 1–15). Early frequenters included natural philosophers and merchants familiar with coffee through contact with the Ottoman Empire, but coffeehouses quickly expanded to serve a wider urban audience. Participation required modest means, literacy, and conversational competence, excluding much of the urban poor. Women were largely absent as customers, though present as proprietors or workers (Cowan, 2001). Before the 1650s, scientists had limited ties to merchants, and historical evidence suggests that the emergence of coffeehouses was not driven by changes in scientific or intellectual life.⁴ Coffeehouses thus connected commercial, intellectual, and technical communities, explaining why merchant-run venues became spaces where scientific problems, methods, and reputations could circulate beyond formal institutions.

2.4. RELATION TO SCIENCE

In January 1684, at Garraway’s Coffee House in London, Robert Hooke (1635–1703), Christopher Wren (1632–1723), and Edmond Halley (1656–1742) met to discuss the problem of planetary motion, specifically the nature of the force that keeps planets in orbit (Mount, 2021, pp. 145–146). Hooke claimed to have discovered the inverse-square law but did not provide a demonstration. Wren offered a prize for a proof, which Hooke failed to deliver. Halley, seeking a resolution, visited Isaac Newton (1643–1727) in Cambridge later that year. Newton replied he had already solved the problem and subsequently sent Halley a manuscript, *De motu corporum in gyrum*. This work became the foundation for his *Philosophiæ Naturalis Principia Mathematica*, one of the most influential books in the history of science.

This episode illustrates how coffeehouses supported scientific dialogue by providing a comparatively accessible setting alongside the Royal Society, private households, workshops, and instrument shops. Access was neither universal nor entirely egalitarian, but coffeehouses widened the range of settings in which natural philosophers and skilled practitioners could meet. Hooke’s diary is filled with references to London coffeehouses, which he visited on nearly two of every three days. As Iliffe (1995) demonstrates, Hooke used these spaces to build networks, exchange technical information, examine machines and instruments, and conduct business. After one such meeting with Wren at Man’s Coffee House, on August 21, 1678, Hooke sketched in his diary the four devices the two had described to each other, among them a “poysd weather glasse” and his own “philosophicall spring scales” (Robinson and Adams, 1935, pp. 372–373). Two of the four were Wren’s.

Coffeehouses complemented print: pamphlets and periodicals supplied material for reading and conversation, while face-to-face exchange allowed claims to be discussed, demonstrated, and combined with hands-on expertise (Higgitt et al., 2024; Johns,

⁴Among 30 so-called mathematical practitioners described in Section 3 for whom detailed network data are available in the *Oxford Dictionary of National Biography* (Oxford Dictionary of National Biography, 2004), only 9 of 348 recorded connections are to merchants, and only 6 practitioners had any such connection. By contrast, early coffeehouse proprietors were embedded in mercantile networks from the outset. Thomas Garraway and Edward Lloyd established coffeehouses near the Royal Exchange and transformed them into centers for auctions, shipping intelligence, and financial exchange, while Pasqua Rosée’s first coffeehouse emerged directly from Levant Company connections. This contrast suggests that commercial and technical communities initially belonged to distinct social cultures, and that coffeehouses helped bridge, rather than reflect, an existing integration between them.

2006). More broadly, scientific practice in early modern London developed through overlapping networks of natural philosophers, mathematical practitioners, and artisans operating across coffeehouses, workshops, shops, and institutions (Higgitt et al., 2024, pp. 8–9, 19–20).

Instrument production remained largely commission based: guild apprenticeships transmitted craft and commercial knowledge, while advertising connected makers with professional users (Bryden, 1992; Crawford, 1987; Stewart, 2005). The guild itself met in coffeehouses. The Clockmakers' Company held its Feast at the George in Ironmonger Lane in 1672 and its ordinary meetings there until 1698, and its Search Court sat at the Angel and Crown in Threadneedle Street and at the Crown in West Smithfield (Lillywhite, 1963, entries 44, 317, 1751).⁵ One especially concrete sequence connects coffeehouse exchange to instrument production. The clockmaker Thomas Tompion (1639–1713) already possessed substantial practical skill when he began working with Hooke. At Child's Coffee House on June 27, 1674, Hooke described new dividing compasses and other mechanisms to him. At Garraway's Coffee House on March 8, 1675, Hooke showed Tompion how to fix double springs inside a balance wheel. Tompion brought clockwork to Hooke six days later, and on April 7 they showed Hooke's new spring watch to Charles II, who praised it and encouraged its further development (Robinson and Adams, 1935, pp. 109, 151–157). Their work continued through testing and revision. Iliffe (1995, p. 311) emphasizes that the collaboration became a mutual exchange of techniques and skills. The episode shows how coffeehouses could make existing skills more productive by connecting designs, practical knowledge, collaborators, patrons, and buyers.

Coffeehouses also hosted disputes over experimental philosophy. Shapin and Schaffer (1985, pp. 292–293) document that Tillyard's Coffee-House in Oxford served as a venue for an experimental group and chemistry lectures, and that a London coffeehouse dispute over barometric phenomena was discussed in a 1662 pamphlet. These episodes fit their broader account of experimental knowledge as a product of material instruments, credible witnessing, and communication. Coffeehouses did not replace laboratories or the Royal Society; they extended the social settings around this experimental system.

This exchange rested on the broader knowledge economy of London. Beeley (2019, p. 225) describes London's mathematical practitioners as an interdependent community spanning teachers, instrument makers, printers, booksellers, merchants, and even mathematical clubs, known contemporaneously as "the London mathematicians" in distinction to their academic counterparts at Oxford and Cambridge. One of these London mathematicians was John Collins (1625–1683), who exemplifies the brokers within this community. He circulated manuscripts, arranged publication, identified relevant expertise, and connected practical problems with mathematicians and instrument makers (Beeley, 2017). Though Collins held no university position, he and others like him could repeatedly interact through coffeehouses with those more closely tied to elite intellectual networks. Hooke's diary places Collins in coffeehouse company throughout the 1670s. At the Spanish Coffee House on April 18, 1674, Hooke

⁵Four other companies appear in the same entries: the Pewterers at Kiftell's in Cheapside from 1677, the earliest mention of coffee in their accounts (entry 670); the Ironmongers and the Basket Makers at North's in King Street (entry 914); and the Wax Chandlers at the Half Moon (entry 519). Committees of the Hudson's Bay Company met at ten of these houses over the same decades, and the vestry of St Stephen Walbrook sat at Maddison's and at Powell's while Wren rebuilt its church (entries 775 and 1011).

noted, “Collins shewd me the 3rd book of Kerseys algebra,” the treatise the London teacher John Kersey had just published (Robinson and Adams, 1935, pp. 67, 98, 392). Coffeehouses expanded the decentralized, face-to-face infrastructure through which such matching could occur.

Stewart (1999) more broadly documents that coffeehouses set aside rooms for meetings, lectures, and discussions, serving as important sites for exchanging knowledge and print materials. They hosted lectures and demonstrations in natural philosophy, sometimes with a formality comparable to that of formal schools, the Royal Society, or Gresham College. Stewart characterizes coffeehouses as “other centres of calculation,” frequently attended by key figures of the Royal Society, underscoring their role in knowledge production and circulation.

3. DATA

We assemble records of coffeehouses, mathematical practitioners, and scientific instrument shops, each with an address and dates. We geocode every address and assign it to one of the 162 jurisdictions into which London was divided in 1660: 113 City of London parishes, 97 within the walls; 9 Westminster parishes; 20 parishes in Middlesex and Surrey; and 20 liberties or extra-parochial areas, including the Inns of Court and the Tower. We refer to all 162 as parishes, and we count records by parish and decade from the 1600s through the 1690s.

Coffeehouses. To determine where and when coffeehouses opened in London, we compile data from *London Signs* (Lillywhite, 1972). The work draws on primary and secondary sources at the Guildhall Library in London, covering coffeehouses and other establishments from the seventeenth through the nineteenth centuries. These include tokens, records of house signs, lost-and-reward notices, business directories, newspapers, advertisements, letters, reference works, and diaries.⁶ Lillywhite (1972) consistently reports the names and years of mention of each coffeehouse and provides addresses for 2,124 of the 2,254 coffeehouses.

To our knowledge, Lillywhite (1972) remains the most comprehensive source on early modern London coffeehouses, documenting far more than the prominent ones. Its more widely cited predecessor, Lillywhite (1963), accounts for 2,034 of the 2,254 coffeehouses listed in Lillywhite (1972).⁷ Ellis (2006) treats Lillywhite (1963) as the major reference work, Pincus (1995, p. 809) describes it as an “encyclopedia of coffeehouses,” and Cowan (2007, p. 1197) characterizes it as an “enormously valuable catalogue of references to London coffeehouses.”⁸

⁶Before the London Streets Act of 1762, buildings were identified by house signs rather than street numbers. These signs often indicated the type of social institution on the premises, and historical records of them form the basis of Lillywhite (1972).

⁷Appendix Figure D2 shows an excerpt from Lillywhite (1963). Entry 656 details “Jonathan’s Coffee House,” located at “Exchange Alley, Cornhill, No. 20,” first mentioned in “1680–1684,” along with its background. Appendix Figure D3 shows an excerpt from Lillywhite (1972). Entry 13056 refers to the “Sandford Coffee House” on Lawrence Lane in Cheapside ward, a seventeenth-century coffeehouse not listed in Lillywhite (1963). Entry 1163 corresponds to a coffeehouse already documented in Lillywhite (1963).

⁸Lillywhite worked from London house signs, the addresses in use before the London Streets Act of 1762, and came to coffeehouses as one class of sign-bearing house (Lillywhite, 1972, p. xv). Much of his

In constructing the dataset, we address two sources of potential measurement error. First, we consolidate duplicate entries. When the same coffeehouse appears under multiple names, [Lillywhite \(1963, 1972\)](#) lists each separately and notes overlaps, which we use to merge records into a single entry.⁹ Second, we identify and remove a small number of non-coffeehouse public houses, primarily taverns and inns, that [Lillywhite \(1963\)](#) includes for contextual completeness rather than as coffeehouses.¹⁰ Appendix Section D.10 then evaluates the completeness of the resulting dataset by benchmarking it against contemporaneous administrative records and independent archival searches. We find that coverage is strong for the late seventeenth century but more limited in the 1650s and 1660s, consistent with the broader loss of historical records in the Great Fire of 1666.

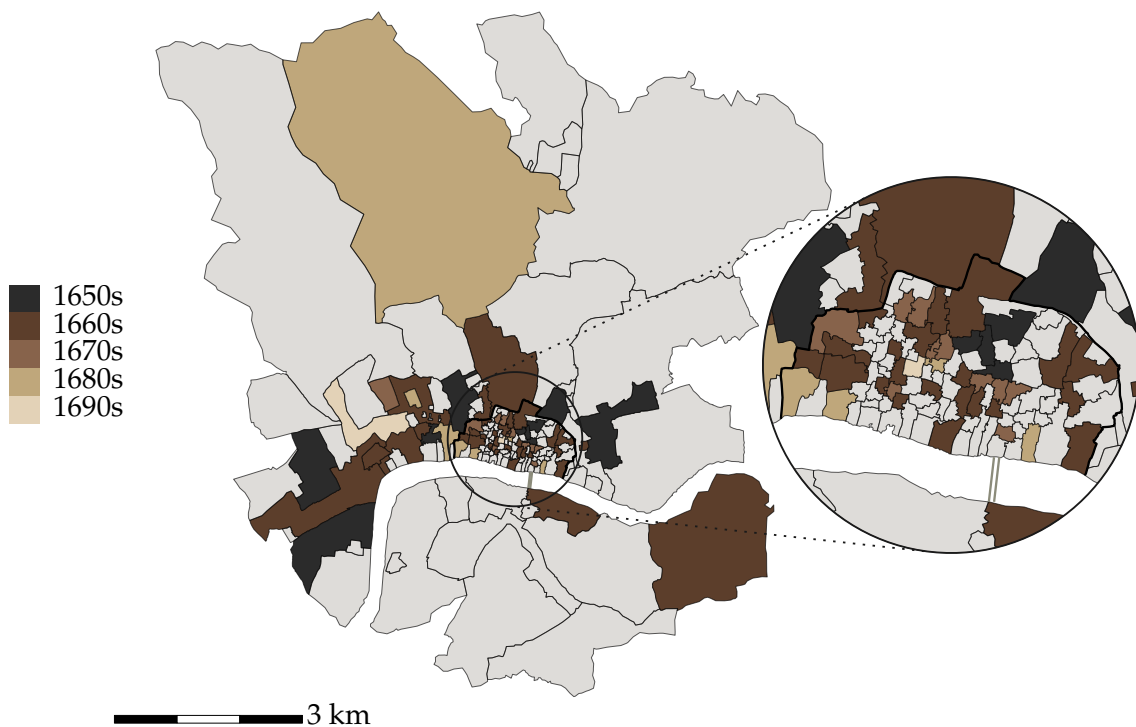


FIGURE 3. MAPPING THE FIRST COFFEEHOUSES IN LONDON

Notes: Map of London showing the decade of first recorded coffeehouse by parish, with an inset of the area inside the wall. Gray parishes indicate no recorded coffeehouses in the 1600s. [Return to page 13]

In total, the panel contains 286 coffeehouses across 61 parishes in seventeenth-century London.¹¹ Figure 3 maps the decade of first coffeehouse entry in each parish. Of the 61 treated parishes, 49 lie in the City of London and 12 outside it. Openings in

material reached him as postal history, letters addressed to those signs ([Lillywhite, 1963](#), p. 10). He enters a house when a document fixes its sign, its address, and a year, and reports whatever else his sources give. What he found about the clientele fills out an entry; it does not create one.

⁹For example, “The New Bagnio” (entry 1845) also appears as “Turk’s Head Bagnio” (entry 1957). In general, coffeehouses were known by various names based on location (e.g., “St. Dunstan”), sign (e.g., “Turk’s Head”), or proprietor (e.g., “Bowman”), and [Lillywhite \(1963\)](#) cross-references these variants.

¹⁰For example, “Pope’s Head Tavern” (entry 1870) is “included by reason of its site and close connexion with coffee-houses” ([Lillywhite, 1963](#), p. 724).

¹¹The source data contain 287 coffeehouses first recorded before 1700. Don Saltero’s coffeehouse in Chelsea, first recorded in 1695, lies outside the 162-parish study area and is therefore excluded. Appendix Table D3 summarizes the historical sources for the 286 included coffeehouses.

the 1650s cluster around the Royal Exchange and Gresham College, the financial and intellectual centers of the period, with a second cluster in Westminster. Entry then spreads decade by decade across the walled city and surrounding suburbs, without persistent concentration in a small set of parishes.

Mathematical Practitioners. We measure applied scientists by counting mathematical practitioners, the closest historical analogue: individuals who applied mathematics in teaching, writing, instrument making, and related technical occupations. The data draw on [Taylor \(1954\)](#), which pioneered the study of early English applied mathematicians and instrument makers and introduced the concept of the mathematical practitioner.¹² [Taylor \(1954\)](#) compiled a collective biography of 584 mathematical practitioners active in England between 1507 and 1714,¹³ drawing on contemporary works and diaries, bibliographical compilations, museum catalogs, and archival manuscripts. Since its publication, “[f]ew new [mathematical] practitioners have been added, and rarely has the category itself been seriously challenged” ([Walton, 2017](#), p. 89).

Each biography reports first and last names, dates of flourishing (i.e., activity), workplaces, types of activity (e.g., instrument making, teaching, research), and connections to other practitioners.¹⁴ Because we focus on the seventeenth-century panel, we restrict the sample to the 244 mathematical practitioners with a precisely geocoded London workplace whose recorded flourishing overlaps 1600–1699. Of these, 223 began flourishing during the century, while 21 began earlier and remained active into the century.¹⁵ Each practitioner enters every decade spanned by their recorded flourishing, in every parish where they have a workplace in that decade, so the outcome is a stock of active practitioners. These practitioners rarely appear in the leading scientific journal of the time, the *Philosophical Transactions*, founded in 1665 by Henry Oldenburg, secretary of the Royal Society. Of the 176 practitioners active between 1665 and 1699, only 12 are ever named in its pages.¹⁶

[Taylor \(1954\)](#) records social and intellectual connections among practitioners, of many kinds and without dates. We extend this network with the same types of connections, inferred by machine learning from biographies in the *Oxford Dictionary of National Biography* available through the *Six Degrees of Francis Bacon* database of [Warren et al. \(2016\)](#), which covers 75 of the 244 practitioners (Appendix D.3). Two practitioners

¹²Appendix Figure D4 presents an excerpt from [Taylor \(1954\)](#), and Appendix Figure D21 shows examples of practitioners across cohorts.

¹³Tudor England begins in 1485; the earliest practitioner recorded by [Taylor \(1954\)](#) flourishes from 1507.

¹⁴Dates of flourishing generally indicate the period during which a mathematical practitioner was active. However, for the 65 practitioners for whom [Taylor \(1954\)](#) provides birth dates instead of dates of flourishing, we define flourishing as spanning from the year the individual turned 21 to the year of death or the last year of flourishing, following the implicit chronological order of biographies in [Taylor \(1954\)](#).

¹⁵Appendix Table D4 reports the complete set of harmonized occupations for the 244 practitioners. Occupational categories may overlap: 130 practitioners are classified as scientific instrument makers, 54 as mathematical teachers, and 69 as other mathematical practitioners. Appendix Figure D18 shows that about 60 percent of practitioners were active in London rather than elsewhere in England in a typical year.

¹⁶We match practitioner names against person mentions extracted from the full text of all 1,325 seventeenth-century articles, digitized in the Royal Society Corpus ([Fischer et al., 2020](#)). Matching runs on surnames, but a practitioner counts as named only when first names or textual context confirm the identity. Thus, 12 is a lower bound: 54 practitioners share a surname with someone named in the journal. Hooke, Newton, and Halley account for 313 of the 406 mentions of these 12 practitioners. Six were Fellows of the Royal Society, compared with 13 of the 164 never named.

are ever connected if either source records a connection between them, and they are connected in decade t if they are ever connected and both are active in t . The resulting network contains 475 undirected ever-connections among 204 of the 244 practitioners, with a mean of 4.7 and a median of 3 connections; the remaining 40 practitioners are unconnected. In a decade in which a practitioner is active, they have 3.1 connections on average, with a median of 2.

Scientific Instrument Shops. To capture applied scientific activity in both human and physical capital, we compile data on scientific instrument shops, where instruments were manufactured, repaired, and sold. These data draw on [Hay et al. \(2025\)](#), which is based on the 1991 computerized database of scientific instrument makers compiled by [Clifton \(1991\)](#) and collaborators, as well as on the later published extract in [Clifton \(1995\)](#).¹⁷ Much of the underlying material was assembled at the National Maritime Museum and the Royal Observatory, Greenwich, in the 1980s and 1990s. The 2025 version extends this work through the project *Tools of Knowledge: Modelling the Creative Communities of the Scientific Instrument Trade, 1500–1914*, which brought together curators, historians, and digital humanities experts from five institutions.

This dataset documents all known scientific instrument makers and shops with London addresses up to the 1690s, including each maker’s name, shop address, and active dates.¹⁸ The panel contains 252 makers operating 345 shops that were active at some point during 1600–1699; all but one of these shops were first recorded during the century. To place London in a European rather than a British context, we draw separately on the *Webster Signature Database* ([Webster and Webster, 2007](#)), a catalogue of makers who signed scientific instruments now held in collections worldwide.^{19,20} Shops

¹⁷Much of our data originates from [Clifton \(1995\)](#), which is incorporated in [Hay et al. \(2025\)](#). We additionally consulted the published extract directly, primarily to correct entries with missing addresses. Appendix Figure D5 shows an excerpt from [Clifton \(1995\)](#), including, for example, a record of “ALLEN Elias” opening a shop in 1606 in “Blacke Horse-ally, neere Fleetbridge, London.”

¹⁸Appendix Figures D18 and D20 show that in a typical year about three-quarters of the scientific instrument makers recorded in the British Isles were active in London, above 80 percent from the 1660s, and that production was dominated by practical instruments such as dividers, sectors, and compasses until the 1660s, when experimental instruments including microscopes and air pumps began to expand rapidly.

¹⁹Counts are of recorded makers with a located signature rather than of all instrument makers, and coverage reflects which instruments survive and which collections the Websters visited. Restricting to signatures whose recorded years of activity overlap 1600–1699 and that report an identifiable European city leaves 1,201 makers, of whom 309 worked in London, ahead of Paris (101), Nuremberg (89), Augsburg (55), and Amsterdam (45); nine further located records name a city outside Europe. Restricting this source to the British Isles instead of Europe gives 309 of 437 makers in London, or 71 percent, in the same range as the British share reported above, which comes from an independent compilation. In London we use this source for context only; it enters none of the parish estimates.

²⁰Our sample of mathematical practitioners from [Taylor \(1954\)](#) includes only 130 scientific instrument makers, against the 252 makers recorded by [Clifton \(1995\)](#). The gap reflects timing: [Taylor \(1954\)](#) is the earlier work and served as a source for [Clifton \(1995\)](#), which can be regarded as a partial follow-up. Not all instrument makers in [Taylor \(1954\)](#) appear in [Clifton \(1995\)](#), likely due to minor differences in classification. For instance, Christopher Wren (1632–1723) is listed as an instrument maker in [Taylor \(1954\)](#) but not in [Clifton \(1995\)](#). [Taylor \(1954\)](#) records dates of flourishing, whereas [Clifton \(1995\)](#) emphasizes dates of activity and shop operations, so our analysis of scientific instrument shops draws only on [Hay et al. \(2025\)](#) and [Clifton \(1995\)](#). The model of Section 7 counts people rather than shops: it completes the 244 practitioners of [Taylor \(1954\)](#) with the instrument makers that these later compilations record and [Taylor \(1954\)](#) does not, 403 people in all.

enter every decade spanned by their recorded years of operation, so this outcome is a stock as well.

Other. Additional variables enter the balance comparisons, robustness checks, and mechanism tests. Appendix D defines them, each with its source.

Geocoding. Our sources give historical addresses rather than coordinates. We map each address by hand onto a contemporary Google Map of Greater London, using digital historical maps and gazetteers (Appendix D.11), and spatially join the resulting coordinates with a shapefile from Cummins et al. (2016), which represents the parochial geography of London in 1660 and delineates its 162 parishes.

Hooke’s Diary, 1672–1683. To observe what happened on the days a mathematical practitioner went to a coffeehouse, we manually transcribe the diary of Robert Hooke (1635–1703), curator of experiments at the Royal Society and the best-connected mathematical practitioner of the coffeehouse era: in our network data he has more connections than any other practitioner in every decade from the 1650s to the 1690s (Appendix Figure D13). Historians have studied him closely (Hunter and Schaffer, 1989; Iliffe, 1995), and his life is the subject of several biographies (Inwood, 2002; Jardine, 2003). Hooke recorded where he went and whom he met from March 1672 to May 1683 (Henderson, 2007; Robinson and Adams, 1935), which gives us diary entries for 3,464 dated days. For every entry, we record whether Hooke visited a coffeehouse, another public house such as a tavern, or neither: coffeehouses appear on 2,244 days, other public houses on 337, and neither on 1,122, with 239 days having both. For every person an entry names, we record whether Hooke met them and identify them in the index of the printed edition, which yields 16,325 meetings, counting each person once a day, with 1,426 distinct people (Appendix D.5). The words Hooke wrote after naming a coffeehouse characterize it as a place of discourse. Against clauses that name no public house, the most overrepresented word pairs are “discoursed about,” “discoursd with,” “told me,” and “shewed his” (Figure 4).

European Cities. For the extension to Europe in Section 6, we use data on 200 European cities from Bosker et al. (2013), those with at least 10,000 inhabitants in 1600 and a recorded population in 1500 in the parts of Europe never under Ottoman rule, observed by decade from the 1600s to the 1690s. We add new data on their first coffeehouses in the seventeenth century, compiled through a systematic review of the historical scholarship (Appendix D.6 and Table D1); 37 cities record a coffeehouse before 1700. To the best of our knowledge, no European source lists mathematical practitioners the way Taylor (1954) lists them for England, so we count applied mathematics titles printed in a city in the Universal Short Title Catalogue [USTC] (2026), classified by the fourteen subject headings that Kelly and Ó Gráda (2022, p. 849) use to measure English practical mathematics, and use that series in its place.²¹ The substitution is natural,

²¹The headings are those of the *English Short Title Catalogue*. We apply them as keyword rules, in seven languages, to every title the catalog classes as Science or Mathematics, which yields 3,415 titles printed in the 200 cities between 1600 and 1699: arithmetic (1,223), astronomical instruments (461), bookkeeping (55), compasses (148), geometry (882), gunnery (69), logarithms (155), mathematics (1,215), mathematical

untreated to treated. Later openings increase treatment intensity without creating a new transition, so the estimates capture the total effect of entry, including subsequent growth.²² Coffeehouse openings also bundle multiple underlying treatments: the room, the coffee, the newsprint, the debate, and the clientele all arrive together. Thus, what we measure is the compound effect of a coffeehouse institution rather than the contribution of any single ingredient.

The historical record indicates that entry was governed primarily by commercial considerations (Appendix A.1). Proprietors were middling merchants and traders who chose sites for foot traffic and custom, and their clientele extended well beyond scientists to merchants, aristocrats, and politicians. Before 1650, recorded commercial ties among mathematical practitioners were rare (Appendix Table A4). Coffee supply, excise, and licensing changed citywide, in the same decade across parishes.²³ The Great Fire of 1666 was a common shock with uneven local exposure, so as a robustness check we allow exposed parishes to follow separate trends (Appendix C.4).

As descriptive evidence, Table 1 reports normalized differences between parishes with and without a coffeehouse. Treated parishes were poorer in 1582 but matched controls by 1638 in the share of substantial households and before 1652 in scientific letters and in goldsmiths, a proxy for financial capital. They were less densely populated. The largest differences were in social capital and commerce: treated parishes had more tradesmen’s tokens, commercial premises, taverns and inns, and booksellers. They also lay closer to major centers such as the Royal Exchange, Gresham College, and Paternoster Row. Below, we test whether these imbalances also predict how scientific activity would have evolved without coffeehouse entry, using the same comparison applied to pre-1652 changes (Appendix Table C1) and adjustment for the observed predictors of entry.

Entry need not be random, and treated and control parishes may differ in levels. Identification requires instead that parishes did not change behavior in anticipation of a first opening, and that absent entry, scientific activity in treated parishes would have followed the path observed in never-treated parishes. Concretely, we estimate an unconditional staggered difference-in-differences model with multiple time periods that is robust to heterogeneous treatment effects across groups and time:²⁴

$$y_{it} = \alpha + \sum_{g=1650}^{1690} \sum_{\substack{s=1600 \\ s \neq g-10}}^{1690} \beta_{gs} \mathbb{1}[s = t] \mathbb{1}[g = E_i] + \gamma_i + \delta_t + \epsilon_{it}. \quad (1)$$

Here, y_{it} denotes the number of mathematical practitioners or scientific instrument shops in parish i in decade t . γ_i and δ_t denote parish and decade fixed effects that control for local time-invariant differences and London-wide shocks and trends. E_i

²²Appendix B.6 groups treated parishes by the number of coffeehouses open in the decade of entry and estimates each group against the never-treated.

²³Applied mathematics publishing and progress-oriented language rose nationally over the same decades (Appendix Figure D19).

²⁴This addresses problems arising from treatment effect heterogeneity in staggered adoption designs. By contrast, two-way fixed effects estimators can place non-convex or negative weights on group-time treatment effects, while dynamic two-way fixed effects event study estimators can suffer from contamination across relative periods (de Chaisemartin and D’Haultfoeuille, 2020; Goodman-Bacon, 2021; Sun and Abraham, 2021).

TABLE 1—PRE-TREATMENT SUMMARY STATISTICS

	Parishes			Normalized difference
	Nonmissing	Treatment	Control	
<i>Panel A. Demographics</i>				
Baptisms, 1600s–1640s	130	84.72 (120.74)	57.44 (113.62)	0.23
Baptisms per km ² , 1600s–1640s	130	493.76 (349.55)	634.63 (503.77)	−0.32
Deaths, 1600s–1640s	110	117.25 (174.38)	86.08 (159.55)	0.19
Deaths per km ² , 1600s–1640s	110	740.21 (564.89)	872.94 (665.37)	−0.22
Households, 1638	107	249.00 (329.67)	157.87 (255.37)	0.31
Households per km ² , 1638	107	3419.34 (1368.22)	4114.09 (2010.83)	−0.40
<i>Panel B. Wealth</i>				
Median tax paid, 1582	98	93.20 (87.91)	124.53 (132.58)	−0.28
Substantial households (%), 1638	92	26.45 (15.99)	25.97 (20.17)	0.03
<i>Panel C. Social capital and commerce</i>				
Apothecaries, 1617–1640s	162	0.12 (0.32)	0.03 (0.11)	0.39
Booksellers, 1600s–1640s	162	1.53 (3.29)	0.19 (0.37)	0.57
Clay-pipe finds, 1600s–1640s	162	0.50 (2.50)	0.39 (1.98)	0.05
Commercial premises, 1600s–1640s	162	0.50 (0.90)	0.06 (0.18)	0.68
Dissenting meetings, 1600s–1640s	162	0.33 (0.85)	0.06 (0.34)	0.41
Goldsmiths, 1600–1651	162	0.18 (1.18)	0.05 (0.41)	0.15
Luxury shops, 1600s–1640s	162	0.07 (0.20)	0.05 (0.17)	0.13
Pubs, 1600s–1640s	162	3.01 (3.63)	0.53 (1.21)	0.92
Theaters, 1600s–1640s	162	0.08 (0.23)	0.04 (0.19)	0.15
Tradesmen’s tokens, 1648–1651	162	1.16 (1.88)	0.23 (1.02)	0.62
<i>Panel D. Science before 1652</i>				
Instrument shops, 1600s–1640s	162	0.15 (0.40)	0.05 (0.20)	0.32
Non-scientific letters, 1600s–1640s	162	3.87 (10.41)	1.09 (3.98)	0.35
Practitioners, 1600s–1640s	162	0.40 (0.95)	0.16 (0.48)	0.32
Scientific letters, 1600s–1640s	162	0.38 (1.08)	0.23 (1.38)	0.12
<i>Panel E. Geography</i>				
Area (km ²)	162	0.99 (4.23)	1.28 (4.37)	−0.07
City of London	162	0.80 (0.40)	0.63 (0.48)	0.38
Distance to Gresham College (km)	162	1.18 (1.02)	1.37 (1.20)	−0.18
Distance to Paternoster Row (km)	162	1.01 (0.88)	1.24 (1.12)	−0.24
Distance to the Royal Exchange (km)	162	1.01 (1.01)	1.22 (1.21)	−0.19
Inside the London Wall	162	0.66 (0.48)	0.56 (0.50)	0.19
Parishes	162	61	101	

Notes: Table compares London parishes with a first recorded coffeehouse opening during the 1650s–1690s (treatment) with parishes with no recorded opening by the end of the 1690s (control). The nonmissing column reports the total number of treatment and control parishes with a recorded value for each characteristic. Treatment and control columns report means, with standard deviations in parentheses. All characteristics predate the first London coffeehouse opening in 1652. Values labeled 1600s–1640s are averages of observed decadal counts from 1600 through 1640, except scientific letters, non-scientific letters, and dissenting meetings, which are period totals. The normalized difference is $(\bar{x}_T - \bar{x}_C) / \sqrt{(s_T^2 + s_C^2)/2}$, where T and C denote treatment and control parishes. [Return to pages [18](#), [22](#), [20](#), [26](#), [27](#), [41](#), [47](#)]

denotes the decade in which the first coffeehouse is recorded in parish i .²⁵ For each cohort, the omitted reference period is the decade before entry, $s = g - 10$. For $s \geq g$, β_{gs} is the average effect of coffeehouse entry for cohort g in decade s , relative to that decade and to the never-treated counterfactual, $E_i = \infty$. Earlier coefficients are pre-treatment contrasts, not effects. ϵ_{it} is an error term. Standard errors are clustered by parish.

Dynamic treatment effects are our estimands of interest. Following Wooldridge (2023), we estimate equation (1) by OLS and predict outcomes using all estimated coefficients and fixed effects, yielding $\hat{y}_{it}$. We impute the untreated potential outcome, $y_{it}(\infty)$, by predicting from equation (1) using only the estimated intercept and fixed effects, yielding $\hat{y}_{it}(\infty)$. The observation-specific treatment effect is therefore $\hat{\tau}_{it} \equiv \hat{y}_{it} - \hat{y}_{it}(\infty)$. We estimate the dynamic treatment effect by averaging $\hat{\tau}_{it}$ across parishes observed h years after the first coffeehouse opens:

$$\hat{\tau}_h \equiv \frac{1}{|I_h|} \sum_{i \in I_h} \hat{\tau}_{i, E_i + h}, \quad (2)$$

where I_h denotes the set of parishes observed in period $E_i + h$.²⁶ We plot these effects, measured in levels relative to the decade immediately before treatment.

The resulting dynamic treatment effect estimates and standard errors are numerically equivalent to those from the long-difference estimators of Callaway and Sant'Anna (2021). We adopt the approach of Wooldridge (2023) because, unlike long differences, its regression form extends to nonlinear models. Both outcomes are counts with many zeros, so as a robustness check we re-estimate the model by Poisson quasi-maximum likelihood with an exponential conditional mean.²⁷

$$\mathbb{E}[y_{it} \mid i, t] = \exp \left(\alpha + \sum_{g=1650}^{1690} \sum_{\substack{s=1600 \\ s \neq g-10}}^{1690} \beta_{gs} \mathbb{1}[s = t] \mathbb{1}[g = E_i] + \gamma_i + \delta_t \right). \quad (3)$$

Identification now requires that, absent a coffeehouse, mean outcomes in treated and control parishes would have grown by the same factor, not the same amount.

4.2. MATHEMATICAL PRACTITIONERS AND SCIENTIFIC INSTRUMENT SHOPS

Figure 5 reports estimates of equation (1) for both outcomes. Flat pre-trends support their validity.²⁸ Panel A shows a persistent increase in practitioners. The 61 treated parishes averaged 0.62 practitioners in the decade before entry, and the number rises by 0.68 to 1.30 thereafter, roughly doubling.²⁹ Panel B shows a persistent increase in

²⁵Appendix Figure D15 shows the rollout of treatment across parishes. Appendix Figure D17 shows coffeehouse counts by event time relative to the first coffeehouse entry.

²⁶Because the panel ends in the 1690s, fewer cohorts contribute at longer horizons: $\hat{\tau}_{40}$ is identified from the 10 parishes treated in the 1650s (Appendix Figure D15).

²⁷The exponential mean rules out negative predictions and avoids the need for transformations such as the inverse hyperbolic sine (Chen and Roth, 2024; Roth and Sant'Anna, 2023; Wooldridge, 2023).

²⁸We fail to reject the null that the pre-treatment contrasts two to five decades before entry are jointly zero ($p = 0.99$ for practitioners and $p = 0.29$ for instrument shops).

²⁹Because the outcome is a stock, the increase could in principle reflect longer recorded activity per practitioner rather than more practitioners. It does not: practitioners in parishes that ever received a coffeehouse flourish for 25.3 years per location spell against 22.8 in parishes that never did, a difference of 2.5 years with a standard error of 2.4.

instrument shops, from a baseline of 0.33 to 1.57 after entry, an increase of 1.24 and a nearly fivefold rise.

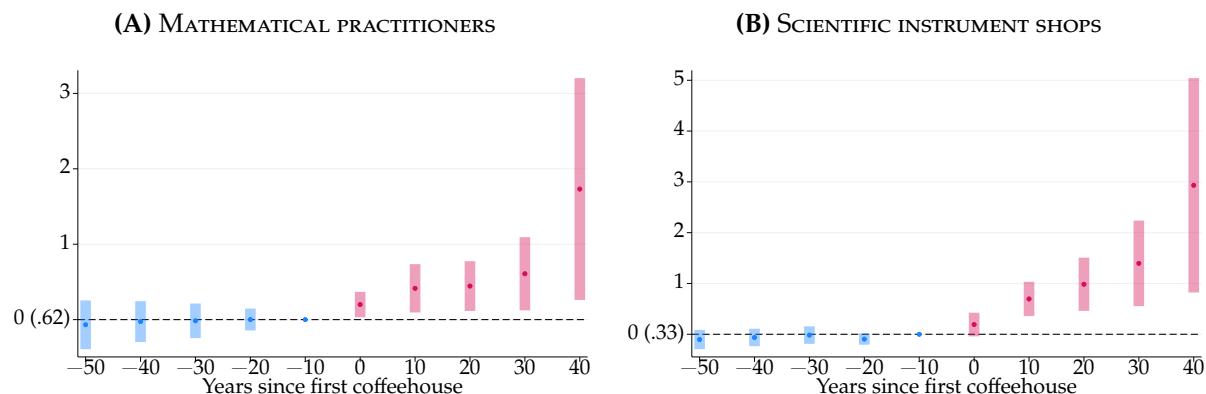


FIGURE 5. EFFECTS ON SCIENTIFIC ACTIVITY

Notes: Figure shows event study estimates and 95 percent confidence intervals from equation (1) of the effect of first coffeehouse opening on scientific activity. Outcomes are the decadal number of mathematical practitioners and scientific instrument shops in a parish. Sample includes all 162 parishes from the 1600s to the 1690s. Unit of observation is parish-decade. The parenthetical beside 0 reports the sample outcome mean one decade before treatment. Standard errors clustered by parish. [Return to pages 20, 22, 99]

Despite data scarcity, the effects are precise, statistically significant at conventional levels, and large. Between the 1640s and the 1690s, London gained 68 mathematical practitioners and 136 instrument shops. Aggregated across treated parishes, the estimated effects in the 1690s account for 49 percent of the increase in practitioners and 63 percent of the increase in instrument shops.³⁰ Relative to total activity rather than growth, coffeehouse entry accounts for 20 percent of all 502 practitioner-decades and 46 percent of all 412 instrument shop-decades recorded between the 1650s and the 1690s.³¹ These results survive the robustness checks reported below.

The treatment and the two outcomes come from three distinct compilations, assembled by different scholars working from different archives and for different purposes. No single source generates both the coffeehouse locations and the scientific activity we measure, so the consistent effects across outcomes cannot be an artifact of one compilation's coverage. This also bears on observability bias, the concern that those who frequented coffeehouses were more likely to enter the historical record at all. The concern has some force for practitioners, who are drawn from a collective biography of notable individuals. It has little force for instrument shops, which are identified from signatures on surviving instruments in museum collections, guild records, and trade cards, none of which plausibly depends on whether a coffeehouse stood nearby.

³⁰The standard errors of these shares are 18 and 16 percentage points, respectively.

³¹The numerator sums the estimated dynamic effects across treated parishes and post-treatment decades; the denominators are the 502 practitioner-decades and 412 shop-decades recorded across all 162 parishes from the decade of first entry onward. The shares give the fraction of observed activity statistically attributable to coffeehouse entry, with standard errors of 7 and 10 percentage points. Both sets of shares sum within-parish effects only, so they miss any rise in activity nearby but count as gains the activity that merely moved into treated parishes (Section 7).

4.3. ROBUSTNESS

Entry could still have coincided with unobserved local shocks to scientific activity. We assess this concern by adjusting for the characteristics that predict where coffeehouses opened, by comparing the estimates against placebo assignments of treatment, and by testing for effects on outcomes that coffeehouses should not affect; Appendix C develops each check in more depth.

Coffeehouses did not open at random. Fourteen of twenty-eight pre-1652 measures in Table 1 show normalized differences above the 0.25 threshold of Imbens and Rubin (2015, p. 277), and treated parishes begin with more practitioners and more instrument shops, at 0.32 each. Level imbalances threaten parallel trends when the imbalanced characteristics also predict how scientific activity would have evolved without a coffeehouse (Baker et al., 2025). Two checks suggest they do not. First, balance in changes: between the 1600s–1610s and the 1630s–1640s, both before the first London coffeehouse in 1652, both outcomes are balanced in changes, at -0.05 and 0.08 against 0.32 in levels, and only pubs and commercial premises exceed 0.25 , two outcomes we examine as placebos below (Appendix Table C1). Second, adjustment: effects under regression adjustment (0.57 practitioners and 1.15 shops) and inverse probability weighting (0.56 and 0.69) remain close to the unadjusted 0.68 and 1.24 and statistically significant, with pre-treatment estimates jointly indistinguishable from zero (Appendix C.1).

We next compare the estimated effects against a benchmark in which parish labels and opening decades are random. Across 1,000 placebo samples matching the size of the treated group, post-treatment effects based on historical timing lie in the extreme upper tail of the placebo distributions for both outcomes, while pre-treatment effects are centered near the medians (Appendix C.2). Because that benchmark places coffeehouses without regard to where they actually went, we repeat the exercise drawing placebo locations to match the estimated probability of entry and preserving the observed entry decades; the post-treatment estimates remain in the upper tail under this more demanding assignment.

The third check turns to outcomes that entry should not move. Pubs were too numerous for one coffeehouse to displace, at 400 active in the 1640s; luxury shops would have risen had a growing taste for imported goods driven both entry and mathematical practice; non-scientific letters would have risen had established correspondents arrived with the coffeehouse; and baptism and death densities would have risen had entry drawn population. None responds. Commercial premises, which register commerce at large, are the one exception: commercial activity was rising before entry, a drift absent from the scientific outcomes, and the Poisson estimates allow a slight, insignificant rise after it (Appendix C.3).

We noted above that observability bias is less likely for instrument shops, identified from surviving objects and guild records, than for practitioners, who enter the record through diaries and published work (Taylor, 1954); for both, survival and registration are not independent of local prosperity. On the outcome side, we restrict the sample to the 175 of 252 instrument makers for whom a guild is recorded, with and without separate trends for parishes exposed to the Great Fire, which destroyed many guild records. On the treatment side, we re-estimate after excluding, in turn, each broad and each finer category of source through which a coffeehouse is first recorded. The estimates are unchanged throughout (Appendix C.5). Matching the two rosters person by person links 93 of the 244 practitioners to one of the 252 makers, and estimating

equation (1) separately on the two disjoint groups and on the overlap yields a positive and significant effect in each (Appendix C.6).

The remaining checks vary one ingredient of the design at a time. We allow parishes exposed to the Great Fire of 1666, and parishes inside the London Wall, to follow separate nonparametric trends (Appendix C.4). We change how treatment is defined and timed, moving from decades to two-, three-, and five-year intervals, allowing treatment to switch off, and estimating the effect of closures, which lower the number of mathematical practitioners (de Chaisemartin and d’Haultfoeuille, 2024) (Appendix C.7). We change the unit and the standard errors, re-estimating on a hexagonal partition of the city and computing Conley (1999) standard errors (Appendix C.8). We change the control group and the estimator, adding not-yet-treated parishes (Callaway and Sant’Anna, 2021) and re-estimating under Poisson and inverse hyperbolic sine specifications (Appendix C.9). Finally, we remove differential trends and influential observations, detrending within treatment cohorts (Goodman-Bacon, 2021), dropping each of the 162 parishes in turn, and bounding how far parallel trends could fail before the effects lose significance (Rambachan and Roth, 2023) (Appendix C.10). None changes the conclusion.

4.4. SPILLOVERS, EXPANSION, AND RELOCATION

The estimates above measure the effect of a coffeehouse on its own parish: activity rose more in parishes that received a coffeehouse than in parishes that did not, beginning when the first one opened and not before, and the checks just reported defend that comparison. What the comparison does not give is the effect on London. Two things separate the two: coffeehouses may have raised activity beyond the parish housing them, and some of the activity we count may have moved rather than appeared.

Coffeehouses may also raise activity nearby: actors in neighboring parishes could reach a coffeehouse on foot, and spillovers onto control parishes, if positive, bias the parish-level estimate downward (Butts, 2023). Parishes are too coarse to locate effects at that scale. We therefore re-estimate on a partition of London into hexagons 200 meters across, comparing the hexagon that receives the coffeehouse, and each 200 meter ring around it, with hexagons more than a kilometer from any opening (Appendix Figure B1 and Appendix Table B1). Mathematical practitioners roughly double, rising by 0.37 in the opening hexagon and 0.14 in its immediate neighbors on pre-entry means of 0.26 and 0.12. Scientific instrument shops rise severalfold, by 0.72 and 0.12 on means of 0.12 and 0.05, echoing the nearly fivefold parish-level rise in Section 4.2. The estimates are flat at every greater distance, and both series are level for five decades before the opening and rise after it (Appendix Figure B3).³² Varying the grid delivers the same range without the rings or the distant controls: rebuilt at widths from 50 to 1,200 meters, the effect per coffeehouse rises with cell size up to about 400 meters and is flat thereafter, as expected once a cell contains the range over which the effect operates (Appendix Figure B2).

Two implications follow. The parish-level estimates are lower bounds on the total effect of an opening: summed across the opening hexagon and every ring, the effect is 0.82 mathematical practitioners and 1.06 scientific instrument shops, so about half of

³²Two shop coefficients beyond the first ring reach the 10 percent level individually but are jointly indistinguishable from zero ($p = 0.09$), on pre-entry means of 0.025 and 0.000. The short range is consistent with the mobility of Robert Hooke: of the 2,497 coffeehouse visits he recorded between March 1672 and May 1683, 2,096 were within 1 km of his lodgings (Appendix Figure B5).

the practitioner response and a third of the shop response accrue to neighbors coded as controls.³³ And the aggregate shares in Section 4.2, which sum only within-parish effects, miss these spillovers but count as gains the activity that merely moved into treated parishes from elsewhere in London. Section 7 models London as a whole, adding the spillovers and taking out the moves.

Three patterns favor new activity over a reshuffling of the stock already in London. Scientific activity expanded both in London and elsewhere in England over the period (Appendix Figure D18). Decomposing the post-treatment effects into entry and expansion components shows both margins responding, and for shops entry dates from the year a maker is first recorded, so it cannot come from moves between recorded London locations (Appendix Figure B4). The positive spillovers documented above are likewise difficult to reconcile with displacement from neighboring parishes. Moves from unrecorded locations, or from outside London, would still register as entry.

5. MECHANISMS

The observed effect on scientific activity could operate through several channels. Coffeehouses turned geographic proximity into knowledge exchange, lowering the cost of entering the public sphere of applied science while widening access beyond established elites. As open commercial venues, they also brought natural philosophers with a demand for experimental instruments into contact with instrument makers. These channels predict stronger effects among practitioners with weak institutional ties, more links between theoretical and practical knowledge, and greater production of experimental instruments. Ordinary sociability would not generate these patterns.

5.1. LOWERING BARRIERS TO ENTRY

Coffeehouses may have raised scientific activity by making knowledge that others already held cheaper to reach (Mokyr, 2005, p. 312). Contemporaries called them “penny universities” (Ellis, 1956); before their arrival, scientific interaction was largely confined to aristocratic households and college chambers (Cowan, 2005, pp. 102ff.). If coffeehouses relaxed these access constraints, effects should be stronger among practitioners with weaker inherited ties to elite institutions. Following Clark and Cummins (2014), we proxy such ties by the relative representation of a practitioner’s surname s among Oxford and Cambridge (Oxbridge) alumni before 1650:

$$z(s) \equiv \frac{\text{share of } s \text{ among Oxbridge alumni before 1650}}{\text{share of } s \text{ among English couples married 1538–1649}}. \quad (4)$$

The denominator proxies the share of s in the adult English population before 1650. Higher values of $z(s)$ indicate stronger ties. Surnames are observed for all 244 practitioners.³⁴

³³The sums weight each ring coefficient in Appendix Table B1 by the average number of hexagons of that ring per opening, with standard errors of 0.14 and 0.20 computed from the same variance matrix. Openings cluster in the City, so a typical opening has fewer never-opening neighbors than the average.

³⁴We are able to gather data on paternal occupation, the more direct measure, for only 56 practitioners. The data are gathered from the Wikidata-linked biographies described in Section 3 and, for a small number of practitioners, directly from Taylor (1954), whose brief entries rarely record parentage. For a

Figure 6 shows the predicted pattern. Average post-treatment effects, measured relative to the mean in the decade before entry, are concentrated among practitioners below the median of $z(s)$ and markedly smaller above it (Panel A). Panel B estimates effects for the bottom x percent of practitioners ranked by $z(s)$, across values of x : the effect declines from approximately 5 for the 16 percent of practitioners whose surnames never appear in the matriculation registers of either university to about 1.1 once the most represented are included. Panel C states the result in shares, which need no pre-entry mean for each group: the bottom half by $z(s)$ held 39 percent of the practitioners recorded in treated parishes in the decade before entry but accounts for 61 percent of the post-treatment effect. Thus, the expansion of applied scientific activity was concentrated among practitioners outside established elite networks, the group with the most to gain from a venue with low barriers to entry.

5.2. CONNECTING THEORY AND PRACTICE

Coffeehouses presumably not only facilitated the diffusion of knowledge across social and occupational groups (Mokyr, 2005, 2016), but also functioned as a decentralized institution that matched previously segmented communities: craftsmen trained in the guild system and natural philosophers coming into science from Oxbridge. Coffeehouses were thus likely to connect individuals holding prescriptive knowledge (knowledge how) with those holding propositional knowledge (knowledge what).³⁵ This view is supported by Ellis (2006, pp. x–xi), who describes coffeehouses as spaces where scientific virtuosi conducted business with instrument makers and where the experimental method was debated by the wider public.

The first prediction is that coffeehouses increased social connections among scientific practitioners, with particularly strong effects involving instrument makers and theoretically oriented practitioners. Figure 7 compares average treatment effects before and after coffeehouse entry. Panel A shows that connections increase for practitioners overall and separately for theorists and makers. Panel B decomposes makers' connections: those among makers and those to theorists both increase after entry, while the corresponding pre-treatment effects are close to zero. The rise in cross-type connections is especially informative, since it joins practitioners who produced instruments with those most likely to use them in theoretical and experimental work. Read together with the heterogeneity in Figure 6, this is the matching pattern the second channel predicts, and not one that diffusion alone would produce.

Hooke's diary shows the widening in the company of one natural philosopher. On days he visited a coffeehouse, Hooke recorded meeting 5.69 people, against 2.92 on other days (Figure 8, Panel A). Comparing days in the same year and on the same weekday, a coffeehouse day brought 2.23 more people (s.e. 0.12). Of these, 1.44 (s.e. 0.08), about two-thirds, were outside the correspondence network: no letter they sent from London is recorded in Early Modern Letters Online (Panel B). Among them were mathematical

further 42 practitioners identified via Wikidata, these sources do not identify the father's occupation, and the remaining 146 have no biographical entry beyond Taylor (1954) to search. The cost of the surname proxy is that $z(s)$ measures representation at a particular set of institutions rather than parental status, and values within a paternal occupation are widely dispersed. Appendix Figure D22 plots the individual values of $z(s)$ by paternal occupation, including those for whom it is unobserved.

³⁵This maps onto the older epistemological distinction between knowing-how and knowing-that. The distinction between propositional and prescriptive knowledge was made popular by Mokyr (2002) and has more recently been studied in the history of science by Koschnick (2025a).

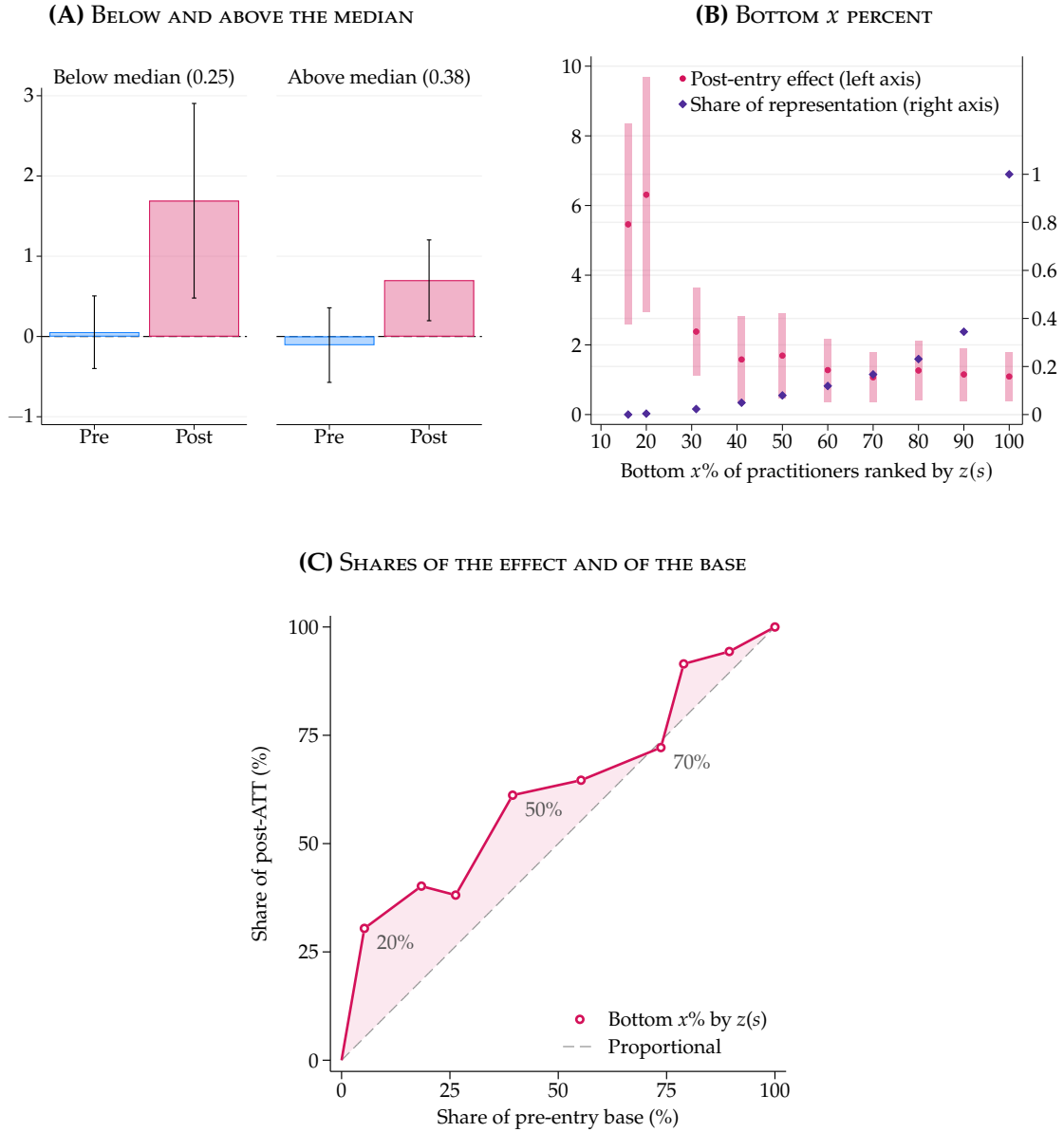
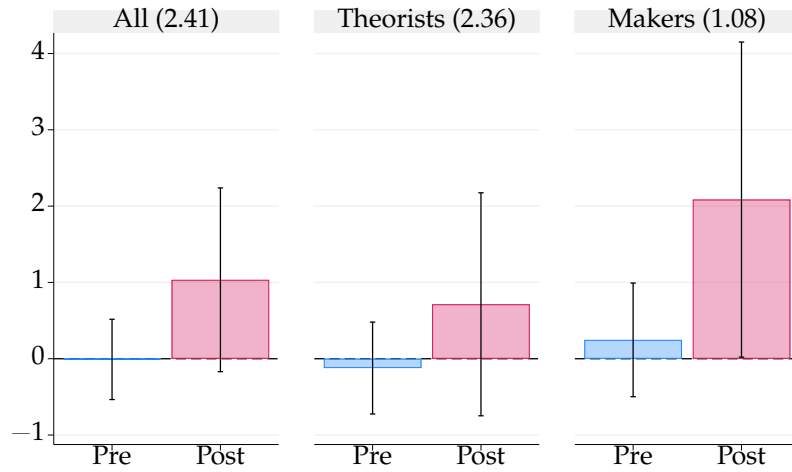


FIGURE 6. EFFECTS ON MATHEMATICAL PRACTITIONERS BY OXBRIDGE REPRESENTATION

Notes: Figure shows ATTs and 95 percent confidence intervals based on event study estimates from equation (1) of the effect of first coffeehouse opening on scientific activity. Outcomes are the decadal number of mathematical practitioners, conditional on the relative representation $z(s)$ of a practitioner's surname s at Oxbridge before 1650, as defined in equation (4). Pre-ATTs average the five to two decades before treatment; post-ATTs average the decade of treatment and the four decades thereafter. Panel A splits practitioners by whether $z(s)$ falls above or below the median of its empirical distribution; the parenthetical after each group gives its mean in the decade before treatment. Panel B restricts to practitioners in the bottom x percent of that distribution, for fixed values of x ; diamonds, read against the right axis, give the share of all Oxbridge representation, summed $z(s)$, that those practitioners hold. In Panels A and B each outcome is divided by its mean in the decade before treatment. Panel C uses the same groups from the bottom 20 percent up, without that division: it plots the share of the average post-treatment effect on all practitioners (0.68) that each group accounts for against its share of the 38 practitioner-decades recorded in treated parishes in the decade before entry. The dashed line marks equal shares, and the shaded area the excess of the first over the second. Sample includes all 162 parishes from the 1600s to the 1690s. Unit of observation is parish-decade. Standard errors clustered by parish. [Return to page 25]

(A) ALL PRACTITIONERS, THEORISTS, AND MAKERS



(B) MAKER-TO-MAKER AND MAKER-TO-THEORIST CONNECTIONS

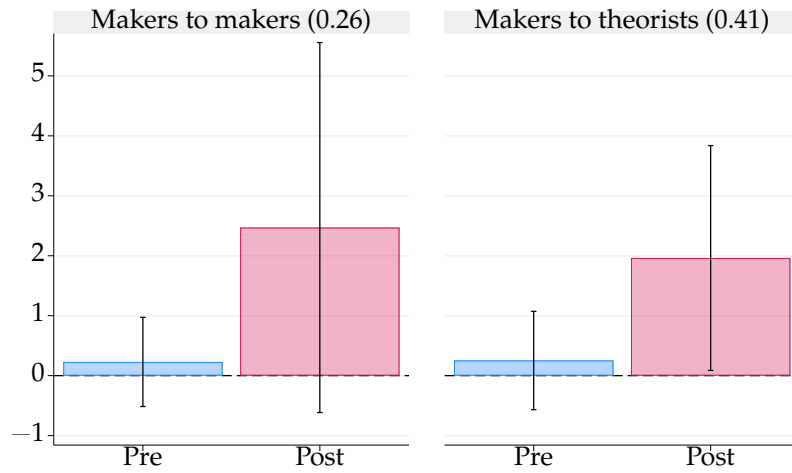


FIGURE 7. EFFECTS ON CONNECTIONS BY PRACTITIONER TYPE

Notes: Figure shows average treatment effects and 95 percent confidence intervals from equation (1) of first coffeehouse entry on practitioners' connections. Panel A reports effects for all mathematical practitioners and separately for theoretically oriented practitioners and scientific instrument makers. Panel B decomposes makers' connections into those among makers and those from makers to theorists. "Pre" averages event study estimates from 50 to 20 years before entry; "Post" averages estimates from the decade of entry through 50 years afterward. Each outcome is divided by its reference mean, the sample mean one decade before the first coffeehouse opened, which the parenthetical beside each label reports. Sample includes all 162 parishes from the 1600s to the 1690s. Unit of observation is parish-decade. Standard errors clustered by parish. [Return to page 25]

practitioners and men in manual trades, two groups that coffeehouse days brought far more often than days at a tavern (Appendix B.4). The diary records days rather than rooms, so these counts include people Hooke met elsewhere on the same day, and one diary cannot say whom he would have met without the coffeehouse.

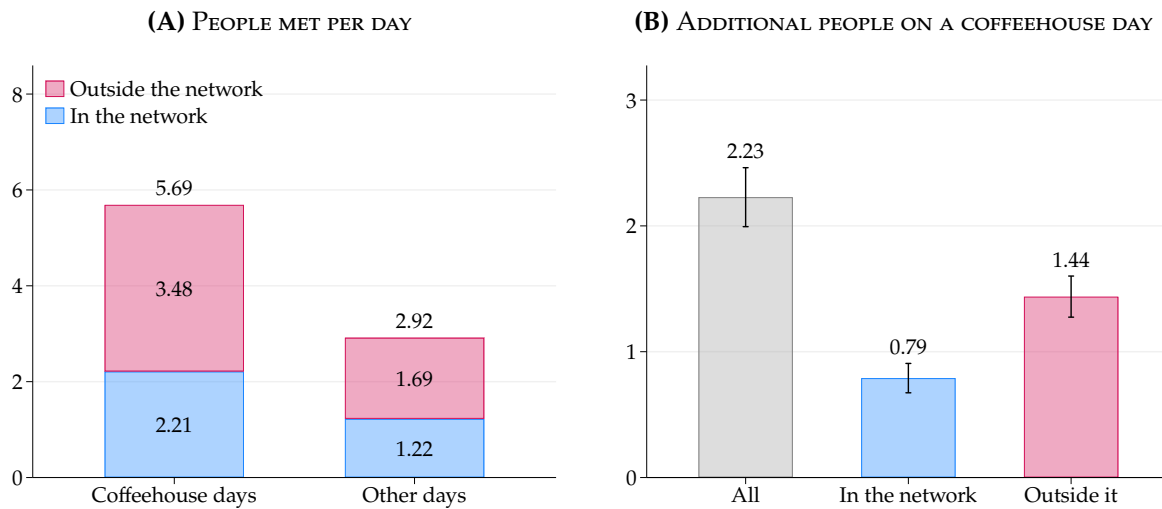


FIGURE 8. ROBERT HOOKE'S COMPANY ON COFFEEHOUSE DAYS

Notes: Figure reports the number of people Robert Hooke records meeting on each day of his diary, 10 March 1672 to 15 May 1683: 3,464 days, 2,244 of them with a coffeehouse visit. Panel A gives means on coffeehouse days and on other days, each split into people in the correspondence network, those linked to a writer of at least one letter sent from London between 1600 and 1699 in Early Modern Letters Online, and people outside it. Panel B reports the difference between coffeehouse days and other days from a regression with year-by-weekday fixed effects, for all people and for each part, with 95 percent confidence intervals from heteroskedasticity-robust (HC3) standard errors. Meetings are dated by day, not placed at the coffeehouse. Data from [Robinson and Adams \(1935\)](#) and [Henderson \(2007\)](#); Appendix D.5 describes the construction. [Return to pages [25](#), [13](#), [14](#)]

5.3. EXPERIMENTAL SCIENCE

Publicly performed experiments stimulated interest in testing and demonstrating scientific theories using the new instruments of experimental inquiry. If coffeehouses brought theorists and makers into more frequent contact, makers could learn about theorists' needs and encounter a larger market for instruments that made new phenomena visible and measurable. This yields a second prediction: coffeehouse entry should increase the supply of experimental instruments, not only established practical tools.

Figure 9 compares average treatment effects on practical and experimental instrument shops before and after coffeehouse entry. The number of shops producing experimental instruments rises sharply after entry, alongside an increase in shops producing mathematical, navigational, and surveying instruments. The estimated increase is larger for experimental instruments, although the difference between the two post-treatment effects is not statistically significant. The pattern is nevertheless consistent with coffeehouses strengthening the link between experimental inquiry and the craftsmen who supplied its material tools.

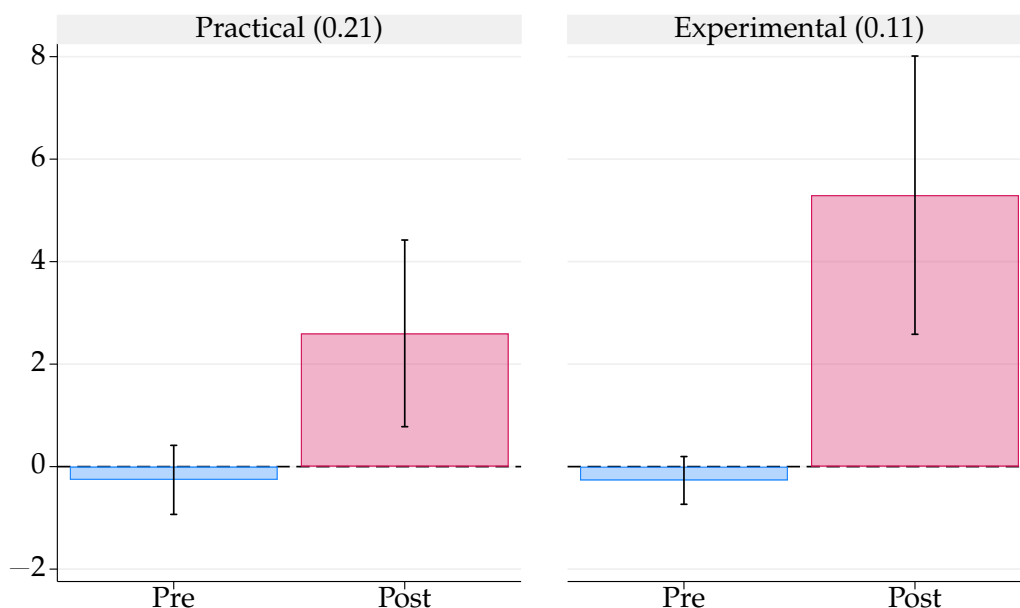


FIGURE 9. EFFECTS ON SCIENTIFIC INSTRUMENT SHOPS BY TYPE

Notes: Figure shows average treatment effects and 95 percent confidence intervals from equation (1) of first coffeehouse entry on practical (mathematical, navigational, and surveying) and experimental (optical and philosophical) instrument shops. “Pre” averages event study estimates from 50 to 20 years before entry; “Post” averages estimates from the decade of entry through 50 years afterward. Each outcome is divided by its reference mean, the sample mean one decade before the first coffeehouse opened. Sample includes all 162 parishes from the 1600s to the 1690s. Unit of observation is parish-decade. Standard errors clustered by parish. [Return to page 28]

5.4. PUBS AND ORDINARY SOCIABILITY

Pubs, understood here as taverns and inns, were long-standing venues for everyday social interaction. Since the Middle Ages, they had supported sociability through the provision of food, drink, lodging, and convivial exchange (Clark, 1983). Coffeehouses, by contrast, were a novel institution. They were similarly low-cost and sociable public venues, but centered on news, debate, and business, attracting civil, curious, and cosmopolitan customers, including virtuosi engaged in learned discourse (Cowan, 2005, p. 89). Scientific activity could and did occur in pubs (Stewart, 1992, p. 142), but historians link coffeehouses more closely to the virtuosi and the culture of experimental inquiry (Cowan, 2005; Ellis, 2004). As Cowan (2005, p. 260) argues, the coffeehouse was “not simply one among many backdrops” but “the product of the cultural world forged by the virtuosi in Britain’s age of scientific revolution.” In this sense, coffeehouses complemented rather than replaced pubs.

This distinction motivates a placebo test using pubs as an alternative class of meeting spaces. If meeting places alone raised scientific activity, pub openings should raise it too. Because pub and coffeehouse openings are correlated once coffeehouses appear, potentially reflecting common factors such as commercial rents (Appendix Table C2), the most informative sample is the 1550s–1640s, before coffeehouses existed. Figure 10 reports the estimated effects of the first seventeenth-century pub opening on this sample. Pre-trends are flat for scientific instrument shops but not for mathematical practitioners, whose numbers were already drifting upward where pubs opened, so a causal reading

is not warranted there. The dashed line carries that drift across the decades after entry, and the practitioner estimates are read against the line rather than against zero. Practitioners stop rising at the opening and fall further below the extended path with every decade, ending 0.60 beneath it after forty years, against a pre-entry mean of 0.43. Instrument shops move the other way, roughly doubling from a pre-entry mean of 0.11 to about 0.21 in the decade of entry, although the estimate is imprecise and the rise is far short of the nearly fivefold increase that follows a first coffeehouse. Subtracting the estimated group-specific pre-trends rather than overlaying them gives the same conclusion (Appendix B.5), where full-period estimates, which strong pre-trends render uninformative, are also reported.

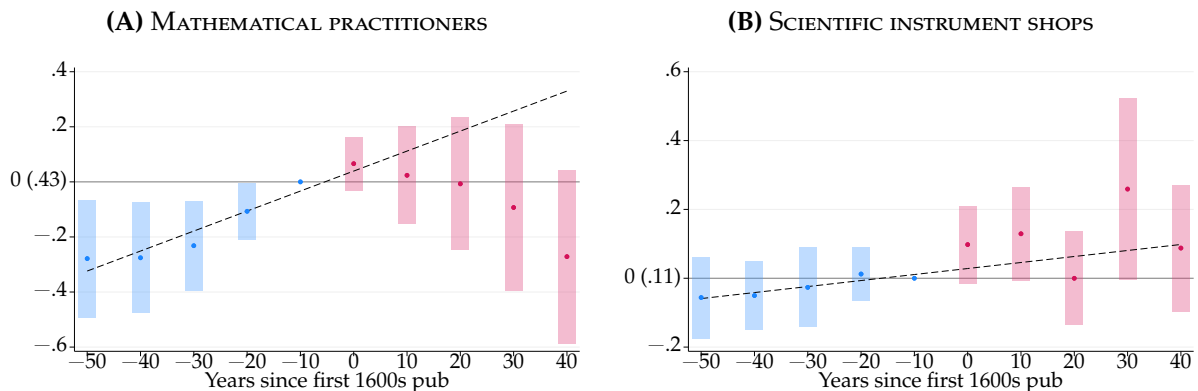


FIGURE 10. PUB EFFECTS ON SCIENTIFIC ACTIVITY

Notes: Figure shows event study estimates and 95 percent confidence intervals from equation (1) of the effect of first pub opening on scientific activity, estimated on the 1550s–1640s, before the first coffeehouse. Effects are allowed to vary with time since the first pub opened but not across cohorts. Pubs are defined as taverns and inns. The dashed line is a least squares fit to the estimates from 50 to 10 years before the first pub, extended across the post-entry window; nothing is subtracted from the estimates, so the distance between a post-entry estimate and the line is what an adjustment for a linear pre-trend would remove. Outcomes are the decadal number of mathematical practitioners and scientific instrument shops in a parish, in levels. The parenthetical beside 0 reports the sample outcome mean one decade before the first pub opened. Sample includes all 162 parishes. Unit of observation is parish-decade. Standard errors clustered by parish. [Return to page 29]

Hooke’s diary draws the same line between the two venues. Days on which he visited a coffeehouse brought him mathematical practitioners and men in manual trades far more often than days at a tavern, and the word pairs most overrepresented after he names a tavern concern claret and payment (Appendix B.4).

5.5. PUTTING PRACTITIONERS INTO CIRCULATION

Historical scholarship portrays the coffeehouse as a meeting technology that turned proximity into knowledge exchange (Cowan, 2005; Stewart, 1999). On this view, the coffeehouse did not necessarily create the skills needed to be later defined as a “London mathematician,” but raised the chance that a potential practitioner was set into the circulation of knowledge around applied mathematics and the instrument trade, present in a room a stranger could enter for a penny. As Ellis (2008) argues, coffeehouse sociability was open and egalitarian in principle, but participation was governed by implicit norms that channeled conversation toward rational, critical, polite, and evidence-based discussion. Within those norms, coffeehouses became sites for the circulation of codified knowledge, pamphlets and periodicals along with the commercial

intelligence traded in the rooms, while demonstrations and learning from men who already had the relevant skills supplied tacit knowledge (Higgitt et al., 2024; Johns, 2006).

Figure 11 gathers the evidence that coffeehouses put practitioners into that context. For interpretability, it repeats the main results of Section 4.2 as the scale for the rest: a parish gains 0.68 mathematical practitioners and 1.24 scientific instrument shops. The validity of these results is supported by flat pre-trends.

First, coffeehouse sites suggest a more prolific and more versatile pool of practitioners. Dated works rise by 1.55 on a base of 0.70 per parish-decade, more than tripling.³⁶ The range of what they make widens too. We count the distinct classes of instruments recorded in a parish and decade, mathematical, optical, and philosophical, and entry raises that count by 0.45 on a base of 0.28, more than doubling it. Second, both practitioners and makers take on successors. Instrument makers in coffeehouse parishes take on 0.75 more apprentices on a base of 0.21, and connections from older to younger practitioners rise by 1.28 on a base of 0.93, suggesting that established men were drawing younger ones in. Exposure to public experimental science may have redirected entrants, bound at about age fourteen (Wallis, 2025), into instrument making. Third, the response is not confined to the trades. Booksellers rise by 0.77 on a base of 1.79, and Fellows of the Royal Society associated with the parish by 0.68 on a base of 0.15, the largest effect in the figure relative to its own mean and also the least precise ($p = 0.06$). Fellows are scarce, with 337 records spread across 1,620 parish-decades of which only 133 contain one, and the measure places a fellow where he was buried rather than where he worked, which adds noise to the assignment.

The last two rows ask whether the newcomers did more than the men already there. Before entry a practitioner accounts for 1.12 dated works, while one gained after entry accounts for 2.29, so output rises by 0.79 more than the incumbent rate would imply, a difference the data cannot distinguish from zero, with a standard error of 0.60.³⁷ Instrument classes run the other way. A shop already in the parish before entry accounts for 0.85 distinct classes, while a shop gained after entry accounts for 0.36, so the shops a parish gains concentrate in the kinds of instruments it already makes rather than adding new ones.

6. EXTENSION TO EUROPE

What does scaling the London result up to Europe imply, and does it make sense? Coffeehouses marked a shift toward open sociability at a low cost of access in Europe more broadly (Albrecht, 1989; Habermas, 1991, pp. 30–33). In London, that access clustered mathematical practitioners and instrument makers around the same parishes (Section 4). This section asks whether the relationship between coffeehouses and applied science found in London also appears in European cities, but without claiming

³⁶Dated works count the titles listed by Taylor (1954) together with the distinct ESTC editions recorded under the same author authority string, placed at the workplace of the author rather than at the imprint, which gives the address of the bookseller.

³⁷Each row plots the outcome less the pre-entry rate times the count it is measured against, so zero would mean that the added practitioners, or shops, matched the rate of those already there. Since both series are outcomes, the rate is an association rather than the output of one more practitioner holding everything else fixed.

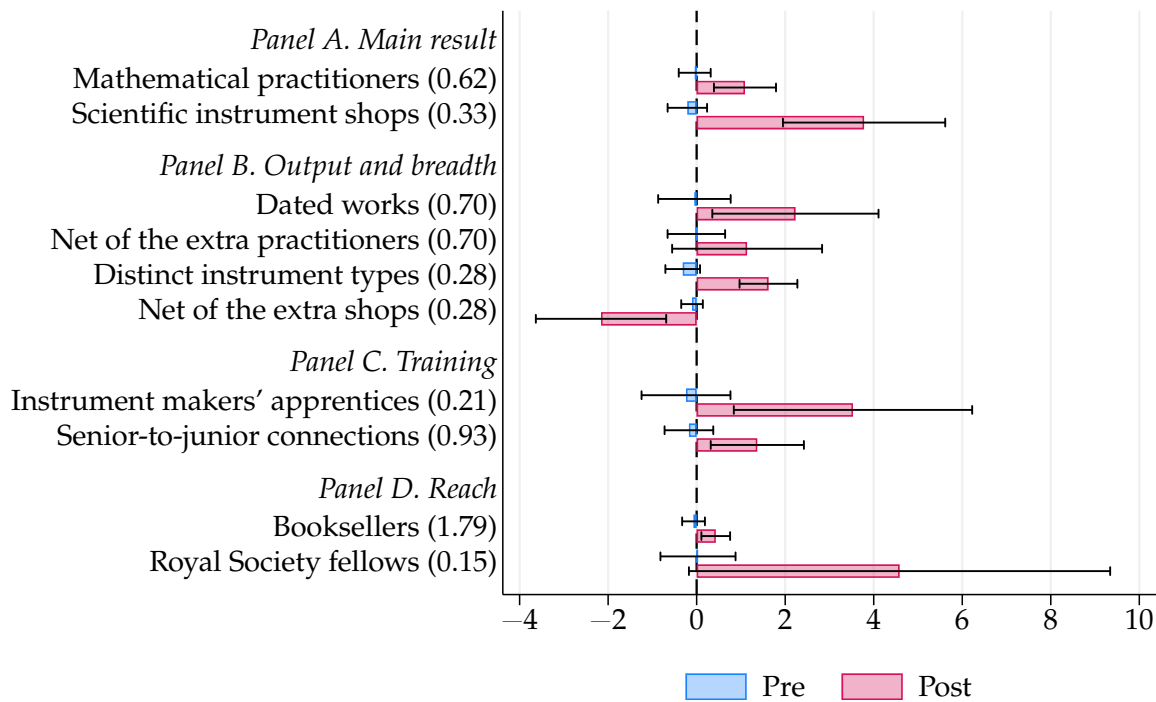


FIGURE 11. EFFECTS ON OUTPUT, BREADTH, TRAINING, AND INSTITUTIONAL REACH

Notes: Figure shows average treatment effects and 95 percent confidence intervals from equation (1) of the effect of first coffeehouse entry on the outcomes listed on the vertical axis, grouped into panels. Panel A repeats the two main outcomes of Section 4.2 as the scale for the rest. The parenthetical after each row label reports that reference mean. The two net rows subtract the pre-entry rate of 1.12 dated works per practitioner and 0.85 instrument classes per shop, so zero indicates a response matching the rate of those already there. Instrument classes are mathematical, optical, and philosophical, so that count is bounded above by three. “Pre” averages event study estimates from 50 to 20 years before entry; “Post” averages estimates from the decade of entry through 50 years afterward. Each outcome is divided by its reference mean, the sample mean one decade before the first coffeehouse opened, and each net row by the reference mean of the outcome it is built from. Sample includes all 162 parishes from the 1600s to the 1690s. Unit of observation is parish-decade. Standard errors clustered by parish. [Return to page 31]

to identify the effect of a coffeehouse. Instead, we take the first coffeehouse in a city as a meaningful marker of a critical juncture in social change: the arrival of an institution open on terms the older venues were not, as “the first place where it was possible to establish a bourgeois-oriented public forum” (Albrecht, 1989, p. 95). Such a forum may, as in London, have benefited practical science more broadly, whose know-how passed more easily across a table than through print (Polanyi, 1966); in 1690s Hamburg one keeper even styled his coffeehouse an “Academy” with “the special character of an assembly of Science” (Robilliard, 2024, p. 159).

To this end, we use data on 200 European cities and new data on their first coffeehouses in the seventeenth century (Section 3); the dates track the spread of coffee outward from the five ports where it entered Europe (Appendix Figure D1). We focus on the parts of Europe never under Ottoman rule in the sixteenth or seventeenth centuries, where the coffeehouse was a private trade rather than an institution of the ruler.³⁸

³⁸In the Ottoman lands the coffeehouse was in the hands of the ruler: building a grand coffeehouse was among the first things he did in a newly conquered city (Ellis, 2004, p. 13), and in 1633 Murad IV, fearing the sedition plotted in coffeehouses, had those of Istanbul torn down; they stayed closed for

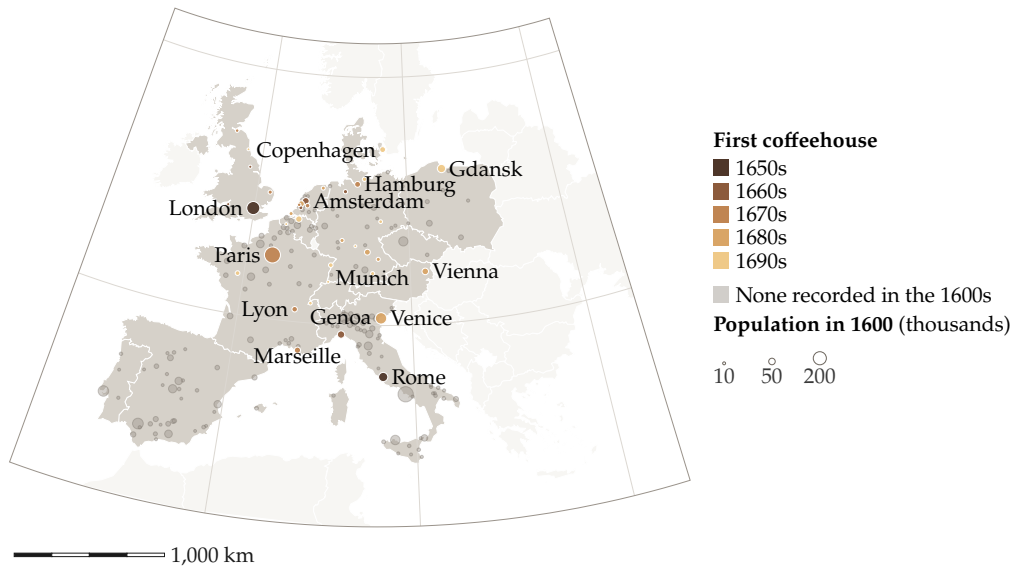


FIGURE 12. MAPPING THE FIRST COFFEEHOUSES IN EUROPE

Notes: Map of Europe showing the 200 populated areas of at least 10,000 inhabitants in 1600 in [Bosker et al. \(2013\)](#), which is that paper's preferred definition of a city. Circle area is population in 1600, and fill is the decade of the first recorded coffeehouse, gray for the 163 cities with no coffeehouse recorded in the 1600s. Dates and sources detailed in Appendix Table D1. [Return to page 33]

Figure 12 maps our sample of 200 cities. Thirty-seven cities record a coffeehouse before 1700, from London in the 1650s to Paris in the 1670s and Munich in the 1690s. The two outcomes stand in for the London ones: applied mathematics titles printed in a city for mathematical practitioners, and instrument makers recorded in a city for instrument shops (Section 3).

Figure 13 compares the 37 cities with a coffeehouse to the 163 without, by decade relative to the decade of the first coffeehouse, estimating equation (1) at the city level with never-treated cities as controls and adding the growth of the city from 1500 to 1600 and its log population in 1600 with a separate coefficient in every decade. Thus, the difference is taken relative to comparison cities with similar trends conditional on size and growth. Flat pre-trends support the comparison.³⁹ Panel A shows a persistent rise in applied mathematics titles. The 37 coffeehouse cities averaged 5.95 titles in the decade before entry, and the number is 4.97 (s.e. 2.30) higher thereafter, at 10.92, close to a doubling. Panel B shows a persistent rise in instrument makers, from a baseline of 3.73 to 8.61 after entry, a difference of 4.89 (s.e. 2.97) and more than a doubling.

We interpret these as conditional correlations, and Appendix C.11 shows that they are largely stable. The titles difference holds whether the cities whose sources admit more than one entry decade are dated as early or as late as the sources allow, or dropped, between 4.59 and 5.45 against a baseline of 4.97 (Appendix Table C4), and it holds within countries, at 4.25 (s.e. 2.38). Nor do the differences rest on London alone. Over the first two decades after entry, the other coffeehouse cities print 2.29 more titles (s.e. 1.29) and

decades ([Hattox, 2014](#), p. 102). The institution arrived in lands that were or soon became Ottoman a century earlier: Mecca (1511), Cairo (1532), and Constantinople (1554) against Western Europe in the 1650s ([Ellis, 2004](#), p. 259). The restriction removes cities in Bulgaria, Greece, Albania, Cyprus, Hungary, Romania, and the lands of the former Yugoslavia.

³⁹We fail to reject the null that the differences two to five decades before entry are jointly zero ($p = 0.46$ for titles and $p = 0.33$ for makers).

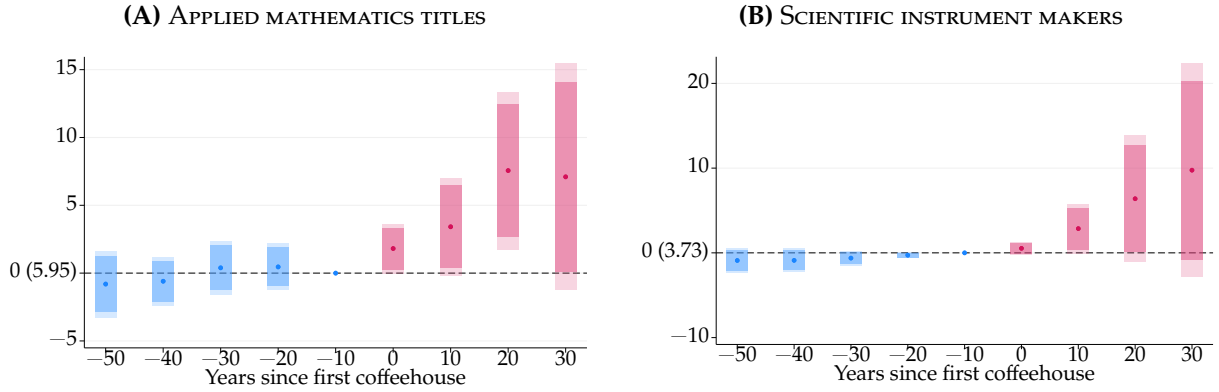


FIGURE 13. COFFEEHOUSE CITIES RELATIVE TO COMPARISON CITIES OF SIMILAR SIZE AND GROWTH

Notes: Figure shows event study estimates and 90 (darker) and 95 (lighter) percent confidence intervals from equation (1) at the city level, by decade relative to the first coffeehouse, with the decade before entry the omitted category. Outcomes are the decadal number of applied mathematics titles printed and scientific instrument makers recorded in a city. Controls are log population in 1600 and growth from 1500 to 1600, each interacted with decade fixed effects. The window stops at thirty years because only London and Rome are observed at forty. Sample includes all 200 cities from the 1600s to the 1690s. Unit of observation is city-decade. The parenthetical beside 0 reports the sample outcome mean one decade before treatment. Standard errors clustered by city. [Return to pages 33, 19, 42]

record 0.91 more makers (s.e. 0.53), about two-fifths of their pre-entry means.⁴⁰

If the association reflects coffeehouses drawing new entrants into science rather than city size alone, it should be strongest where the new science already had readers. We locate them with the census by [Gingerich \(2002\)](#) of every surviving copy of Copernicus's *De revolutionibus* (1543), the heliocentric treatise conventionally taken to open the Scientific Revolution ([Jacob, 1988](#), p. 256). Holding a copy in 1620 has little to do with size: 10 of the 19 coffeehouse cities at or above the median population in 1600 do, against 7 of the 18 below it. Among the large cities, the post-entry difference in titles is 18.20 (s.e. 7.40) where a copy was held and 1.26 (s.e. 1.63) where none was, and among the small neither group shows a rise ([Appendix B.7](#) and [Table B2](#)). The association thus appears where a larger stock of potential entrants into science already existed.

We take these results as evidence that the London pattern is not peculiar to London: the same institution, in 37 cities across ten countries, comes with the same two outcomes moving with entry, with the same timing and magnitudes of the same order. They also bear on the worry that coffeehouses simply relocated scientists: activity shifting between London parishes is easy to imagine, but between Hamburg, Paris, and Venice it is not. [Section 7](#) uses these city-level differences as the aggregate margin that a comparison of London's parishes cannot see.

7. AGGREGATE IMPLICATIONS FOR LONDON

The event studies of [Sections 4](#) and [6](#) measure what a coffeehouse on average did to a London parish or a European city. The parish estimates do not account for moves within London, and the decade fixed effects absorb whatever coffeehouses did for London as a whole. The city estimate gives the average effect of a city's first coffeehouse, abstracting

⁴⁰We use the first two decades here because the third rests on the eight cities that opened a coffeehouse by the 1660s, seven once London is left out. [Appendix Table C3](#) reports both windows.

from London's coffeehouse geography. To get at what this geography did to London's science as a whole, we move beyond these estimates with a quantitative urban model, disciplined by both sets of estimates and detailed in Appendix E.

Our model consists of two blocks. The first block conceptualizes the coffeehouse as a meeting technology: each applied scientist in parish n meets each in parish i , d_{ni} kilometers away, at a Poisson rate $\propto e^{\phi D_{nt}} e^{-\delta d_{ni}} e^{\phi D_{it}}$, with $D_{nt} = 1$ if n has a coffeehouse in decade t (Assumption E.1). A coffeehouse multiplies the rate by e^{ϕ} because it puts its parish's scientists into circulation, drawing them socially and intellectually into a public sphere of exchange around the new experimental science (Section 5.5). Thus, a coffeehouse at the other end adds to the first rather than competing with it. If each meeting is documented independently of everything else, the number of connections T_{nit} documented between parishes n and i follows the gravity equation (Theorem E.1)

$$\mathbb{E}[T_{nit}] = \exp(\ln S_{nt} + \ln S_{it} - \delta d_{ni} + \phi(D_{nt} + D_{it}) + \mu_t), \quad (5)$$

with S_{nt} parish n 's scientists, the combined de-duplicated roster of 403 mathematical practitioners and scientific instrument makers of Section 3, and μ_t a decade fixed effect for how often meetings are documented. Poisson pseudo-maximum likelihood on the 244 practitioners' connections, with S_{nt} counting only them, gives $\phi = 0.554$ and $\delta = 0.277$, so a coffeehouse multiplies the rate by 1.74, and 2.5 kilometers halve it.

The second block is a choice model that governs entry into science and location choices of scientists. Here, the return to practicing in parish n is $R_{nt} = \psi_{nt} K_{nt}^{\eta} S_{nt}^{-\gamma}$. It rises at elasticity η with the parish's access $K_{nt} = e^{\phi D_{nt}} \sum_i e^{-\delta d_{ni}} e^{\phi D_{it}} S_{it}$, proportional to the meetings each scientist there expects, and, conditional on access, falls at elasticity γ with those already there, who compete for its premises. The fundamental ψ_{nt} makes the parish attractive apart from access. Given these quantities, (i) whether a potential entrant v practices, p_v , and (ii) where, n_v , are given by

$$p_v = \mathbb{1}[V_t \zeta_{1v} > \bar{u} \zeta_{0v}], \quad n_v = \arg \max_n R_{nt} z_{nv} \text{ if } p_v = 1, \quad (6)$$

with $\bar{u}$ the outside trade's payoff and z_{nv} and (ζ_{0v}, ζ_{1v}) Fréchet tastes of shapes ε_P and ε_S , respectively (Assumption E.2). Solving backward, the parish choice gives the value of practice $V_t = (\sum_k R_{kt}^{\varepsilon_P})^{1/\varepsilon_P}$. Parish n 's share is $\sigma_{nt} \propto R_{nt}^{\varepsilon_P}$ (Theorem E.2).

These blocks lead to two equations, for where scientists work and for how many there are, used for estimation and counterfactuals. The first is given by substituting $S_{nt} = \sigma_{nt} S_t$ and $K_{nt} = \kappa_{nt} S_t$ into $\sigma_{nt} \propto R_{nt}^{\varepsilon_P}$: $\ln \sigma_{nt} = \beta_P \ln \kappa_{nt} + \ln \tilde{\psi}_{nt} + c_t$, with κ_{nt} access in shares σ_{it} , $\tilde{\psi}_{nt} = \psi_{nt}^{\varepsilon_P/(1+\gamma\varepsilon_P)}$, $\beta_P = \varepsilon_P \eta / (1 + \gamma \varepsilon_P)$, and c_t common to all parishes (Theorem E.3). The event study's decade fixed effects absorb c_t , so it sees only shares moving with access at elasticity β_P . Indirect inference picks $\beta_P = 0.560$, at which the event study on the model's parish counts, built from London's observed totals and coffeehouses, with each $\tilde{\psi}_{nt}$ inverted from its 1600–1640 share and frozen, best matches the post-entry coefficients on the combined roster (Appendix Figure E1). The second is given by $V_t = \bar{K}_t^{\eta} S_t^{-\gamma}$, which the shares imply, and, with a large pool of potential entrants, $S_t \propto V_t^{\varepsilon_S}$: $\ln S_t = \beta_S \ln \bar{K}_t + \alpha_t$, with $\bar{K}_t = S_t (\sum_n \tilde{\psi}_{nt} \kappa_{nt}^{\beta_P})^{1/\beta_P}$ London's average access, $\beta_S = \varepsilon_S \eta / (1 + \gamma \varepsilon_S)$, and α_t set by the pool and the outside payoff (Corollary E.1).

To quantify the aggregate implications, we consider a counterfactual, marked by primes, that changes each decade's coffeehouse geography but not its fundamentals. In

ratios to the observed decade, $\tilde{\psi}_{nt}$ and α_t cancel and the equations become

$$\sigma'_{nt}/\sigma_{nt} = (\kappa'_{nt}/(M_t \kappa_{nt}))^{\beta_P}, \quad \ln(S'_t/S_t) = \frac{\beta_S}{1-\beta_S} \ln M_t, \quad (7)$$

with $M_t = (\sum_n \sigma_{nt} (\kappa'_{nt}/\kappa_{nt})^{\beta_P})^{1/\beta_P}$ the change in $\bar{K}_t/S_t$, average access in shares (Theorems E.8 and E.10). As κ'_{nt} depends on the new shares, the first equation is a fixed point, solved uniquely from the observed counts, ϕ , δ , and β_P (Theorem E.5), and the second the total from β_S . No comparison within London sees β_S , but a city's first coffeehouse can, absent migration between cities. If it reaches the whole city, it multiplies every pair's rate by e^ϕ at each end and every κ_{nt} by $M_t = e^{2\phi}$, so by (7) the city's log count rises by $2\phi\beta_S/(1-\beta_S)$. Section 6's treated cities rise from a pre-entry mean base to base + b , and disciplining the model with these moments, $2\phi\beta_S/(1-\beta_S) = \ln([b + \text{base}]/\text{base})$, gives $\beta_S = 0.35$ for applied mathematics titles and 0.43 for instrument makers.

At the average $\beta_S = 0.39$, removing every coffeehouse in the 1650s to 1690s would have cost London 243 of 797 scientists counted decade by decade, 31 percent. Coffeehouse parishes would have lost 182, 23 percent to other parishes and 77 percent to the outside trade, and the others, losing access too, a net 61 (Theorem E.11). Had Charles II's proclamation of 29 December 1675, forbidding the retail sale of coffee from 10 January 1676, not been suspended on 8 January (Cowan, 2005, pp. 196–198), London would have had 106 scientists in the 1680s and 150 in the 1690s, 35 and 33 percent fewer (Figure 14).

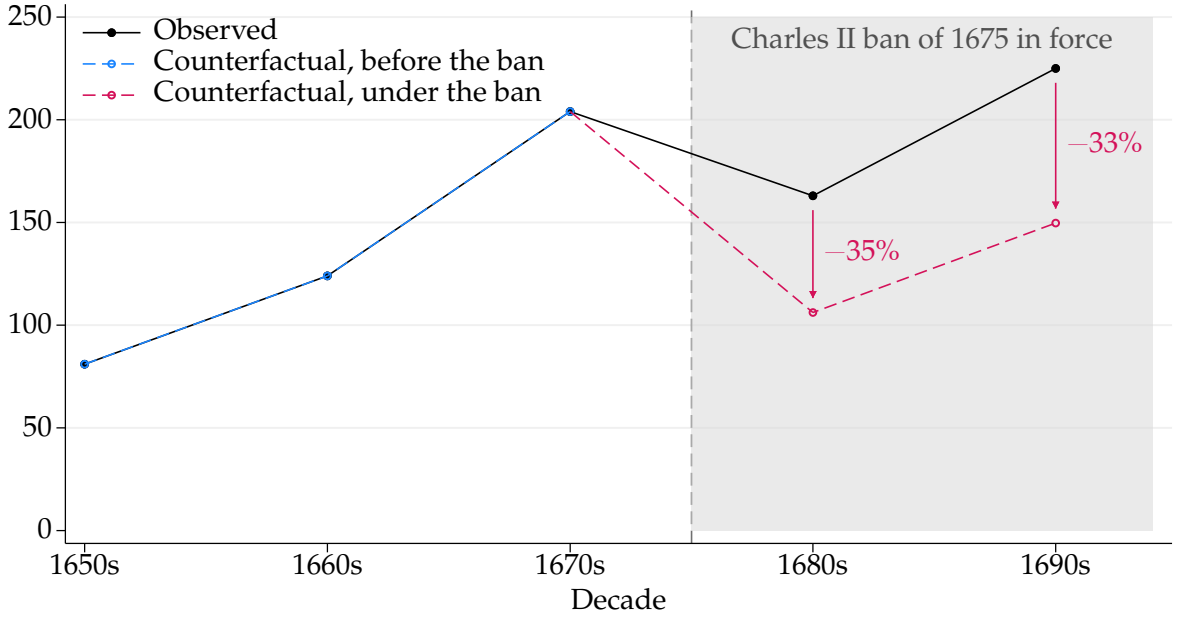


FIGURE 14. LONDON'S APPLIED SCIENTISTS WITH AND WITHOUT THE BAN OF 1675

Notes: Figure plots London's observed applied scientists by decade (Clifton, 1995; Hay et al., 2025; Taylor, 1954) and the counterfactual number the model gives with every coffeehouse closed in the 1680s and 1690s, at $\beta_S = 0.39$, the average of the two city estimates (Procedure E.4). [Return to page 36]

8. CONCLUSION

This paper asks whether a low-cost venue for face-to-face exchange raised the production of useful knowledge. Coffeehouses spread parish by parish across seventeenth-century

London after 1652, while the historical record tracks practitioners, instrument shops, connections, apprentices, and formal scientific membership. Using geocoded records and a staggered event study across 162 parishes, we find that the first recorded coffeehouse roughly doubled the number of mathematical practitioners and raised scientific instrument shops nearly fivefold, with effects that persisted. The results are not explained by differential pre-trends, observed predictors of entry, or coincident local booms, and survive placebo treatment assignments, alternative exposure definitions, and checks on how coffeehouses and scientists entered the historical record. Earlier pub openings produce no comparable increase. The historical setting supports this interpretation: coffeehouse founders and natural philosophers came from distinct social worlds, but philosophers were early customers who helped turn coffeehouses into venues for empirical discussion.

The magnitudes are large. Estimated effects account for 20 percent of mathematical practitioner-decades and 46 percent of scientific instrument shop-decades recorded in London from the 1650s through the 1690s, shares that are statistically attributable to entry under parallel trends. Because these estimates sum effects within treated parishes, they describe the response where coffeehouses opened rather than a London-wide equilibrium. The quantitative urban model of Section 7, disciplined by these estimates and by the European cities, puts the London-wide share at 31 percent of practitioner-decades once the roster of [Taylor \(1954\)](#) is completed with the instrument makers found by later research. Coffeehouses did not merely reflect London's growing scientific culture. They helped organize it. Where the response came from helps explain why. Effects are largest among practitioners whose surnames were least represented at Oxford and Cambridge. Connections between instrument makers and theoretically oriented practitioners rise, experimental-instrument shops become more common, makers take more apprentices, and Fellows associated with treated parishes increase in later decades. These patterns do not isolate a single channel, but together they are consistent with coffeehouses lowering the cost of reaching knowledge that others already held. The mathematical practitioners and instrument makers we count are the skilled implementers that accounts of Britain's later technological change place at its center ([Meisenzahl and Mokyr, 2012](#); [Mokyr, 2016](#)). Our panel ends in 1699 and cannot reach that change; what it shows is that a venue costing a penny raised the local supply of such men.

That mechanism speaks to a question modern settings cannot yet answer. Wider access to a frontier technology can be observed immediately, but whether outsiders enter, relationships persist, and training and formal institutions follow takes decades to learn. Seventeenth-century London provides that horizon. Coffeehouses did not create the Scientific Revolution. But they made science more open, connected, and useful.

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ONLINE APPENDIX

“Coffeehouses and the Rise of Science”

Monir Bounadi & Jens Oehlen

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A. HISTORICAL BACKGROUND

A.1. WHY COFFEEHOUSES OPENED

Identification in Section 4 requires the timing and location of first openings to be unrelated to trends in local scientific activity. The concern is that keepers anticipated scholarly demand and opened where scientific activity was already poised to grow. The documentary record cannot rule out such selection, but it points instead to non-scientific demand. Every recorded motive is commercial. Table A2 documents the middling economic status of coffeehouse keepers, while Table A4 shows that commercial ties among mathematical practitioners were rare before 1650.

The first coffeehouse in London illustrates this decision rule. Pasqua Rosée was a Greek servant whom the Smyrna merchant Daniel Edwards brought to London in 1651 to prepare coffee (Cowan, 2017; Ellis, 2004b). When visitors seeking coffee overran his household, Edwards sponsored Rosée in a shed in the churchyard of St. Michael Cornhill because membership in the Levant Company barred Edwards from retailing coffee himself. Alehouse keepers then petitioned the Lord Mayor to stop a non-freeman from trading. Rosée overcame this obstacle in 1654 by taking in Christopher Bowman, a freeman of the Grocers' Company who brought no knowledge of coffee but did bring the freedom of the City. In 1656, the partners leased premises for four pounds a year, and Rosée advertised the drink with a handbill invoking Galenic health claims (Rosée, 1652). Coffee thus entered London as a commodity with a supply chain, a licensing problem, a landlord, and competitors.

The other documented proprietors differed only in the commercial niches they served. Thomas Garraway built his Exchange Alley house around auctions, share dealing, and commercial news (Cowan, 2004a). Edward Lloyd made his house the resort of shipbrokers and marine underwriters and began publishing shipping intelligence in 1692 (Palmer, 2007). The house run by William Urwin attracted Dryden and the wits (Cowan, 2004c), while Alexander Man opened near Charing Cross in 1666 to capture Court trade (Lillywhite, 1963). Trade, finance, faction, and the Court exhaust the recorded motives. Even the hard case confirms the pattern. The Grecian, the house most closely associated with the Royal Society, was founded by George Constantine, a Greek immigrant and former mariner who was selling coffee in Threadneedle Street by 1665. Its initial appeal lay in the classical associations of his name and its location beside the Temple (Harris, 2000). A Royal Society clientele is documented only from 1709, and Newton did not settle in London until 1696. The scientific custom arrived four decades after the founding decision it would need to explain. The uses recorded for the 286 coffeehouses of the panel point the same way (Figure A1): 82 carry a record of commerce, 74 of news and literature, and 60 of politics or a club, against 11 with any record of science or learning, while 141 carry none.

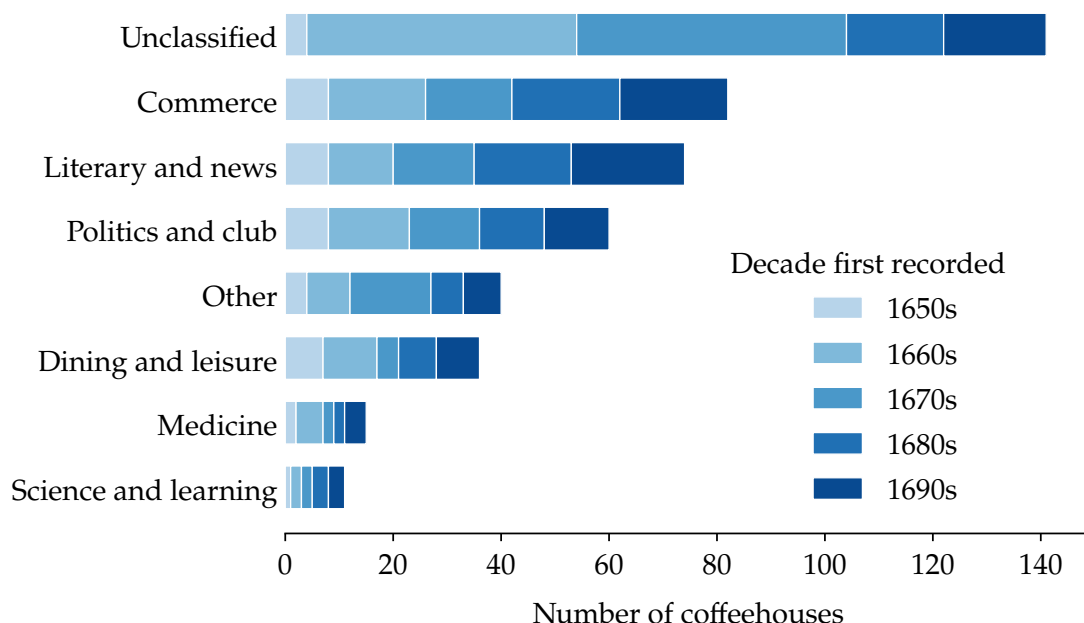


FIGURE A1. RECORDED USES OF COFFEEHOUSES BY DECADE OF FIRST RECORD

Notes: Figure counts the 286 seventeenth-century coffeehouses of the panel by the uses their records mention, with each bar split by the decade in which the house is first recorded. Uses are coded from keyword searches of the entries in [Lillywhite \(1963\)](#) dated before 1700 and from the activity tags of [Mackie \(2013\)](#) for years before 1700, and grouped into seven categories; science and learning covers records of virtuosi, the Royal Society, experiments, mathematics, and instrument makers. A house can carry several uses, so the seven classified bars count 318 uses of 145 houses. The 141 houses with no recorded use are unclassified. [Return to page 1]

Timing was shaped by shocks common to the whole city: bulk imports by the East India Company from 1657 and its first sale in 1660 (Table [A3](#)), the Excise Act of 1660, the licensing requirement introduced by the excise reform of 1663, and the Fire of 1666 ([Cowan, 2005](#); [Ellis, 2004a](#)). None varied with scientific activity in a particular parish. Repeated attempts by the Crown to suppress coffeehouses, from 1666 through the proclamation of December 1675, accused them of sedition and licentious discussion of affairs of state, never of natural philosophy ([Cowan, 2004b, 2005](#)).

The record cannot, however, rule out co-location. Coffeehouses concentrated in the wards around the Exchange, as did printers, instrument makers, and mathematical practitioners ([Stewart, 1999](#)). The historical evidence therefore addresses whether keepers deliberately targeted anticipated scientific demand, not whether coffeehouses and scientific activity shared the same commercial geography. Section 4 addresses the latter concern using pre-treatment dynamics, covariate adjustment, randomization inference, and placebo outcomes.

Selection into the historical record is also unlikely to explain the evidence. The houses discussed above are famous partly because historians later connected them to science, which makes them the houses most likely to have been sited on the outcome, yet their documented origins still lie in insurance, auctions, literary custom, and Court trade. The anonymous establishments comprising most of the sample are even less likely to have been founded in response to scholarly demand. This does not deny that the virtuosi shaped the reception of coffee or that coffeehouses later became consequential for

science (Cowan, 2005; Pincus, 1995). The narrower claim is that the available evidence identifies coffee supply, commercial geography, immigrant expertise, taxation and licensing, and non-scientific clientele as the main determinants of where and when coffeehouses opened.

A.2. COFFEE, COFFEEHOUSES, AND SCIENCE

Table A1 traces coffee, coffeehouses, and science from the first Ottoman coffeehouses to the end of the seventeenth century. Two patterns bear on what follows. English natural philosophers wrote about coffee as a botanical and medical curiosity decades before the first London coffeehouse opened. And once the houses existed, what passed through them was not only natural philosophy but mathematical practice and instrument making: designs handed from theorists to clockmakers, calculating devices shown across a table, quadrants and watches commissioned and revised, printers and makers carried into the Royal Society.

TABLE A1—CHRONOLOGY OF COFFEE, COFFEEHOUSES, AND SCIENCE

COFFEE BECOMES AN OBJECT OF SCIENTIFIC INQUIRY

- 1555 • The Ottoman chronicler Ibrahim Pecevi dates Istanbul's first coffeehouses to about this year.
- 1573 • Leonhard Rauwolf, a German physician and botanist, visits a coffeehouse in Aleppo and writes the earliest European description of coffee, which he circulates to naturalists including Carolus Clusius.
- 1592 • Prospero Alpini, an Italian physician and botanist, publishes the first European botanical description and illustration of the coffee plant in *De plantis Aegypti*.
- 1623 • Francis Bacon, the philosopher of the experimental method, discusses coffee in *Historia Vitae et Mortis*.
- 1627 • Bacon returns to coffee in *Sylva Sylvarum*, classing it with tobacco and opium among drugs that "condense the spirits" and sharpen the mind.
- 1640 • John Parkinson, apothecary to James I, introduces coffee to English botanical literature in *Theatrum Botanicum*, describing it as the "Turkes berry drinke."
- 1650s • William Harvey, who established the circulation of the blood, drinks coffee privately with his merchant brother Eliab "before Coffee-houses were in fashion in London," according to Aubrey; the family's Levant and East India trade was the likely channel.

THE COFFEEHOUSE BECOMES A SITE OF SCIENTIFIC WORK

- 1652 • Pasqua Rosée, a Greek servant of the Smyrna merchant Daniel Edwards, opens the first London coffeehouse near the Old Exchange.

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- 1654 • Samuel Hartlib, the intelligencer at the center of England's scientific correspondence network, notes in his *Ephemerides* the erection of "a cuphye-house or a Turkish, as it were ale-house [. . .] neere the Old Exchange," serving "a Turkish kind of drink made of water and some berry or Turkish beane."
- 1655 • Robert Boyle's circle, including Christopher Wren and John Wilkins, meets in the coffee-room of the Oxford apothecary Arthur Tillyard, the earliest documented gathering of natural philosophers in an English coffee-room. Wood dates the club at Tillyard's to about this year.
- 1657 • The East India Company orders bulk imports, ending the Levant Company's monopoly on English coffee.
- 1657 • Walter Rumsey, a Welsh judge and virtuoso, describes coffee as a "wakefull and civil drink."
- 1657 • The Baconian projector John Beale writes to Hartlib that "the Turkish Coffa, which is soe much praysed by my Lord Bacon [. . .] is nowe brought into public sale."
- 1659 • Edward Pococke, Laudian Professor of Arabic at Oxford, publishes *The Nature of the Drink Kahui, or Coffe*, a translation of an Arabic treatise; Hartlib circulates it in manuscript among the virtuosi and John Wallis later renders it into Latin.
- 1659 • The Rota Club, a debating society founded by the political theorist James Harrington, meets at Miles's Coffeehouse at the Turk's Head; [Hunter \(1989\)](#) estimates that eleven of its twenty-seven identifiable members later became Fellows of the Royal Society.
- 1659 • Peter Stahl, a German chemist brought to Oxford by Robert Boyle and lodged in his own house, becomes the first public teacher of chemistry in the university, instructing a circle tied to Tillyard's coffee-room; Richard Lower, John Locke, and others attend, and several later become Fellows of the Royal Society ([Turnbull, 1953](#)).
- 1660 • The Royal Society is founded; the East India Company holds its first official coffee sale; the Excise Act taxes coffee at the keeper.
- 1660–62 • Boyle's air-pump experiments, and Thomas Hobbes's attack on them, make the experimental method and the instruments that embody it a matter of public controversy.
- 1662 • Robert Hooke becomes Curator of Experiments at the Royal Society.
- 1662 • A pamphlet defending the *fuga vacui* account of the Torricellian experiment follows a dispute in a Wilde Street coffeehouse.

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- 1663 • Samuel Pepys, the naval administrator and diarist, records “good discourse about physique and chymistry” at a coffeehouse, including news of a projector’s design for “an instrument to sink ships.”
- 1666 • The Great Fire destroys and reorders the premises of the City.
- 1670s • Contemporaries begin to speak of “coffee-house philosophy” as a recognized form of debate.
- 1672 • Thomas Willis, the Oxford physician and anatomist, analyzes the physiological effects of coffee in *De Anima Brutorum*, describing it as counteracting sleep.
- 1672–83 • Hooke’s diary records near-daily visits to at least 56 London coffeehouses, where he exchanged information, displayed instruments, and debated scientific questions.
- 1674 • Hooke tells the clockmaker Thomas Tompion “at Childs coffee house of new Dividing compasses screw upon a Rule”; in October he is at Child’s again with Henry Oldenburg, John Collins, Moses Pitt, and Joseph Moxon, who shows a calculating device made from Napier’s bones.
- 1675 • Hooke and Tompion meet at Garraway’s to discuss Hooke’s proposed arrangement of springs for a watch; the design passes through inspection and revision to presentation before the king.
- 1675 • Charles II, who had first conferred with Clarendon on suppressing the houses in 1666, proclaims their suppression; the keepers petition at Whitehall and the order is rescinded before it takes effect. The charge is sedition, never natural philosophy.
- 1678 • Joseph Moxon, a printer who moved into globes, instruments, and mathematical authorship through Hooke’s coffeehouse circle, is elected to the Royal Society, an exceptional election for a tradesman.
- 1679 • Hooke and Wren work through the “Planetary catena and coyled cone for Celestiall theory” at Bruin’s coffeehouse, five years before the Garraway’s meeting.
- 1680s • European physicians publish numerous treatises on the physiological and medicinal effects of coffee.
- 1681 • John Aubrey frets over “what the academiques say at the coffee-houses” about his life of Hobbes, an early sign of scholarly reputation being contested in the coffeehouses.
- 1684 • Hooke, Wren, and Edmond Halley debate planetary motion at Garraway’s Coffee House, prompting Halley’s visit to Newton and the manuscript *De motu corporum in gyrum*, the basis of the *Principia*.
- 1686 • John Ray, the naturalist, describes the coffee plant in *Historia Plantarum*.

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1687	•	Antonie van Leeuwenhoek studies coffee seeds under the microscope and reports his observations to the Royal Society.
1689	•	Naturalists begin to club together at the Temple Coffee House, among them Hans Sloane and the apothecary James Petiver, meeting on Friday evenings to exchange specimens and botanical intelligence until about 1706. The membership of forty often attributed to the club is an error repeated from a misreading of the sources (Riley, 2006).
1690s	•	Public lecture courses in mathematics, chemistry, and natural philosophy are advertised in London coffeehouses, often with demonstrations using experimental apparatus.
1697–1702	•	James Brydges moves deliberately among coffeehouses and taverns according to the company he expects to find, and leaves when the clientele proves unpromising: patrons treated venues as differentiated search technologies.
1698	•	Henri Misson de Valbourg, a French traveler, records that London's coffeehouses are "extreamly convenient": "You have all Manner of News there [. . .] you meet your Friends for the Transaction of Business, and all for a Penny, if you don't care to spend more" (Misson de Valbourg, 1719 , p. 40).
1698	•	John Harris begins lectures on mathematics and natural philosophy at the Marine Coffee House in Birchin Lane.
1699	•	John Houghton, an apothecary, coffee merchant, and Fellow of the Royal Society, publishes <i>A Discourse of Coffee</i> in the <i>Philosophical Transactions</i> , giving a history of the coffee trade and a natural history of the plant, and concluding that the "coffee-house makes all sorts of people sociable, they improve arts, and merchandize, and all other knowledge."
1701	•	Houghton returns to the theme in his <i>Collection for the Improvement of Husbandry and Trade</i> : an inquisitive man "may find out such coffee-houses, where men frequent, who are studious in such matters as his enquiry tends to," and there "gain the pith and marrow of the others reading and studies," getting "more in an evening than he shall by books in a month."

Notes: Table summarizes key dates linking coffee, coffeehouses, natural philosophy, mathematical practice, and instrument making in Europe, England, and London to the end of the seventeenth century based on [Clark \(2000\)](#); [Cowan \(2005\)](#); [Ellis \(2006\)](#); [Hattox \(2014\)](#); [Lillywhite \(1963\)](#); [Mount \(2021\)](#); [Shapin and Schaffer \(1985\)](#), on the diaries of [Hooke \(1935\)](#) and [Pepys \(2026\)](#), and on [Houghton \(1699, 1701\)](#). [Return to page 3]

TABLE A2—RELATIVE WEALTH IN THE LONDON VICTUALLING TRADES, 1692–93

Trade	Individuals	Householders	Mean Rent, £	Mean Stock, £
Victuallers	773	768	21	26
Coffeemen and Women	137	135	30	20
Vintners	131	123	80	149
Innholders or Keepers	91	91	80	74
Tapsters	23	3	–	–
Alehouse-keepers	10	10	21	25
Strongwater-men	10	10	16	42
Total	1,175	1,140		

Notes: Data from Cowan (2005, table 7). [Return to page 1]

TABLE A3—COFFEE TRADE OF THE EAST INDIA COMPANY, 1650s–80s

Period	Activity	Pounds
1657	1st coffee order	22,400
1659	2nd coffee order	44,800
1660	1st official coffee sale	–
1660s	Coffee imports	45,000–56,000 per year
1680s	Coffee imports	280,000 per year

Notes: Avoirdupois pounds. Data from Smith (2001, pp. 264–5).
[Return to page 2]

TABLE A4—COMMERCIAL CONNECTIONS OF MATHEMATICAL PRACTITIONERS BEFORE 1650

Practitioner network size	Practitioners	With commercial tie	Commercial ties / all ties
1–9 ties	7	0	0/36 $\approx$ 0.0%
10–19 ties	15	3	6/197 $\approx$ 3.0%
$\geq$ 20 ties	9	6	12/348 $\approx$ 3.4%
Total	31	9	18/562 $\approx$ 3.2%

Notes: The sample is the mathematical practitioners whose recorded flourishing ends before 1650. Network size is the number of SDFB relationships touching each practitioner; relationships are treated as undirected. Commercial contacts comprise merchants, traders, mercers, drapers, grocers, haberdashers, clothiers, shopkeepers, innkeepers, tavern keepers, victuallers, vintners, brewers, and publicans. Booksellers, printers, stationers, publishers, bankers, and financiers are excluded. The third column counts practitioners with at least one such contact. The final column counts each relationship once within a network-size group and once in the total. A relationship joining practitioners in different size groups appears in both relevant rows, so row denominators need not sum to the total.

B. ADDITIONAL RESULTS

B.1. SPILLOVER ESTIMATION

Section 4 measures spillovers on a partition of London into 2,830 hexagons 200 meters across (Appendix Figure B1). Let y_{it} be the number of mathematical practitioners or scientific instrument shops in hexagon i in decade t , and E_i the decade of the first coffeehouse recorded there, with $E_i = \infty$ where none opens and $D_i = \mathbb{1}[E_i < \infty]$. Each hexagon with $E_i = \infty$ falls in the 200 meter ring $r \in \mathcal{R} = \{200, 400, 600, 800, 1000\}$ holding the distance to its nearest opening hexagon, and is exposed from S_i , the earliest opening in that ring; since $R_i^r = 1$ for at most one r , S_i needs no r index. Hexagons beyond 1,000 meters have $R_i^r = 0$ throughout and serve as controls.

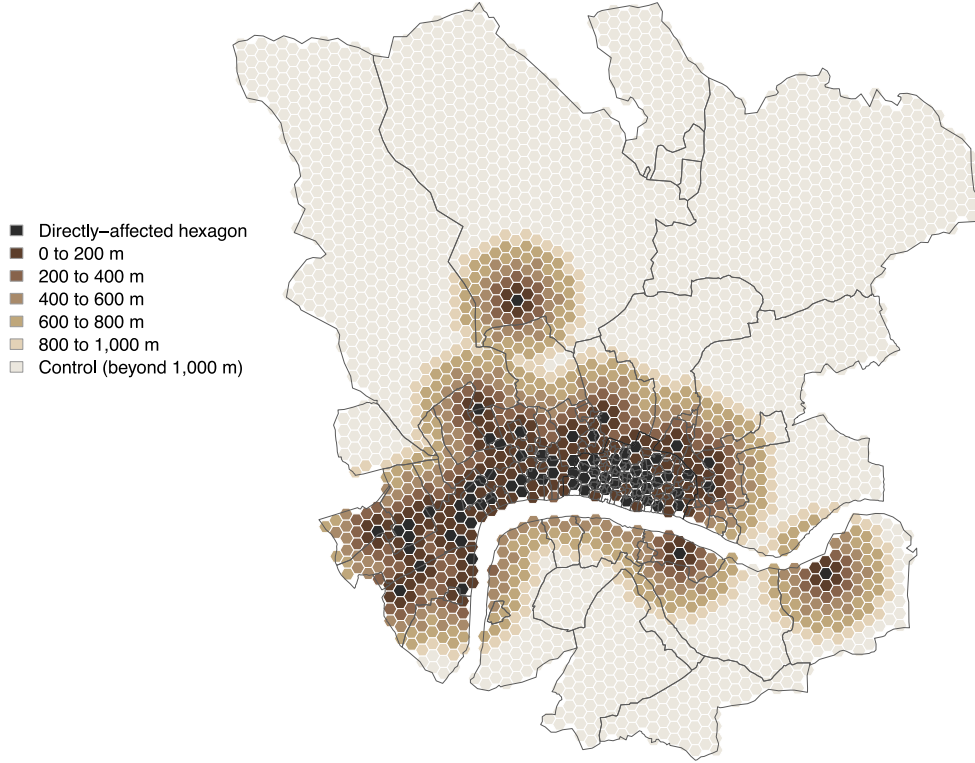


FIGURE B1. SPILLOVER RINGS AROUND FIRST COFFEEHOUSE OPENINGS

Notes: Map of London divided into 2,830 hexagons of 25,981 m² each, 200 meters corner to corner, with black lines for the 162 parish boundaries. Hexagons with a first coffeehouse in the 1600s are directly affected. Every other hexagon is assigned to a 200 meter distance ring around its nearest opening hexagon; the 1,906 hexagons more than 1,000 meters from any opening serve as controls in Appendix Table B1. [Return to pages 23, 8, 9, 11]

Table B1 reports estimates of

$$y_{it} = \alpha + \tau D_i \mathbb{1}[t \geq E_i] + \sum_{r \in \mathcal{R}} \theta_r R_i^r \mathbb{1}[t \geq S_i] + \gamma_i + \delta_t + \epsilon_{it}, \quad (\text{B.1})$$

where γ_i and δ_t are hexagon and decade fixed effects, τ the effect on the hexagon receiving the coffeehouse, and θ_r the effect at distance r . Columns (1) and (3) impose $\theta_r = 0$, returning exposed hexagons to the control group. Identification requires that spillovers are local, so hexagons beyond a kilometer are unaffected, and that absent an opening within a kilometer, hexagons in each ring would have followed the path of hexagons beyond it (Butts, 2023).

TABLE B1—DIRECT AND SPILLOVER EFFECTS OF COFFEEHOUSE OPENINGS

	Hexagons	Mathematical practitioners			Scientific instrument shops		
		Mean	(1)	(2)	Mean	(3)	(4)
Opening hexagon	76	0.263	0.355*** (0.084)	0.365*** (0.084)	0.118	0.708*** (0.168)	0.715*** (0.168)
0 to 200 m	172	0.116		0.136*** (0.034)	0.047		0.119*** (0.026)
200 to 400 m	121	0.091		0.025 (0.021)	0.025		0.019* (0.011)
400 to 600 m	172	0.006		0.016 (0.011)	0.000		0.012* (0.006)
600 to 800 m	225	0.027		0.012 (0.008)	0.004		−0.000 (0.005)
800 to 1000 m	158	0.006		0.014* (0.009)	0.000		0.006 (0.007)
Beyond 1,000 m	1906	0.001			0.001		
Hexagons in control group			2754	1906		2754	1906
Hexagon fixed effects			✓	✓		✓	✓
Decade fixed effects			✓	✓		✓	✓
Within R^2			0.032	0.043		0.081	0.086
Hexagons			2830	2830		2830	2830
Observations			28,300	28,300		28,300	28,300

Notes: Table reports estimates from equation (B.1) of the effect of first coffeehouse opening on scientific activity, on the hexagonal partition shown in Appendix Figure B1. Outcomes are the decadal number of mathematical practitioners and scientific instrument shops in a hexagon. Each hexagon is assigned to a 200 meter distance ring around its nearest opening hexagon, following Butts (2023); exposure begins with the earliest opening in that ring. Columns (1) and (3) omit the ring indicators, so exposed hexagons remain in the control group; columns (2) and (4) estimate the effect on the directly-affected hexagon jointly with the effect on each ring. Mean reports the sample outcome mean one decade before exposure, and in the 1640s for hexagons beyond 1,000 meters. Sample includes all 2,830 hexagons from the 1600s to the 1690s. Unit of observation is hexagon-decade. Standard errors follow Conley (1999) with a 1 kilometer cutoff and serial correlation over the full panel; significance at the 10, 5, and 1 percent levels is denoted by *, **, and ***. [Return to pages 23, 24, 8]

Appendix Figure B3 estimates the same two effects dynamically. Let $N_i = R_i^{200}$ mark hexagons immediately neighboring an opening and $W_{it} = D_i \mathbb{1}[t \geq E_i] + N_i \mathbb{1}[t \geq S_i]$ indicate exposure of either kind. The first stage estimates the fixed effects where $W_{it} = 0$,

$$y_{it} = \alpha + \gamma_i + \delta_t + \epsilon_{it} \quad \text{for } (i, t) \text{ with } W_{it} = 0, \quad (\text{B.2})$$

and imputes the untreated potential outcome $\hat{y}_{it}(\infty) = \hat{\alpha} + \hat{\gamma}_i + \hat{\delta}_t$, as in equation (1). The second stage regresses the imputed deviation on event time,

$$y_{it} - \hat{y}_{it}(\infty) = \sum_{h=-50}^{40} \tau_h D_i \mathbb{1}[t - E_i = h] + \sum_{h=-50}^{40} \theta_h N_i \mathbb{1}[t - S_i = h] + v_{it}, \quad (\text{B.3})$$

where h advances in steps of ten years and τ_h and θ_h are τ and θ_r indexed by event time rather than distance. No period is omitted, because the dependent variable is already measured against the imputed counterfactual: coefficients are levels, and pre-exposure values test the identifying assumption rather than being normalized to zero. Standard errors follow [Conley \(1999\)](#) with a 1 kilometer cutoff in equation (B.1) and are clustered by hexagon in equation (B.3), accounting for first-stage estimation of the fixed effects.

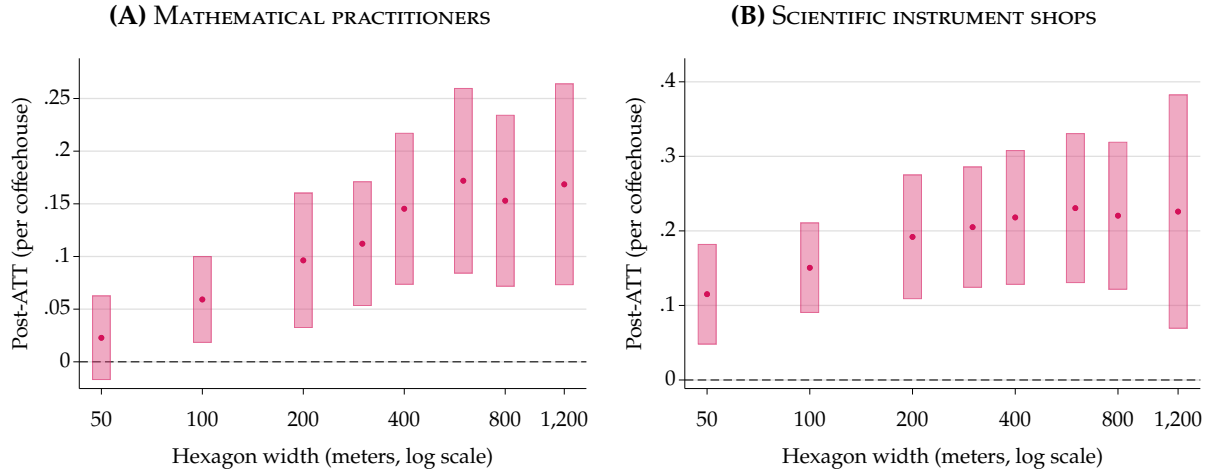


FIGURE B2. EFFECTS ON SCIENTIFIC ACTIVITY: SIZE OF GRID CELL

Notes: Figure shows average post-treatment effects of first coffeehouse opening on scientific activity, divided by the number of coffeehouses in the treated hexagon, across hexagonal partitions of increasing cell width. Each partition is built anew and every coffeehouse, practitioner, and instrument shop is reassigned to it, so the eight estimates come from eight separate panels. Estimation follows equation (1) with never-treated hexagons as controls. Coffeehouses per treated hexagon rise from 2.8 at 50 meters to 19.5 at 1,200 meters, so unnormalized effects confound spillovers absorbed within the cell with treatment intensity. Widths are on a logarithmic axis. Unit of observation is hexagon-decade. Standard errors clustered by hexagon. [Return to page 23]

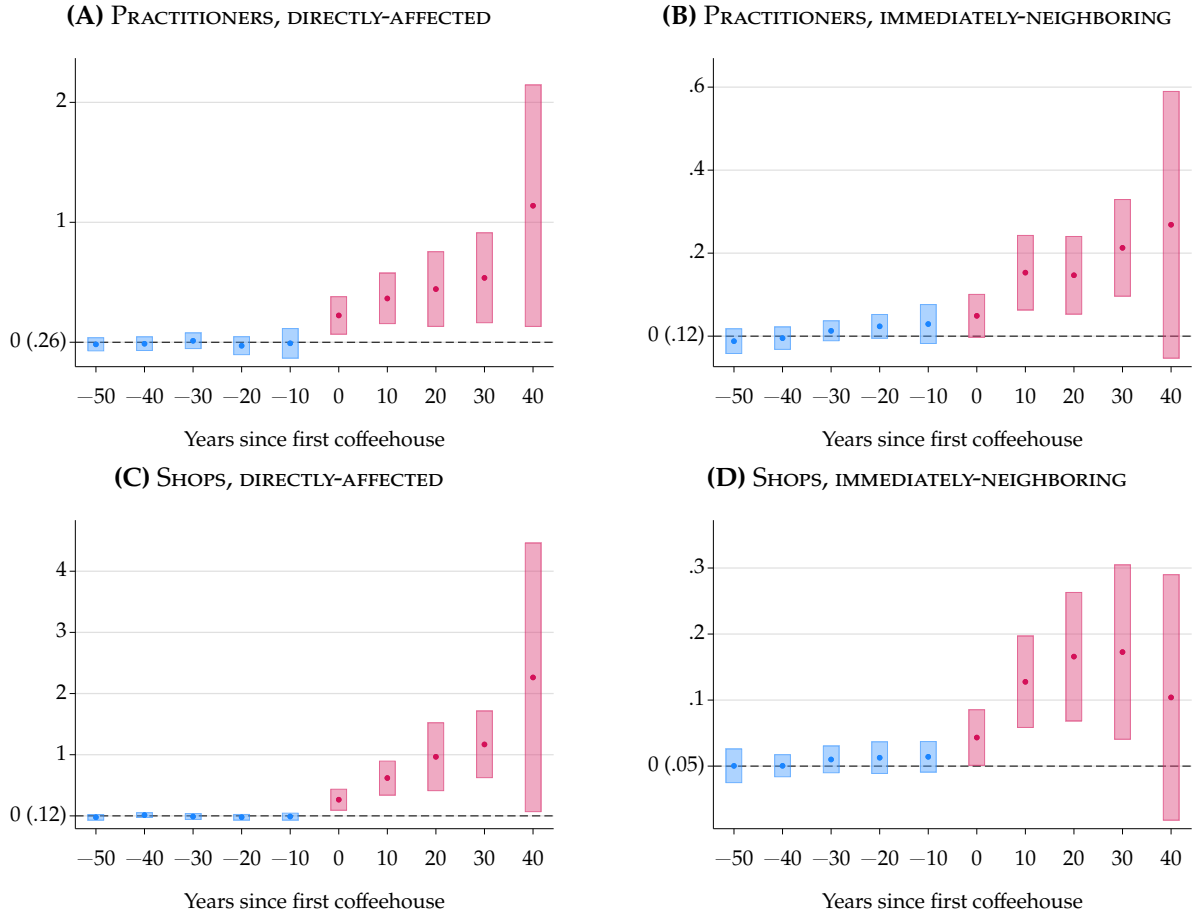


FIGURE B3. EFFECTS ON SCIENTIFIC ACTIVITY: DIRECT AND SPILLOVER EFFECTS

Notes: Figure shows event study estimates and 95 percent confidence intervals of the effect of first coffeehouse opening on scientific activity, from equations (B.2) and (B.3), following Butts (2023), on the hexagonal partition shown in Appendix Figure B1. Hexagon and decade effects are estimated on observations neither directly affected nor within 200 meters of an opening; residuals are then regressed on event time separately for directly-affected hexagons, in the left panels, and immediately-neighboring hexagons, in the right panels. Exposure begins with the earliest opening within 200 meters. No period is omitted, so pre-treatment estimates are levels against the imputed counterfactual. Vertical scales differ across panels: effects on neighboring hexagons are roughly a quarter the size of direct effects. Sample includes 76 directly-affected, 172 immediately-neighboring, and 2,582 remaining hexagons from the 1600s to the 1690s. Unit of observation is hexagon-decade. The parenthetical beside 0 reports the sample outcome mean in the relevant group one decade before treatment. Standard errors clustered by hexagon. [Return to pages 23, 9]

B.2. ENTRY AND EXPANSION

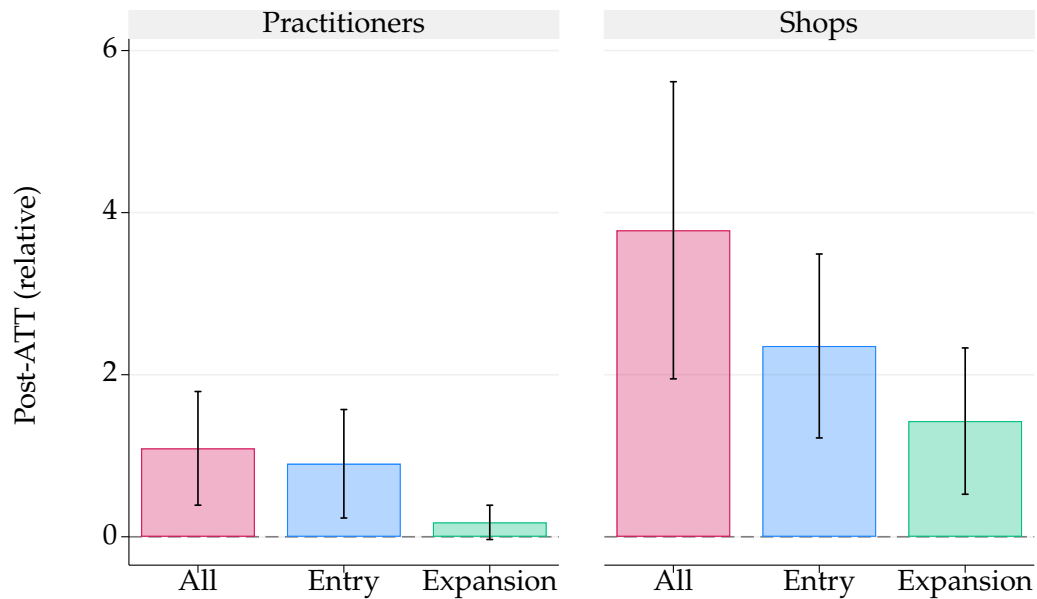


FIGURE B4. EFFECTS ON SCIENTIFIC ACTIVITY: ENTRY AND EXPANSION

Notes: Figure reports average post-treatment effects and 95 percent confidence intervals from equation (1), relative to the outcome mean one decade before first coffeehouse opening. It separates overall effects into entry and expansion components for mathematical practitioners and scientific instrument shops. For practitioners, entry counts active person-location spells at a person's first recorded London location and expansion counts spells at later recorded London locations. For shops, entry counts establishments beginning in a maker's first recorded shop year and expansion counts establishments beginning later. Post-treatment effects average event study estimates from opening through 50 years after. Sample includes all 162 parishes from the 1600s to the 1690s. Unit of observation is parish-decade. Standard errors clustered by parish. [Return to page 24]

B.3. HOOKE'S COFFEEHOUSE VISITS WERE LOCAL

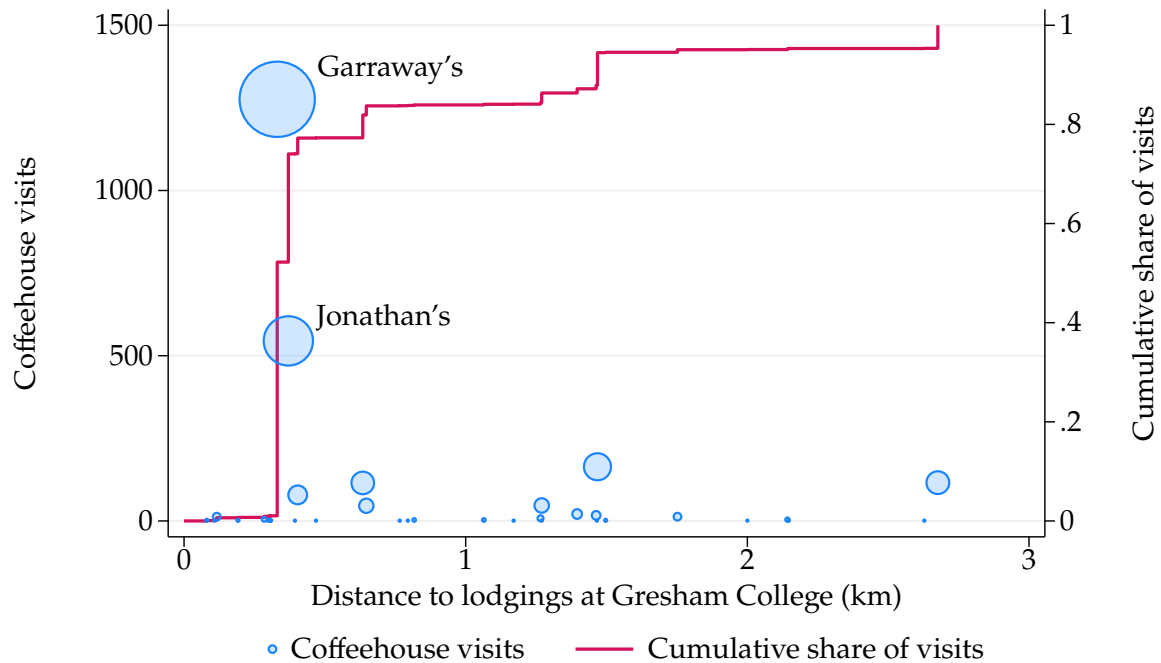


FIGURE B5. ROBERT HOOKE'S COFFEEHOUSE VISITS, 1672–1683

Notes: Figure shows the number of visits Robert Hooke (1635–1703) made to different coffeehouses between March 1672 and May 1683, grouped by their distance from his lodgings at Gresham College. Circle size is proportional to the share of his 2,497 recorded coffeehouse visits that a house received, and the step function read against the right axis gives the cumulative share of those visits made within a given distance. Of the 2,497 visits, 2,096 occurred within 1 km of his lodgings. Garraway's Coffee House was the most frequently visited, with 1,276 visits, followed by Jonathan's Coffee House with 545 visits, both located within 1 km. Data from [Brown et al. \(2021\)](#) and drawn from Hooke's diary. [Return to pages 23, 51]

B.4. HOOKE'S COMPANY ON COFFEEHOUSE AND TAVERN DAYS

Figure B6 splits the additional company of a coffeehouse day in Figure 8 by occupation. Outside the correspondence network, the 1.44 additional people comprise 0.22 mathematical practitioners, 0.21 men in other manual trades, 0.07 physicians and mathematicians, 0.31 people in the book trade, the church, law, trade, office, or Hooke's household, and 0.62 people whose occupation the index does not record. Inside the network, the 0.79 additional people are mostly in the book trade, the church, law, trade, and office (0.50), with 0.17 practitioners.

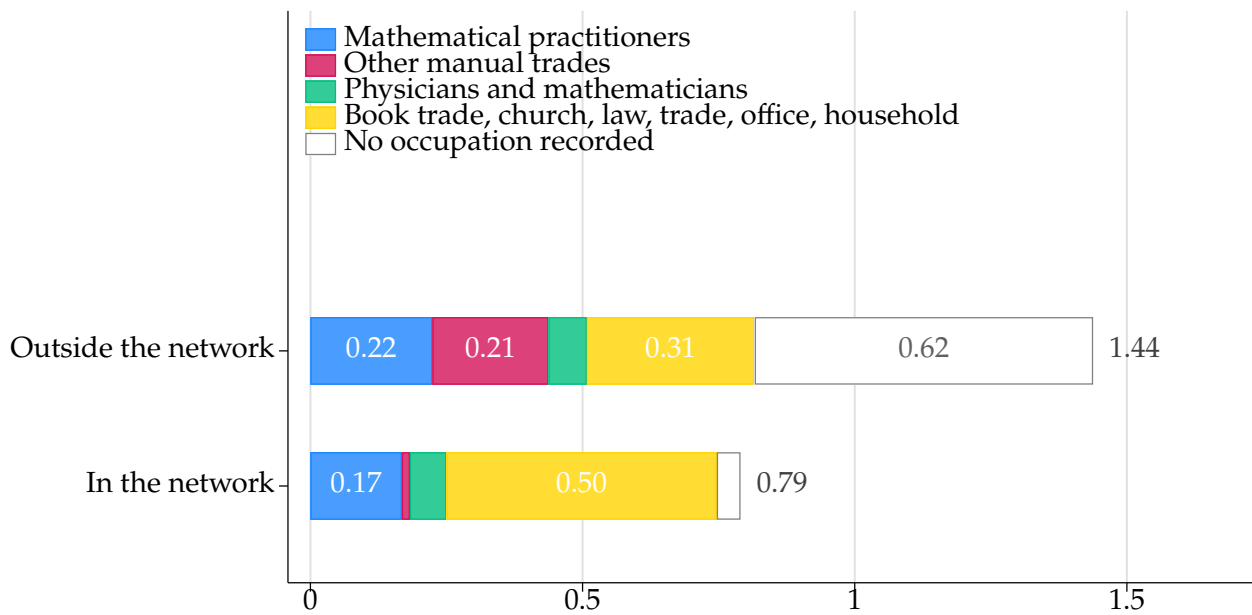


FIGURE B6. HOOKE'S ADDITIONAL COMPANY ON COFFEEHOUSE DAYS BY OCCUPATION

Notes: Figure splits the difference in people met between coffeehouse days and other days in Figure 8, Panel B, by occupation, for people outside the correspondence network and in it. Each segment is the coefficient on a coffeehouse day from a regression of the number of people in that occupation met per day on that indicator and year-by-weekday fixed effects, over the 3,464 days of the diary. Mathematical practitioners are the people matched by name to [Taylor \(1954\)](#) or to SIMON; the match is partly circular, since Taylor cites the diary in 58 entries. Other occupations are read from the notes in the index of the printed editions. Values below 0.12 are not printed. Appendix [D.5](#) describes the data. [Return to pages [13](#), [15](#)]

Coffeehouse days and tavern days brought different company (Figure [B7](#)). In one regression per occupation, with an indicator for each kind of day and year-by-weekday fixed effects, a coffeehouse day raises the number of mathematical practitioners Hooke meets by 79 percent of its rate on days with neither, against 8 percent for a day at a tavern or other drinking house ($p < 0.001$ for the difference), and the number of men in other manual trades by 109 percent, against 25 percent ($p = 0.002$). For the church, law, trade, and office, the two kinds of day do not differ ($p = 0.28$). The word pairs most overrepresented after Hooke names a drinking house concern claret, business, and payment (Figure [B8](#)), although they rest on only 261 passages.

B.5. PUB OPENINGS OVER THE FULL PERIOD

Section 5 restricts the pub placebo to the 1550s–1640s because pub and coffeehouse openings are correlated once coffeehouses appear (Table C2). Over the full 1550s–1690s sample, strong pre-trends in mathematical practitioners indicate that the placebo design is invalid there (Figure B9). Controlling for group-specific linear trends removes these pre-trends and yields negative effects on practitioners (Figure B10); estimated effects on instrument shops are positive over the full sample but small and insignificant before the 1650s.

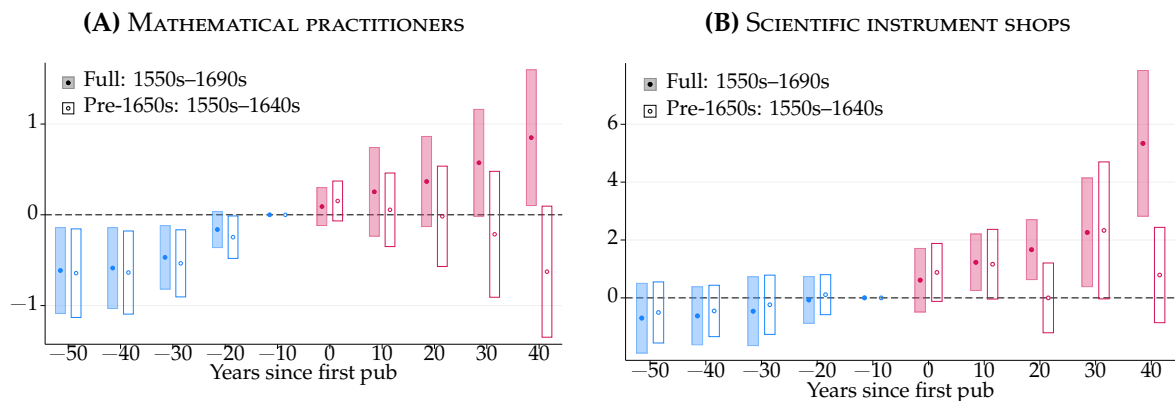


FIGURE B9. PUB EFFECTS ON SCIENTIFIC ACTIVITY OVER THE FULL PERIOD

Notes: Figure shows event study estimates and 95 percent confidence intervals from equation (1) of the effect of first pub opening in the 1600s on scientific activity, over the full period, filled markers and shaded intervals, and before the 1650s, hollow markers and outlined intervals. Effects are allowed to vary with time since the first pub opened but not across cohorts. Public houses comprise taverns, inns, alehouses, wine houses, victuallers, and mum houses. Outcomes are the decadal number of mathematical practitioners and scientific instrument shops in a parish. Each outcome is divided by its reference mean, i.e., the sample mean of the outcome one decade before the first pub opened, so the axis carries no mean in parentheses. Sample includes all 162 parishes from the 1550s to the 1690s (full) or the 1550s to the 1640s (before the 1650s). Unit of observation is parish-decade. Standard errors clustered by parish. [Return to page 16]

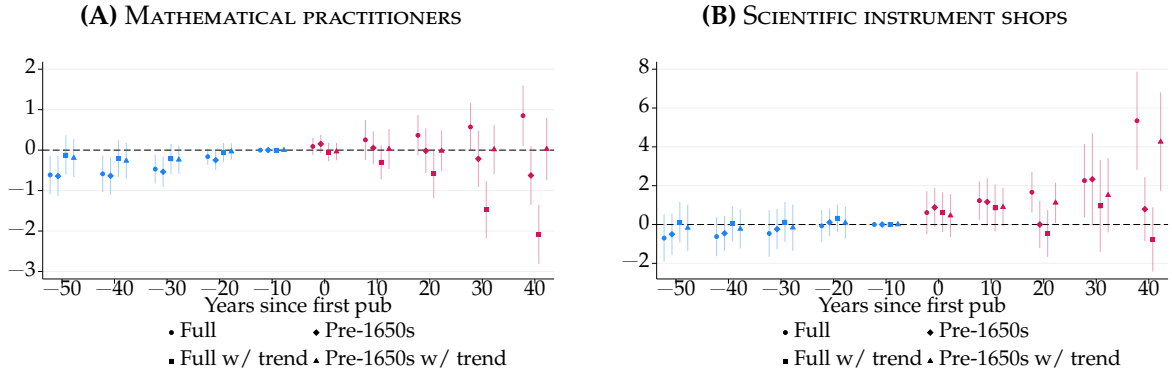


FIGURE B10. PUB EFFECTS ON SCIENTIFIC ACTIVITY: GROUP-SPECIFIC TRENDS

Notes: Figure shows event study estimates and 95 percent confidence intervals from equation (1) of the effect of first pub opening in the 1600s on scientific activity. Effects are allowed to vary with time since the first pub opened but not across cohorts. Public houses comprise taverns, inns, alehouses, wine houses, victuallers, and mum houses. Outcomes are the decadal number of mathematical practitioners and scientific instrument shops in a parish. Each outcome is divided by its reference mean, i.e., the sample mean of the outcome one decade before the first pub opened. Sample includes all 162 parishes from the 1550s to the 1690s (full) or the 1550s to the 1640s (before the 1650s). Results “w/ trend” first adjust for group-specific linear trends, following a suggestion in [Goodman-Bacon \(2021\)](#): we estimate a linear outcome trend by treatment cohort using only pre-treatment observations and then subtract the estimated trend from the outcome. Unit of observation is parish-decade. Standard errors clustered by parish. [Return to page 16]

B.6. TREATMENT INTENSITY

Treated parishes differ in how many coffeehouses arrive. Grouping them by the number open in the decade of entry and comparing each group with the never-treated, both groups roughly double their practitioners, while instrument shops rise by 3.2 times their pre-entry mean in the 40 parishes that begin with one coffeehouse and 4.7 times in the 21 that begin with two or more, from the same pre-entry mean of 0.33 (Figure B11). These are group-specific average effects rather than a dose-response curve. Identifying the response to an additional coffeehouse would require the stronger parallel trends condition that rules out selection on gains into intensity ([Callaway et al., 2026](#)), and with 40 parishes at one coffeehouse and five above two, the intensity observed at entry has too little variation to estimate it.

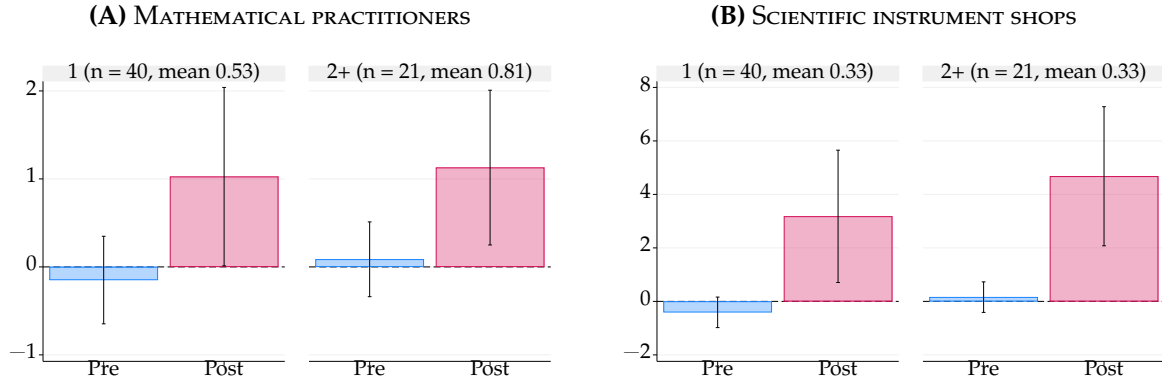


FIGURE B11. EFFECTS ON SCIENTIFIC ACTIVITY: COFFEEHOUSES AT ENTRY

Notes: Figure shows average effects before and after first coffeehouse opening, with 95 percent confidence intervals, from equation (1). Treated parishes are grouped by the number of coffeehouses open in the parish in the decade of first entry, and each group is estimated separately against all 101 never-treated parishes. Pre-entry effects average event study estimates from 50 to 20 years before opening; post-entry effects average them from opening through 50 years after. Both are divided by the mean of the outcome in that group one decade before entry, which the panel heading reports beside the number of treated parishes in the group, n , so 1 denotes a doubling; standard errors are divided by the same constant, so the intervals carry no sampling error in the group mean. Intensity is measured in the decade of entry rather than as the peak over the panel, which is realized after treatment. Sample includes all 162 parishes from the 1600s to the 1690s. Unit of observation is parish-decade. Standard errors clustered by parish. [Return to page 17]

B.7. WHERE THE EUROPEAN ASSOCIATION CONCENTRATES

The association concentrates where the new science already had a readership. [Gingerich \(2002\)](#) traces the provenance of every surviving copy of Copernicus's *De revolutionibus* (1543), so a copy held in a city in 1620 marks early engagement with the new science at large, not with astronomy alone. Read for dated, city-specific provenance, the 593 census entries place a copy in 17 of the 37 coffeehouse cities, and holding one has little to do with size: 10 of the 19 large coffeehouse cities record a copy against 7 of the 18 small. In large cities holding a copy, the post-entry difference in applied mathematics titles is 18.20 (s.e. 7.40) on a pre-entry mean of 10.30; in large cities without one it is 1.26 (s.e. 1.63) on a mean of 7.33, and among small cities neither group shows a rise (Table B2). Instrument makers run the same way, 16.68 (s.e. 11.39) against 0.99 (s.e. 0.62), without the precision to carry the point alone. The rise appears where there was already a sizable pool of potential entrants into science, which is what the London mechanism predicts and equally what one expects if coffeehouses and science went to the same cities.

TABLE B2—THE POST-ENTRY DIFFERENCE BY CITY SIZE AND COPERNICAN STOCK

	Copy recorded	No copy	Copy minus no copy
<i>Panel A. Applied mathematics titles</i>			
Large cities	18.20 (7.40) [10.30]	1.26 (1.63) [7.33]	16.94 (7.44)
Small cities	−2.36 (1.60) [5.00]	0.33 (1.45) [1.45]	−2.69 (0.97)
Large minus small	20.56 (7.55)	0.93 (0.38)	19.63 (7.51)
<i>Panel B. Scientific instrument makers</i>			
Large cities	16.68 (11.39) [10.70]	0.99 (0.62) [1.33]	15.69 (11.40)
Small cities	0.32 (0.58) [0.71]	0.77 (0.53) [1.27]	−0.45 (0.41)
Large minus small	16.36 (11.40)	0.23 (0.15)	16.14 (11.40)

Notes: Table reports post-entry differences from equation (1) with the covariates of Figure 13, averaged over entry through thirty years, the horizons every cell is observed at. Coffeehouse cities are split at their median population in 1600 and by whether any entry in the census of [Gingerich \(2002\)](#) places a copy of *De revolutionibus* in the city in 1620 on dated, city-specific provenance: 10 large and 7 small coffeehouse cities hold a copy, and 9 large and 11 small do not. The last column subtracts the no-copy cell from the copy cell, the last row subtracts the small cities from the large, and the corner, the triple difference, is the copy contrast among large cities minus that among small cities. Brackets give the mean of the outcome in the decade before entry. Within a panel all entries share one variance matrix. Sample includes all 200 cities from the 1600s to the 1690s. Unit of observation is city-decade. Standard errors clustered by city in parentheses. [Return to pages [34](#), [18](#)]

C. ROBUSTNESS CHECKS

This section develops the robustness checks summarized in Section 4, in the same order. None changes the conclusions of the main text.

C.1. COVARIATE BALANCE AND ADJUSTMENT

Coffeehouses did not open at random. Table 1 and Appendix Figure C1 compare parishes that ever received a coffeehouse with parishes that never did: fourteen of twenty-eight pre-1652 measures show normalized differences above the 0.25 threshold of Imbens and Rubin (2015, p. 277), with the largest gaps in pubs, commercial premises, tradesmen’s tokens issued between 1648 and 1651, and booksellers. Treated parishes also begin with more practitioners and more instrument shops, at 0.32 each. The 1582 tax difference is likely minor, since a temporally closer indicator of wealth, the share of substantial households in 1638, is balanced. Level imbalances of this kind threaten parallel trends only if the imbalanced characteristics also predict how scientific activity would have evolved without a coffeehouse.

Balance in changes provides the first check. Table C1 reports normalized differences in the change in each characteristic between the 1600s–1610s and the 1630s–1640s; both halves of the window precede the first London coffeehouse in 1652, so no change can reflect treatment. Both outcomes are balanced in changes, at -0.05 and 0.08 against 0.32 in levels, and booksellers fall from 0.57 in levels to -0.02 in changes. Only pubs and commercial premises exceed 0.25 , and both are examined directly as placebo outcomes below (Figure C7, Panels A and D).

Adjustment provides the second. We estimate the propensity score of ever being treated with a probit on pre-1652 pubs, tradesmen’s tokens, booksellers, and an indicator for the City of London. Weighting on this score cuts the mean absolute normalized difference from 0.30 to 0.18 (Figure C2). Effects under regression adjustment, which interacts the same covariates with decade fixed effects, are 0.57 practitioners and 1.15 shops (Figure C4); effects under inverse probability weighting are 0.56 and 0.69 (Figure C3). Both remain close to the unadjusted 0.68 and 1.24 and statistically significant, and the pre-treatment estimates are jointly indistinguishable from zero.

C.2. RANDOMIZATION INFERENCE

We benchmark the estimates against a null in which parish labels and opening decades carry no information. We construct 1,000 placebo samples by randomly assigning treatment to 61 of the 162 parishes, matching the size of the treated group, and drawing the decade of first entry for each placebo parish from the 1650s to the 1690s, coding the remainder as never treated. We then re-estimate equation (1) for each draw. Figure C5 plots the resulting placebo distributions by event time. For both outcomes, post-treatment effects based on historical timing lie in the extreme upper tail of the placebo distributions, while pre-treatment effects are centered near the medians. The association between historical entry and scientific activity is therefore unusual relative to random parish labels and random timing.

Reassigning treatment uniformly discards two features of the historical rollout: entry followed commercial geography, and it was concentrated in the 1660s, when 31 of the 61 treated parishes received their first coffeehouse. We therefore repeat the exercise

under a more demanding assignment. Using the propensity score estimated above, we group parishes into quintiles and draw the observed number of treated parishes within each, so that placebo entry is as likely as observed characteristics imply; treatment rates across the quintiles range from 0.09 to 0.78. We then deal the observed entry decades among the drawn parishes, preserving the number of parishes contributing at each event time. Because placebo entry now concentrates where actual entry was most likely, and those parishes gained scientific activity over the century, the placebo distributions center above zero: the test asks whether the observed estimates exceed what a commercially selected but causally inert assignment would produce. They do. Across 1,000 draws, the post-treatment averages lie in the upper tail ($p = 0.011$ for mathematical practitioners and $p = 0.002$ for scientific instrument shops), while the pre-treatment averages sit closer to zero than the typical placebo draw (Figure C6). This assignment reproduces where coffeehouses went and how many arrived each decade, not the association between the two; it is a benchmark, not the entry process.

C.3. PLACEBO OUTCOMES

Coffeehouse entry should leave a set of observable outcomes unmoved, and each placebo outcome targets a distinct alternative to the causal reading. Figure C7 reports each placebo under the linear specification of equation (1), the specification of the main results.

Pubs test whether entry moved the wider market for drinking houses. That market was already saturated: the panel records 400 public houses active in the 1640s, the decade before the first coffeehouse opened, so one new establishment should leave the stock unmoved. It does (Figure C7, Panel A); movement here would instead have signaled that coffeehouses entered parishes on a rising commercial tide.

Luxury shops test for a taste channel. The measure counts retailers of the imported and fashionable goods that coffee itself belonged among: grocers, confectioners, tobacconists, silk traders, and their like. If mathematical practice flourished wherever interest in the wider world was growing, an interest that practitioners served through navigation and trade, coffeehouse entry would mark that interest rather than cause the increase, and luxury shops should rise with the outcomes. They do not (Figure C7, Panel B).

Non-scientific letters test for elite arrival. The measure counts letters sent or received at addresses in a parish, and correspondence within the Republic of Letters ran through networks built over years, so even a genuine coffeehouse effect should leave general correspondence flat over the decades the panel observes. Rising letter counts soon after entry would indicate instead that established correspondents were moving into treated parishes alongside the coffeehouse. No rise appears (Figure C7, Panel C).

Baptism and death densities test for population inflows. They are the only measures of population that vary by parish and decade, and if entry drew residents, the effects on scientific activity could reflect more people rather than more science. Neither density responds to entry (Figure C7, Panels E and F).

Commercial premises test whether entry rode, or produced, a quickening of commerce at large. Before street numbering, premises of every trade hung a sign, and the count covers signed premises other than drinking houses. This is the one placebo without a clean null (Figure C7, Panel D). The linear estimates fall below zero before

entry, so commercial activity was rising in the decades leading up to the first coffeehouse; the scientific outcomes show no corresponding pre-trends (Figure 5), so the commercial drift does not extend to the outcomes and does not threaten parallel trends. After entry, the linear estimates continue the climb. The Poisson estimates of equation (3), not shown, are flat before entry, stay near zero through the first post-entry decade, and turn positive only two to three decades out, with none reaching the five percent level. Read generously, the estimated coffeehouse effect may bundle some general commercial activity; the late and insignificant Poisson response suggests any bundling is modest.

C.4. LOCAL CONFOUNDERS: THE GREAT FIRE AND THE LONDON WALL

Two features of London’s geography could jointly determine coffeehouse locations and scientific activity. The Great Fire of 1666 reshaped the city’s economic geography, displacing poorer residents, concentrating wealth in rebuilt areas, and shifting commerce toward wealthier districts (Ager et al., 2025), and the loss of records in the fire may bias observed treatment timing. The London Wall marks a second divide: the guilds controlled nearly all trade within the walls, no guild existed for scientific instrument making (Cormack, 2017, p. 79), and residential rents were higher inside (Baer, 2010). We additionally allow for separate nonparametric time trends in parishes exposed and not exposed to the fire (Figure C10) and for locations inside and outside the wall (Figure C11). The main results are robust to both controls.

C.5. SELECTION INTO THE HISTORICAL RECORD

Our outcome and treatment measures both depend on survival into the historical record, and coffeehouse activity could in principle affect that survival. On the outcome side, prominent sources such as the diaries of Samuel Pepys and Robert Hooke often mention coffeehouses in connection with leading scientists, reflecting idiosyncratic reporting. Data on mathematical practitioners, which rely largely on diaries and published materials, are more exposed to this concern; data on scientific instrument shops come largely from museum catalogs of signed instruments, guild records, and trade cards. The difference is one of degree, not of kind: survival and registration are not independent of local prosperity either. To address the remaining exposure, we restrict the sample to the 175 of 252 instrument makers active in the seventeenth-century panel for whom a London guild is recorded (Figure C15, Panel A), and, because many guild records were destroyed in the Great Fire of 1666, we additionally allow separate trends for parishes exposed to the fire (Figure C15, Panel B). Record loss in the Fire could also bias the estimates directly, by thinning the outcome records before 1666 where they burned and by dating entry late where the record of an early coffeehouse burned. The post-treatment effects stay positive when the sample keeps only the parishes outside the burnt area, 0.50 practitioners (s.e. 0.32) and 1.15 shops (0.32), the first imprecise on 21 treated parishes; when the panel ends in the 1660s, so that every record faced the same loss, 0.44 (0.20) and 0.41 (0.20); and when treated parishes are limited to those with a coffeehouse recorded after the Fire, 0.77 (0.24) and 1.34 (0.32) (Figure C16). On the treatment side, the decade in which a coffeehouse is first recorded could depend on the type of source that records it. We therefore re-estimate the baseline after excluding, in turn, coffeehouses first recorded in published or private sources, administrative or official sources, material or visual sources, and sources of unknown type, and again

after excluding each detailed source category (Figures C8 and C9). The estimates are broadly unchanged throughout.

C.6. OVERLAP BETWEEN THE OUTCOME ROSTERS

Practitioners and instrument shops are compiled independently, four decades apart, from different archives, but from overlapping populations, since makers prominent enough for Taylor (1954) are also makers whose signed instruments survive. Section 4.2 argues that the consistent effects across outcomes cannot come from one compilation's coverage; partitioning the two rosters tests that argument directly.

We match the 244 practitioners flourishing in London in the 1600s against the 252 instrument makers active in the same century. Candidate pairs come from two machine passes, a zero-shot record linker (Dasanaïke, 2026) and the language model Claude Opus 5, each of which proposed a list of sensible matches; we then reviewed every proposal by hand against the printed entries, so the crosswalk is a manual match built on machine-generated candidates rather than a machine match. We accept a link only when the forenames agree, the surnames agree on a spelling one of the sources prints, and the two activity windows are compatible. Taylor prints his own spelling variants, so the surname test reads them as alternatives rather than as one string. The two rosters disagree about the dates of the same person in 80 of the 93 cases, so a matched person is dated from the earlier of the two first-flourishing years to the later of the two last, and is placed at every address either roster records: Taylor supplies 40 parish assignments SIMON does not and SIMON supplies 54 that Taylor does not.

The match links 93 people, leaving 151 practitioners with no maker record and 159 makers with no practitioner record (Figure C17). Two Taylor entries name a father and son at once and admit no one-to-one link, so 93 is a lower bound. Counting people rather than shops, so that the three series are built alike, the effect is 0.18 (standard error 0.08) among practitioners with no maker record, 0.43 (0.12) on the overlap, and 0.45 (0.11) among makers with no practitioner record, with pre-treatment estimates near zero throughout and post-entry averages of 0.99, 1.62, and 5.77 times the pre-entry mean of each group (Figure C18). Under the strictest reading of the name evidence, which accepts only an identical forename and surname with intersecting windows, 73 links survive and the three effects are 0.22, 0.39, and 0.49. Neither compilation carries the result alone.

C.7. TREATMENT DEFINITION AND TIMING

Event timing is based on the first recorded mention of a coffeehouse, which may differ from its actual opening year; the decade-level design absorbs this error at the cost of coarser aggregation. Re-estimating equation (1) at two-, three-, and five-year intervals shows no evidence of pre-trends and leaves the main results unchanged (Figure C12).⁴¹ Treatment need not be absorbing: applying the method of de Chaisemartin and d'Haultfoeuille (2024) to coffeehouse presence rather than first entry, using the variation shown in Figure D16, yields results that closely match the baseline (Figure C20). The same method estimates the effect of losing the last coffeehouse in a parish. If the entry

⁴¹Annual specifications are infeasible due to limited outcome variation after controlling for parish and year fixed effects. For scientific instrument shops, the same constraint also precludes biennial specifications.

effects reflected differential trends rather than coffeehouses, closures should not reverse them. Twenty-two parishes lose their last coffeehouse inside the panel, ten of them in the 1690s, so few remain exposed at long horizons. Mathematical practitioners decline by 0.17 two decades after closure, against a pre-closure mean of 0.59, while instrument shop estimates are too imprecise to sign; pre-closure contrasts are jointly indistinguishable from zero for both outcomes (Figure C21). Closures are estimated on those parishes rather than on the 61 that gain a coffeehouse, so the two effects are not two halves of one quantity, but the decline is small beside the increase at entry, which is consistent with activity that outlasts the venue.

C.8. SPATIAL AGGREGATION AND INFERENCE

The parish is the natural unit of analysis because many historical addresses identify locations only by their nearest parish church, but many records are finer. Re-estimating equation (1) on a partition of London into 15,351 hexagons of 5,000 m² (Figure C13) leaves the findings qualitatively unchanged (Figure C14). The cell area is small enough that parishes contain several cells: 132 parishes hold more than one hexagon and the median parish holds four. The parish-by-decade specification is therefore identified from contrasts within parishes, and 56 of the 59 parishes containing a hexagon that receives a coffeehouse also contain hexagons that never do. Adding parish-by-decade fixed effects preserves the patterns for instrument shops while reducing the magnitude and significance of the effect on practitioners, plausibly because church-level addresses absorb much of the relevant variation. Because observations are densely clustered in space, we also compute Conley (1999) standard errors at cutoffs of 0.25, 0.5, 1, and 2 kilometers. Post-treatment effects are significant at every cutoff from ten years after opening on (Figure C27). The two widest cutoffs link 42 and 69 percent of all pairs of parishes, and at 2 kilometers the estimated variance matrix is no longer positive semidefinite, so one pre-treatment interval cannot be computed. The spillover analysis in Appendix B.1 uses a coarser partition into hexagons 200 meters across; the two grids differ in cell area by a factor of five, so levels are not comparable across them.

C.9. CONTROL GROUPS AND ESTIMATORS

Equation (1) uses never-treated parishes as controls. The event study design of Callaway and Sant’Anna (2021), which also draws on not-yet-treated parishes, leaves the main results unchanged (Figure C19). The results are likewise robust to an inverse hyperbolic sine transformation of the outcome (Figure C26). The Poisson specification in equation (3), plotted alongside the baseline estimates, reproduces the shape of the response for both outcomes with wider confidence intervals: effects on mathematical practitioners reach the ten percent level throughout the post-treatment window, and effects on instrument shops the five percent level two and three decades after entry (Figure C25). Estimation is restricted to event times within fifty years of first opening, since cohort-decade cells containing no shops otherwise separate the likelihood. The exponential model is estimated by pooled quasi-maximum likelihood with cohort rather than parish effects, because unit fixed effects in a nonlinear model raise an incidental parameters problem.

C.10. TRENDS AND INFLUENCE

Following [Goodman-Bacon \(2021\)](#), we detrend outcomes within treatment cohorts using only pre-treatment observations, removing differential linear pre-trends; the results remain similar in magnitude and significance, with wider confidence intervals (Figure [C22](#)). Re-estimating equation (1) 162 times, excluding one parish at a time, shows that no single parish drives the findings, though the four-decade effect loses significance under some exclusions (Figure [C23](#)). Following [Rambachan and Roth \(2023\)](#), we also ask how far parallel trends could fail after entry before the average post-treatment effect loses significance. It remains significant at the 5 percent level as long as the decade-to-decade change in the violation after entry is at most 0.8 times the largest such change before entry for practitioners and 1.7 times for shops (Figure [C24](#)).

C.11. EUROPEAN CITIES

Section 6 reports conditional correlations. Table [C3](#) adds country-by-decade fixed effects, removes London, Paris, and every capital, and restricts the panel to France, Germany, the Netherlands, and the UK, which hold 27 of the 37 coffeehouse cities; Table [C4](#) recodes the cities whose sources admit more than one entry decade to the earliest and the latest of them, and drops them. The titles difference holds within countries, at 4.25 (s.e. 2.38), and under every recoding of the dates; the makers difference is smaller and imprecise within countries, at 3.15 (s.e. 3.21), and 1.55 (s.e. 0.69) once London is left out.

The association does not rest on London alone. Without London, the post-entry difference is 1.67 titles (s.e. 1.29) and 0.98 makers (s.e. 0.49), roughly a third of the pre-entry mean for titles and two-fifths for makers (Table [C3](#)). Over the first two decades after entry, since the third rests on the seven cities other than London that opened a coffeehouse by the 1660s, the differences are 2.29 titles (s.e. 1.29) and 0.91 makers (s.e. 0.53). London carries much of the magnitude. Without any of the ten capitals, the differences are smaller, 1.26 titles (s.e. 0.93) and 0.23 makers (s.e. 0.33), but the remaining 27 coffeehouse cities also start from a lower base, 3.56 titles and 1.56 makers per decade before entry against 5.95 and 3.73 in the full panel, and for titles the difference is still about a third of that base. Taken together, the checks describe a substantive correlation that keeps its sign in every comparison, which lends credence to the notion that the association between coffeehouses and applied science holds across European cities and not in London alone.

C.12. FIGURES

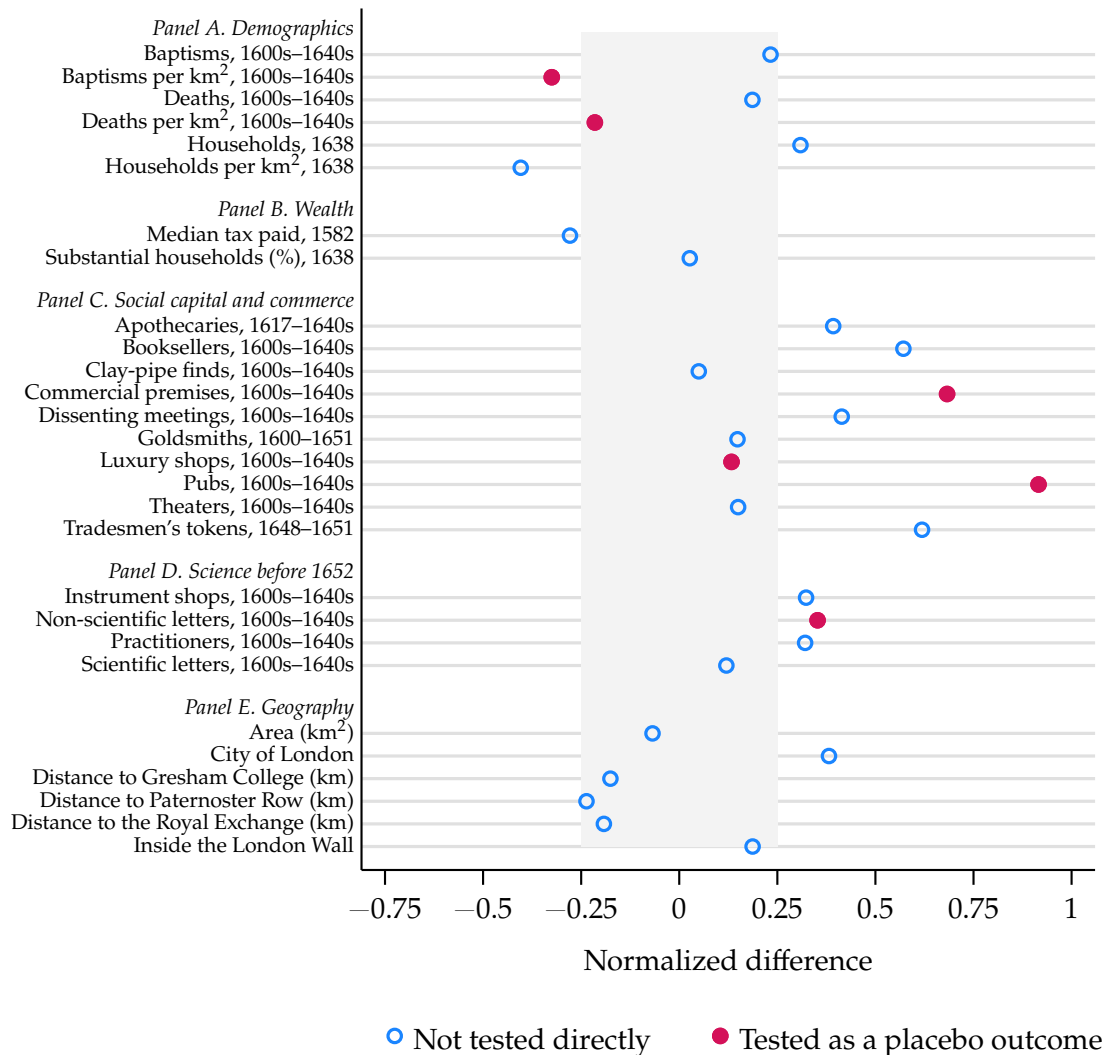


FIGURE C1. NORMALIZED DIFFERENCES IN BASELINE CHARACTERISTICS

Notes: Figure plots the normalized differences in a set of pre-treatment characteristics between ever-treated and never-treated parishes, where treatment is defined as the decade of first coffeehouse opening in seventeenth-century London. The normalized difference for a covariate x is $\frac{\bar{x}_t - \bar{x}_c}{\sqrt{(s_t^2 + s_c^2)/2}}$, where $\bar{x}_t$ and s_t^2 denote the sample mean and variance of x in the ever-treated group, and $\bar{x}_c$ and s_c^2 the corresponding quantities in the never-treated group. The shaded region marks absolute differences below 0.25, the threshold suggested by [Imbens and Rubin \(2015, p. 277\)](#). Filled markers indicate covariates that also appear as placebo outcomes in Appendix Figure C7. Rows and their measurement follow Table 1. [Return to pages 20, 27]

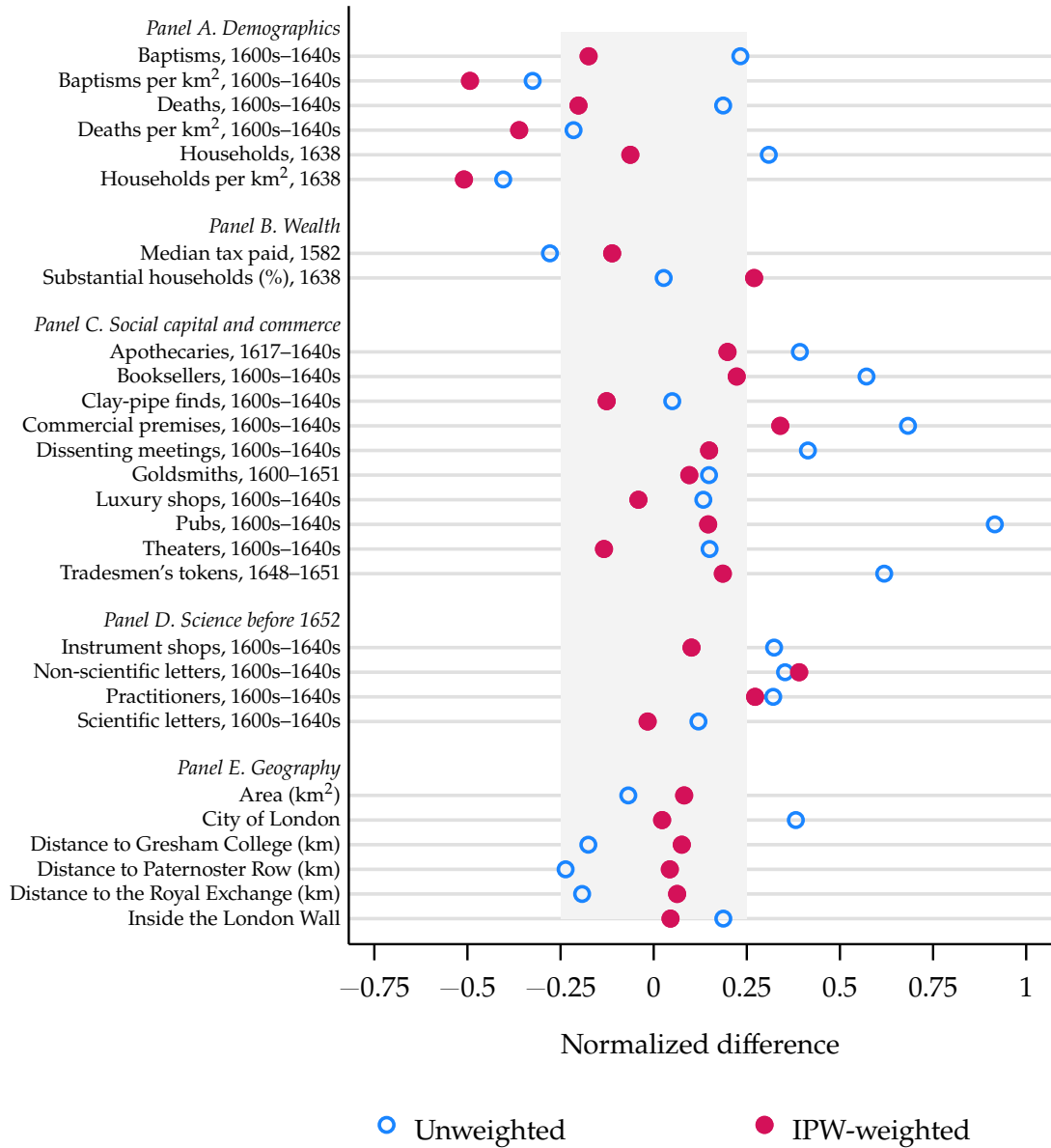


FIGURE C2. NORMALIZED DIFFERENCES BEFORE AND AFTER WEIGHTING

Notes: Figure plots the normalized differences of Figure C1 without weights and with the inverse-probability weights (IPWs) of Figure C3. The IPWs are based on a probit-estimated propensity score for ever being treated, using pre-1652 pubs, tradesmen's tokens (1648–1651), and booksellers, together with an indicator for the City of London. Weights are stabilized and trimmed at the first and ninety-ninth percentiles. Weighting cuts the mean absolute normalized difference across the 28 rows from 0.30 to 0.18. The shaded region marks absolute differences below 0.25. Rows and their measurement follow Table 1. [Return to page 20]

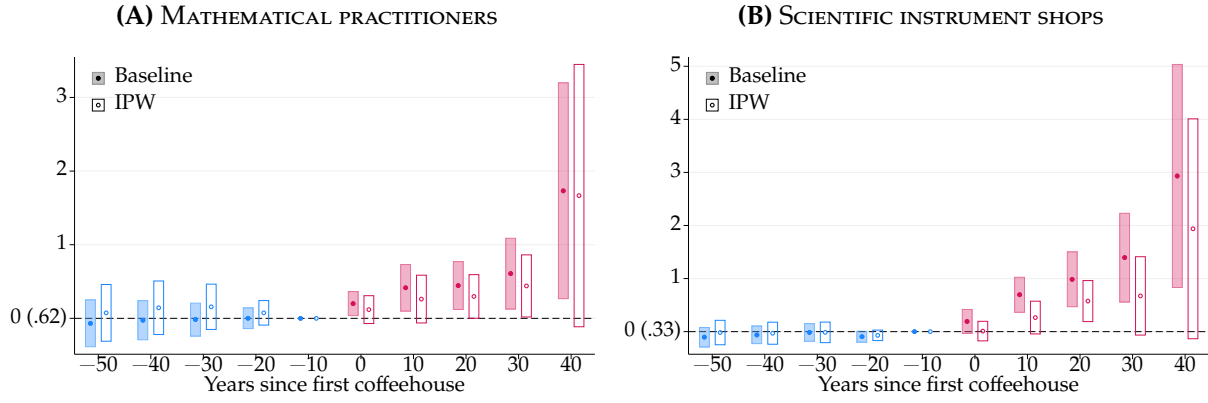


FIGURE C3. EFFECTS ON SCIENTIFIC ACTIVITY: IPW ADJUSTMENT

Notes: Figure shows event study estimates and 95 percent confidence intervals of the effect of first coffeehouse opening on scientific activity, from the baseline specification in equation (1), filled markers and shaded intervals, and the same equation weighted with inverse-probability weights (IPWs), hollow markers and outlined intervals. The IPWs are based on a probit-estimated propensity score for ever being treated, using pre-1652 pubs, tradesmen's tokens (1648–1651), and booksellers, together with an indicator for the City of London. Weights are stabilized and trimmed at the first and ninety-ninth percentiles. Outcomes are the decadal number of mathematical practitioners and scientific instrument shops in a parish. Sample includes all 162 parishes from the 1600s to the 1690s. Unit of observation is parish-decade. The parenthetical beside 0 reports the sample outcome mean one decade before treatment. Standard errors clustered by parish. [Return to pages 20, 27]

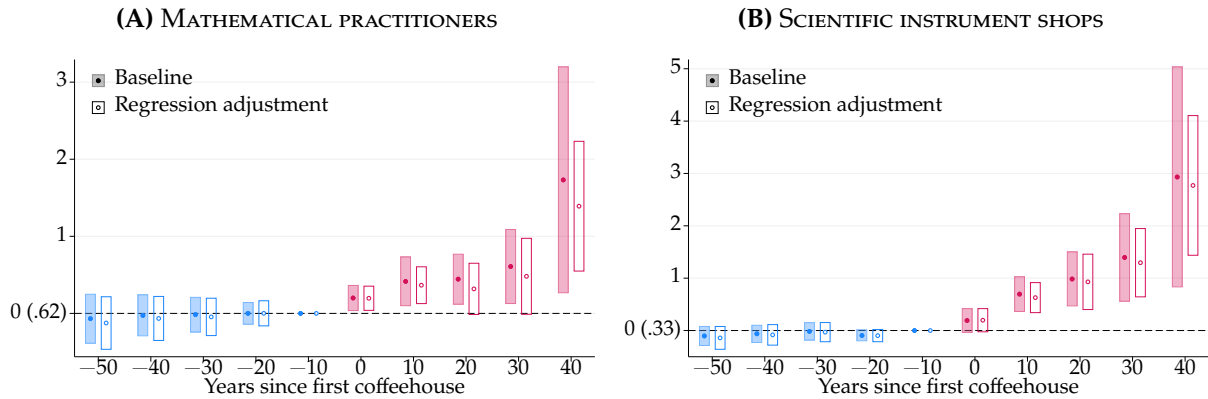


FIGURE C4. EFFECTS ON SCIENTIFIC ACTIVITY: REGRESSION ADJUSTMENT

Notes: Figure shows event study estimates and 95 percent confidence intervals of the effect of first coffeehouse opening on scientific activity, from the baseline specification in equation (1), filled markers and shaded intervals, and the same equation including covariates, hollow markers and outlined intervals. The adjusted specification includes pre-1652 pubs, tradesmen's tokens (1648–1651), and booksellers, together with an indicator for the City of London, each interacted with decade fixed effects. Outcomes are the decadal number of mathematical practitioners and scientific instrument shops in a parish. Sample includes all 162 parishes from the 1600s to the 1690s. Unit of observation is parish-decade. The parenthetical beside 0 reports the sample outcome mean one decade before treatment. Standard errors clustered by parish. [Return to page 20]

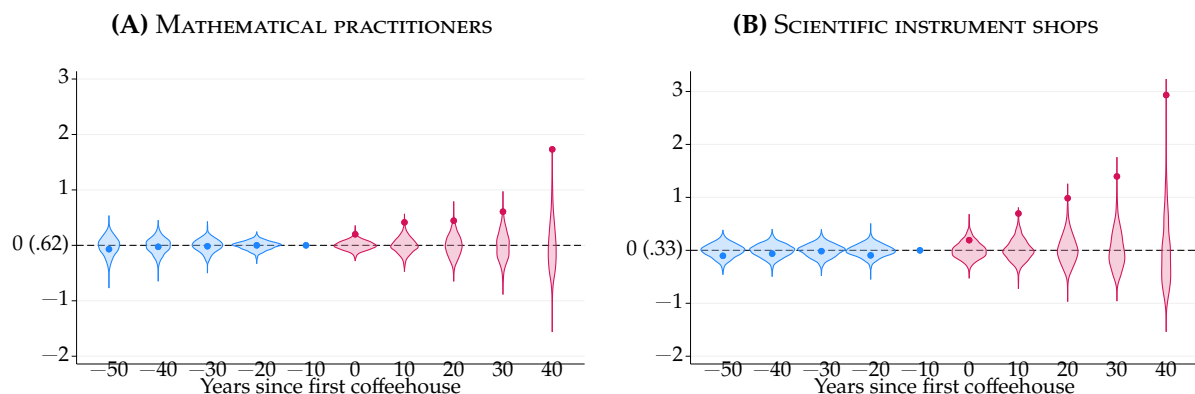


FIGURE C5. EFFECTS ON SCIENTIFIC ACTIVITY: RANDOMIZATION INFERENCE

Notes: Figure shows violin plots of event study estimates from a series of 1,000 placebo estimates of equation (1) of the effect of the first coffeehouse opening on scientific activity. Outcomes are the decadal number of mathematical practitioners and scientific instrument shops in a parish. In each placebo dataset, we randomly assign the timing of the first coffeehouse opening to one of the decades from the 1650s to the 1690s for 61 of the 162 parishes, preserving the size of the treated group in the main sample. Violin plots show the distribution of placebo treatment effects by event time, with dots indicating the actual event study estimates from the true dataset. Sample includes all 162 parishes from the 1600s to the 1690s. Unit of observation is parish-decade. The parenthetical beside 0 reports the sample outcome mean one decade before treatment. [Return to page 20]

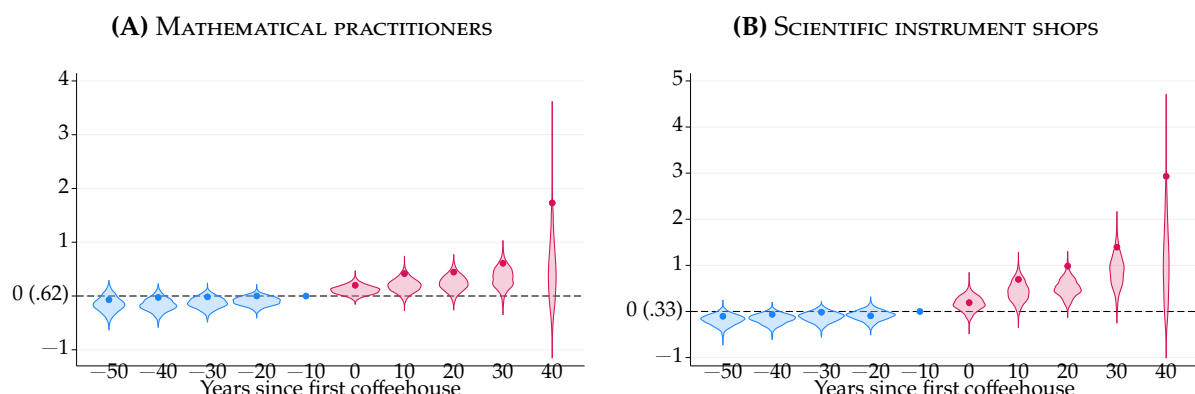


FIGURE C6. EFFECTS ON SCIENTIFIC ACTIVITY: CONDITIONAL RANDOMIZATION INFERENCE

Notes: Figure shows violin plots of event study estimates from 1,000 placebo estimates of equation (1) of the effect of the first coffeehouse opening on scientific activity. Outcomes are the decadal number of mathematical practitioners and scientific instrument shops in a parish. In each placebo dataset, treated parishes are drawn without replacement within quintiles of the propensity score of ever receiving a coffeehouse, estimated by probit on pre-1652 pubs, tradesmen's tokens, and booksellers, together with an indicator for the City of London, preserving the number treated in each quintile, and the observed decades of first entry are then assigned at random among the drawn parishes, so that both the commercial geography of entry and the number of parishes observed at each event time match the historical rollout. Violin plots show the distribution of placebo treatment effects by event time, with dots indicating the actual event study estimates from the true dataset. Randomization p -values report the share of placebo draws at least as large as the observed estimate, counting the observed assignment in the reference set. Sample includes all 162 parishes from the 1600s to the 1690s. Unit of observation is parish-decade. The parenthetical beside 0 reports the sample outcome mean one decade before treatment. [Return to page 21]

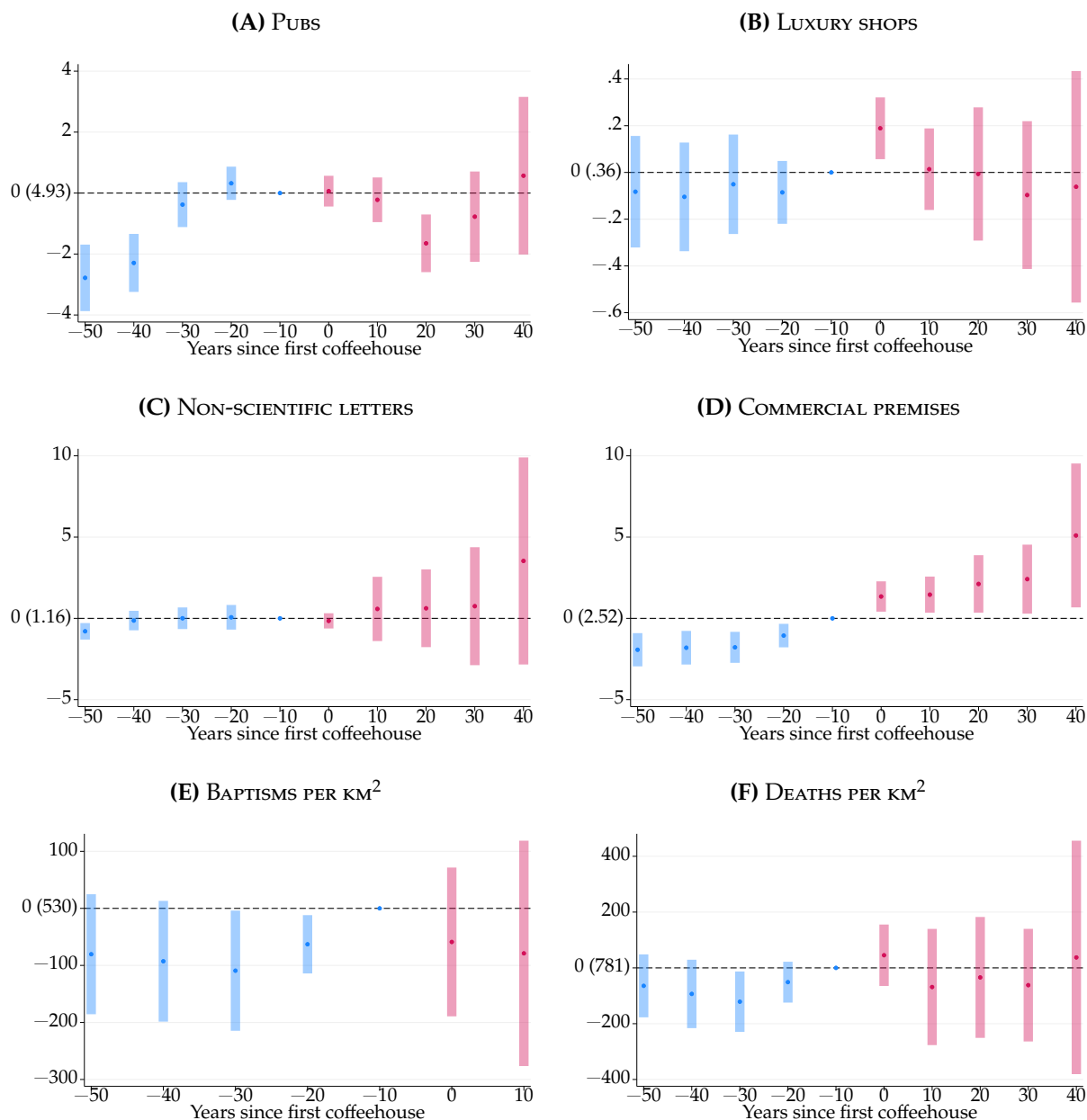


FIGURE C7. EFFECTS ON PLACEBO OUTCOMES

Notes: Figure shows event study estimates and 95 percent confidence intervals from equation (1) of the effect of first coffeehouse opening on outcomes it should leave unmoved. Outcomes are the decadal number in a parish of public houses, meaning taverns, inns, alehouses, wine houses, victuallers, and mum houses (Panel A), luxury shops (Panel B), non-scientific letters (Panel C), and commercial premises, signed premises other than drinking houses (Panel D), and the decadal number of baptisms and deaths per square kilometer (Panels E and F). Sample includes all 162 parishes from the 1600s to the 1690s, except for baptisms, 130 of 162 parishes from the 1600s to the 1660s, and deaths, 116 of 162 parishes from the 1600s to the 1690s; deaths are recorded in 119 parishes, but three of them are observed in a single decade and drop out of a design with parish and decade effects. Unit of observation is parish-decade. The parenthetical beside 0 reports the sample outcome mean one decade before treatment. Standard errors clustered by parish. [Return to pages 20, 21, 26]

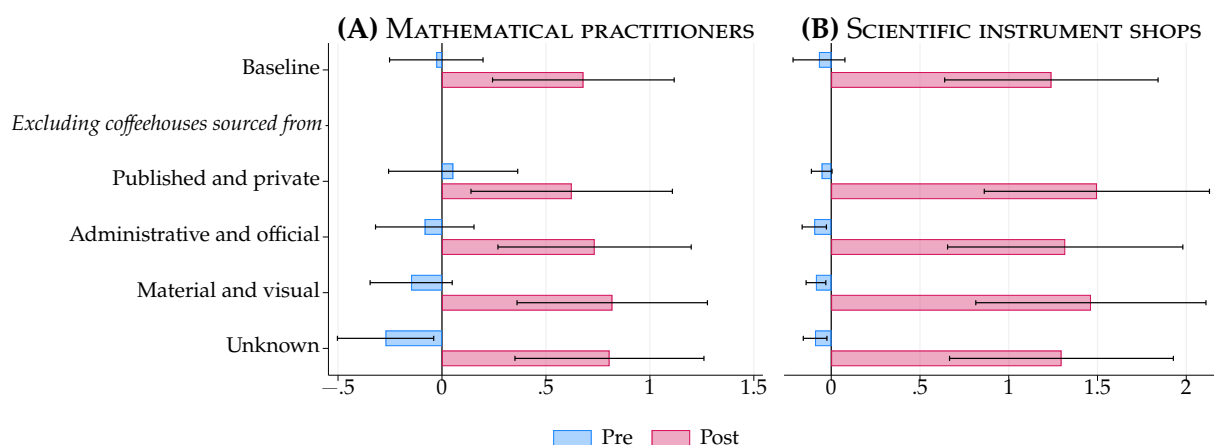


FIGURE C8. EFFECTS ON SCIENTIFIC ACTIVITY: DROPPING COFFEEHOUSE SOURCE CATEGORIES

Notes: Figure reports average pre-treatment and post-treatment effects, with 95 percent confidence intervals, from equation (1), on mathematical practitioners (Panel A) and scientific instrument shops (Panel B). In each alternative specification, we exclude coffeehouses first recorded in the indicated source category: published or private, administrative or official, material or visual, or unknown. The baseline uses all coffeehouse sources. Pre-treatment effects are averages of event study estimates from 50 to 20 years before opening; post-treatment effects are averages from opening through 50 years after. Sample includes all 162 parishes from the 1600s to the 1690s. Unit of observation is parish-decade. Standard errors clustered by parish. [Return to page 23]

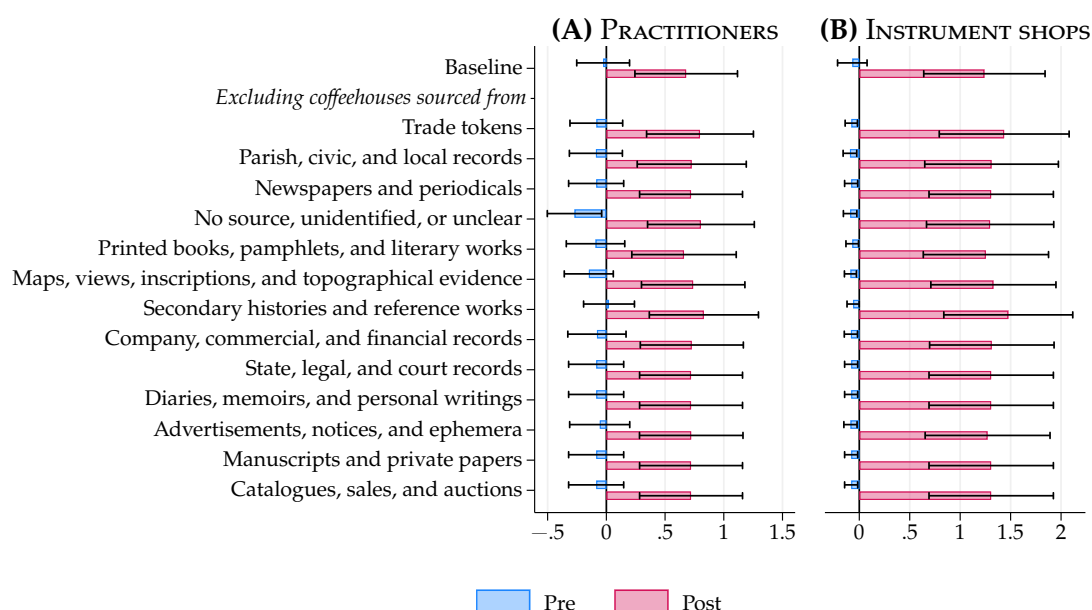


FIGURE C9. EFFECTS ON SCIENTIFIC ACTIVITY: DROPPING DETAILED COFFEEHOUSE SOURCE CATEGORIES

Notes: Figure reports average pre-treatment and post-treatment effects, with 95 percent confidence intervals, from equation (1), on mathematical practitioners (Panel A) and scientific instrument shops (Panel B). In each alternative specification, we exclude coffeehouses first recorded in the indicated detailed source category. The baseline uses all coffeehouse sources. Pre-treatment effects are averages of event study estimates from 50 to 20 years before opening; post-treatment effects are averages from opening through 50 years after. Sample includes all 162 parishes from the 1600s to the 1690s. Unit of observation is parish-decade. Standard errors clustered by parish. [Return to page 23]

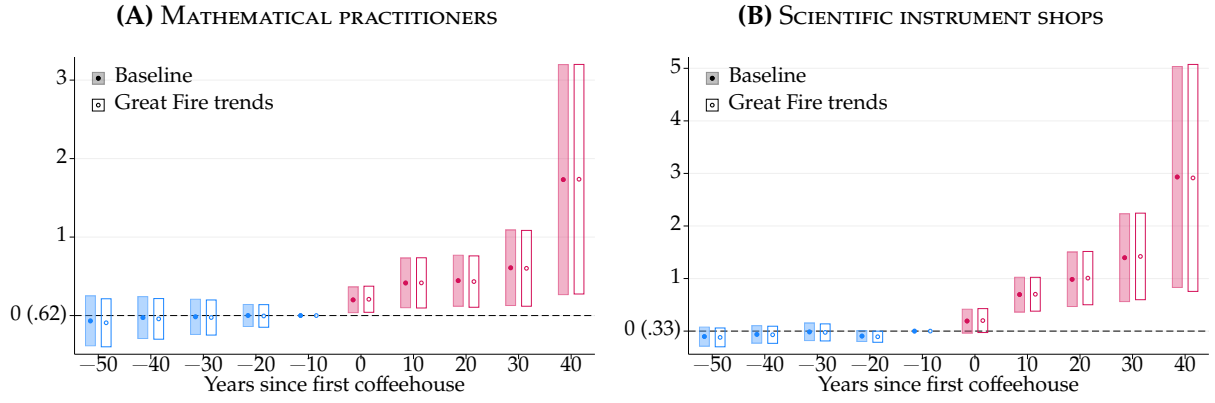


FIGURE C10. EFFECTS ON SCIENTIFIC ACTIVITY: CONTROL FOR EXPOSURE TO THE GREAT FIRE

Notes: Figure shows event study estimates and 95 percent confidence intervals of the effect of first coffeehouse opening on scientific activity, from the baseline specification in equation (1), filled markers and shaded intervals, and the same equation including fixed effects for the interaction between an indicator for exposure to the Great Fire and each decade, hollow markers and outlined intervals. Outcomes are the decadal number of mathematical practitioners and scientific instrument shops in a parish. Sample includes all 162 parishes from the 1600s to the 1690s. Unit of observation is parish-decade. The parenthetical beside 0 reports the sample outcome mean one decade before treatment. Standard errors clustered by parish. [Return to page 22]

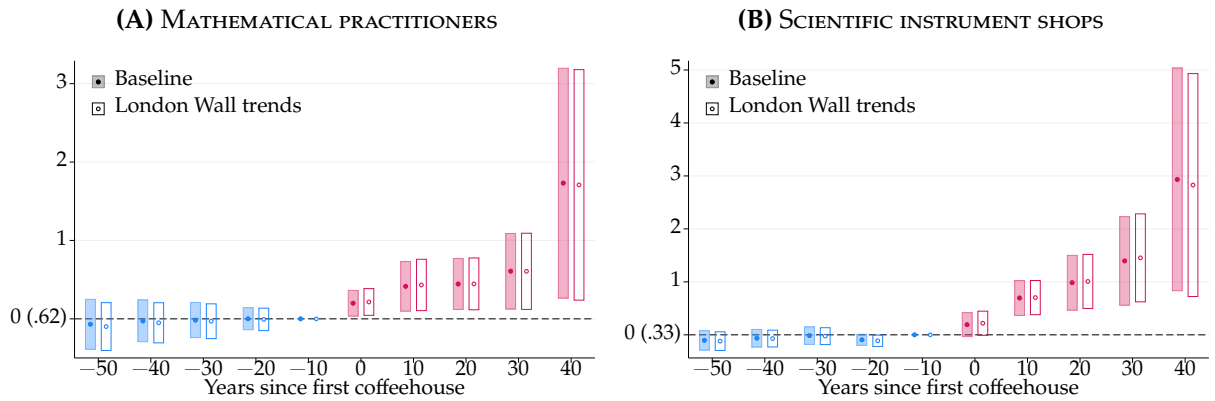


FIGURE C11. EFFECTS ON SCIENTIFIC ACTIVITY: CONTROL FOR WITHIN LONDON WALL

Notes: Figure shows event study estimates and 95 percent confidence intervals of the effect of first coffeehouse opening on scientific activity, from the baseline specification in equation (1), filled markers and shaded intervals, and the same equation including fixed effects for the interaction between an indicator for whether a parish lies within the London Wall and each decade, hollow markers and outlined intervals. Outcomes are the decadal number of mathematical practitioners and scientific instrument shops in a parish. Sample includes all 162 parishes from the 1600s to the 1690s. Unit of observation is parish-decade. The parenthetical beside 0 reports the sample outcome mean one decade before treatment. Standard errors clustered by parish. [Return to page 22]

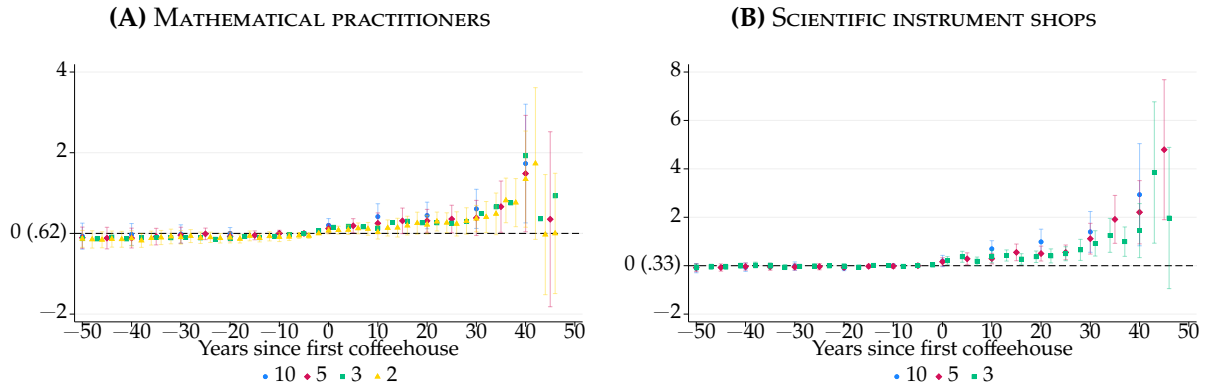


FIGURE C12. EFFECTS ON SCIENTIFIC ACTIVITY BY TIME-FRAME LENGTH

Notes: Figure shows event study estimates and 95 percent confidence intervals from equation (1) of the effect of first coffeehouse opening on scientific activity. Outcomes are the number of mathematical practitioners and scientific instrument shops in a parish, estimated across varying time-frame lengths. The unit of observation is a parish-by-time-frame-length cell (decadal, quinquennial, triennial, or biennial). The parenthetical beside 0 reports the sample outcome mean one decade before treatment. Standard errors clustered by parish. [Return to page 23]

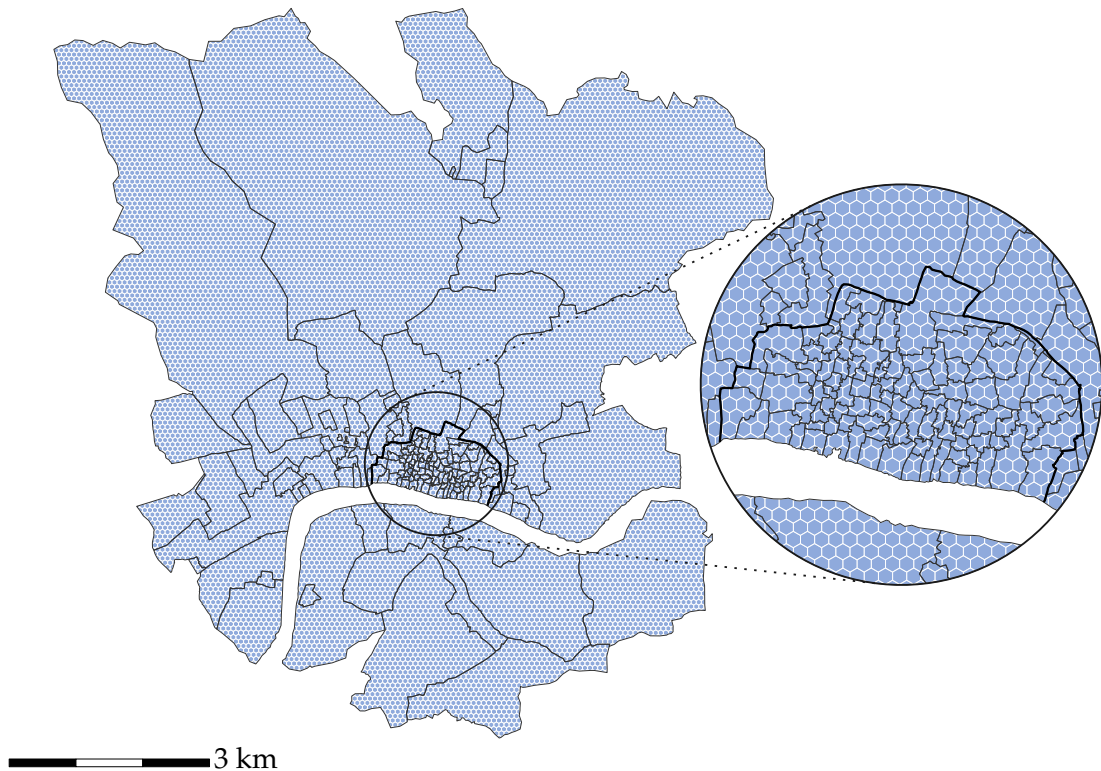


FIGURE C13. HEXAGONAL PARTITION OF LONDON

Notes: Map of London divided into 15,351 hexagons of 5,000 m² each, with white lines for hexagon boundaries and black lines for the 162 parish boundaries. [Return to pages 24, 34]

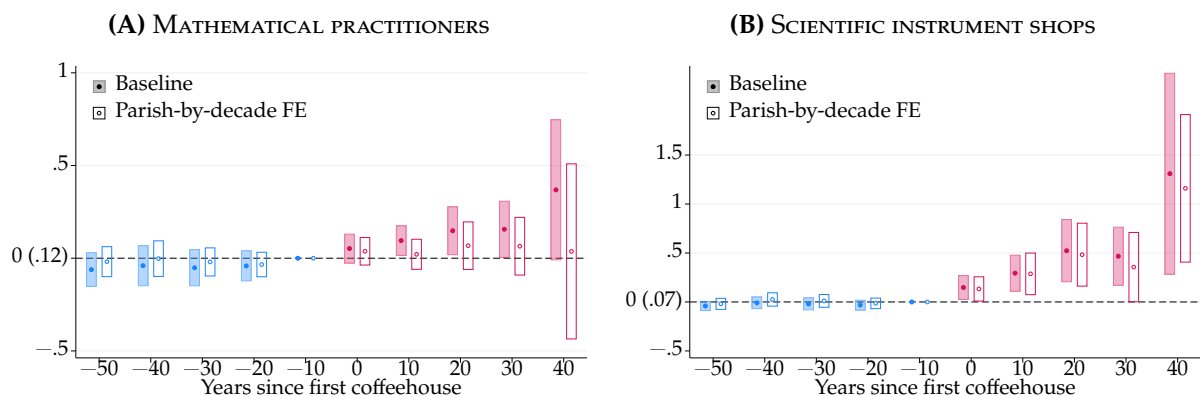


FIGURE C14. EFFECTS ON SCIENTIFIC ACTIVITY: GRID CELL LEVEL

Notes: Figure shows event study estimates and 95 percent confidence intervals from equation (1) of the effect of first coffeehouse opening on scientific activity, without and with parish-by-decade fixed effects. Outcomes are the decadal number of mathematical practitioners and scientific instrument shops in a 5,000 m² hexagon. Sample includes all 15,351 hexagons of size 5,000 m² (see Appendix Figure C13) from the 1600s to the 1690s. Unit of observation is hexagon-decade. The parish-by-decade specification drops the 593 hexagons whose centroid falls outside every parish, none of them treated; 155 of the 162 parishes contain at least one hexagon and 132 at least two, and 56 of the 59 parishes containing a treated hexagon also contain untreated hexagons. The parenthetical beside 0 reports the sample outcome mean one decade before treatment. Standard errors clustered by parish. [Return to page 24]

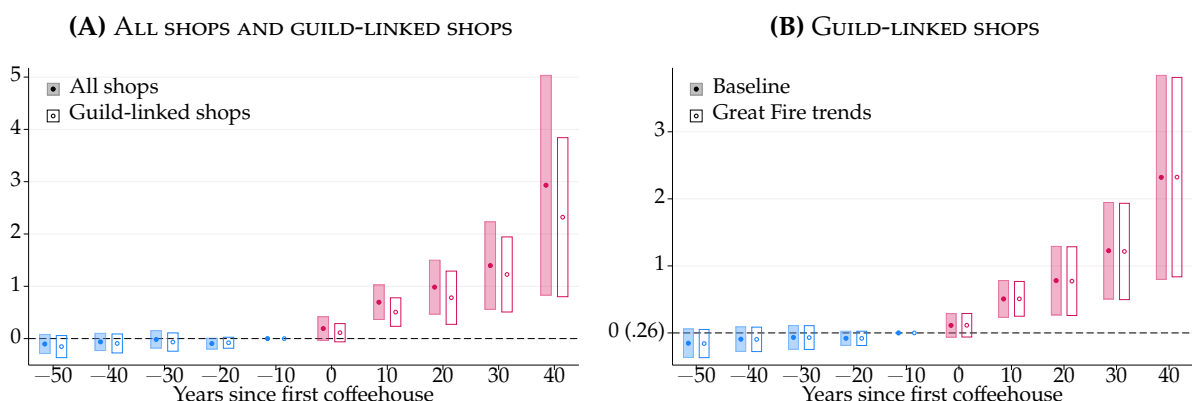


FIGURE C15. EFFECTS ON SCIENTIFIC INSTRUMENT SHOPS IN GUILD

Notes: Figure shows event study estimates and 95 percent confidence intervals from equation (1) of the effect of first coffeehouse opening on scientific instrument shops. Panel A counts all shops, filled markers and shaded intervals, and only those operated by instrument makers for whom a guild is recorded, hollow markers and outlined intervals; the two series count different sets of shops, so the axis carries no reference mean: the sample mean one decade before treatment is 0.33 for all shops and 0.26 for guild-linked shops. Panel B counts guild-linked shops, without controls, filled markers and shaded intervals, and including fixed effects for the interaction between an indicator for exposure to the Great Fire and each decade, hollow markers and outlined intervals; the parenthetical beside 0 reports the sample outcome mean one decade before treatment. The guild-linked shops are the subset of makers whose records are best preserved. Sample includes all 162 parishes from the 1600s to the 1690s. Unit of observation is parish-decade. Standard errors clustered by parish. [Return to page 22]

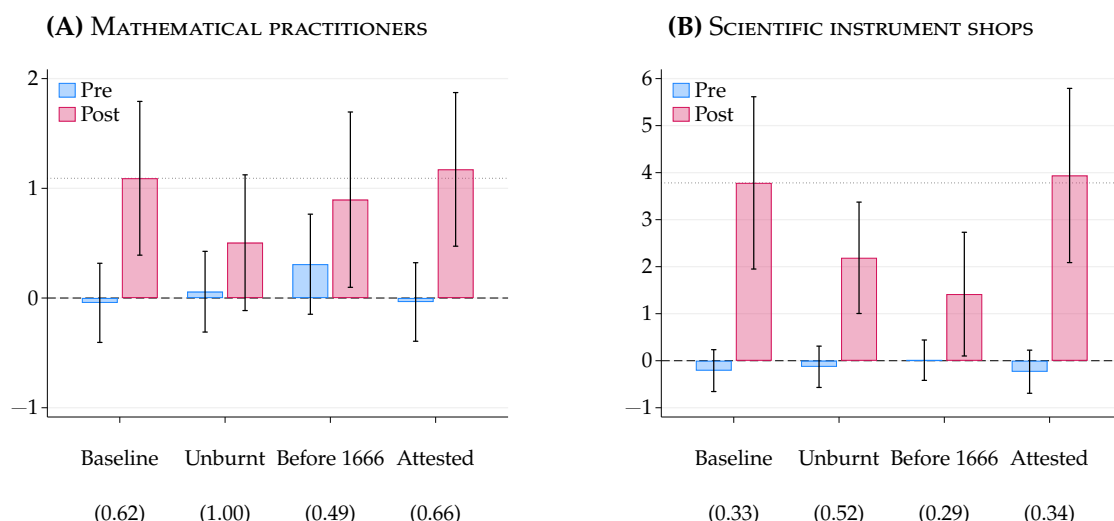


FIGURE C16. EFFECTS ON SCIENTIFIC ACTIVITY: RECORDS LOST IN THE GREAT FIRE

Notes: Figure reports average pre-treatment and post-treatment effects of first coffeehouse opening, with 95 percent confidence intervals, from equation (1), each divided by the mean of the outcome among treated parishes one decade before treatment in its own sample, which the parenthetical under each label reports. Baseline uses all 162 parishes from the 1600s to the 1690s. Unburnt keeps the 67 parishes outside the area burnt in the Great Fire of 1666, where no records burned. Before 1666 ends the panel in the 1660s, so that every record in the sample faced the same loss; parishes first treated from the 1670s serve as controls there. Attested drops the five treated parishes with no coffeehouse recorded after 1666 and keeps the other 56 at their baseline entry dates. Pre-treatment effects are averages of event study estimates from 50 to 20 years before opening; post-treatment effects are averages from opening through 50 years after. The dotted line marks the baseline post-treatment effect. Unit of observation is parish-decade. Standard errors clustered by parish. [Return to page 22]

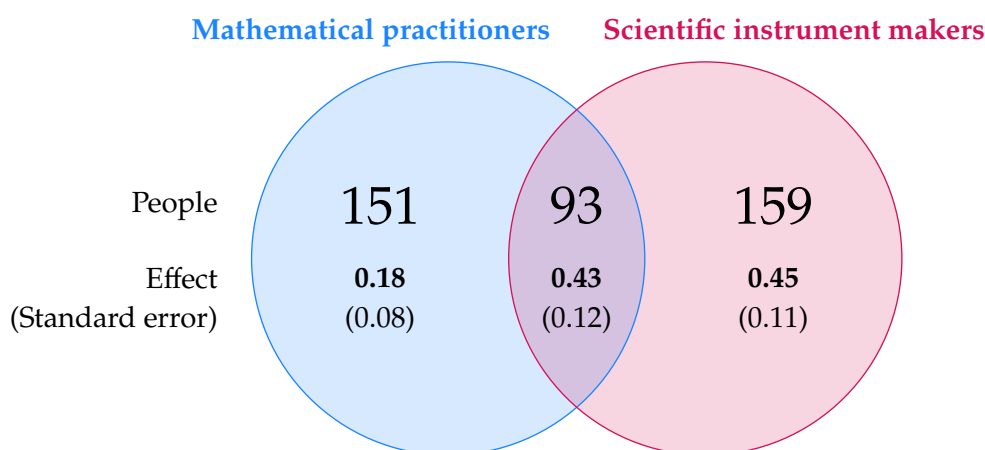


FIGURE C17. OVERLAP BETWEEN THE OUTCOME ROSTERS

Notes: Figure partitions the two outcome rosters: the left circle is the 244 mathematical practitioners of Taylor (1954) flourishing in London in the 1600s, the right circle the 252 scientific instrument makers recorded in SIMON and active in the same century, and the intersection the 93 people a match on name, generation, and dates links across the two. Beneath each count is the aggregate effect of first coffeehouse opening from equation (1) estimated on that region alone, with its standard error, clustered by parish, in parentheses. A person recorded in both rosters is dated from the earlier of the two first-flourishing years to the later of the two last and is placed at every address either roster records. Every region counts people rather than shops, so a person enters once per parish-decade and the maker series differs from the shop count used in the main text. Two Taylor entries name a father and son at once and admit no one-to-one link, so the intersection is a lower bound. Sample includes all 162 parishes from the 1600s to the 1690s. Unit of observation is parish-decade. [Return to pages 23, 36]

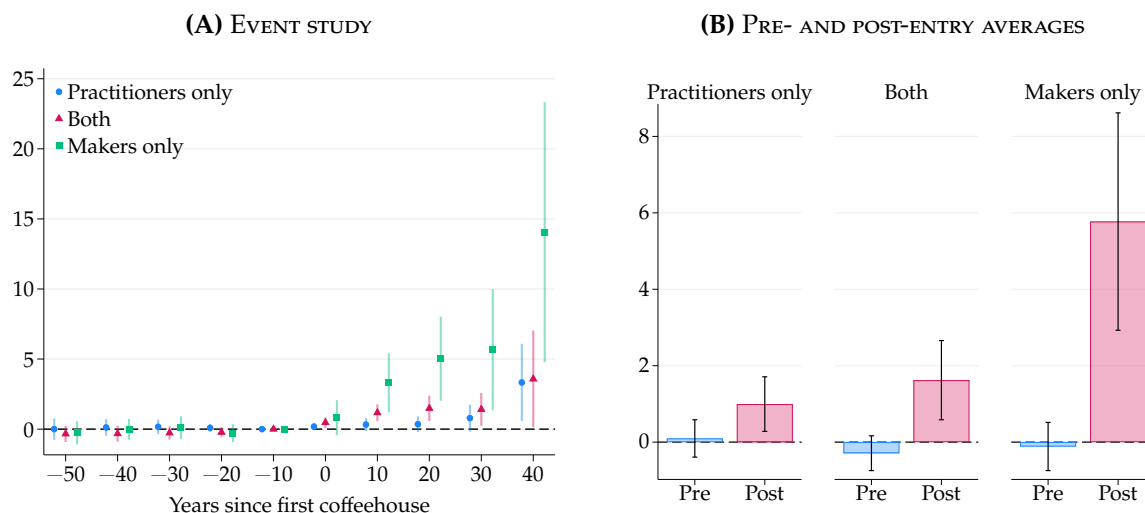


FIGURE C18. EFFECTS BY CELL OF THE ROSTER PARTITION

Notes: Figure shows the effect of first coffeehouse opening from equation (1) estimated separately on each region of the partition in Figure C17: practitioners recorded only by Taylor (1954) (Practitioners only), people recorded in both rosters (Both), and instrument makers recorded only by SIMON (Makers only). The three regions start from pre-entry means of 0.36, 0.36, and 0.11, so levels are not comparable across them and both panels are expressed relative to the pre-entry mean of the region they belong to. Panel A divides each event study estimate and interval by that mean. Panel B divides the outcome itself by that mean before estimation, so the standard errors scale with it, and averages the resulting coefficients over the pre-entry and post-entry windows, giving 0.10, -0.29 , and -0.11 before entry and 0.99, 1.62, and 5.77 after it. Whiskers and shaded intervals are 95 percent intervals. Every region counts people rather than shops, so a person enters once per parish-decade and the maker series differs from the shop count used in the main text. Sample includes all 162 parishes from the 1600s to the 1690s. Unit of observation is parish-decade. Standard errors clustered by parish. [Return to page 23]

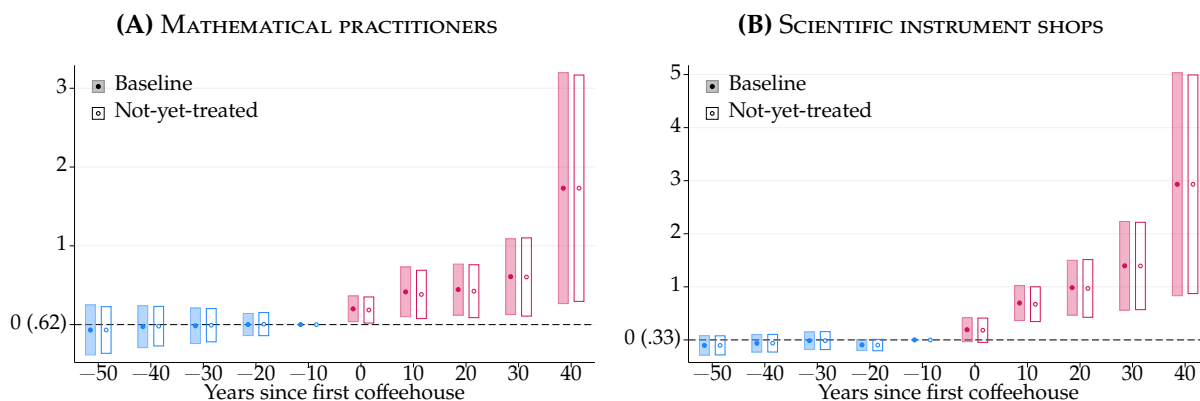


FIGURE C19. EFFECTS ON SCIENTIFIC ACTIVITY: NEVER- AND NOT-YET-TREATED CONTROLS

Notes: Figure shows event study estimates and 95 percent confidence intervals from our baseline specification, filled markers and shaded intervals, and the estimator of Callaway and Sant'Anna (2021), hollow markers and outlined intervals, where the control group consists of both never- and not-yet-treated parishes. That estimator omits the reference period rather than reporting it as a zero, so it is drawn here at zero to keep the two series aligned. Outcomes are the decadal number of mathematical practitioners and scientific instrument shops. Sample includes all 162 parishes from the 1600s to the 1690s. Unit of observation is parish-decade. The parenthetical beside 0 reports the sample outcome mean one decade before treatment. Standard errors clustered by parish. [Return to page 24]

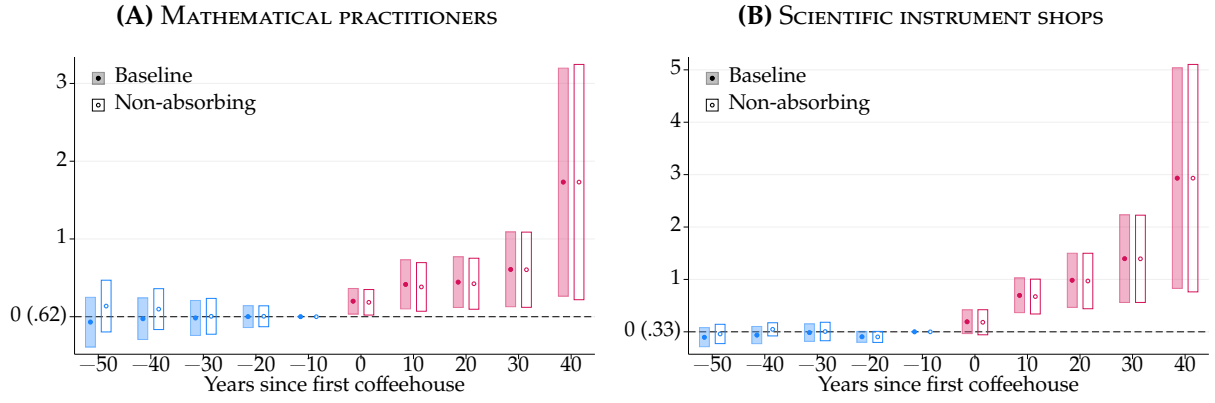


FIGURE C20. EFFECTS ON SCIENTIFIC ACTIVITY: NON-ABSORBING TREATMENT

Notes: Figure shows event study estimates and 95 percent confidence intervals from our baseline specification, filled markers and shaded intervals, and the estimator of [de Chaisemartin and d'Haultfoeuille \(2024\)](#) for the effect of coffeehouse presence on scientific activity, hollow markers and outlined intervals. Outcomes are the decadal number of mathematical practitioners and scientific instrument shops. Sample includes all 162 parishes from the 1600s to the 1690s. Unit of observation is parish-decade. The parenthetical beside 0 reports the sample outcome mean one decade before treatment. Standard errors clustered by parish. [Return to page 23]

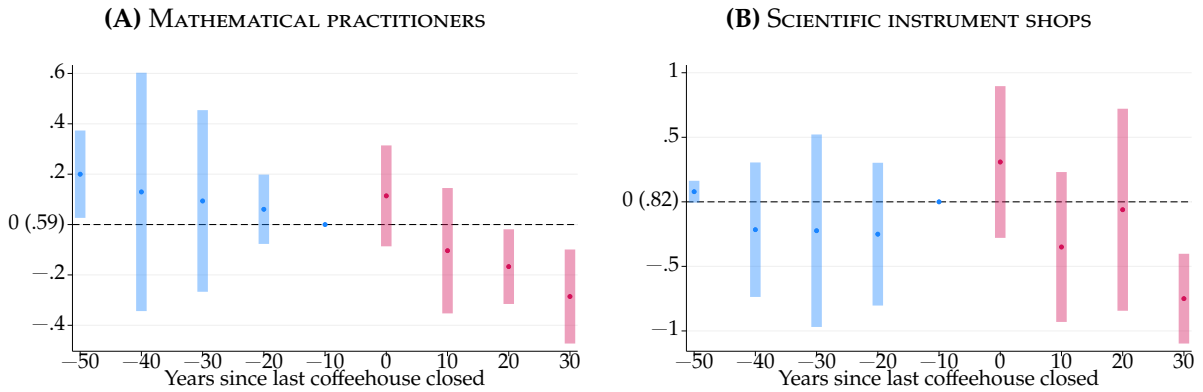


FIGURE C21. EFFECTS ON SCIENTIFIC ACTIVITY: CLOSING DOWN THE LAST COFFEEHOUSE

Notes: Figure shows event study estimates and 95 percent confidence intervals based on [de Chaisemartin and d'Haultfoeuille \(2024\)](#) for the effect of closing the last coffeehouse on scientific activity. Outcomes are the decadal number of mathematical practitioners and scientific instrument shops. Sample includes all 162 parishes from the 1600s to the 1690s. Unit of observation is parish-decade. The parenthetical beside 0 reports the sample outcome mean one decade before treatment. Standard errors clustered by parish. [Return to page 24]

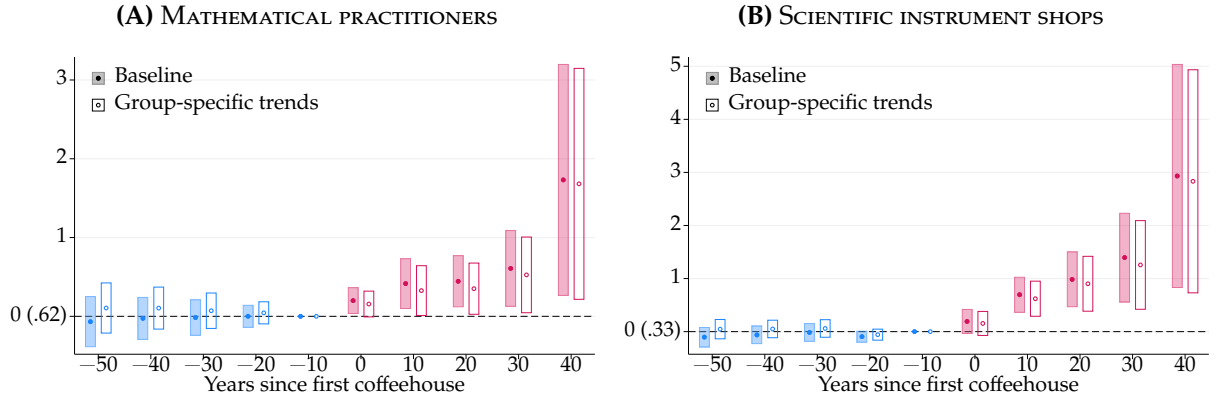


FIGURE C22. EFFECTS ON SCIENTIFIC ACTIVITY: GROUP-SPECIFIC TRENDS

Notes: Figure shows event study estimates and 95 percent confidence intervals from equation (1) of the effect of first coffeehouse opening on scientific activity, without adjustment, filled markers and shaded intervals, and after adjusting for group-specific trends, hollow markers and outlined intervals. Outcomes are the decadal number of mathematical practitioners and scientific instrument shops in a parish, and the parenthetical beside 0 reports the mean of the unadjusted outcome, so the two series share a reference. Following Goodman-Bacon (2021), we estimate a linear outcome trend by treatment cohort using only pre-treatment observations and then subtract the estimated trend from the outcome. Sample includes all 162 parishes from the 1600s to the 1690s. Unit of observation is parish-decade. The parenthetical beside 0 reports the sample outcome mean one decade before treatment. Standard errors clustered by parish. [Return to page 25]

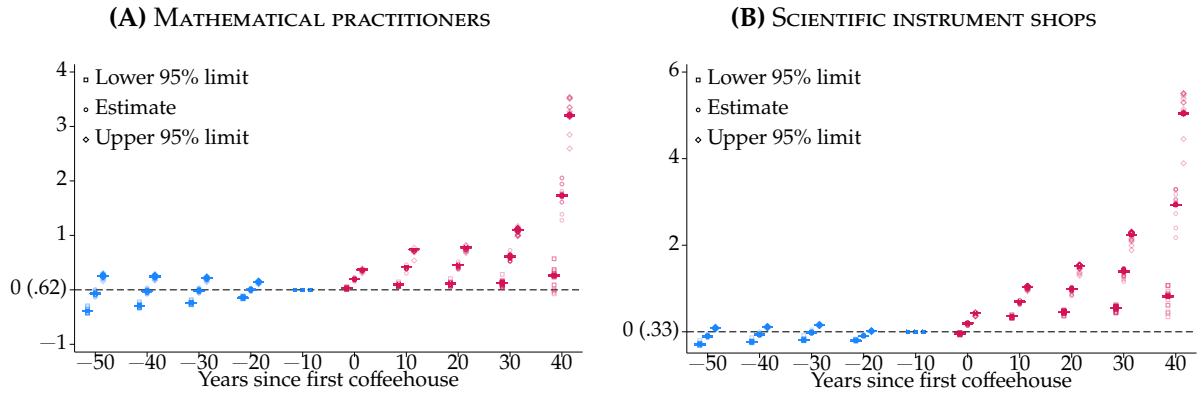


FIGURE C23. EFFECTS ON SCIENTIFIC ACTIVITY: LEAVE-ONE-OUT SENSITIVITY CHECK

Notes: Figure shows boxplots, including outliers, of event study estimates and their 95 percent confidence limits from equation (1) for the effect of first coffeehouse opening on scientific activity. Outcomes are the decadal number of mathematical practitioners and scientific instrument shops in a parish. The boxplots use 162 leave-one-out samples from the 1600s to the 1690s, with each parish omitted once. Unit of observation is parish-decade. The parenthetical beside 0 reports the sample outcome mean one decade before treatment. Standard errors clustered by parish. [Return to page 25]

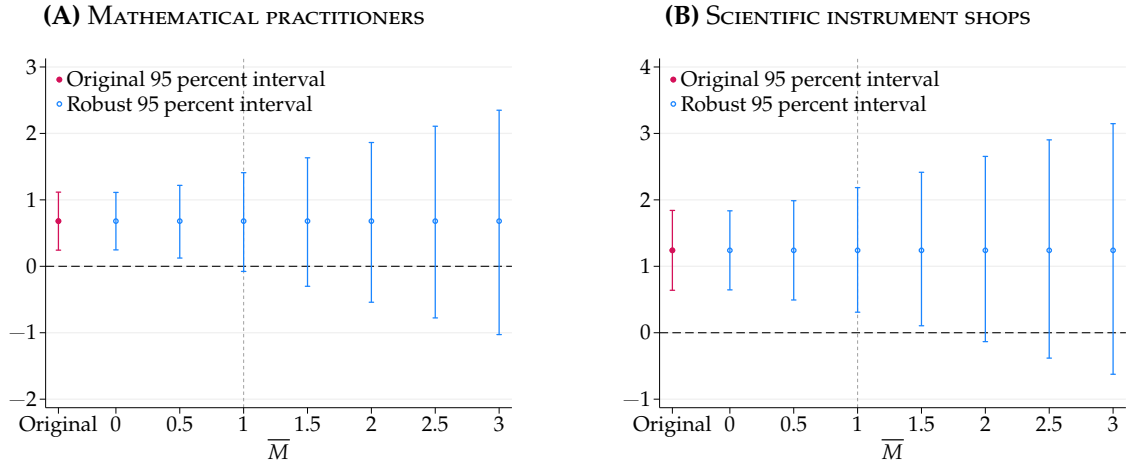


FIGURE C24. EFFECTS ON SCIENTIFIC ACTIVITY: VIOLATIONS OF PARALLEL TRENDS

Notes: Figure shows 95 percent confidence intervals for the average post-treatment effect of first coffeehouse opening from equation (1) that allow parallel trends to fail, following [Rambachan and Roth \(2023\)](#). Under their relative-magnitudes restriction, the change in the violation of parallel trends from one decade to the next after entry is at most $\bar{M}$ times the largest such change before entry; the intervals use their conditional least-favorable hybrid. The interval labeled Original is the conventional one, with its point estimate, and the vertical line marks $\bar{M} = 1$. On a grid of 0.1, the lower bound stays above zero up to $\bar{M} = 0.8$ for mathematical practitioners and $\bar{M} = 1.7$ for scientific instrument shops. Post-treatment effects are averages of event study estimates from opening through 50 years after. Sample includes all 162 parishes from the 1600s to the 1690s. Unit of observation is parish-decade. Standard errors clustered by parish. [Return to page 25]

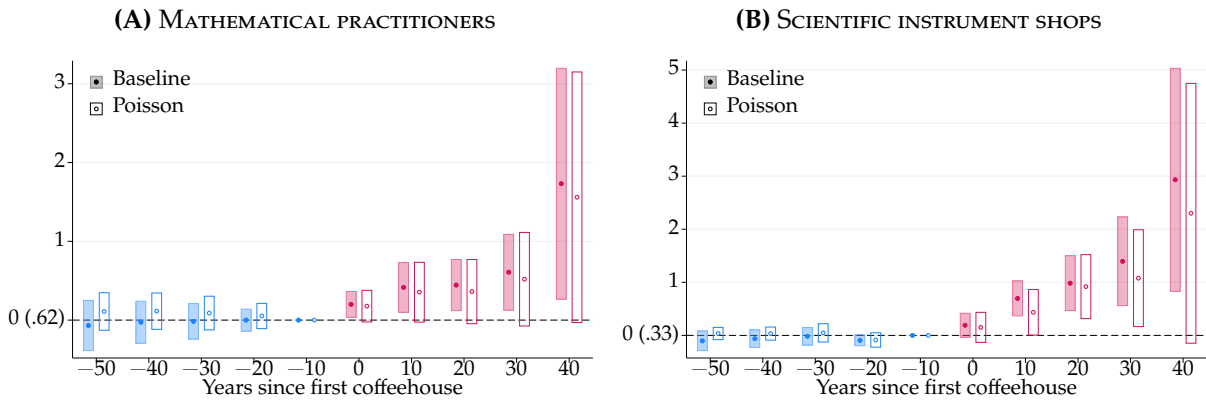


FIGURE C25. EFFECTS ON SCIENTIFIC ACTIVITY: POISSON SPECIFICATION

Notes: Figure shows event study estimates and 95 percent confidence intervals of the effect of first coffeehouse opening on scientific activity, from the baseline linear specification in equation (1), filled markers and shaded intervals, and the Poisson specification in equation (3), hollow markers and outlined intervals. Outcomes are the decadal number of mathematical practitioners and scientific instrument shops in a parish. Poisson estimation is restricted to event times from 50 years before to 50 years after first opening, retaining all never-treated parishes, and replaces parish with cohort effects. Sample includes all 162 parishes from the 1600s to the 1690s. Unit of observation is parish-decade. The parenthetical beside 0 reports the sample outcome mean one decade before treatment. Standard errors clustered by parish. [Return to page 24]

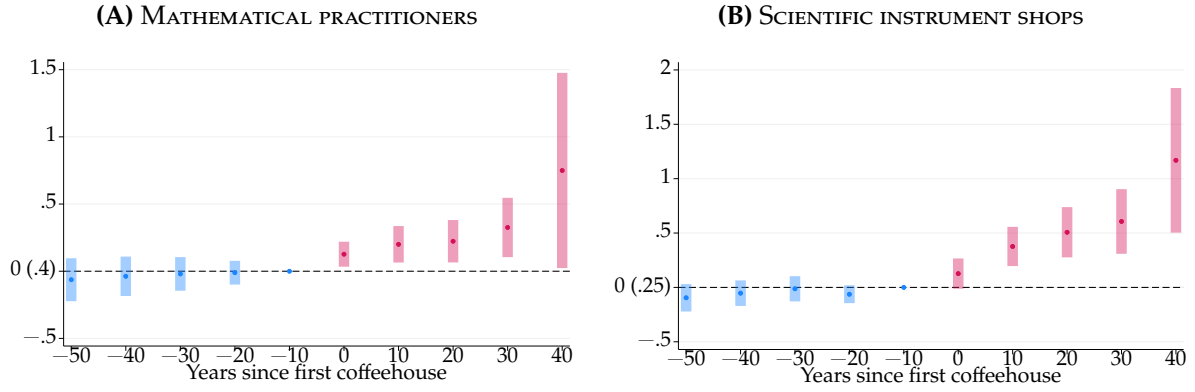


FIGURE C26. EFFECTS ON SCIENTIFIC ACTIVITY: INVERSE HYPERBOLIC SINE

Notes: Figure shows event study estimates and 95 percent confidence intervals for the effect of first coffeehouse opening on scientific activity, obtained by estimating equation (1) with the inverse hyperbolic sine of the outcome: $\text{IHS}(y_{it}) = \alpha + \sum_{g=1650}^{1690} \sum_{s=1600}^{1690} \beta_{gs} \mathbb{1}[s = t] \mathbb{1}[g = E_i] + \gamma_i + \delta_t + \epsilon_{it}$. Outcomes are the decadal number of mathematical practitioners and scientific instrument shops in a parish. Sample includes all 162 parishes from the 1600s to the 1690s. Unit of observation is parish-decade. The parenthetical beside 0 reports the sample outcome mean one decade before treatment. Standard errors clustered by parish. [Return to page 24]

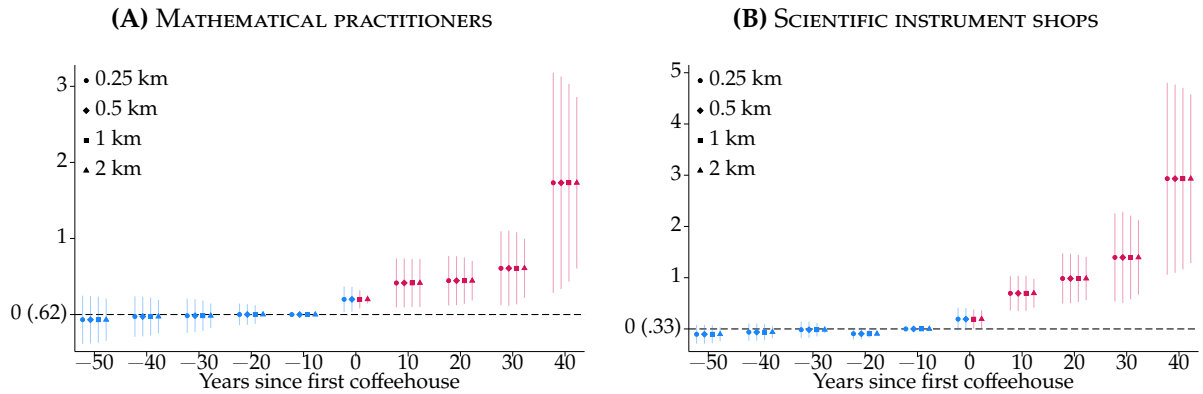


FIGURE C27. EFFECTS ON SCIENTIFIC ACTIVITY: CONLEY (1999) STANDARD ERRORS

Notes: Figure shows event study estimates and 95 percent confidence intervals from equation (1) of the effect of first coffeehouse opening on scientific activity. Outcomes are the decadal number of mathematical practitioners and scientific instrument shops in a parish. Sample includes all 162 parishes from the 1600s to the 1690s. Unit of observation is parish-decade. The parenthetical beside 0 reports the sample outcome mean one decade before treatment. Standard errors follow Conley (1999): they allow for heteroskedasticity, for correlation between any two decades of the same parish, and for correlation within a decade between parishes whose centroids lie within the cutoff, with weights declining linearly in distance; the cutoffs are 0.25, 0.5, 1, and 2 kilometers. At 2 kilometers the estimated variance matrix is not positive semidefinite, and the interval 20 years before opening cannot be computed. [Return to page 24]

C.13. TABLES

TABLE C1—BALANCE OF PRE-TREATMENT CHANGES

	Observed parishes	Treatment parishes	Control parishes	Normalized difference
<i>Panel A. Demographics</i>				
Baptisms	130	31.49 (73.83)	17.93 (76.60)	0.18
Baptisms per km ²	130	104.57 (196.59)	107.73 (239.48)	−0.01
Deaths	91	26.44 (100.91)	7.08 (83.22)	0.21
Deaths per km ²	91	10.61 (183.58)	37.26 (286.36)	−0.11
<i>Panel B. Social capital and commerce</i>				
Booksellers	162	−0.04 (2.15)	−0.01 (0.56)	−0.02
Clay-pipe finds	162	0.16 (0.91)	0.26 (1.28)	−0.09
Commercial premises	162	0.50 (1.02)	0.10 (0.37)	0.52
Dissenting meetings	162	0.08 (0.41)	0.03 (0.17)	0.17
Luxury shops	162	0.11 (0.33)	0.09 (0.37)	0.06
Pubs	162	4.16 (4.67)	0.67 (1.63)	1.00
Theaters	162	0.02 (0.24)	−0.03 (0.16)	0.25
<i>Panel C. Scientific activity before 1652</i>				
Instrument shops	162	0.09 (0.32)	0.06 (0.36)	0.08
Non-scientific letters	162	0.68 (2.52)	0.24 (1.47)	0.21
Practitioners	162	0.07 (0.81)	0.10 (0.52)	−0.05
Scientific letters	162	0.03 (0.40)	−0.03 (0.68)	0.11
Parishes	162	61	101	

Notes: Table compares the change in each pre-treatment characteristic between the 1600s–1610s and the 1630s–1640s, for London parishes that received a coffeehouse and parishes that never did. Entries are mean changes with standard deviations in parentheses. Characteristics that are fixed or recorded once, reported in Table 1, have no change to measure. Apothecaries are omitted because the source begins in 1617. The normalized difference is $(\bar{\Delta x}_T - \bar{\Delta x}_C) / \sqrt{(s_T^2 + s_C^2)/2}$. [Return to pages 18, 22, 20, 47]

TABLE C2—CORRELATION BETWEEN COFFEEHOUSES AND PUBS

	Coffeehouses (standardized)				Coffeehouses > 0			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Pubs (standardized)	0.408 (0.067)	0.264 (0.094)	0.412 (0.068)	0.251 (0.103)				
Pubs > 0					0.169 (0.025)	0.073 (0.035)	0.154 (0.024)	0.017 (0.032)
Mean dep. var.	0	0	0	0	0.107	0.107	0.107	0.107
Observations	1620	1620	1620	1620	1620	1620	1620	1620
Parish FE		✓		✓		✓		✓
Decade FE			✓	✓			✓	✓

Notes: Table shows the relationship between coffeehouses and pubs. Columns 1–4 show effect sizes from regressions of the number of coffeehouses on the number of pubs, standardized to mean 0 and standard deviation 1. Beginning with the raw correlation in Column 1, the specifications add parish fixed effects, decade fixed effects, and both parish and decade fixed effects. Columns 5–8 show the corresponding extensive-margin regressions of indicators for coffeehouses and pubs with the same sequence of fixed effects. The sample includes all 162 parishes from the 1600s to the 1690s. The unit of observation is the parish-decade. Standard errors clustered by parish. [Return to pages 29, 16]

TABLE C3—COFFEEHOUSE CITIES RELATIVE TO COMPARISON CITIES, ALTERNATIVE SAMPLES

	Pre-entry mean		Post-entry difference	
	Titles	Makers	Titles	Makers
Baseline	5.95	3.73	4.97 (2.30)	4.89 (2.97)
Country-by-decade fixed effects	5.95	3.73	4.25 (2.38)	3.15 (3.21)
Without London	5.22	2.42	1.67 (1.29)	0.98 (0.49)
Without London, first two decades	5.22	2.42	2.29 (1.29)	0.91 (0.53)
Country-by-decade fixed effects, without London	5.22	2.42	2.23 (1.48)	1.55 (0.69)
Without Paris	5.56	3.17	4.26 (2.26)	4.73 (3.05)
Without any capital	3.56	1.56	1.26 (0.93)	0.23 (0.33)
Only France, Germany, the Netherlands, and the UK	6.96	4.67	5.18 (1.42)	5.92 (0.38)

Notes: Table reports the equal-weighted post-entry difference from entry through thirty years after, from equation (1) estimated at the city level with never-treated controls and log population in 1600 and growth from 1500 to 1600 by decade, the average of the estimates in Figure 13. The row marked first two decades averages from entry through twenty years after instead, since the third decade rests on the seven coffeehouse cities other than London that opened by the 1660s. Country-by-decade fixed effects add a separate intercept for every country in every decade. Without any capital drops the ten coffeehouse cities and eleven comparison cities that were seats of government. The last row keeps only the cities of France, Germany, the Netherlands, and the UK, which hold 27 of the 37 coffeehouse cities. Pre-entry means are the mean of the outcome among coffeehouse cities in the decade before entry, in the sample of the row. Titles are applied mathematics titles; makers are scientific instrument makers. Sample includes all 200 cities from the 1600s to the 1690s, less those a row drops. Unit of observation is city-decade. Standard errors clustered by city in parentheses. [Return to pages 34, 25, 43]

TABLE C4—COFFEEHOUSE CITIES RELATIVE TO COMPARISON CITIES, ALTERNATIVE ENTRY DECADES

	Pre-entry mean		Post-entry difference	
	Titles	Makers	Titles	Makers
Baseline	5.95	3.73	4.97 (2.30)	4.89 (2.97)
Earliest admissible decade	5.74	3.56	4.75 (2.24)	4.76 (2.92)
Latest admissible decade	6.47	3.79	4.59 (1.42)	6.09 (0.53)
Without the recoded cities	5.76	3.76	5.45 (1.58)	6.69 (0.61)

Notes: Table reports post-entry differences as in Table C3 under alternative entry decades. The sources admit more than one entry decade for Rome (1650s–1690s), Venice (1680s or dropped), Utrecht (1670s or 1680s), Nuremberg (1680s or 1690s), Bremen (1660s or 1670s), Haarlem (1680s or never), Alkmaar (1680s or 1690s), and Antwerp (1680s or 1690s); the earliest and latest rows assign every one of them to the end of its interval, and the last row, without the recoded cities, drops them together with The Hague and Leiden, whose dates were recoded on source grounds. Haarlem and Alkmaar are comparison cities in the baseline. Pre-entry means are the mean of the outcome among coffeehouse cities in the decade before entry, in the sample of the row. Titles are applied mathematics titles; makers are scientific instrument makers. Sample includes all 200 cities from the 1600s to the 1690s, less those the last row drops. Unit of observation is city-decade. Standard errors clustered by city in parentheses. [Return to pages 33, 25]

D. DATA APPENDIX

D.1. TREATMENT VARIABLE

Treatment is the opening of the first coffeehouse in a parish, so the treatment date is constructed from the coffeehouse counts defined here. Appendix Section D.10 assesses their completeness against independent records.

Coffeehouses, 1652–1699 (Lillywhite, 1963, 1972). The number of coffeehouses established between 1652 and 1699 in a given parish is based on public houses identified as coffeehouses by Lillywhite (1963, 1972), excluding entries explicitly marked as duplicates. When Lillywhite (1963, 1972) is uncertain whether two mentions refer to the same establishment, both are retained as separate entries. For any year t , the count includes all coffeehouses whose recorded years of activity encompass t , as determined by their earliest and latest mentions. In cases where only a decade is known (9 of the 286 coffeehouses), activity years are set to the full decade range (e.g., 1660s as 1660–1669). The count also includes chocolate houses, which Lillywhite (1963, 1972) treats as one subject with coffeehouses and at times cannot separate, typing a house “Chocolate or Coffee-house?”. Five of the 286 are recorded as chocolate houses, four of them in St. James Westminster, a parish treated from 1659 on Harper’s Coffee House. The fifth, in Queen’s Head Alley, is the earliest record in St. Botolph Bishopsgate, at 1657; without it that parish enters in the 1660s.

D.2. OUTCOME VARIABLES

The two main outcomes, mathematical practitioners and scientific instrument shops, measure applied scientific activity in a parish and decade; they draw on Taylor (1954) and Clifton (1995); Hay et al. (2025) respectively. The remaining variables support the connection and mechanism analyses. All are counts, and a practitioner or maker recorded at addresses in more than one parish contributes to each of them.

Mathematical Practitioners, 1600–1699 (Taylor, 1954). The number of mathematical practitioners in a given parish and decade is based on Taylor (1954), restricted to practitioners with at least one workplace located precisely enough to be geocoded within London and whose recorded flourishing overlaps 1600–1699. Each practitioner contributes a record for every decade from the decade containing the first year of flourishing to the decade containing the last, and is assigned to the parish of each recorded workplace. Within a parish-decade, a practitioner is counted once, however many workplaces Taylor (1954) records there; a practitioner with workplaces in more than one parish contributes to each.

Connections Between Mathematical Practitioners, 1600–1699 (Taylor, 1954; Warren et al., 2016). The number of connections in a given parish and decade counts the undirected pairs g_{ijt} defined in Appendix D.3, assigned to the parish of practitioner i and restricted to decades in which both i and j are flourishing. Connections are further split by whether each endpoint is a maker of scientific instruments or a theorist, defined

as any other mathematical practitioner, so that a connection from a theorist in one parish to a maker elsewhere is counted in the theorist's parish.

Scientific Instrument Shops, 1600–1699 (Clifton, 1995; Hay et al., 2025). The number of scientific instrument shops in a given parish and decade is based on Clifton (1995); Hay et al. (2025), which record a shop address and years of operation for each maker. Addresses are geocoded and assigned to parishes as described in Appendix D.11. A shop contributes a record for every decade encompassed by its recorded years of operation and is counted once per parish-decade.

Practical Instrument Shops, 1600–1699 (Clifton, 1995; Hay et al., 2025). Shops whose makers are recorded as producing mathematical, navigational, or surveying instruments, counted as above. These are instruments for measuring, calculating, and setting out: sectors, dividers, and slide rules for calculation; quadrants, nocturnals, and compasses for navigation; theodolites, circumferentors, levels, and waywisers for survey. Surviving objects in Hay et al. (2025) include a brass sector attributed to Elias Allen, a horary quadrant by Henry Sutton dated 1658, and a surveying compass by the same maker dated 1661.

Experimental Instrument Shops, 1600–1699 (Clifton, 1995; Hay et al., 2025). Shops whose makers are recorded as producing optical or philosophical instruments, counted as above. These are instruments for observation and for producing effects under controlled conditions: microscopes, refracting and reflecting telescopes, and lenses on the optical side; air pumps, barometers, and thermometers on the philosophical side. A maker recorded in both groups contributes a shop to each, so the practical and experimental counts need not sum to the total number of shops.

Guild-Linked Instrument Shops, 1600–1699 (Clifton, 1995; Hay et al., 2025). Scientific instrument shops restricted to makers for whom Clifton (1995); Hay et al. (2025) record a London guild, counted as above. Of the 252 makers active in the panel, 175 have a recorded guild. No guild was dedicated to instrument making, so makers joined existing companies, and most London makers did (Cormack, 2017, p. 79). The restriction therefore selects makers whose guild affiliation survives in the record, not makers who belonged to a guild, and it enters Section 4 as a check on selective visibility in the historical record rather than as a subgroup of independent interest.

The four outcomes that follow trace entry, transmission, and institutional reach in Section 5 rather than applied scientific activity directly.

Royal Society Fellows, 1600–1699 (The Royal Society, 2024). The number of Fellows of the Royal Society associated with a parish in a given decade, drawn from the Society's own catalogs and restricted to fellows whose recorded burial place or place of death lies in London. A fellow is assigned to the parish containing their geocoded burial site and contributes a record for every decade from the decade of election through the last decade of recorded fellowship. Burial is the only locator available at this resolution, so the variable places a fellow where their life ended rather than where they worked. It is therefore evidence on which parishes the Society's membership was eventually drawn

from, not on where any particular piece of scientific work was done, and the lag between working and being buried should be kept in mind when reading the post-treatment horizon. The measure counts fellows associated with a parish in a decade rather than fellows newly elected in it.

Instrument Makers' Apprentices, 1600–1699 (Clifton, 1995). The number of apprentice bindings recorded to scientific instrument makers in a parish and decade. Clifton (1995) records each master's apprentices as a free-text list of names and years, which we transcribe and parse into one record per named apprentice and year. Apprentices recorded without a name are dropped, so a master whose entry notes an unnamed cohort contributes only the apprentices it names and the count is a lower bound. An apprentice named in two years, for instance at binding and again at turnover, contributes to both, so the variable counts binding-years rather than distinct individuals. The source records the master an apprentice was bound to, not the shop at which he served. Where a master kept more than one shop in a year, the binding is therefore counted in each of that master's parishes, since the evidence does not identify which one; parish totals consequently sum to 280 rather than to the 252 distinct binding-years recorded.

Booksellers, 1600–1699 (Isaac et al., 2024). The number of members of the London book trades recorded at an address in a parish and decade, from the British Book Trade Index. Addresses are geocoded and assigned to parishes as described in Appendix D.11. A trader recorded more than once within the same parish is counted once, since repeated entries reflect alternative descriptions of a single address, and each trader contributes a record for every decade between the first and last recorded trading dates. Booksellers enter Section 4 as a covariate and Section 5 as an outcome.

Older-to-Younger Practitioner Connections, 1600–1699 (Taylor, 1954; Warren et al., 2016). Connections as defined above, restricted to those in which the practitioner assigned to the parish began flourishing in an earlier decade than the practitioner they are connected to. Each such connection is counted in the earlier practitioner's parish, so the measure captures ties running from established practitioners to younger ones rather than the reverse.

D.3. PRACTITIONER CONNECTIONS

Connections recorded by Taylor (1954) take various forms, including “friends,” “master,” “received advice from,” “associated with,” “visited,” “employed or consulted,” “corresponded with,” and “mentioned by.” Practitioners are recorded as connected whenever historical evidence indicates a social or intellectual association. Connections therefore suggest a general likelihood of influence or interaction. To extend this network, we turn to the *Six Degrees of Francis Bacon* (SDFB) database of Warren et al. (2016), which uses machine learning to infer influence or interaction among early modern individuals from mentions in *Oxford Dictionary of National Biography* (ODNB) biographies (Oxford Dictionary of National Biography, 2004). We identify ODNB entries for 75 of the 244 practitioners in our sample, all of whom are recorded in the SDFB.⁴² This procedure

⁴²We identify ODNB entries by linking practitioners to Wikidata items that carry an ODNB index number. Of the 244 practitioners, 90 are linked to a Wikidata item and 75 of those items carry the

identifies additional connections of the same type not recorded by [Taylor \(1954\)](#). The two sources together record 1,200 directed ties among 293 individuals. Collapsing reciprocated pairs leaves 631 undirected connections, and restricting both endpoints to the 244 practitioners in the panel leaves 475 among 204 of them: 326 recorded by [Taylor \(1954\)](#) alone, 83 by the SDFB alone, and 66 by both. Both distributions are skewed: the most connected practitioner has 54 connections, and 18 percent of active practitioner-decades carry none. Fifty of the 475 pairs are never active in the same decade and so never enter g_{ijt} , among them the father and son pairs [Taylor \(1954\)](#) records. The remaining 425 pairs are jointly active in 1,053 pair-decades across the panel, against 670 practitioner-decades of activity.

The data do not record when individuals became connected, and the direction of influence is uncertain. We therefore treat all connections as undirected, interpreting them as evidence that two individuals were socially or intellectually associated at some point. Formally, we set $g_{ij} = 1$ if practitioners i and j are ever connected in this sense, and 0 otherwise. To capture variation over time, we define $g_{ijt} = 1$ if $g_{ij} = 1$ and both i and j are active in period t , and 0 otherwise. This measure represents the maximum possible connectivity under the assumption that connections require both individuals to be active.⁴³

D.4. COVARIATES

The covariates are pre-treatment parish characteristics. They enter the balance comparisons in Tables 1 and C1, the inverse-probability weighted and regression-adjusted specifications in Section 4, and, for several of them, the placebo outcomes reported in the appendix figures. Table D2 gives the source for each. Definitions follow the order of the rows in Table 1 and close with exposure to the Great Fire, which is not a row in that table but enters Section 4 as a separate decade trend for exposed parishes. Counts recorded as an interval of activity are assigned to every decade the interval spans, and addresses are geocoded and assigned to parishes as described in Appendix D.11.

Households and Baptisms and Deaths ([Cummins et al., 2016](#)). Parish counts of households in 1638 and of baptisms and deaths by decade, taken from the parish register data assembled by [Cummins et al. \(2016\)](#). Densities divide each count by parish area in square kilometers. Baptisms and deaths are observed for 130 and 119 of the 162 parishes respectively, and the balance table reports the number of parishes with a nonmissing value in each row.

Median Tax Paid, 1582 ([Cummins et al., 2016](#)). The parish median assessment in the lay subsidy of 1582, in the units reported by [Cummins et al. \(2016\)](#).

identifier, so 75 is a lower bound on ODNB coverage in our sample.

⁴³The set of practitioner dyads (i, j) for which $g_{ij} = 1$ includes biological ties such as “father” and “son” recorded by [Taylor \(1954\)](#). Restricting the analysis to within-time connectivity, as defined by g_{ijt} , implicitly excludes such ties, since $g_{ij} = 1$ does not necessarily imply $g_{ijt} = 1$ for some t , and the fathers and sons in our dataset are not active in the same period. Hence, we restrict attention to non-familial connections.

***Substantial Households, 1638* (Cummins et al., 2016).** The share of households living in properties with an annual rent of at least £20 in 1638.

***Tradesmen's Tokens, 1648–1651* (Burn, 1855).** The number of tradesmen's tokens issued in a parish between 1648 and 1651, from the catalog of Burn (1855). Tokens were issued by individual traders as small change and are used here as a measure of the density of retail commerce immediately before the first coffeehouse opened.

***Goldsmiths, 1600–1651* (Lillywhite, 1972).** The number of goldsmiths recorded at an address in a parish before 1652.

***Commercial Premises* (Lillywhite, 1972).** The number of signed premises recorded in a parish and decade, excluding taverns, inns, other pubs, and coffeehouses. Before street numbering, premises were identified by a hanging sign, so the residual category counts signed establishments that were not drinking houses. From the 1600s to the 1690s the measure records 1,560 premises-decades. Booksellers account for 452 of them, goldsmiths for 200, grocers for 85, printers for 37, and mealmen for 29; the remaining 757 are spread over 226 other trades.

***Pubs* (Lillywhite, 1972).** The number of public houses active in a parish and decade, counted in every decade between the first and last recorded mention. The category runs wider than taverns and inns: it also takes in alehouses, ale men, wine houses, victuallers, and mum houses. The companion measure counts public houses newly recorded in the decade; the placebo figures and the balance tables use the active count.

***Booksellers* (Isaac et al., 2024).** Defined among the outcomes above, where booksellers also appear in Section 5.

***Dissenting Meetings* (Hine, 1976).** The number of dissenting meeting places first recorded in a parish and decade, from Hine (1976), a manuscript index of dissenting chapels in the City of London held at the Guildhall Library under shelfmark SL 18. The index is closed access and was transcribed on site. Each meeting place is dated by its earliest known date, which the source records either to the year or to the decade. End dates are not recorded, so the variable measures the number of newly recorded meeting places in a decade rather than a stock.

***Clay-Pipe Finds* (Museum of London, 2013).** The number of excavated clay tobacco pipes assigned to a parish and decade. Finds are dated by the typological interval supplied with the record, and each retained find contributes to every decade its interval spans. Records carrying the non-informative interval 1580–1910 are excluded, since they would otherwise contribute to every decade in the panel.

***Luxury Shops* (Lillywhite, 1972).** The number of shops in luxury trades recorded in a parish and decade, active in every decade between the first and last recorded

mention. Trades are read from the description the source records for each sign, and the measure counts grocers, comfit makers, confectioners, tobacconists, and pastry cooks; the silk trades, meaning silk weavers, silk dyers, silk mercers, silkmen, and throwsters; mercers, lacemen and lace shops, milliners, and perfumers; cabinet makers and upholsterers; musical instrument makers and music sellers; and toy shops, wax chandlers, and glass sellers. The measure is meant to capture retailers of the imported and fashionable goods that coffee itself belonged among, so three groups are excluded by construction: drinking and sociability venues, which lie too close to treatment; the book, print, map, and instrument trades together with clockmakers, which overlap the outcomes; and goldsmith-bankers, which are financial rather than retail establishments. Goldsmiths and silversmiths, apothecaries and druggists, hatters, glovers, calico printers, cheesemongers, and fruiterers are coded as borderline and left out. The narrower variant, restricted to the exotic imported consumables that reached London on the same ships as coffee, namely grocers, comfit makers, confectioners, and tobacconists, is constructed but not used in the paper. From the 1600s to the 1690s the measure records 246 premises-decades in parishes of the panel: grocers 128, mercers 32, confectioners and comfit makers 20, tobacconists 19, and milliners 12, with the remaining 35 spread over 13 other trades.

Apothecaries, 1617–1640s ([Wallis and Webb, 2000](#)). The number of distinct masters named in geocoded binding records of the Apothecaries' Company in a parish and decade. The source records apprenticeship bindings rather than a complete stock of practicing apothecaries, so the variable measures masters who took an apprentice in the decade, not apothecaries present. It begins in 1617, so the pre-treatment average excludes the 1600s.

Theaters ([Wikipedia, 2024](#)). The number of playhouses whose operating interval overlaps a given decade. We construct the venue list ourselves, starting from the index of English Renaissance theatres on Wikipedia ([Wikipedia, 2024](#)) and the venue pages it links: we separate the first and second Blackfriars theaters, add venues the index omits, and date each playhouse by the end of ordinary or legal performance rather than by demolition, so operating intervals are shorter than a naive reading of the venue records would give. The list holds 19 venues, 18 of which fall inside the parish geography of the panel; the Hope, on Bankside, geocodes just outside it. Eleven had already closed before the Civil War, the earliest being the Red Lion in 1568 and the last the Swan around 1632. The other eight were still playing when the Long Parliament suppressed stage plays on 2 September 1642, and all eight are dated to that order. Playing continued illicitly at the Fortune, the Red Bull, the Cockpit, and Salisbury Court, and legal playing resumed at the Restoration, but the variable follows ordinary or legal performance, so no venue in the list reaches the 1650s and the count is zero in every decade from the first coffeehouse onward.

Scientific and Non-Scientific Letters ([Hotson and Lewis, 2024](#)). The number of letters sent or received from an address in a parish in a given decade, recorded in Early Modern Letters Online. Each correspondent is recorded in a single field holding a name, dates, and a short descriptive gloss, and a letter is scientific when the field for the author or the field for the recipient contains one of ten strings, matched in lower

case and anywhere in the field. Three are complete words, namely “cartographer”, “meteorologist”, and “pharmacist”. Seven are stems, cut short so as to pick up the variants of a term: “astro” for astronomer and astrologer, “natur” for natural philosopher and naturalist, “experim” for experimental philosopher, “math” for mathematician, “physic” for physician and physicist, “chem” for chemist and alchemist, and “botan” for botanist. All remaining letters are non-scientific. Other learned descriptors are not treated as scientific, among them unqualified philosopher, theologian, historian, philologist, economist, humanist, professor, agriculture, and the book trades. The counts used here are restricted to letters whose location is recorded directly rather than interpolated. Because the classification keys on the correspondents rather than on the contents of the letter, it identifies correspondence involving people described in scientific terms, not correspondence about science. From the 1600s to the 1690s the measure records 1,220 pairs of a non-scientific letter and one of its authors. Grouped by the description of the author, 310 fall under clergy and religious, 266 under scholarship, letters, and education, 166 under nobility, royalty, and court, 154 under government and administration, and 93 under law; 126 fall under five other occupations, and 105 have no occupation recorded.

Mathematical Practitioners and Scientific Instrument Shops Before 1652. Defined among the outcomes above, measured over the decades preceding the first coffeehouse.

City of London and London Wall ([Cummins et al., 2016](#)). The City of London indicator covers all 113 parishes within the City. The Inside the London Wall indicator covers the 97 of these parishes located within the historic London Wall and is therefore narrower. Both are constructed from the parish geography of [Cummins et al. \(2016\)](#).

Distances and Area. Great-circle distance in kilometers from the parish centroid to the Royal Exchange, Gresham College, and Paternoster Row, and parish area in square kilometers. Centroids and areas are taken from the parish geography of [Cummins et al. \(2016\)](#).

Exposure to the Great Fire. We construct this indicator by hand for each parish, recording whether the parish lay in the area destroyed by the Great Fire of 1666. We read the burnt area from the map of the fire’s extent in [Schofield \(1984, pp. 172–173\)](#), the same map digitized by the study of the fire’s economic impact that we cite in Section 4.

D.5. HOOKE’S DIARY

Robert Hooke kept a memorandum book from 10 March 1672 to May 1683, now Guildhall Library MS 1758. [Robinson and Adams \(1935\)](#) print the entries from August 1672 to December 1680, and [Henderson \(2007\)](#) the months they omit, March to July 1672 and January 1681 to May 1683. We transcribe both editions by hand into one row per printed entry and check each row against the text of its printed page: 2,944 of the 2,950 rows from the first edition and 438 of the 509 from the second match a passage on

the page closely.⁴⁴ Entries that cover several days are assigned to each of them, and dates keep the Old Style calendar, with the year starting on 1 January. The result is 3,464 dated days.

Coffeehouse days are those whose entry names a coffeehouse or mentions coffee. We match each entry and its notes against regular expressions built from the spelling variants in the indexes of both editions and from the establishments that [Brown et al. \(2021\)](#) identify in the diary. On 2,193 days the entry names one of 57 coffeehouses, 3,097 times when each house counts once a day; coffee alone is mentioned on 51 further days, of which at most six refer to coffee taken at home. Drinking houses, mostly taverns, are named on 337 days.⁴⁵

We annotated all 3,464 days using large language models run through Codex. For every person an entry names, the annotation records the name as written, whether Hooke met the person, and the matching person in the index of the printed editions, together with the passage that supports each decision. The two indexes were merged first: the 397 people in the index of [Henderson \(2007\)](#) were linked to the 2,043 in the index of [Robinson and Adams \(1935\)](#) through a search for candidates and a check of every link by hand. Of the 22,132 mentions, 17,341 record a meeting, and 16,435 of these name a person the index identifies. Counting each person once a day gives 16,325 meetings with 1,426 distinct people on 3,045 days. Each annotation records the passage and the reasoning behind every decision, and a validator checks each identification against the printed indexes. The annotations of 172 days, 5 percent, remain flagged for manual review; their 1,101 certain meetings are kept, and uncertain encounters and unresolved identities are left out throughout.

People are in the correspondence network if the index links them to a writer of at least one letter sent from London between 1600 and 1699 in Early Modern Letters Online ([Hotson and Lewis, 2024](#)). Candidate links to the 1,581 such writers come from a search on names, their variants, and life dates, and each accepted link was reviewed: 198 people in the index are linked to 199 writers, and 155 of the 1,426 people Hooke met are among them.

We group the people Hooke met by occupation from the notes in the index. Mathematical practitioners are the 55 people matched by name to [Taylor \(1954\)](#) or to SIMON, 29 of them instrument makers; each match was adjudicated individually, and a bare surname never suffices. The match is partly circular, since Taylor cites the diary in 58 entries. The index records no occupation for 902 of the 1,426 people. Coding by an independent language model agrees with these groups for 85 percent of people, and on whether an occupation lies in science for 94 percent.

D.6. FIRST COFFEEHOUSES IN EUROPEAN CITIES

To the best of our knowledge, no previous work attempts a comprehensive listing of the first coffeehouse of every European city. The place of the coffeehouse in general

⁴⁴The check locates each transcribed entry in the machine-read text of the printed page by long runs of identical words, so it confirms the passage rather than every character. The 1935 edition itself is “corrupt in places, where words have been mistranscribed or left out” ([Henderson, 2007](#), p. 133).

⁴⁵Appendix Figure B5 uses the visits that [Brown et al. \(2021\)](#) code instead. The export of their data that we use contains no entries for 1679. Leaving that year out, our mentions at the houses they locate number 2,412, against their 2,497 visits, and 83 percent of our 2,915 located mentions lie within 1 kilometer of Hooke’s lodgings at Gresham College, against 84 percent of their visits.

historical scholarship has, however, meant sustained attention to individual houses and to the first mention of one in a given city, and we rely on that literature in a systematic review of the 200 cities of Section 6. Table D1 reports the year, the decade, and the source for each of the 37 cities in which we find a first coffeehouse before 1700, when the panel ends.

The search made three passes over the 200 cities. The first was a systematic review of the historical scholarship, searched in the language of each country: national and city histories of coffee and of sociability, municipal and guild records in print, digitized archival finding aids, and the secondary literature on coffeehouses. The second was algorithmic. We read every record in the *Universal Short Title Catalogue* whose title mentions coffee in any of the catalog languages, which surfaces ordinances regulating the premises, excise documents, and polemics naming a house, and which is what dates Middelburg from a placard of 1686. The third searched the candidate list for false positives, checking each candidate against the cited text and rejecting dates that record the arrival, sale, or first mention of coffee rather than a coffeehouse.⁴⁶

Cities without a dated coffeehouse enter as comparison cities, but an absent record is not an absent coffeehouse, and Withington (2020) shows that in provincial England coffee was often sold inside inns that left no coffeehouse record.

Two checks support the dates. Coffee entered Europe through Venice, Livorno, Marseille, London, and Amsterdam, and among the 33 cities whose source gives a year, the first coffeehouse opens 3.3 years later for every 100 km of distance from the nearest of them (s.e. 0.8; Figure D1, Panel B). Across all 200 cities, the nearest third by that distance open coffeehouses first, the middle third overtake them in the 1690s, and the farthest third hardly open any before 1700 (Panel A). Seventeen of the 37 also printed a title on coffee before 1700, against 13 of the 163 comparison cities, and in 14 of the 17 the first such title follows the first coffeehouse.⁴⁷

⁴⁶We used Claude and Codex for the third pass. The models located and re-read sources and flagged candidates for rejection; they supplied no dates, and every date in Table D1 rests on the source cited beside it. Barcelona, where coffee is recorded in 1627, is the kind of candidate the pass rejects, and it enters as a comparison city.

⁴⁷The three exceptions are academic: a Frankfurt dissertation of 1666, a Lübeck tea tract of 1694, and Daniel Wencker's Strasbourg dissertation *De potu café* of 1693, two years before the Magistrat's ordinance on the *maisons de café*.

TABLE D1—FIRST COFFEEHOUSES BY CITY IN EUROPE

Country	City	Year	Decade	Source
Austria	Vienna	1685	1680s	Albrecht (1989)
Belgium	Antwerp	1690	1690s	Van Aert (2009)
	Kortrijk	1690	1690s	P. D. B. (1934)
Denmark	Copenhagen		1690s	Hoff (2016)
France	Angers	1693	1690s	Bertoldi (2013)
	Lyon		1670s	Grand-Carteret (1902); Van Damme (2005)
	Marseille	1671	1670s	Ellis (2004a)
	Paris	1671	1670s	Ellis (2004a); Leclant (1951)
	Strasbourg	1695	1690s	Perry (2011)
Germany	Bremen	1669	1660s	Ellis (2004a)
	Frankfurt am Main	1689	1680s	Cowan (2014)
	Hamburg	1677	1670s	Albrecht (1989); Robilliard (2024)
	Leipzig	1694	1690s	Ellis (2004a)
	Lübeck	1695	1690s	Meyer (2011)
	Munich	1699	1690s	Literaturportal Bayern (2019)
	Nuremberg	1686	1680s	Cowan (2014)
	Regensburg	1686	1680s	Cowan (2014)
	Würzburg	1697	1690s	Denzinger (1848)
Italy	Genoa	1661	1660s	Belgrano (1875)
	Rome		1650s	Lombardo (2016)
	Venice	1683	1680s	Ellis (2004a)
Netherlands	Amsterdam	1665	1660s	Stadsarchief Amsterdam (2019); Hell (2017)
	Groningen	1686	1680s	Kuiken (1982)
	Leiden	1685	1680s	van Oostrum (2006)
	Middelburg	1686	1680s	City of Middelburg (1686)
	Rotterdam	1666	1660s	Heise (1987)
	The Hague	1670	1670s	van Oostrum (2006); Betz (1900)
	Utrecht	1678	1670s	van Oostrum (2006)
Poland	Gdańsk	1697	1690s	Gedanopedia (2020)
Switzerland	Basel		1690s	Meier (1973)
	Geneva	c. 1690	1690s	Encyclopédie de Genève (1992)
UK	Bristol	1666	1660s	Pincus (1995)
	Edinburgh	1673	1670s	Pincus (1995)
	London	1652	1650s	Ellis (2004a)
	Newcastle upon Tyne	1692	1690s	Story (1692)
	Norwich	1676	1670s	Withington (2020)
	York	1669	1660s	Pincus (1995)

Notes: Table lists every city recorded with a first coffeehouse inside the panel window, grouped by country and listed alphabetically within it. Year is the year the source establishes and is left blank where the source fixes only a decade, as for Rome, Copenhagen, Basel, and Lyon. Decade is the decade the source establishes and is the cohort the estimation uses, since treatment is decadal and a source establishing only a decade is therefore sufficient. Cities are dated only from sources that name a coffeehouse or regulate the premises, as against recording the arrival or sale of coffee; cities whose entry decade could not be established on that rule are coded as never treated. [Return to pages [16](#), [33](#), [52](#), [54](#)]

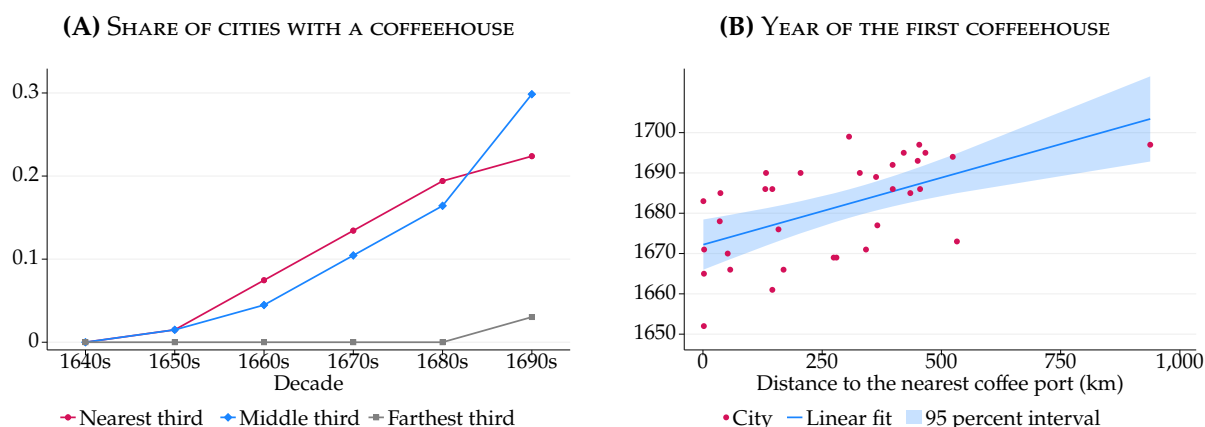


FIGURE D1. THE SPREAD OF COFFEEHOUSES FROM THE COFFEE PORTS

Notes: Figure plots the spread of coffeehouses against the distance from each city to the nearest of the five ports through which coffee entered Europe, Venice (1624), Livorno (1632), Marseille (1644), London (1650), and Amsterdam (1661), with the year coffee is first recorded there. Panel A splits the 200 cities into thirds by that distance, within 223 kilometers, 224 to 555 kilometers, and 576 kilometers or more, and plots for each third the share of its cities with a coffeehouse by each decade, from the decades of Table D1; the thirds hold 15, 20, and 2 of the 37 coffeehouse cities. Panel B plots the year of the first coffeehouse against distance for the 33 coffeehouse cities whose source gives a year, leaving out Rome, Copenhagen, Basel, and Lyon, whose sources fix only a decade, with a linear fit of 3.3 years per 100 kilometers and its 95 percent confidence interval, from a robust standard error of 0.8. [Return to pages 32, 52]

D.7. DATA SOURCES

Appendix Table D2 lists the source of every variable used in the paper.

TABLE D2—DATA SOURCES

Data Source	Online Description	Variable(s)
Burn (1855)	Internet Archive	Tradesmen's Tokens 1648–1651
Taylor (1954)	Internet Archive	Mathematical Practitioners, Connections Between Mathematical Practitioners
Lillywhite (1963)	Internet Archive	Coffeehouses
Lillywhite (1972)	Internet Archive	Coffeehouses, Commercial Premises, Taverns, Inns, Goldsmiths
Clifton (1995)	Internet Archive	Scientific Instrument Shops, Scientific Instrument Makers
Hay et al. (2025)	National Museums Scotland	Scientific Instrument Shops, Scientific Instrument Makers
Clark and Cummins (2014)	Human Nature	Surname Counts Among Alumni at the University of Cambridge and Oxford
Cummins et al. (2016)	Economic History Review	Median Tax Paid 1582, Percentage of Substantial Households 1638, Baptisms, Deaths
Warren et al. (2016)	Six Degrees of Francis Bacon	Connections Between Figures in the <i>Oxford Dictionary of National Biography</i>
FamilySearch (2025)	FamilySearch	Surname Counts Among English Couples Married 1538–1649
Hotson and Lewis (2024)	Early Modern Letters Online	Letters Sent and Received, Hooke's Correspondence Network
Robinson and Adams (1935)	Internet Archive	Hooke's Diary, 1672–1680: Coffeehouse Days, People Met
Henderson (2007)	Notes and Records of the Royal Society	Hooke's Diary, 1672 and 1681–1683: Coffeehouse Days, People Met
Brown et al. (2021)	Intoxicating Spaces	Establishments Named in Hooke's Diary, Hooke's Coffeehouse Visits
Isaac et al. (2024)	University of Oxford	Booksellers
The Royal Society (2024)	The Royal Society	Fellows of the Royal Society
Hine (1976)	Guildhall Library	Dissenting Meeting Places
Wallis and Webb (2000)	LSE Research Online	Apothecaries
Museum of London (2013)	Museum of London	Clay-Pipe Finds
Wikipedia (2024)	Wikipedia	Theaters
Kelly and Ó Gráda (2022)	openICPSR	Applied Mathematics Titles
Almelhem et al. (2026)	Quarterly Journal of Economics	Average Progress Score

Notes: Table lists the historical data sources used to construct the study variables. Online descriptions link to the relevant catalogue, database, or source. [Return to pages [47](#), [54](#)]

D.8. PAGES OF ORIGINAL HISTORICAL DATA

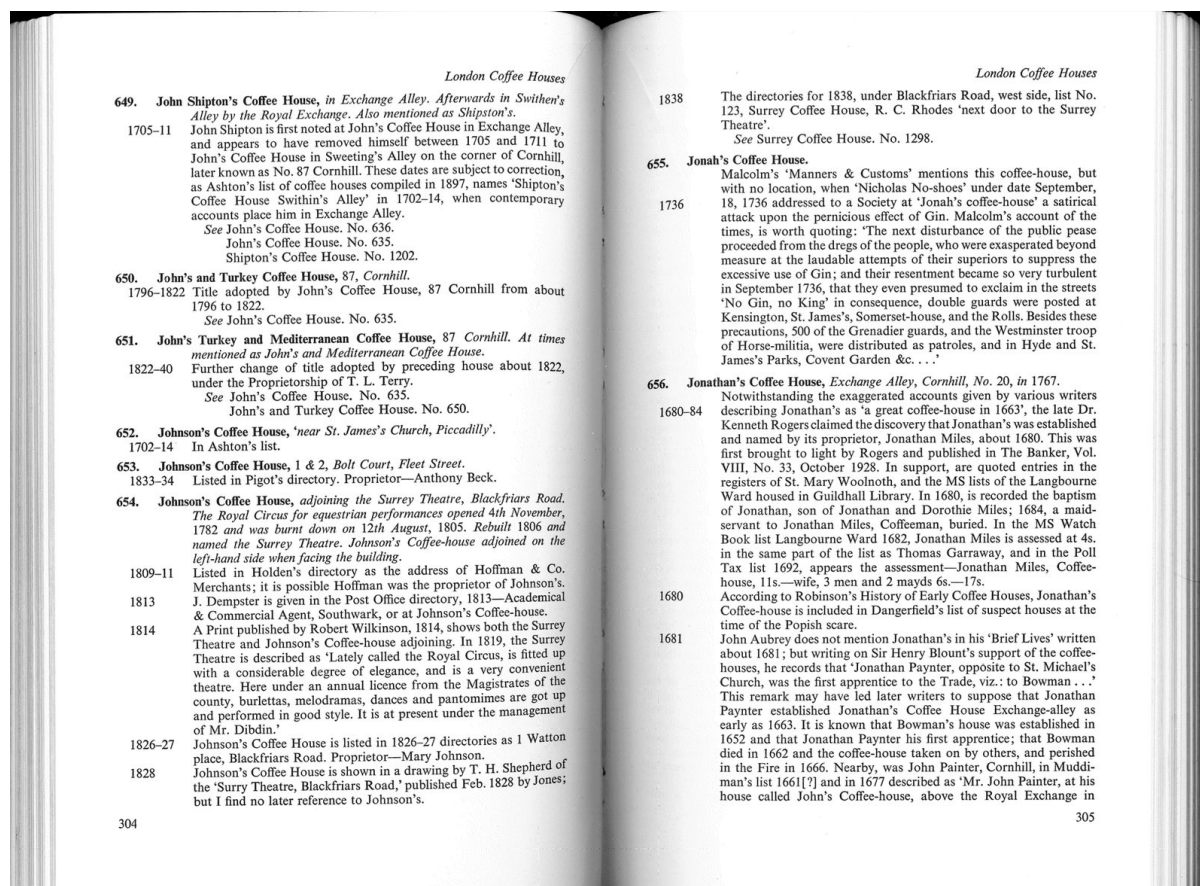


FIGURE D2. COFFEEHOUSES IN LONDON (LILLYWHITE, 1963, PP. 304–5)

Notes: Excerpt from Lillywhite (1963), showing London coffeehouse entries and their historical references.
[Return to page 12]

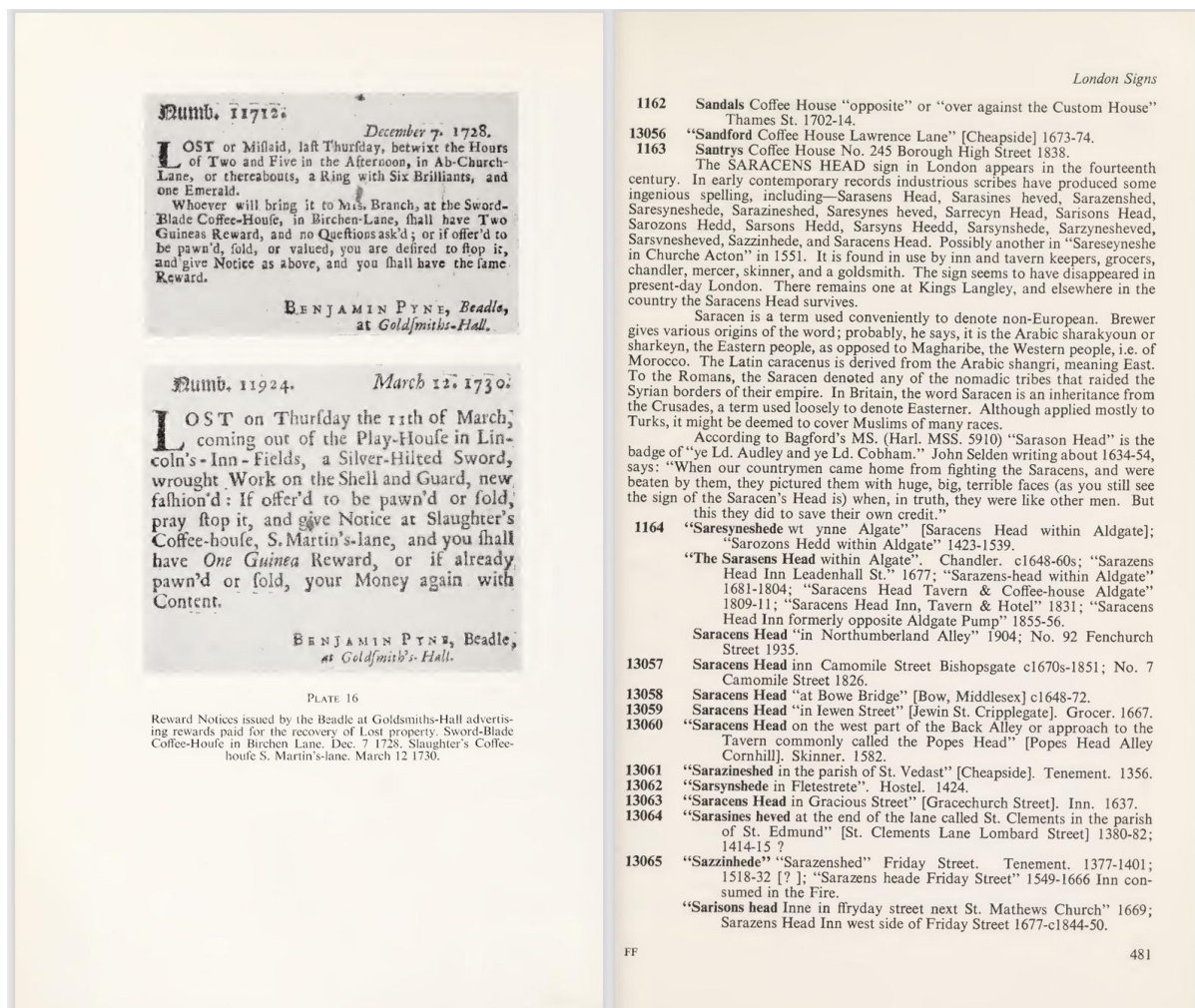


FIGURE D3. SIGNS IN LONDON (LILLYWHITE, 1972, PP. 480–1)

Notes: Excerpt from Lillywhite (1972), showing information about seventeenth-century London signs.
 [Return to page 12]

THE MATHEMATICAL PRACTITIONERS

to a mathematician for reading a mathematical lecture to his Majesty's servants relating to his Navy and instructing them in the art of navigation and other parts of the mathematics'. The syllabus he offered was of the usual type, but under 'Navigation' he added 'the construction and application of instruments for the taking of observations, heights, distances, plotting of countries, working of traverse etc., and the making of maps and charts'. He had read of the lectureship in a work on gauging by Henry Bond (144), who had held it, but there seems no record of whether his request was acceded to.

254. SUGAR, JOHN (fl. 1645-53), was a *globe-maker* at the Sign of the Atlas on Parnassus Hill, near St Michael's, Cornhill, making celestial and terrestrial globes of 14 in. and 15 in. diameter, as appears from his advertisements in *Booker's* and *Wharton's Almanacks* for 1653. The name of Joseph Moxon (228) is here associated as partner with that of Sugar, and as this is the first mention of Moxon, and the last of Sugar, the latter may have been at the end of his career. John Pell (173) writing in 1645 from Amsterdam to John Leake, a London stationer, said he had spoken to 'Mr Sugar' about some papers sent to London, which suggests that Sugar (like Moxon) had business in Holland.

255. ATKINSON, JOHN (ob. 1673), was a *compass-maker* at Harwich, Essex, who is known only by his will.

256. COULTON, JOHN (fl. 1653-4), was a *mathematical practitioner* living at Cobham, Surrey, whose Almanack was printed by R. & W. Leybourne.

257. OSBORNE (or OSBURN), HENRY (fl. 1653-86), a gentleman living in the Strand and later in Ireland, was praised by William Molyneux of Dublin as an excellent *mathematician* and astronomer. He was the correspondent of the young astronomer Jeremy Shakerley (233) who wrote to say that he had observed the solar eclipse of 1652 in Surat, and of Robert Hooke (270) who supplied him with instruments including a quadrant fitted with telescopic sights, a telescope and a micrometer, which were sent out to him in Drogheda. Edmund Halley (401) later recorded Osborne's observations of a solar eclipse in Ireland in his edition of *Streete's Astronomia Carolina* (269).

258. PIGOT, FRANCIS (fl. 1653-66), a *mathematical practitioner* who published an Almanack during the period 1654-62 which was 'rectified' for the meridian of Ludlow, himself lived at Cleobury Mortimer in Shropshire. Here he was ready to instruct any 'heroic spirit or generous gentleman' in the use of all sorts of mathematical and surveying instruments, the use of the globes, and so on.

259. SUTTON, WILLIAM (fl. 1653-63), was presumably the brother or near kinsman of Henry Sutton (188) since their names appear jointly in advertisements over a period of years. 'All manner of Mathematical Instruments either for Sea or Land are exactly made in Brass or Wood by

BIOGRAPHIES

Henry Sutton or by William Sutton'. The latter had premises 'a little beyond the Church' in Upper Shadwell, which is down the Thames below Wapping. No examples of his work are known to have survived.

260. WREN, CHRISTOPHER (1632-1723), was from boyhood fertile in designing and making *mathematical instruments*, including a rain-gauge (1662), the moon globe ordered by Charles II, a new type of level (1667), an instrument for calculating the times of eclipses, an 'instrument for perspective' which Pepys saw when he visited Oldenburg (Secretary of the Royal Society) in May 1669, and many others, while several of his fine sundials survive. John Aubrey says that as a boy he was grounded in arithmetic and geometry by his sister's husband, the Rev. William Holder, while when he went to Oxford he was closely associated with Dr John Wilkins (181), and in London with Robert Hooke (270) who said of him 'Since the time of Archimedes there scarce ever met in one Man in so great a Perfection, such a mechanical Hand and so philosophical a Mind'. Yet designing instruments was only incidental to Wren's main occupations as successively Gresham and Savilian Professor, Surveyor-General after the Fire of London, and architect of the rebuilding. He was knighted in 1673. Work: 277.

261. BRADFORD, JOHN (fl. 1654), is known by a mention of him as a *mathematical instrument-maker* in Foster's *Elliptical Horologigraphy*, published in 1654 by John Twysden (162).

262. COLEY, HENRY (1633-1707), was born at Oxford and, starting as a young clerk, came to London in 1654 to become amanuensis and later successor to the noted astrologer William Lilly. He practised as an astrologer throughout his life and taught this subject as well as the whole range of practical mathematical subjects, besides establishing his own *mathematical school* at a house 'over against the Old Hole in the Wall, Baldwin's Court, Baldwin's Gardens', a Court which ran out of Gray's Inn Lane. Here he taught for over thirty years. Prior to 1663 he had lived in Rose and Crown Court, Gray's Inn Lane. Coley advertised himself widely, in his own almanacks and elsewhere, and had a large circle of friends, including Elias Ashmole and John Kendal (393) of Colchester, both devoted to the mathematical arts. Several of his note-books survive and indicate the wide range of his reading, and his method of work. Work: 441.

263. EYRE, JOHN (fl. 1654), who describes himself as a '*practitioner in the mathematics*' wrote a text-book on surveying, and presumably practised and taught the art. He was associated with Walter Hayes (251) as instrument-maker. Work: 216.

264. SINCLAIR, GEORGE (fl. 1654, ob. 1696), was at first (1654) a schoolmaster at St Andrews, and took part in some experiments on the diving-bell (1655). An M.A. of Glasgow was conferred upon him (1654) and he was appointed to a Chair of Philosophy there which he held until 1666. Returning then to Edinburgh to teach mathematics, he determined

FIGURE D4. MATHEMATICAL PRACTITIONERS IN ENGLAND (TAYLOR, 1954, PP. 240-1)

Notes: Excerpt from Taylor (1954), showing biographical entries of mathematical practitioners. [Return to page 14]

ALLAN John	
ALLEN James* Known to have sold: sextant, repeating circle Sources: T339 (1992/3) item 23; Bn; H; Und; Bennett (1985) p.19,22; Stimson (1985) p.103–04,113 ALLAN John w 1790–1794 Sextant M, Quadrant M London 1790–1794 12 Blewitt's Buildings, Fetter Lane, London Father of ALLAN James* Succeeded by ALLAN James* Known to have sold: sextant Sources: T339 (1992/3) item 23; W; JAC ALLBUT Isaac w 1811–1845 Rule M Alternative name: ALLBUTT 1811–1845 Salop St, Wolverhampton 1818 105 Salop St, Wolverhampton See also ALLBUTT I.* Sources: PB; Pg; Wh ALLBUTT I. w 1845 Rule M 1845 Birmingham See also ALLBUT Isaac* Sources: Roberts (1982) ALLCOCK Thomas w 1851 Chrono M, Watch M 1851 14 South Island Place, Clapham Rd, London Sources: PO ALLEN - w 1820 Alternative name: Possibly Alexander ALLAN 1820 Near the College, Edinburgh Instrument advertised: geometric solids Sources: Larkin (1820) (DJB) ALLEN B. w 1851 Optician 1851 2 Lower Copenhagen St, Islington, London Sources: Wk ALLEN Elias w 1607 d 1653 Math IM (in brass), Book plate engraver Alternative name: ALLIN Blackhorse Alley, Fleet St, London Horseshoe over against St. Clement's Church, Strand, London 1606–1611 Black Horse-ally, neere Fleetbridge, London 1623–1639 Without Temple Bar, over against St.Clement's Church, London 1618 At the Bull's head over against St Clement's Church, Strand, London 1653 Horseshoe without Temple Bar near	St Clement's Church, London 1645 The Blackmoor without Temple Bar, London Guild: Grocers, Clockmakers Apprenticed to WHITWELL Charles c.1602 GC Had apprentice: BLAYTON (BLIGHTON) Edward 1612 GC BLIGHTON John*nd fr 1620 GC SHEWSWELL Thomas fr 1623 [possibly SHASWELL *] GC ALLEN John (I)* 1617 GC SEFTON Henry 1620 GC DAVENPORT Robert* 1623 GC BROOKES Christopher* 1629 GC WINCKFIELD Edward 1629 GC COOKE George 1635 GC JORDAN William 1649/50 GC GREATOREX Ralph* 1639 by t/o CKC CHENEY Withers* 1646 by t/o CKC GRIMES Edward 1640/41 by t/o CKC PRUJEAN (PRIGEON) John* CKC Took over from WHITWELL Charles* Succeeded by GREATOREX Ralph* Associated with THOMPSON John (I)* HAYES Walter* THOMPSON Anthony* fr GC 1612, fr CKC 1633; d 1653; another Elias Allen, who carried out surveys for Charles I was d by 1637 Known to have sold: armillary sphere, compass (magnetic), ellipsograph, marine astrolabe, mural quadrant, rule, sector, sundial Instruments advertised: mathematical instruments Sources: Brown J (1979a) p.24–25; Brown J (1979b) p.11–14,27; Bryden (1992) p.306n,308,318; DG ALLEN James w 1825–1826 Math IM Alternative name: ALLAN? 1825–1826 196 Piccadilly London See also ALLAN James * Same premises as WORTHINGTON & ALLEN* Attended London Mechanics Institute 1825–1826, not known as a master on his own Sources: LMIR (AM) ALLEN James Patishall w 1850–1852 Math IM 1850–1852 20 King St, Stepney, London Guild: Merchant Taylors, fr 1861 Son of ALLEN James [no trade given] Apprenticed to LILLEY John (I)*or (II)* 1841 Sources: PO; Wk; GL: MTC MF 320, 324 (MAC) ALLEN John (I) w 1630–1642 Math IM (in brass) 1630–1631 Near the Savoy, Strand, London Guild: Grocers fr 1631/2, ?Clockmakers 1653 Apprenticed to ALLEN Elias* Had apprentice HAYES Walter* 1631/2 GC Working in own name before free in 1631/2; a John Allen admitted to Clockmakers 1653 Known to have sold: astrolabe, sector Sources: Brown J (1979a) p.26; Brown J (1979b) p.14,27; Bryden (1992) p.309 ALLEN John (II) w 1842–1855 Phil IM, Drawing IM 1842–1845 35 St.Swithin's Lane, London 1847–1855 5 Three King Court, Lombard St, London Succeeded by ALLEN Mrs.A. Sources: PO ALLEN M.H. & J.W. c 1838–1840 [Drawing IS] Dame St, Dublin Known to have sold: beam compass invented and made by SMITH W. & A.* Sources: inst MHS; Pg ALLEN Nathaniel w 1790 w 1793 Ship chandler 1793 Wapping New Stairs, London Guild: Merchant Taylors, fr 1766 Apprenticed to EADE Jonathan (I)* 1754 Son of ALLEN James, Gent , d, of Dulwich, Kent Had apprentice ALLEN Thomas James, his son, 1781 Master MTC 1790 Sources: Wil; GL: MTC MF 319,320,324 (MAC) ALLEN Thomas (I) d 1767 Math IM 1767 Dublin Sources: Burnett & Morrison-Low (1989) p.120 ALLEN Thomas (II) w 1850–1856 Rule M 1850 12 St. Mary's Row, Birmingham 1850 (Residence) Aston Road, Birmingham Took over from ALLEN & ROWE * Associated with ROWE James ROWE Thomas Sources: Roberts (1982); Sl ALLEN William (I) w 1788–1791 Phil IM, Math IM 1787 37 High St, Birmingham 1791 Bull Ring, Birmingham Sources: Pye ALLEN William (II) fl 1838 Math IM, Phil IM 29 Seward St, Goswell St. London Said by Taylor (1966) p.466 to be listed in directories, but not verified; possibly ALLES William* ALLEN & ROWE w 1849–1850 Rule M

FIGURE D5. SCIENTIFIC INSTRUMENT MAKERS IN THE BRITISH ISLES (Clifton, 1995, p. 6)

Notes: Excerpt from [Clifton \(1995\)](#), showing entries on scientific instrument makers in the British Isles.
[Return to page 15]

D.9. PRIMARY AND SECONDARY SOURCES

This section presents examples of primary and secondary sources on which our main data sources are based ([Clifton, 1995](#); [Hay et al., 2025](#); [Lillywhite, 1963, 1972](#); [Taylor, 1954](#)).

The works by [Lillywhite \(1963, 1972\)](#) draw on a wide range of primary sources. For 1652 to 1672, these mainly include catalogs of tokens issued by London coffeehouses; Appendix Figure [D6](#) provides an example. Another key source is the diary of Samuel Pepys, which covers 1660 to 1669. Lillywhite also relies extensively on contemporary advertisements, such as the one in Appendix Figure [D7](#), and archival materials on London history held in the Guildhall Library.

Taylor (1954) is based on a broad set of primary and secondary sources, detailed in Appendix Figure D8. The 628 works on applied mathematics from about 1400 to 1715 are not listed here, as they are already cataloged in her study.

The underlying bibliography for the data on scientific instrument shops and makers in Hay et al. (2025) is available at <https://toolsofknowledge.org/sources/>. It covers 441 primary and secondary sources, with Clifton (1995) and its cited bibliographies serving as key references. Examples include trade cards, such as Appendix Figure D9, and signed instruments, such as Appendix Figure D10. These are often supplemented by later identification in guild records, since most scientific instrument makers belonged to a guild, although none was dedicated specifically to instrument making (Cormack et al., 2017, p. 4).



FIGURE D6. THE TURK'S HEAD COFFEEHOUSE TOKEN, 1669

Notes: Copper alloy trade token issued by Nicholas Strange in 1669 for his coffeehouse in St. Ann's Lane, now part of Gresham Street, Aldersgate Ward, City of London. The obverse bears the inscription "NICH. STRAINGE. AT. YE. COFFEE" surrounding a Turk's head device. The reverse reads "HOVSE.IN.ST.ANNS.LANE.69" and shows a hand pouring coffee into a cup. London Museum, object ID 96.66/100*. [Return to page 59]

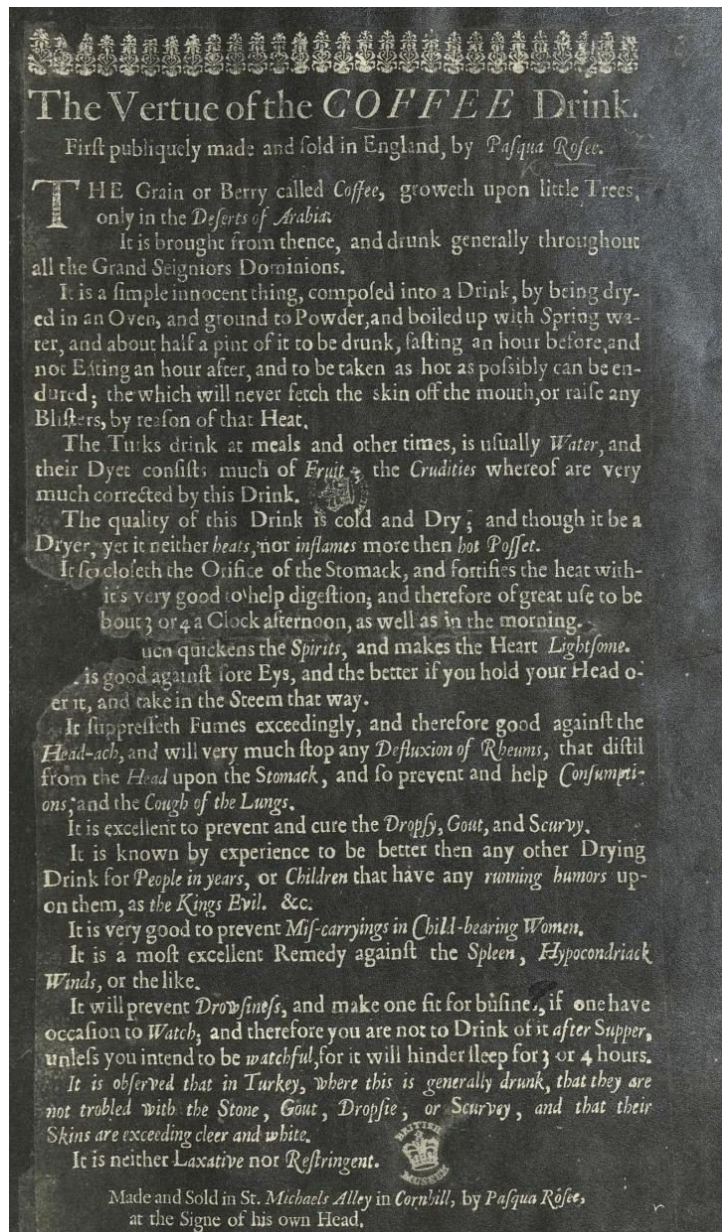


FIGURE D7. THE VERTUE OF THE COFFEE DRINK, 1652

Notes: This 1652 handbill, "The Vertue of the COFFEE Drink," is the earliest known advertisement for coffee in England. The final lines identify the coffeehouse as Pasqua Rosée's Coffee House, located in St. Michael's Alley in Cornhill. Reproduced by Williams (1936) from the original held at the British Museum. [Return to page 59]

BIBLIOGRAPHY OF SECONDARY WORKS

The Works listed in Part III provide the main source material for the book, and in addition a large number of contemporary works (including annual almanacks) compiled by or referred to by the Mathematical Practitioners have been examined for biographical details. The manuscript material used is mainly that in the British Museum, supplemented by the Ashmolean MSS. in the Bodleian Library, and the Minutes and Letter Books (with a few miscellaneous original Papers) of the Royal Society.

- AUBREY, JOHN. *Letters written by Eminent Men . . . and Lives of Eminent Men*. London 1813.
- BIRCH, THOS. *History of the Royal Society of London, 1756-7*.
Calendars of State Papers.
- CLAY, R. S. and COURT, T. H. *History of the Microscope*, 1932.
- CUDWORTH, W. *Life of Abraham Sharp*, 1889.
- EDLESTONE, J. *Correspondence of Newton and Cotes*, 1850.
- GUNTHER, R. T. *Early Science in Oxford*, 1923-35.
- GUNTHER, R. T. *Early Science in Cambridge*, 1937.
- HALLIWELL, J. O. *Collection of Letters illustrative of the Progress of Science in England*, 1841.
- HALLIWELL, J. O. *Rara Mathematica*, 1841.
- HALLIWELL, J. O. *Private Diary of John Dee*, 1842.
- JOHNSON, F. M. *Astronomical Thought in Renaissance England*, 1937.
[With a Bibliography of Works relating to the History of Mathematical Science].
- MACPIKE, E. F. *Correspondence and Papers of Edmund Halley*, 1932.
- PEARCE, E. M. *Annals of Christ's Hospital*, 1920.
- PEPYS, SAMUEL. *Catalogue of Manuscripts* (Navy Records Society, Vols. 27, 36).
- PEPYS, SAMUEL. *Tangier Papers* (Navy Records Society. Vol. 73).
- PEPYS, SAMUEL. *Naval Minutes* (Navy Records Society, Vol. 60).
- RIGAUD, S. P. *Correspondence of Scientific Men of the Seventeenth Century*, 1841.
- ROBINSON, H. W. and ADAMS, W. *The Diary of Robert Hooke*, 1935.
- STONE, EDMUND. *Construction and Use of . . . Mathematical Instruments* (2nd edn.), 1758.
- TAYLOR, E. G. R. *Tudor Geography*, 1930.
- TAYLOR, E. G. R. *Late Tudor and Early Stuart Geography*, 1935.
- TAYLOR, E. G. R. Contributions (since 1928) to the *Geographical Journal*, *Imago Mundi*, *Journal of the Institute of Navigation* and *Mariner's Mirror*.
- TURNBULL, H. W. *James Gregory Tercentenary Memorial Volume*, 1939.
- WARD, JOHN. *Lives of the Professors of Gresham College*, 1740.
- WOLF, A. *History of Science, Technology and Philosophy in the 16th and 17th Centuries*, 1935.

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FIGURE D8. PRIMARY AND SECONDARY SOURCES USED BY [TAYLOR \(1954\)](#)

Notes: Selection of primary and secondary sources cited in [Taylor \(1954\)](#). [Return to page 60]

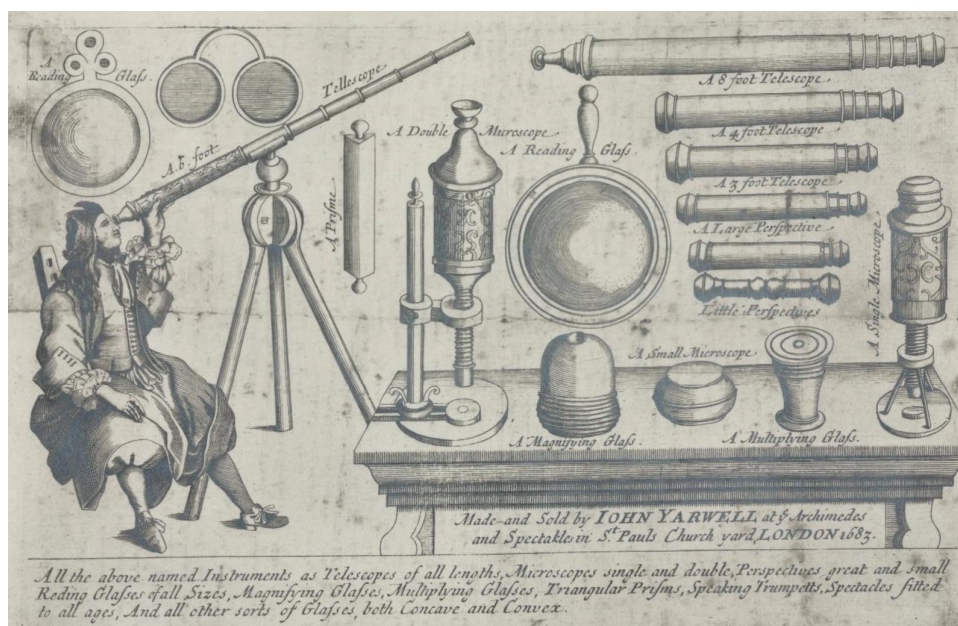


FIGURE D9. TRADE CARD OF JOHN YARWELL, LONDON, 1683

Notes: Trade card of the optical instrument maker John Yarwell, St. Paul's Churchyard, London, 1683. Science Museum Group Collection, object no. 1951-687/45. [Return to page 60]



FIGURE D10. COMPENDIUM BY ELIAS ALLEN, LONDON, 1630–1653

Notes: Compendium consisting of a sundial, compass, and calendar indices, made by Elias Allen in London, 1630–1653, near St. Clement's Church close to the Strand. Inscribed "Elias Allen fecit," with the outer lid engraved with the royal coat of arms, attributed to the reign of Charles I. Science Museum Group Collection, object no. 1883-135. [Return to page 60]

D.10. DATA VALIDATION

Coffeehouses. First, we assess the completeness of the coffeehouse dataset by benchmarking the number of recorded coffeehouses against counts from the [City of London Magistrates \(1663\)](#) survey and the 1692–1698 poll tax returns reported by [Cowan \(2005\)](#). The 1663 survey lists 82 coffeehouses in the City of London, while [Lillywhite \(1963, 1972\)](#) records 31.⁴⁸ The 1692–1698 poll tax lists 90 coffeehouses in 13 of 26 wards for which data are available, while 1701 administrative records note 34 more in 7 of the 13 previously missing wards, giving a total of about 124 coffeehouses in the late seventeenth century, slightly above our estimate of 113 for 1692–1698 based on [Lillywhite \(1963, 1972\)](#). This pattern suggests that [Lillywhite \(1963, 1972\)](#) covers late seventeenth-century coffeehouses more comprehensively than those from mid-century, likely reflecting the loss of records associated with the 1666 Great Fire. Observation delays also contribute. “Henry Willington” in “Dowgate” is listed in the 1663 survey, with “H. Wellington” in “Dowgate” first noted in a 1665 token by [Lillywhite \(1963, 1972\)](#). Similarly, [Lillywhite \(1963, 1972\)](#) may have ceased observing a coffeehouse before 1663. To facilitate a broader 1663 comparison, we consider a margin of error: allowing a margin of ± 1 year around 1663 results in 37 coffeehouses recorded by [Lillywhite \(1963, 1972\)](#), ± 2 years in 42, and ± 3 years in 71.⁴⁹

Second, [Mackie \(2013\)](#) attempts to extend [Lillywhite \(1963\)](#) using modern searchable databases, including Electronic Enlightenment, Old Bailey Online, and British Periodicals, among others.⁵⁰ This effort yields only a small number of additional and often ambiguous cases. We conduct a parallel search using two Gale resources, the *Seventeenth and Eighteenth Century Nichols Newspapers Collection* and the *Seventeenth and Eighteenth Century Burney Newspapers Collection*, focusing on mentions of “coffee” and its variants. These sources reference coffeehouses less frequently than would be expected based on [Lillywhite \(1963, 1972\)](#). Furthermore, when “coffee” is associated with a location not listed in those sources, it is often unclear whether the venue is a coffeehouse, a tavern, or an inn. This ambiguity underscores the difficulty of identifying additional establishments from digitized archives and reinforces the conclusion that it is hard to improve upon [Lillywhite \(1963, 1972\)](#). Moreover, claims that thousands of coffeehouses operated in London by the late seventeenth or early eighteenth century (e.g., [Iliffe, 1995](#)) rely primarily on secondary sources and lack primary documentation. In fact, the first systematically documented count appears in 1739, when William Maitland recorded 551 coffeehouses in his *History of London*, compiled over an eleven-month period ([Ellis, 2006](#), pp. xxix–xxx).

Third, Appendix Figure [D11](#) plots the number of coffeehouses in seventeenth-century London in the dataset. The series exhibits discrete shifts that align with major historical events. The first recorded opening occurs in 1652, and counts increase sharply from 14 in 1659 to 26 in 1660, coinciding with the Stuart Restoration. The Great Fire of 1666 is associated with a temporary disruption in recorded coffeehouses and documentation, and the series does not display a pronounced decline in 1666,

⁴⁸[Lillywhite \(1963, 1972\)](#) does not mention [City of London Magistrates \(1663\)](#), likely because the survey lists coffeehouses only by owner and ward, without names or specific addresses.

⁴⁹The sharp increase under the ± 3 margin reflects a spike in 1666 shown in Appendix Figure [D11](#), likely driven by the extensive documentation of properties affected by the Great Fire, which destroyed nearly all of London within the city walls.

⁵⁰[Mackie \(2013\)](#) does not draw on [Lillywhite \(1972\)](#).

consistent with incomplete observation. The number of new coffeehouses increases sharply following the Rebuilding Acts of 1666 and 1670, and the upward trend persists through the end of the century.

Fourth, Appendix Figure D12 plots the frequency of coffeehouse-related terms in the British English corpus from the Google Books Ngram Viewer. Because most coffeehouses in the British Isles were located in London (Cowan, 2005), movements in the corpus likely reflect developments in London. The close alignment between the two series suggests that Lillywhite (1963, 1972) provides a reasonably representative account of the historically observed universe of coffeehouses. Mentions in the early seventeenth century, prior to the first London coffeehouse opening in 1652, likely reflect the emergence of discourse about coffee and coffeehouses rather than physical establishments (Smith, 2001).

Mathematical Practitioners. To our knowledge, no independent source exists to cross-verify the set of so-called mathematical practitioners compiled by Taylor (1954), beyond the consensus noted in Section 3 that subsequent research has added only a small number of additional practitioners and that the list is largely considered definitive (Walton, 2017, p. 89). A simple internal validity exercise nonetheless lends credibility to the dataset. Appendix Figure D13 reports, by decade, the number of connections held by the most connected practitioner. These central figures align closely with established accounts of seventeenth-century London science.

Before the prominence of the polymath Robert Hooke (1635–1703), widely known for frequent coffeehouse visits and highly sociable engagement with fellow scientists, the leading instrument maker Elias Allen (c. 1588–1653) consistently ranks among the most connected practitioners. Allen had an extensive clientele, including Kings James I and Charles I, and maintained ties with prominent mathematicians such as Edmund Gunter (1581–1626) and William Oughtred (1574–1660), inventor of the slide rule and the multiplication sign. Alongside these figures appears William Leybourne (1626–1716), an active mathematician, land surveyor, author, printer, and bookseller.

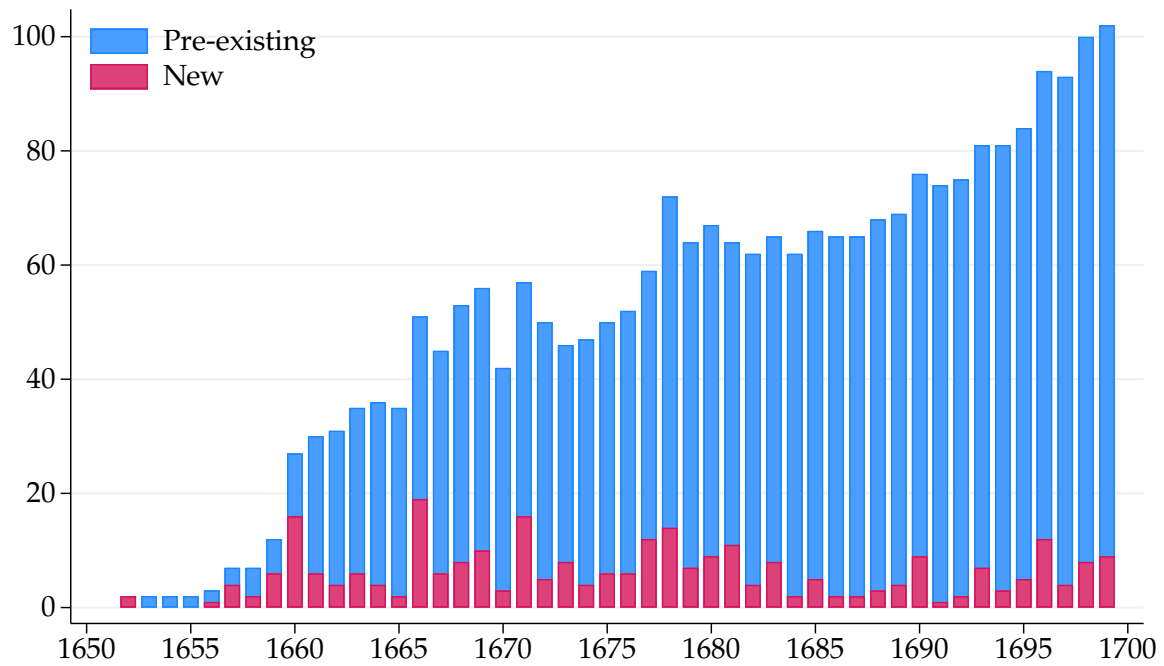


FIGURE D11. COFFEEHOUSES IN LONDON OVER TIME

Notes: Figure plots the annual number of coffeehouses in London from 1652, when the first opened, to 1699, distinguishing between those already established and those newly opened. Data from [Lillywhite \(1963, 1972\)](#). [Return to page 64]

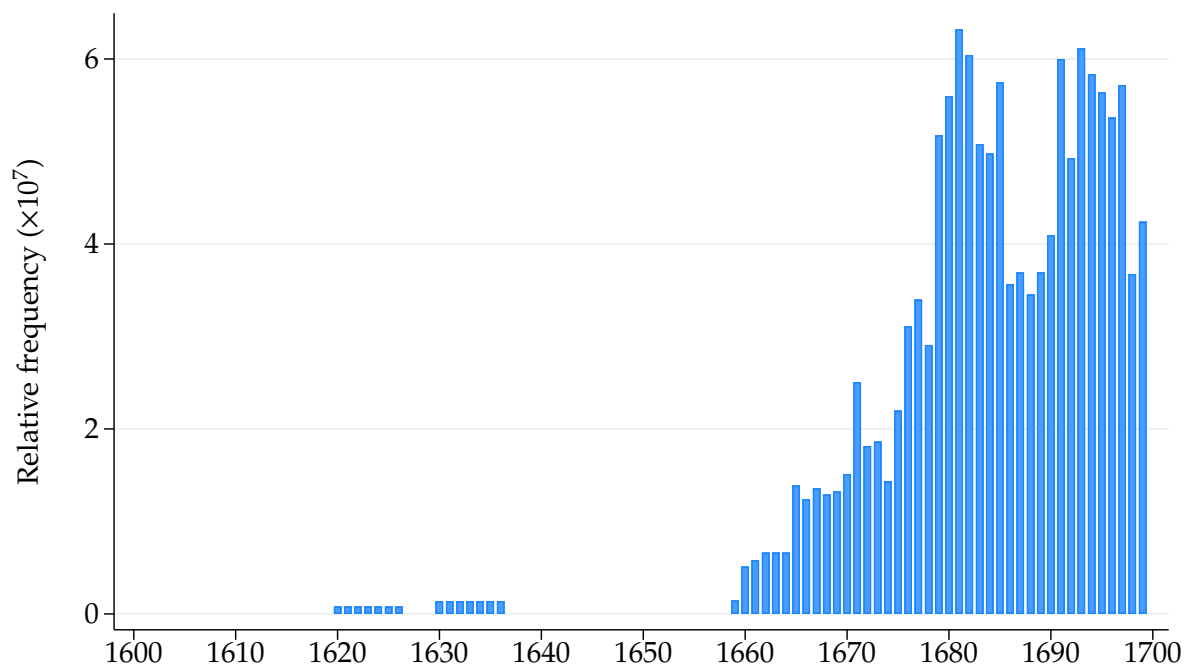


FIGURE D12. FREQUENCY OF N-GRAMS CONTAINING COFFEEHOUSE TERMS OVER TIME

Notes: Figure plots the annual relative case-insensitive frequency of n-grams containing the terms "coffee house" and "coffeehouse" from 1600 to 1699. Data from Google Books Ngram Viewer, British English corpus, accessed November 5, 2025. [Return to page 65]

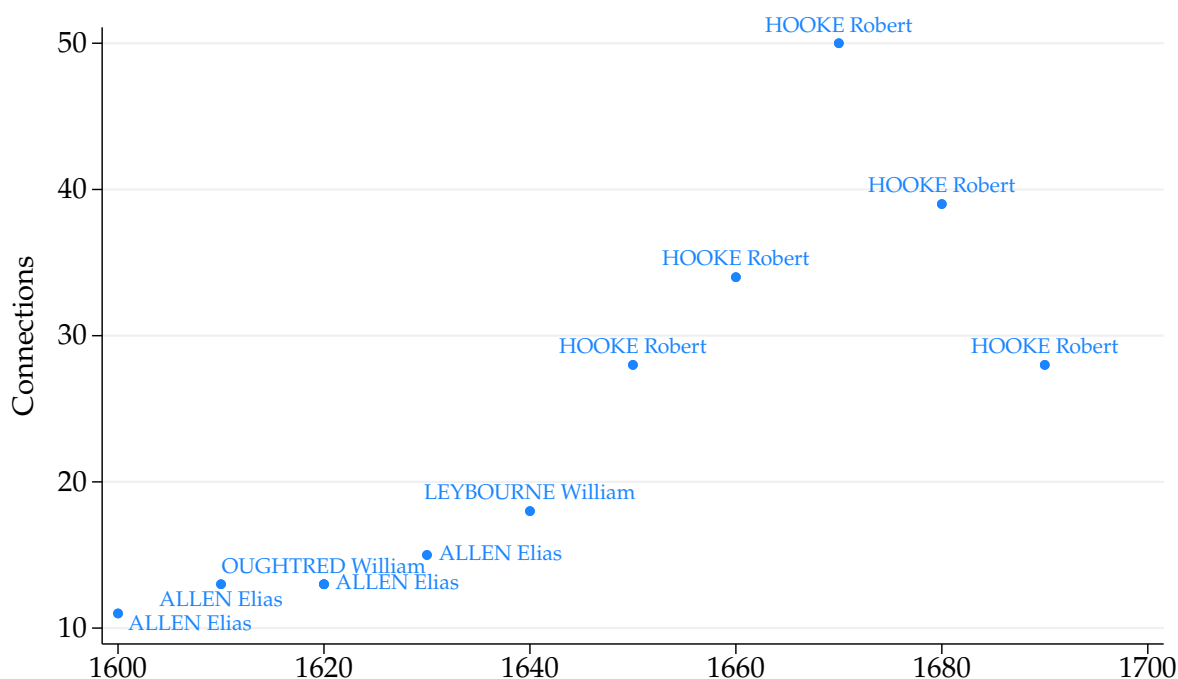


FIGURE D13. MOST CONNECTED MATHEMATICAL PRACTITIONER OVER TIME

Notes: Figure shows, for each decade, the number of connections of the most connected mathematical practitioner. In the 1620s, William Oughtred and Elias Allen are tied with 13 connections each. [Return to pages 16, 65]

D.11. GEOCODING

D.11.1 Procedure

Each historical address is assigned a geographic coordinate through a two-step, manually performed process aided by digital tools:

1. The address is located on a historical map of London.
 - Digitized historical maps of London are available on the Grub Street Project website (<https://www.grubstreetproject.net/>). Several of these maps support searchable places. See Figure D14.⁵¹
 - If the address cannot be found on a historical map of London, we use contextual information from the source or additional resources (e.g., Wikipedia, historical gazetteers, historical works on London) to identify either a nearby address that can be located on a historical map of London or, failing that, the smallest containing geography that can be so located, typically the street. We prefer the nearby address where one is available, and fall back on the containing geography only when the exact position within it cannot be recovered.

⁵¹A digital gazetteer of locations from the Woodcut Map of London (early 1560s) has been created on the Map of Early Modern London website (<https://mapoflondon.uvic.ca/index.htm>), which was occasionally used for hard-to-geocode addresses.

2. The geographic coordinate of the location identified in Step 1 is estimated by visually overlaying the historical map onto a contemporary Google map of London, aligning key geographic features (e.g., rivers, streets, and churches) that survive in both. This comparison is inherently imperfect, owing to the limited accuracy of seventeenth-century maps of London. John Rocque's 1746 map constitutes the earliest topographic survey of the city precise enough to allow comprehensive georeferencing against modern maps, so for the seventeenth century the alignment rests on individual features whose present-day position is known independently rather than on the geometry of the historical map itself.⁵²

Additionally, though infrequently, we may deviate from this process and rely on other credible sources. For example, we use a map of about 100 London workplaces from Taylor (1954), whose plotted points are themselves transferred to coordinates by the overlay described in Step 2, and coordinates for roughly 600 addresses already geocoded in the Early Modern Letters Online project (Hotson and Lewis, 2024), which supplies them directly. All coordinates are manually verified.

D.11.2 Worked Examples

Below we geocode four historical addresses, ordered from the most to the least straightforward, to demonstrate the procedure.

Direct match, no modern name. Geocode "Jewen St," recorded as the address of a scientific instrument maker active in 1671.

1. Strype's 1720 survey and accompanying map, reproduced in Figure D14, record Jewen (Jewin) Street running east from Aldersgate Street to Redcross Street, immediately north of the churchyard of St. Giles-without-Cripplegate.⁵³
2. Unlike the cases below, the street has no modern counterpart that can be matched by name: Jewin Street and the neighboring Redcross Street were demolished in the twentieth century for the construction of the Barbican Estate.⁵⁴ The coordinate is therefore obtained by overlaying the historical map onto a contemporary Google map and aligning two features whose present-day position is known, the church of St. Giles-without-Cripplegate and the line of Aldersgate Street. The midpoint of the street yields the following coordinate: (51.51955, -0.09577).

Direct match, surviving name. Geocode "St. Katharine's Stairs."⁵⁵

⁵²Cf. *Locating London's Past: Recreating Rocque, London 1746*, available at <https://locatinglondonspast.wordpress.com/2011/06/22/recreating-rocque-london-1746>, accessed September 5, 2025.

⁵³Strype describes the yard of the Castle Inn on Aldersgate Street as having "a passage into Jewen street" (Strype, 1720). See *The Grub Street Project :: Places :: Castle and Falcon*, available at <https://www.grubstreetproject.net/places/945/>, accessed July 27, 2026.

⁵⁴*Jewin Street*, London Picture Archive, The London Archives, [available online](#), accessed July 27, 2026.

⁵⁵Spelling varies across sources between "Katharine" and "Katherine"; we retain each source's form when quoting it.

1. The Grub Street Project provides points on digitized maps of London from 1553 to 1736 indicating the location of “St. Katherine’s Stairs.”⁵⁶
2. The 1736 map clearly shows that the location lies just southeast of the Tower of London, next to the Thames. Because the name survives, the corresponding modern location is identified directly on Google Maps as “St. Katharine Docks,” and the overlay of Step 2 serves only to confirm the match against the Tower and the river frontage. This yields the following coordinate: (51.50640, –0.07187).

Nearby landmark. Geocode “Pasqua Rosée’s Coffee House” in “St. Michael’s Alley, Cornhill.”

1. The location refers to a medieval alley next to St. Michael’s Church in Cornhill.⁵⁷ The original church, destroyed in the Great Fire of London (1666), was replaced in 1670 by the present church on the same site.⁵⁸
2. The exact position of the coffeehouse within the alley cannot be recovered, so we substitute the adjacent church, whose site is unchanged and which Google Maps locates directly. This yields the following coordinate: (51.51325, –0.08556).⁵⁹

Containing street. Geocode “Pilkington Court.”

1. The location refers to a court in Little Britain, according to a contemporary obituary register kept by the London bookseller and antiquary Richard Smyth.⁶⁰
2. We are unable to identify the exact location of Pilkington Court within Little Britain, and no nearby address is recoverable, so we fall back on the containing street. Little Britain is a sufficiently precise reference, being a short and well-known street in seventeenth-century London, and its Wikipedia entry provides the following coordinate directly: (51.51699, –0.09780).⁶¹ Accordingly, these coordinates are assigned to “Pilkington Court.”

⁵⁶The Grub Street Project :: Places :: St. Katherine’s Stairs, available at <https://www.grubstreetproject.net/places/4318/>, accessed September 4, 2025.

⁵⁷The Grub Street Project :: Places :: St. Michael’s Alley, available at <https://www.grubstreetproject.net/places/4421/>, accessed August 5, 2025.

⁵⁸St Michael, Cornhill, Wikipedia, available at https://en.wikipedia.org/wiki/St_Michael,_Cornhill, accessed August 5, 2025.

⁵⁹St Michael Cornhill Saint Michael’s Alley, London on Google Maps, available at <https://maps.app.goo.gl/9DJKJm9cwTVkz1fN7>, accessed August 5, 2025.

⁶⁰Smyth’s 1661 entry records “Mr. Franck, a juryman at Pilkington Court in Little Britain” (Smyth, 1849, p. 53). The register was published posthumously in 1849 and was discovered through a Google search for “Pilkington Court” “London” filetype:pdf.

⁶¹Little Britain, London, Wikipedia, available at https://en.wikipedia.org/wiki/Little_Britain, accessed September 5, 2025.



FIGURE D14. 1720 LONDON (STRYPE): [EXCERPT FROM THE GRUB STREET PROJECT WEBSITE](#)

Notes: Digitized 1720 map of London with a geocoded layer of streets, roads, thoroughfares, public markets, bodies of water, and places, from the Grub Street Project at the University of Saskatchewan, based on *A New Plan of the City of London, Westminster, and Southwark* in [Strype \(1720\)](#). [Return to pages [67](#), [68](#)]

D.12. FIGURES

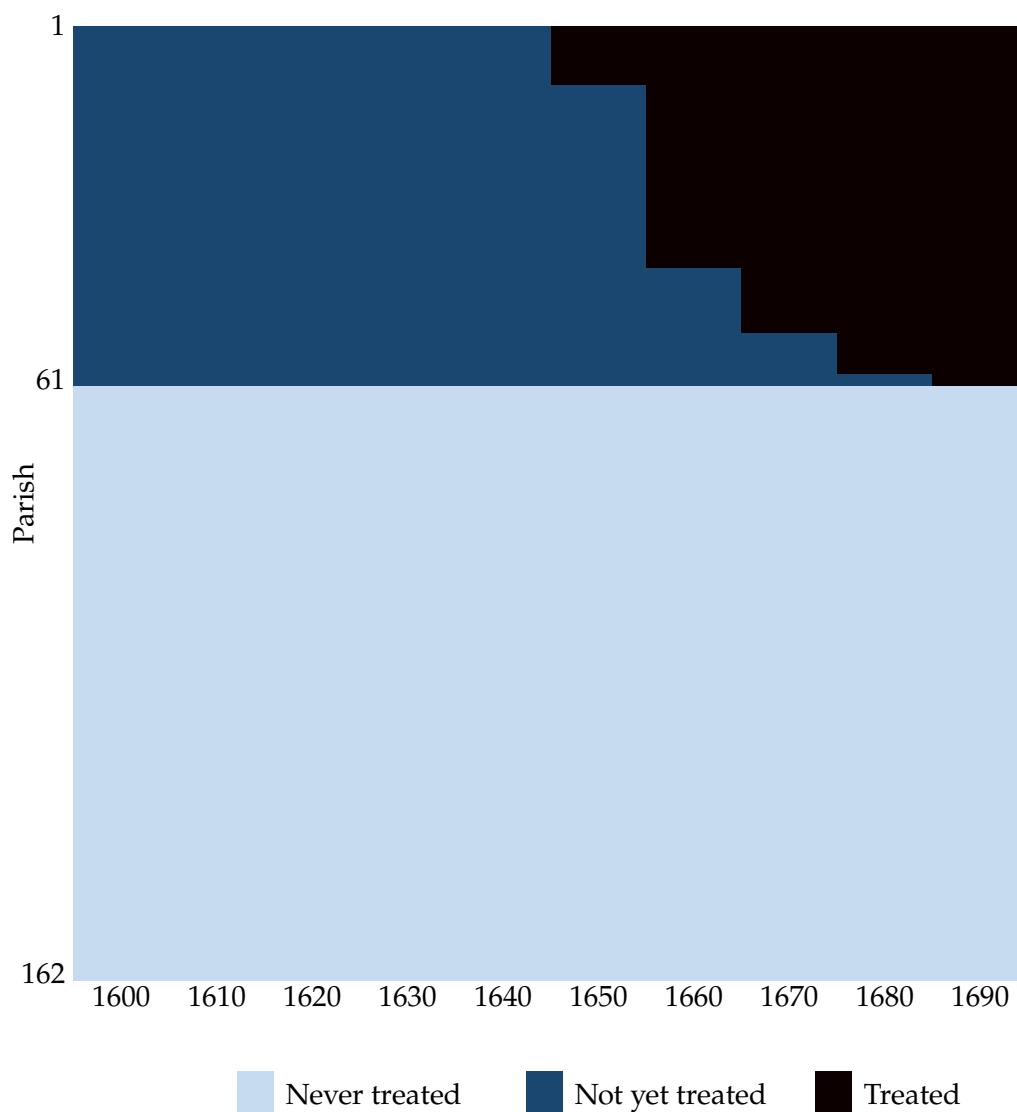


FIGURE D15. TREATMENT ROLLOUT: FIRST COFFEEHOUSES

Notes: Figure shows the decades in which the first coffeehouse opened across all 162 parishes, grouped by decade of first opening. Parishes are ordered from earliest to latest opening decade, followed by those with no coffeehouse openings. [Return to page 20]

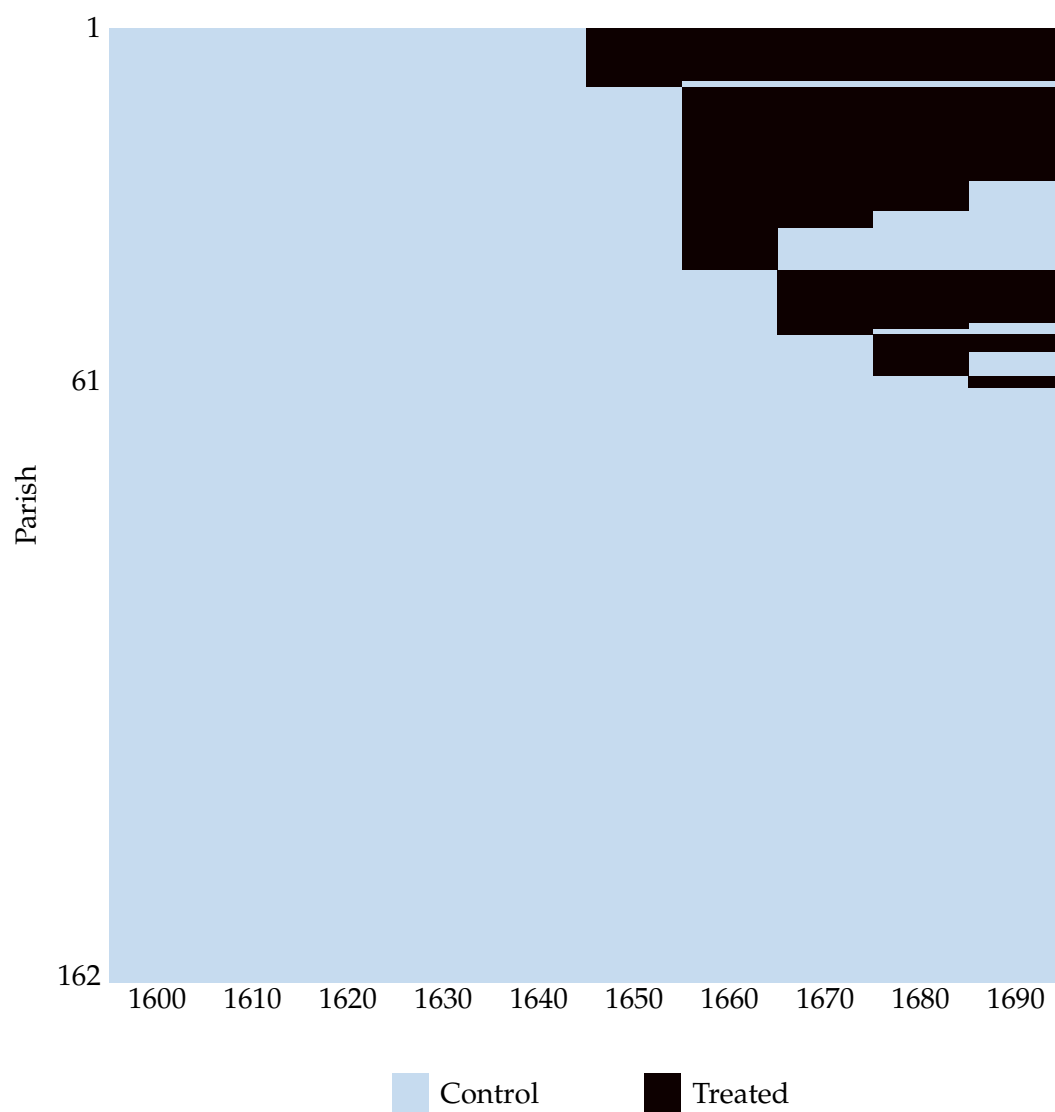


FIGURE D16. TREATMENT ROLLOUT: COFFEEHOUSE PRESENCE

Notes: Figure shows the decades in which a coffeehouse is open across all 162 parishes, grouped by decade of first opening. Parishes are ordered from earliest to latest opening decade, followed by those with no coffeehouse openings. [Return to page 23]

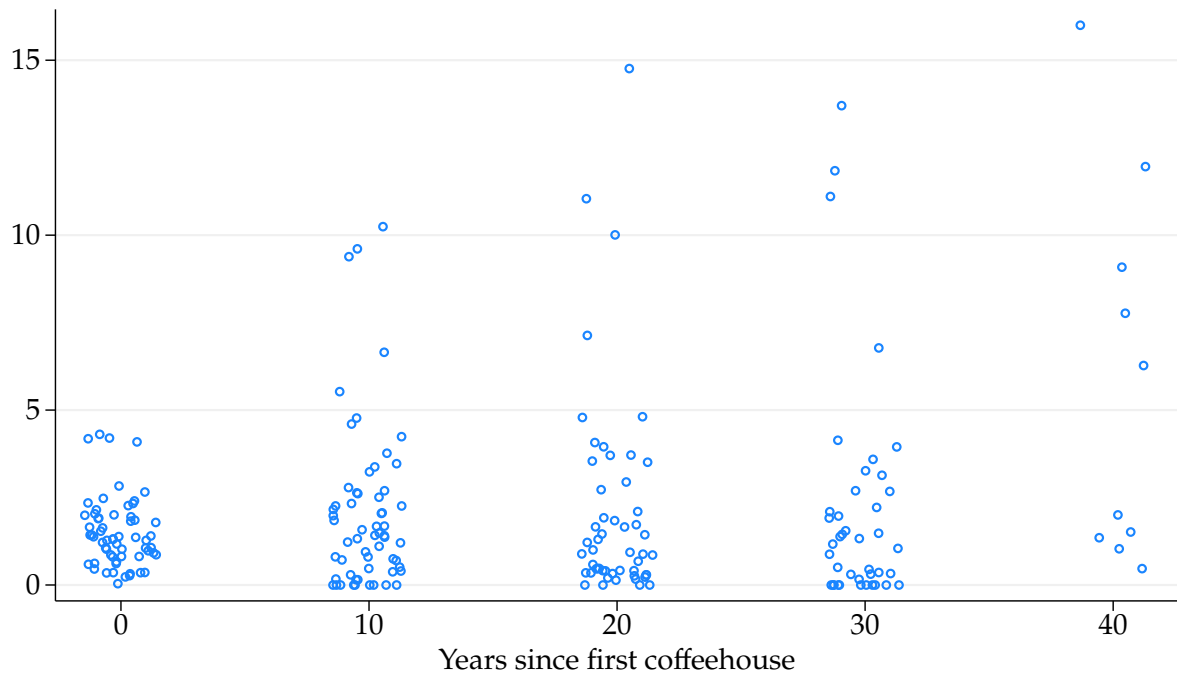


FIGURE D17. COFFEEHOUSES BY YEARS SINCE FIRST OPENING

Notes: Figure shows a jittered strip plot of the number of coffeehouses by years since the first coffeehouse opened. Sample includes the 61 parishes that ever have a coffeehouse from the 1600s to the 1690s. [Return to page 20]

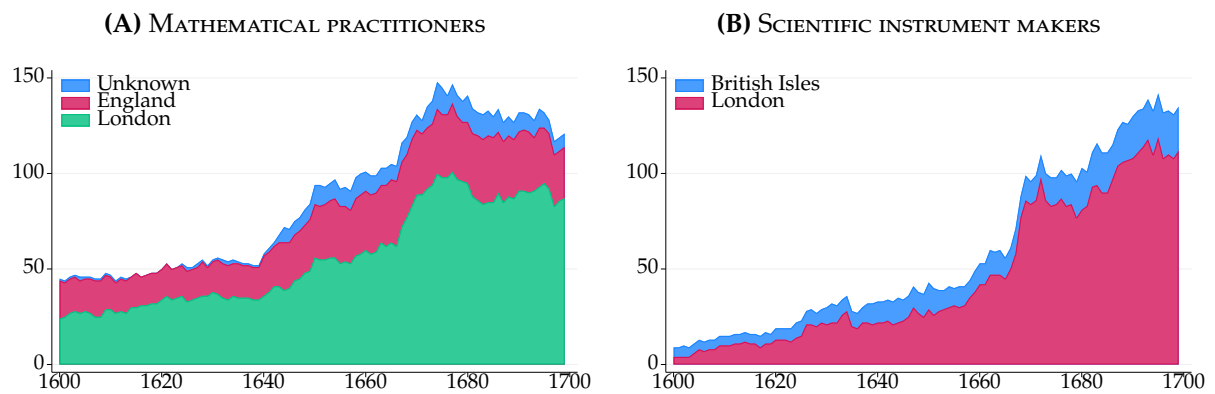


FIGURE D18. SCIENCE IN LONDON RELATIVE TO ENGLAND AND THE BRITISH ISLES OVER TIME

Notes: Panel A plots the annual number of mathematical practitioners in seventeenth-century England, separated by whether they were ever active in London or only outside it but still within England. Unknowns were active in England, but their London activity is unknown. Panel B plots the annual number of scientific instrument makers in the seventeenth-century British Isles and London. Data from Clifton (1995); Hay et al. (2025); Taylor (1954). [Return to pages 14, 15, 24]

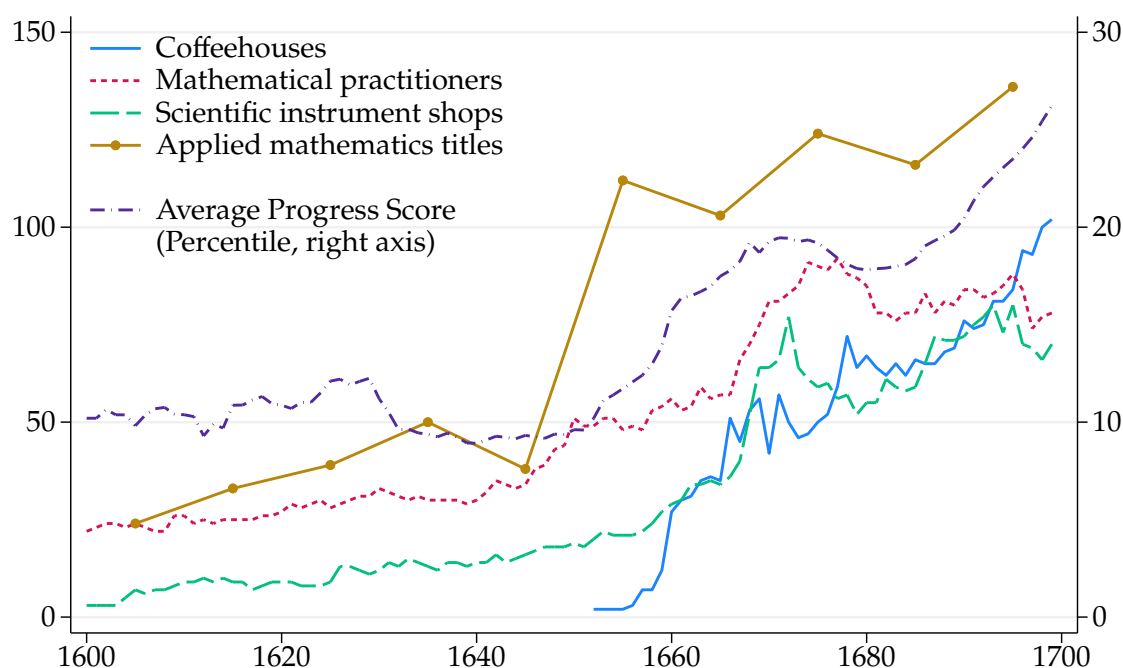


FIGURE D19. COFFEEHOUSES, SCIENCE, MATHEMATICAL TITLES, AND BELIEF IN PROGRESS

Notes: Figure repeats the three series of Figure 1, the annual number of coffeehouses, mathematical practitioners, and scientific instrument shops in seventeenth-century London, and overlays two national series from published work. Applied mathematics titles count the books in English published in each decade, excluding reprints, in the replication files of [Kelly and Ó Gráda \(2022\)](#), plotted at decade midpoints as in their published figure. These are decade totals, so their level is not comparable to the annual counts plotted beside them. The average progress score, read against the right axis, follows the construction of [Almelhem et al. \(2026\)](#): the score of each volume on their main progress dictionary is converted to its percentile rank across their corpus of 264,433 English-language volumes, and the line plots, for each year, the mean percentile among volumes published within ten years of it. Data from [Almelhem et al. \(2026\)](#); [Clifton \(1995\)](#); [Hay et al. \(2025\)](#); [Kelly and Ó Gráda \(2022\)](#); [Lillywhite \(1963, 1972\)](#); [Taylor \(1954\)](#). [Return to page 18]

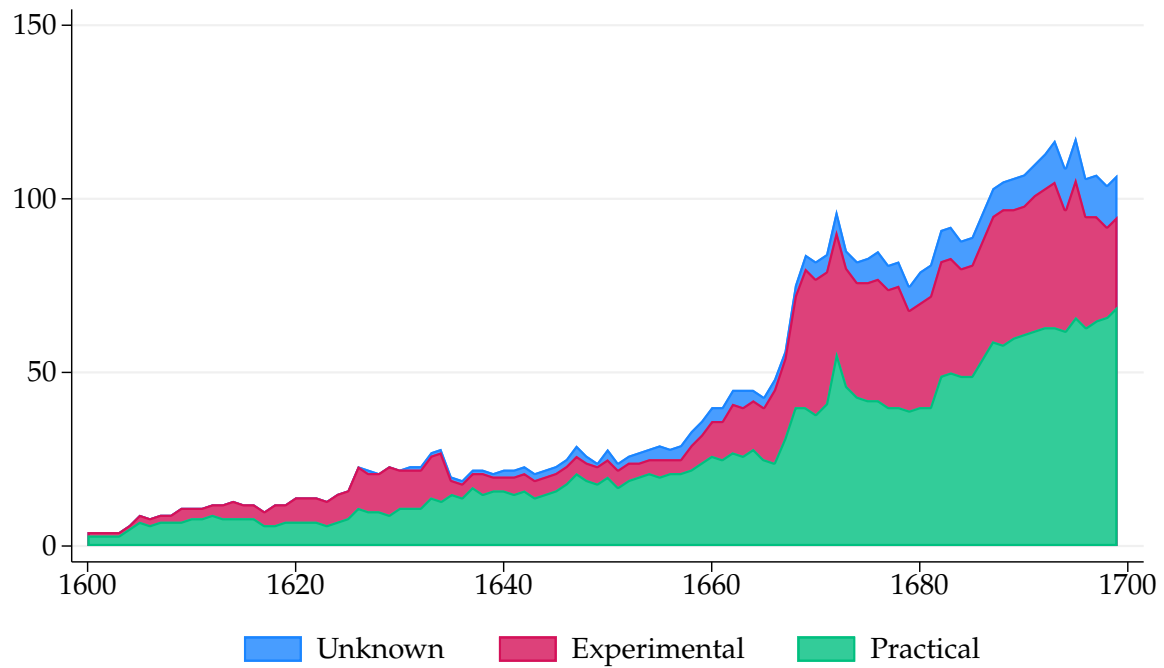


FIGURE D20. SCIENTIFIC INSTRUMENT MAKERS IN LONDON OVER TIME BY TYPE

Notes: Figure plots the annual number of scientific instrument makers in seventeenth-century London by type. Experimental makers produced optical (e.g., telescopes, microscopes) and philosophical (e.g., air pumps, barometers, electric machines) instruments. Practical makers produced mathematical (e.g., dividers, sectors, scales), navigational (e.g., compasses, astrolabes), and surveying (e.g., theodolites, circumferentors) instruments. Data from [Clifton \(1995\)](#); [Hay et al. \(2025\)](#). [Return to page 15]

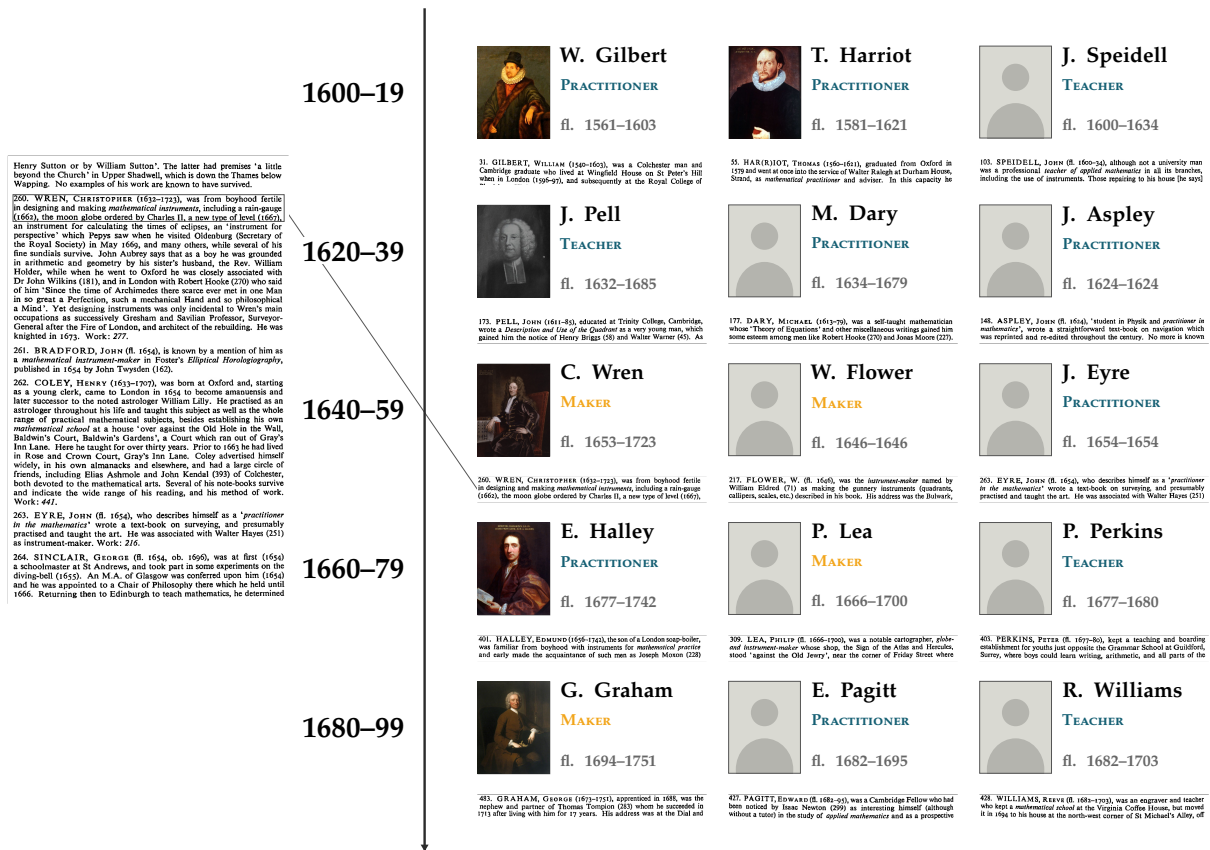


FIGURE D21. EXAMPLES OF MATHEMATICAL PRACTITIONERS (TAYLOR, 1954)

Notes: Figure shows three mathematical practitioners for each twenty-year cohort of first flourishing in the London sample described in Section 3, labeled by the years the cohort spans. Portrait slots go to practitioners matched to an entry in the *Oxford Dictionary of National Biography* (*Oxford Dictionary of National Biography*, 2004) and use images from Wikipedia; their number rounds the cohort's matched share to thirds, and the remaining slots name a practitioner and stand in a silhouette. Beneath each practitioner is a facsimile of the opening lines of their entry in Taylor (1954), and Christopher Wren's entry is boxed on the page reproduced at left. Orange marks instrument makers. [Return to page 14]

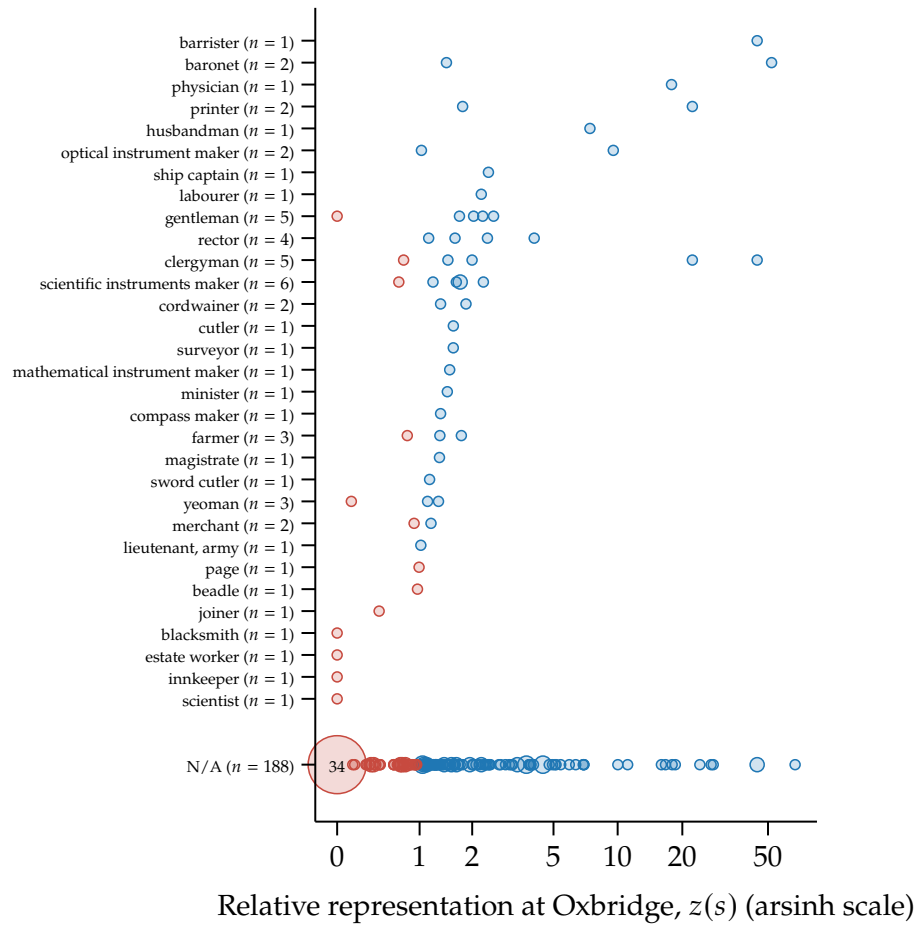


FIGURE D22. RELATIVE OXBRIDGE REPRESENTATION BY FATHER'S OCCUPATION

Notes: Sample includes all 244 mathematical practitioners whose London activity interval overlaps 1600–1699. The horizontal axis shows relative surname representation among pre-1650 Oxbridge alumni, $z(s)$, on an inverse hyperbolic sine (arsinh) scale. Each circle is one value of $z(s)$ within a category, with area proportional to the number of practitioners sharing that value, and counts are printed inside circles drawn large enough to hold them. Blue circles have $z(s) > 1$; red circles have $z(s) \leq 1$. Numbers in parentheses give the number of practitioners in each category, and the 31 observed occupations are ordered by their median $z(s)$. The bottom row holds the 188 practitioners whose paternal occupation is unobserved, 34 of whom carry a surname that never appears among pre-1650 Oxbridge alumni. The median $z(s)$ in that row is 1.14, against a median of 1.51 across the observed occupation medians. Occupations are coded from fathers' occupational titles using the PST system developed by [Wrigley \(2010\)](#). Data from the authors' dataset. [Return to page 25]

D.13. TABLES

TABLE D3—COFFEEHOUSE SOURCE CATEGORIES

	Any source		Only source	
	Coffeehouses	Per coffeehouse	Coffeehouses	Per coffeehouse
<i>Panel A. Source categories</i>				
Published and private	158	0.55	67	0.23
Printed books, pamphlets, and literary works	47	0.16	9	0.03
Newspapers and periodicals	44	0.15	7	0.02
Secondary histories and reference works	40	0.14	8	0.03
Manuscripts and private papers	36	0.13	13	0.05
Diaries, memoirs, and personal writings	32	0.11	6	0.02
Advertisements, notices, and ephemera	31	0.11	9	0.03
Catalogues, sales, and auctions	8	0.03	2	0.01
Material and visual	98	0.34	44	0.15
Trade tokens	65	0.23	34	0.12
Maps, views, inscriptions, and topographical evidence	36	0.13	10	0.03
Administrative and official	90	0.31	27	0.09
Parish, civic, and local records	44	0.15	13	0.05
Company, commercial, and financial records	34	0.12	7	0.02
State, legal, and court records	33	0.12	6	0.02
Unknown	52	0.18	50	0.17
No source, unidentified, or unclear	52	0.18	50	0.17
<i>Panel B. Summary statistics</i>				
Coffeehouses	286		1	
Sources	617		2.16	
Sources by finer categories	502		1.76	
Sources by broader categories	398		1.39	

Notes: Panel A reports source-category coverage for distinct coffeehouses. The first pair of columns counts coffeehouses with at least one source in the category and divides that count by the total number of distinct coffeehouses; the second pair does the same for coffeehouses whose sources all fall in the given category, meaning no other broader category for broad rows and no other finer category for indented rows. Fine rows need not sum to broad rows because a coffeehouse can appear in multiple finer categories. Panel B reports summary totals and averages per coffeehouse. [Return to page 13]

TABLE D4—DETAILED TYPES OF MATHEMATICAL PRACTITIONERS

Occupation	Distinct practitioners
<i>Any</i>	244
<i>Other mathematical practitioners</i>	69
Mathematical practitioner	25
Surveyor	5

Continued on next page

Table D4 continued

Occupation	Distinct practitioners
Hydrographer	4
Mathematician	4
Land surveyor	3
Amateur mathematician	2
Engineer	2
Almanac maker; mathematics teacher; land surveyor; dial maker	1
Applied mathematics student; mathematics teacher	1
Astrologer	1
Astrologer; dial maker; surveyor	1
Astronomer; astrologer; mathematical practitioner	1
Bookseller	1
Chart maker; surveyor	1
Dial maker; mathematical practitioner	1
Magnetism researcher	1
Marine hydrographer	1
Marine surveyor	1
Master of the mint	1
Master printer	1
Mathematical bookseller	1
Mathematics promoter	1
Mathematics student	1
Navigational instrument seller	1
Navigator	1
Operator to the Royal Society	1
Physician	1
Professional surveyor	1
Surveyor; cartographer; instrument designer	1
Surveyor; instrument maker	1
Surveyor; mathematical practitioner	1
<i>Teachers</i>	54
Mathematics teacher	21
Mathematics professor	6
Mathematics schoolmaster	6
Navigation teacher	4
Practical mathematics teacher	3
Astronomy professor	2
Geometry professor	2
Mathematics lecturer	2
Almanac maker; mathematics teacher; land surveyor; dial maker	1
Applied mathematics student; mathematics teacher	1
Applied mathematics teacher	1
Instrument maker; mathematics teacher	1
Mathematics school founder	1
Mathematics teacher or lecturer	1
Mathematics tutor	1
Practical mathematics and navigation teacher	1

Continued on next page

Table D4 continued

Occupation	Distinct practitioners
<i>Makers</i>	130
Instrument maker	44
Mathematical instrument maker	25
Optical instrument maker	19
Clockmaker	5
Chart maker	4
Compass maker	4
Globe maker	4
Glass grinder	3
Dial maker	2
Engraver	2
Globe maker; mathematical instrument maker	2
Almanac maker	1
Almanac maker; mathematics teacher; land surveyor; dial maker	1
Astrologer; dial maker; surveyor	1
Astronomical instrument maker	1
Chart maker; surveyor	1
Craftsman	1
Dial maker; calendar maker	1
Dial maker; mathematical practitioner	1
Engraver; mathematical instrument maker	1
Instrument maker; mathematics teacher	1
Instrument repairer	1
Master craftsman; smith	1
Mechanic	1
Nautical instrument maker	1
Surveyor; cartographer; instrument designer	1
Surveyor; instrument maker	1

Notes: Counts are distinct mathematical practitioners with a usable London geocode whose recorded flourishing overlaps 1600–1699, matching the period covered by the empirical panel. A given practitioner can belong to multiple categories; together, the categories are exhaustive. Indented rows report occupations harmonized from raw occupation strings. [Return to page [14](#)]

E. MODEL APPENDIX

E.1. ENVIRONMENT

Decades are indexed by t and parishes by $n = 1, \dots, N$ with $N \geq 2$, and $d_{ni} \geq 0$ is the distance in kilometers between parishes n and i . $S_{nt} > 0$, S for scientists, is the number of applied scientists in parish n in decade t , in the model an expected count whose realization is the observed count.⁶² $S_t = \sum_n S_{nt}$ is London's total and $\sigma_{nt} = S_{nt}/S_t$ parish n 's share of it. A pool of $\bar{L} > 0$ potential entrants into science makes $\pi_t = S_t/\bar{L}$ the science share of the pool. Treatment $D_{nt} \in \{0, 1\}$ indicates a coffeehouse in parish n in decade t , and $m_{nt} = e^{\phi D_{nt}}$ is the factor by which it scales meetings between applied scientists.

The parameters are $\phi > 0$, $\delta > 0$, $\eta > 0$, $\gamma \geq 0$, and the Fréchet shapes $\varepsilon_P, \varepsilon_S$ of Assumption E.2: ϕ is the coffeehouse effect on meetings, δ the decay of meetings with distance, η the return to access, and γ the dispersion force. The fundamentals, taken as given, are $\psi_{nt} > 0$ and $\bar{u} > 0$: ψ_{nt} is what makes parish n attractive apart from access and $\bar{u}$ the payoff of the trade outside science.

Definition E.1 (Access). The access in count form of parish n in decade t is

$$K_{nt} = m_{nt} \sum_i e^{-\delta d_{ni}} m_{it} S_{it}. \quad (\text{E.1})$$

The access in share form of parish n in decade t is

$$\kappa_{nt}(\sigma_t) = m_{nt} \sum_i e^{-\delta d_{ni}} m_{it} \sigma_{it}, \quad (\text{E.2})$$

with $\sigma_t = (\sigma_{1t}, \dots, \sigma_{Nt})$ the vector of shares, written κ_{nt} when the argument is clear.

Access, K for the knowledge within an applied scientist's reach, is the agglomeration force of the quantitative urban model (Redding, 2025, §4). There productivity rises with the employment nearby, discounted by travel time, through production externalities such as knowledge spillovers (Redding, 2025, §4.2, eq. (9)). Here the externality is the meeting rate: access is the distance-discounted sum of applied scientists, with the coffeehouse as the shifter m_{nt} at each end.

Lemma E.1 (The aggregate factors out of access). $K_{nt} = S_t \kappa_{nt}(\sigma_t)$.

⁶²A parish with no applied scientist in a decade is the limit of the model as its fundamental ψ_{nt} goes to zero. By (E.9) its share is then zero in the observed equilibrium and in every counterfactual, and by (E.1) it adds nothing to the access of others, so Procedure E.4 keeps it at zero.

Proof. Using $S_{it} = \sigma_{it} S_t$, we have

$$\begin{aligned}
K_{nt} &= m_{nt} \sum_i e^{-\delta d_{ni}} m_{it} S_{it} \\
&= m_{nt} \sum_i e^{-\delta d_{ni}} m_{it} \sigma_{it} S_t \\
&= S_t m_{nt} \sum_i e^{-\delta d_{ni}} m_{it} \sigma_{it} \\
&= S_t \kappa_{nt}(\sigma_t).
\end{aligned}$$

□

Definition E.2 (Return). The return to practicing science in parish n in decade t is

$$R_{nt} = \psi_{nt} K_{nt}^\eta S_{nt}^{-\gamma}. \quad (\text{E.3})$$

The dispersion force γ is, in [Redding \(2025, §4\)](#), an inelastic supply of land: the applied scientists of a parish compete for its premises, whose supply is fixed, so the price of floor space rises with S_{nt} and the return net of that price falls at elasticity γ , the share of income spent on premises.

Definition E.3 (Composite coefficient). The composite coefficient at elasticity ε is

$$\beta(\varepsilon) = \frac{\varepsilon \eta}{1 + \gamma \varepsilon}. \quad (\text{E.4})$$

Furthermore, set $\beta_P = \beta(\varepsilon_P)$ and $\beta_S = \beta(\varepsilon_S)$.

Lemma E.2 (β is increasing). If $0 < \varepsilon_1 \leq \varepsilon_2$ then $0 < \beta(\varepsilon_1) \leq \beta(\varepsilon_2)$.

Proof. Positivity follows from $\varepsilon \eta > 0$ and $1 + \gamma \varepsilon > 0$. Now,

$$\begin{aligned}
\beta(\varepsilon_1) \leq \beta(\varepsilon_2) &\iff \varepsilon_1 \eta (1 + \gamma \varepsilon_2) \leq \varepsilon_2 \eta (1 + \gamma \varepsilon_1) \\
&\iff \varepsilon_1 \eta + \gamma \eta \varepsilon_1 \varepsilon_2 \leq \varepsilon_2 \eta + \gamma \eta \varepsilon_1 \varepsilon_2 \\
&\iff \eta(\varepsilon_2 - \varepsilon_1) \geq 0,
\end{aligned}$$

which holds because $\eta > 0$ and $\varepsilon_1 \leq \varepsilon_2$. □

E.2. MEETINGS AND GRAVITY

Assumption E.1 (Meetings). Each applied scientist in n meets each applied scientist in i at Poisson rate

$$\lambda_{nit} = \chi m_{nt} e^{-\delta d_{ni}} m_{it}, \quad (\text{E.5})$$

with $\chi > 0$, independently across pairs. Each meeting is documented with probability $\rho_t \in (0, 1]$, independently of everything else.

Assumption [E.1](#) is the gravity equation of quantitative urban models applied to meetings ([Redding, 2025, §2.4, §5.2](#)). The flow between two parishes rises with the mass at either end, as interaction rises with the mass of people nearby in the classical account

(e.g., Glaeser et al., 1992), and falls with distance. In our application, the added part is the coffeehouse, which raises the rate given that mass and enters as the shifter m_{nt} at each end of a pair.

Lemma E.3 (Superposition). *If $X_1, \dots, X_N$ are independent Poisson counts with rates $r_1, \dots, r_N$, then $X_1 + \dots + X_N$ is a Poisson count with rate $r_1 + \dots + r_N$.*

Proof. See the Superposition Theorem in Kingman (1993, p. 16, §2.2). □

Lemma E.4 (Thinning). *If the number of events is Poisson with rate r , and each event is kept with probability p , independently of the other events and of their number, then the number of events kept is Poisson with rate pr .*

Proof. See the Colouring Theorem in Kingman (1993, p. 53, §5.1). □

Theorem E.1 (Gravity). *Under Assumption E.1 the number of documented connections T_{nit} between parishes $n \neq i$ in decade t is Poisson with mean*

$$\mathbb{E}[T_{nit}] = (\chi \rho_t) (m_{nt} S_{nt}) e^{-\delta d_{ni}} (m_{it} S_{it}). \quad (\text{E.6})$$

Proof. Since the $S_{nt} S_{it}$ pairs of an applied scientist in n and one in i meet independently, Lemma E.3 gives that the total number of meetings is Poisson with rate $\lambda_{nit} S_{nt} S_{it}$. By Lemma E.4, documentation with probability ρ_t thins it to a Poisson count with rate $\rho_t \lambda_{nit} S_{nt} S_{it}$, which is the display once (E.5) is substituted. □

E.3. THE FRÉCHET DISTRIBUTION

The choice model of the next subsection rests on three standard properties of the Fréchet distribution. We found no textbook that proves them in the form it uses, so we prove them here.

Definition E.4 (Fréchet distribution). A random draw z is Fréchet with shape $\varepsilon > 0$ and scale $A > 0$ if

$$\mathbb{P}[z \leq x] = F_{\varepsilon, A}(x) = \exp(-Ax^{-\varepsilon}), \quad x > 0,$$

with density

$$f_{\varepsilon, A}(x) = A \varepsilon x^{-\varepsilon-1} e^{-Ax^{-\varepsilon}}, \quad x > 0.$$

Definition E.5 (Unit Fréchet distribution). A random draw z is unit Fréchet with shape ε if it is Fréchet with shape ε and scale $A = 1$.

Lemma E.5 (Scaling). *If z is unit Fréchet with shape ε , then Rz with $R > 0$ is Fréchet with shape ε and scale R^ε .*

Proof. For $x > 0$,

$$\begin{aligned}\mathbb{P}[Rz \leq x] &= \mathbb{P}[z \leq x/R] \\ &= \exp(-(x/R)^{-\varepsilon}) \\ &= \exp(-R^\varepsilon x^{-\varepsilon}) = F_{\varepsilon, R^\varepsilon}(x).\end{aligned}$$

□

Lemma E.6 (Max-stability). *If $u_1, \dots, u_N$ are independent Fréchet draws with common shape ε and scales $A_1, \dots, A_N$, then $\max_n u_n$ is Fréchet with shape ε and scale $\sum_{n=1}^N A_n$.*

Proof. For $x > 0$,

$$\begin{aligned}\mathbb{P}\left[\max_n u_n \leq x\right] &= \mathbb{P}[u_1 \leq x, \dots, u_N \leq x] \\ &= \prod_{n=1}^N \mathbb{P}[u_n \leq x] && \text{(independence)} \\ &= \prod_{n=1}^N \exp(-A_n x^{-\varepsilon}) \\ &= \exp\left(-\sum_{n=1}^N A_n x^{-\varepsilon}\right) = F_{\varepsilon, \sum_n A_n}(x).\end{aligned}$$

□

Lemma E.7 (Choice probability). *Let $u_1, \dots, u_N$ be independent Fréchet with common shape ε and scales $A_1, \dots, A_N$, and $S = \sum_k A_k$. Then $\mathbb{P}[u_n = \max_k u_k] = A_n/S$.*

Proof. Let $M = \max_{k \neq n} u_k$. M is a function of draws independent of u_n , so M and u_n are independent. By Lemma E.6, the distribution function of M is $F_{\varepsilon, S-A_n}$. Thus,

$$\begin{aligned}\mathbb{P}[u_n = \max_k u_k] &= \mathbb{P}[M \leq u_n] = \mathbb{E}[\mathbb{1}\{M \leq u_n\}] \\ &= \mathbb{E}\left[\mathbb{E}[\mathbb{1}\{M \leq u_n\} \mid u_n]\right] && \text{(iterated expectations)} \\ &= \mathbb{E}[F_{\varepsilon, S-A_n}(u_n)] && (M \text{ independent of } u_n) \\ &= \int_0^\infty F_{\varepsilon, S-A_n}(x) f_{\varepsilon, A_n}(x) dx \\ &= \int_0^\infty e^{-(S-A_n)x^{-\varepsilon}} \cdot A_n \varepsilon x^{-\varepsilon-1} e^{-A_n x^{-\varepsilon}} dx \\ &= \frac{A_n}{S} \int_0^\infty S \varepsilon x^{-\varepsilon-1} e^{-Sx^{-\varepsilon}} dx \\ &= \frac{A_n}{S} \underbrace{\int_0^\infty f_{\varepsilon, S}(x) dx}_{=1} \\ &= \frac{A_n}{S}.\end{aligned}$$

□

E.4. CHOICE

Assumption E.2 (Choice model). In decade t each of $\bar{L}$ potential entrants v makes two choices in sequence.

1. *Whether to practice.* The outside trade yields entrant v the payoff $\bar{u}\zeta_{0v}$ and practicing the payoff $V_t\zeta_{1v}$, where (ζ_{0v}, ζ_{1v}) are independent unit Fréchet draws with shape ε_S . The entrant practices if $V_t\zeta_{1v} > \bar{u}\zeta_{0v}$.
2. *Where to practice.* Practicing in parish n yields the payoff $u_{nv} = R_{nt}z_{nv}$, the parish's return scaled by an idiosyncratic draw, where $(z_{1v}, \dots, z_{Nv})$ are independent unit Fréchet draws with shape ε_P . The entrant picks the parish with the largest payoff, $\arg \max_n u_{nv}$.

Draws are independent across entrants and across the two choices, and $0 < \varepsilon_S \leq \varepsilon_P$.

Assumption E.2 is the Fréchet choice model of quantitative urban models (Redding, 2025, §4.1), in which a worker chooses where to live and work. Here the entrant makes two choices, whether to practice and where, each with its own dispersion parameter. The restriction $0 < \varepsilon_S \leq \varepsilon_P$ is the consistency condition of a nested choice (McFadden, 1978; Train, 2009): the two choices together are one nested maximization with inclusive-value coefficient $\varepsilon_S/\varepsilon_P$, which must lie in the unit interval for the probabilities to come from utility maximization. This means that entering science is no easier than moving parish. With Lemma E.2, the restriction gives $0 < \beta_S \leq \beta_P$.

Theorem E.2 (Shares and total). *Under Assumption E.2,*

$$\sigma_{nt} = \frac{R_{nt}^{\varepsilon_P}}{\sum_k R_{kt}^{\varepsilon_P}}, \quad V_t = \left(\sum_k R_{kt}^{\varepsilon_P} \right)^{1/\varepsilon_P}, \quad S_t = \bar{L} \frac{V_t^{\varepsilon_S}}{\bar{u}^{\varepsilon_S} + V_t^{\varepsilon_S}}, \quad \frac{\partial \ln S_t}{\partial \ln V_t} = \varepsilon_S(1 - \pi_t).$$

Proof. By Lemma E.5 the payoffs $u_{nv} = R_{nt}z_{nv}$ are independent Fréchet with shape ε_P and scales $R_{nt}^{\varepsilon_P}$. An entrant who practices picks the parish with the largest payoff, so by Lemma E.7 parish n is chosen with probability

$$\sigma_{nt} = \frac{R_{nt}^{\varepsilon_P}}{\sum_k R_{kt}^{\varepsilon_P}}. \quad (\text{E.7})$$

By Lemma E.6 the best payoff $\max_n u_{nv}$ is Fréchet with shape ε_P and scale $\sum_k R_{kt}^{\varepsilon_P} = V_t^{\varepsilon_P}$, that is, by Lemma E.5 read backwards, distributed as $V_t\tilde{z}$ with $\tilde{z}$ unit Fréchet with shape ε_P . The entrant practices when $V_t\zeta_{1v} > \bar{u}\zeta_{0v}$. By Lemma E.5 these two are independent Fréchet with shape ε_S and scales $V_t^{\varepsilon_S}$ and $\bar{u}^{\varepsilon_S}$, so by Lemma E.7 with two alternatives the entrant practices with probability

$$\pi_t = \frac{V_t^{\varepsilon_S}}{\bar{u}^{\varepsilon_S} + V_t^{\varepsilon_S}}. \quad (\text{E.8})$$

Each of the $\bar{L}$ entrants practices with probability π_t , so the expected number who practice is $S_t = \bar{L}\pi_t$, and by the independence of the two choices each practices in parish n with probability $\sigma_{nt}\pi_t$, so the expected number there is $S_{nt} = \sigma_{nt}S_t$. Finally, taking

logarithms of S_t and differentiating with respect to $\ln V_t$,

$$\begin{aligned} \ln S_t &= \ln \bar{L} + \varepsilon_S \ln V_t - \ln(\bar{u}^{\varepsilon_S} + V_t^{\varepsilon_S}) \\ \implies \frac{\partial \ln S_t}{\partial \ln V_t} &= \varepsilon_S - \frac{\varepsilon_S V_t^{\varepsilon_S}}{\bar{u}^{\varepsilon_S} + V_t^{\varepsilon_S}} = \varepsilon_S(1 - \pi_t). \end{aligned} \quad \square$$

E.5. BLOCK RECURSION

Theorem E.3 (Block recursion).

$$\ln \sigma_{nt} = \beta_P \ln \kappa_{nt}(\sigma_t) + \ln \tilde{\psi}_{nt} + c_t, \quad (\text{E.9})$$

$$\ln V_t = (\eta - \gamma) \ln S_t + \ln W_t, \quad (\text{E.10})$$

$$\ln S_t = \ln \bar{L} + \varepsilon_S \ln V_t - \ln(\bar{u}^{\varepsilon_S} + V_t^{\varepsilon_S}), \quad (\text{E.11})$$

where $\tilde{\psi}_{nt} = \psi_{nt}^{\varepsilon_P/(1+\gamma\varepsilon_P)}$ and

$$c_t = \frac{\varepsilon_P(\eta - \gamma) \ln S_t - \ln D_t}{1 + \gamma\varepsilon_P}, \quad D_t = \sum_k R_{kt}^{\varepsilon_P}, \quad W_t = \left(\sum_k (\psi_{kt} \kappa_{kt}^\eta \sigma_{kt}^{-\gamma})^{\varepsilon_P} \right)^{1/\varepsilon_P}.$$

Proof. By Lemma E.1 and $S_{nt} = \sigma_{nt} S_t$,

$$\begin{aligned} R_{nt} &= \psi_{nt} K_{nt}^\eta S_{nt}^{-\gamma} \\ &= \psi_{nt} (S_t \kappa_{nt}(\sigma_t))^\eta (\sigma_{nt} S_t)^{-\gamma} \\ &= \psi_{nt} \kappa_{nt}^\eta \sigma_{nt}^{-\gamma} S_t^{\eta-\gamma}. \end{aligned}$$

Taking logarithms of the share equation of Theorem E.2, with $D_t = \sum_k R_{kt}^{\varepsilon_P}$, and collecting the $\ln \sigma_{nt}$ terms,

$$\begin{aligned} \sigma_{nt} &= \frac{R_{nt}^{\varepsilon_P}}{D_t} \\ \implies \ln \sigma_{nt} &= \varepsilon_P \ln R_{nt} - \ln D_t \\ &= \varepsilon_P (\ln \psi_{nt} + \eta \ln \kappa_{nt} - \gamma \ln \sigma_{nt} + (\eta - \gamma) \ln S_t) - \ln D_t \\ \implies (1 + \gamma\varepsilon_P) \ln \sigma_{nt} &= \varepsilon_P \eta \ln \kappa_{nt} + \varepsilon_P \ln \psi_{nt} + (\varepsilon_P(\eta - \gamma) \ln S_t - \ln D_t) \\ \implies \ln \sigma_{nt} &= \frac{\varepsilon_P \eta}{1 + \gamma\varepsilon_P} \ln \kappa_{nt} + \frac{\varepsilon_P}{1 + \gamma\varepsilon_P} \ln \psi_{nt} + c_t, \quad (\varepsilon_P \eta / (1 + \gamma\varepsilon_P) = \beta_P) \\ c_t &= \frac{\varepsilon_P(\eta - \gamma) \ln S_t - \ln D_t}{1 + \gamma\varepsilon_P}, \end{aligned}$$

which is (E.9), as $\frac{\varepsilon_P}{1+\gamma\varepsilon_P} \ln \psi_{nt} = \ln \tilde{\psi}_{nt}$. For (E.10), by the same factoring,

$$\begin{aligned} V_t^{\varepsilon_P} &= \sum_k R_{kt}^{\varepsilon_P} \\ &= \sum_k (\psi_{kt} \kappa_{kt}^\eta \sigma_{kt}^{-\gamma})^{\varepsilon_P} S_t^{(\eta-\gamma)\varepsilon_P} \\ &= S_t^{(\eta-\gamma)\varepsilon_P} W_t^{\varepsilon_P} \\ \implies \ln V_t &= (\eta - \gamma) \ln S_t + \ln W_t. \end{aligned}$$

Equation (E.11) is the total of Theorem E.2 in logs. □

Corollary E.1 (The total in levels). Let $\bar{\kappa}_t = (\sum_k \tilde{\psi}_{kt} \kappa_{kt}^{\beta_P})^{1/\beta_P}$, and $\bar{K}_t = S_t \bar{\kappa}_t$, London's average access. In equilibrium $W_t = \bar{\kappa}_t^\eta$, so that

$$\ln V_t = \eta \ln \bar{K}_t - \gamma \ln S_t, \quad \ln S_t = \beta_S \ln \bar{K}_t + \alpha_t, \quad \alpha_t = \frac{\ln \bar{L} - \varepsilon_S \ln \bar{u} + \ln(1 - \pi_t)}{1 + \gamma \varepsilon_S},$$

and α_t is fixed by $\bar{L}$ and $\bar{u}$ as $\pi_t \rightarrow 0$. If a counterfactual holds the fundamentals fixed, then $\bar{\kappa}'_t / \bar{\kappa}_t = (\sum_k \sigma_{kt} (\kappa'_{kt} / \kappa_{kt})^{\beta_P})^{1/\beta_P}$.

Proof. By (E.9), $\sigma_{kt} = \tilde{\psi}_{kt} \kappa_{kt}^{\beta_P} e^{c_t}$, and since the shares sum to one, $e^{c_t} = 1 / \sum_k \tilde{\psi}_{kt} \kappa_{kt}^{\beta_P} = \bar{\kappa}_t^{-\beta_P}$. Since $\beta_P(1 + \gamma \varepsilon_P) = \varepsilon_P \eta$,

$$\begin{aligned} \psi_{kt}^{\varepsilon_P} \kappa_{kt}^{\eta \varepsilon_P} &= (\tilde{\psi}_{kt} \kappa_{kt}^{\beta_P})^{1 + \gamma \varepsilon_P} = (\sigma_{kt} \bar{\kappa}_t^{\beta_P})^{1 + \gamma \varepsilon_P} \\ \Rightarrow W_t^{\varepsilon_P} &= \sum_k \psi_{kt}^{\varepsilon_P} \kappa_{kt}^{\eta \varepsilon_P} \sigma_{kt}^{-\gamma \varepsilon_P} = \bar{\kappa}_t^{\eta \varepsilon_P} \sum_k \sigma_{kt} = \bar{\kappa}_t^{\eta \varepsilon_P}, \end{aligned}$$

so (E.10) reads $\ln V_t = (\eta - \gamma) \ln S_t + \eta \ln \bar{\kappa}_t = \eta \ln \bar{K}_t - \gamma \ln S_t$. By Theorem E.2, $1 - \pi_t = \bar{u}^{\varepsilon_S} / (\bar{u}^{\varepsilon_S} + V_t^{\varepsilon_S})$, so (E.11) reads

$$\begin{aligned} \ln S_t &= \ln \bar{L} - \varepsilon_S \ln \bar{u} + \ln(1 - \pi_t) + \varepsilon_S \ln V_t \\ &= \ln \bar{L} - \varepsilon_S \ln \bar{u} + \ln(1 - \pi_t) + \varepsilon_S \eta \ln \bar{K}_t - \varepsilon_S \gamma \ln S_t, \end{aligned}$$

and solving for $\ln S_t$ gives the second display equation, with $\varepsilon_S \eta / (1 + \gamma \varepsilon_S) = \beta_S$. Last, with $\tilde{\psi}_{kt}$ fixed and $\sigma_{kt} = \tilde{\psi}_{kt} \kappa_{kt}^{\beta_P} / \bar{\kappa}_t^{\beta_P}$,

$$\left(\frac{\bar{\kappa}'_t}{\bar{\kappa}_t} \right)^{\beta_P} = \sum_k \frac{\tilde{\psi}_{kt} \kappa_{kt}^{\beta_P}}{\bar{\kappa}_t^{\beta_P}} \left(\frac{\kappa'_{kt}}{\kappa_{kt}} \right)^{\beta_P} = \sum_k \sigma_{kt} \left(\frac{\kappa'_{kt}}{\kappa_{kt}} \right)^{\beta_P}. \quad \square$$

Theorem E.4 (The aggregate fixed point). If $\beta_P < 1$, then for given shares σ_t equations (E.9)–(E.11) give exactly one solution in $\ln S_t$.

Proof. Write $z = \ln S_t$. Substituting (E.10), $\ln V_t = (\eta - \gamma)z + \ln W_t$, into (E.11),

$$\begin{aligned} z &= \ln \bar{L} + \varepsilon_S ((\eta - \gamma)z + \ln W_t) - \ln(\bar{u}^{\varepsilon_S} + e^{\varepsilon_S(\eta - \gamma)z} W_t^{\varepsilon_S}) \\ \Leftrightarrow 0 &= (1 - a)z - b + \ln(c + d e^{az}) =: h(z), \end{aligned}$$

with constants

$$a = \varepsilon_S(\eta - \gamma), \quad b = \ln \bar{L} + \varepsilon_S \ln W_t, \quad c = \bar{u}^{\varepsilon_S}, \quad d = W_t^{\varepsilon_S},$$

so the solutions are the zeros of h . Suppose first that $a < 1$. With $\xi = d e^{az} / (c + d e^{az}) \in (0, 1)$,

$$h'(z) = (1 - a) + \frac{a d e^{az}}{c + d e^{az}} = 1 - a(1 - \xi) > 0,$$

so h is strictly increasing. Since $h(z) \geq z + \min(\ln c, \ln d) - b$ for $z \geq 0$, which tends to $+\infty$ as $z \rightarrow +\infty$, and $h(z) \leq \min(1, 1-a)z + \ln(c+d) - b$ for $z \leq 0$, which tends to $-\infty$ as $z \rightarrow -\infty$, the intermediate value theorem gives a zero and strict monotonicity its uniqueness. It remains to see that $\beta_P < 1$ gives $a < 1$. Since $\beta_S = \varepsilon_S \eta / (1 + \gamma \varepsilon_S)$,

$$\beta_S < 1 \iff \varepsilon_S \eta < 1 + \gamma \varepsilon_S \iff \varepsilon_S(\eta - \gamma) < 1 \iff a < 1,$$

and $\beta_S \leq \beta_P < 1$. □

E.6. EQUILIBRIUM IN LEVELS

Write $\kappa_t = (\kappa_{1t}, \dots, \kappa_{Nt})$ for the vector of access in share form, beside σ_t .

Lemma E.8 (The fixed point in κ_t). *The share block (E.9) is*

$$\sigma_{nt} = C_t \tilde{\psi}_{nt} \kappa_{nt}(\sigma_t)^{\beta_P}, \quad C_t = e^{c_t}, \quad (\text{E.12})$$

and $\kappa_t = \kappa_t(\sigma_t)$ then solves a fixed point in κ_t alone,

$$\kappa_{nt} = \sum_i Q_{nit} \kappa_{it}^{\beta_P}, \quad Q_{nit} = C_t m_{nt} e^{-\delta d_{ni}} m_{it} \tilde{\psi}_{it} > 0. \quad (\text{E.13})$$

Proof. Exponentiating (E.9) gives (E.12). Substituting it into the access share (E.2),

$$\begin{aligned} \kappa_{nt}(\sigma_t) &= m_{nt} \sum_i e^{-\delta d_{ni}} m_{it} \sigma_{it} \\ &= m_{nt} \sum_i e^{-\delta d_{ni}} m_{it} C_t \tilde{\psi}_{it} \kappa_{it}^{\beta_P} \quad (\text{by (E.12)}) \\ &= \sum_i Q_{nit} \kappa_{it}^{\beta_P}. \end{aligned} \quad \square$$

Theorem E.5 (Existence and uniqueness). *Let $Q_{nit} > 0$ and $0 \leq \beta_P < 1$. Then (E.13) has exactly one strictly positive solution, and iterating the system in log coordinates from any start converges to it at rate β_P^k after k iterations. The constant C_t is pinned by $\sum_n \sigma_{nt} = 1$: scaling Q by c scales the solution and the shares by $c^{1/(1-\beta_P)}$, so exactly one $c > 0$ normalizes them.*

Proof. *The fixed point and the rate.* In log coordinates $x_n = \ln \kappa_{nt}$ the system is $\mathbf{x} = T(\mathbf{x})$ with $T(\mathbf{x})_n = \ln \sum_i Q_{nit} e^{\beta_P x_i}$. If $x_i \leq y_i + d$ for all i , then

$$\begin{aligned} T(\mathbf{x})_n &= \ln \sum_i Q_{nit} e^{\beta_P x_i} \\ &\leq \ln \sum_i Q_{nit} e^{\beta_P (y_i + d)} \quad (\beta_P \geq 0, \ln \text{ increasing}) \\ &= \ln \left(e^{\beta_P d} \sum_i Q_{nit} e^{\beta_P y_i} \right) \quad (e^{\beta_P d} \text{ common to every term}) \\ &= \beta_P d + \ln \sum_i Q_{nit} e^{\beta_P y_i} = T(\mathbf{y})_n + \beta_P d, \end{aligned}$$

and with the roles exchanged, T is β_P -Lipschitz for the sup metric, so a contraction on the complete space $\mathbb{R}^N$ since $\beta_P < 1$. Banach's theorem gives a unique fixed point $\mathbf{x}^*$, so $\kappa_t = e^{\mathbf{x}^*}$ is the only strictly positive solution, and $\text{dist}(T^k \mathbf{x}_0, \mathbf{x}^*) \leq \beta_P^k \text{dist}(\mathbf{x}_0, \mathbf{x}^*)$.

The constant. For $c > 0$,

$$\begin{aligned} (c^{1/(1-\beta_P)} \kappa_{it})^{\beta_P} &= c^{\beta_P/(1-\beta_P)} \kappa_{it}^{\beta_P} \\ \Rightarrow \sum_i (c Q_{nit}) (c^{1/(1-\beta_P)} \kappa_{it})^{\beta_P} &= c \cdot c^{\beta_P/(1-\beta_P)} \sum_i Q_{nit} \kappa_{it}^{\beta_P} = c^{1/(1-\beta_P)} \kappa_{nt}, \end{aligned}$$

so $c^{1/(1-\beta_P)} \kappa_t$ solves the system with kernel cQ , and, with C_t absorbed into c , the shares $c \tilde{\psi}_{nt} (c^{1/(1-\beta_P)} \kappa_{nt})^{\beta_P}$ sum to $c^{1/(1-\beta_P)} \sum_n \tilde{\psi}_{nt} \kappa_{nt}^{\beta_P}$, which equals one for exactly one $c > 0$. $\square$

Theorem E.6 (Inversion). For any strictly positive data (S_{nt}, K_{nt}) with $S_t = \sum_n S_{nt}$ and any constant c_t ,

$$\ln \psi_{nt} = \frac{1 + \gamma \varepsilon_P}{\varepsilon_P} \ln S_{nt} - \eta \ln K_{nt} + c_t \quad (\text{E.14})$$

defines a strictly positive ψ_{nt} that satisfies the share equation of Theorem E.2 exactly, and every ψ_{nt} that satisfies it is of this form. The constant scales every ψ_{nt} of the decade by the same factor e^{c_t} , so the fundamentals are recovered up to a scale per decade.

Proof. The formula works. The ψ_{nt} it defines is positive, being an exponential. Exponentiating (E.14) gives $\psi_{nt} = e^{c_t} S_{nt}^{(1+\gamma \varepsilon_P)/\varepsilon_P} K_{nt}^{-\eta}$, so in the return (E.3) the access cancels,

$$\begin{aligned} R_{nt} &= \psi_{nt} K_{nt}^\eta S_{nt}^{-\gamma} \\ &= e^{c_t} S_{nt}^{(1+\gamma \varepsilon_P)/\varepsilon_P} K_{nt}^{-\eta} K_{nt}^\eta S_{nt}^{-\gamma} \\ &= e^{c_t} S_{nt}^{(1+\gamma \varepsilon_P)/\varepsilon_P - \gamma} = e^{c_t} S_{nt}^{1/\varepsilon_P}. \end{aligned} \quad (\text{by (E.14)})$$

Hence,

$$\begin{aligned} R_{nt}^{\varepsilon_P} &= e^{\varepsilon_P c_t} S_{nt} \\ \Rightarrow \frac{R_{nt}^{\varepsilon_P}}{\sum_k R_{kt}^{\varepsilon_P}} &= \frac{e^{\varepsilon_P c_t} S_{nt}}{e^{\varepsilon_P c_t} \sum_k S_{kt}} = \frac{S_{nt}}{S_t} = \sigma_{nt}, \end{aligned} \quad (\sum_k S_{kt} = S_t)$$

which is the share equation of Theorem E.2, whatever the value of c_t .

Nothing else works. Conversely, let $\psi_{nt} > 0$ satisfy that share equation. Then $R_{nt}^{\varepsilon_P} = \sigma_{nt} \sum_k R_{kt}^{\varepsilon_P} = (S_{nt}/S_t) \sum_k R_{kt}^{\varepsilon_P}$, and solving (E.3) for the fundamental in logarithms,

$$\begin{aligned} \ln \psi_{nt} &= \ln R_{nt} - \eta \ln K_{nt} + \gamma \ln S_{nt} \quad (\text{by (E.3)}) \\ &= \frac{1}{\varepsilon_P} \ln \left(\frac{S_{nt}}{S_t} \sum_k R_{kt}^{\varepsilon_P} \right) - \eta \ln K_{nt} + \gamma \ln S_{nt} \quad (\text{the share equation}) \\ &= \frac{1 + \gamma \varepsilon_P}{\varepsilon_P} \ln S_{nt} - \eta \ln K_{nt} + \frac{1}{\varepsilon_P} \ln \left(\sum_k R_{kt}^{\varepsilon_P} / S_t \right), \end{aligned}$$

which is (E.14) with $c_t = \frac{1}{\varepsilon_P} \ln(\sum_k R_{kt}^{\varepsilon_P} / S_t)$, the same number for every parish of the decade. Two constants thus give fundamentals differing by the common factor $e^{c_t - c'_t}$, which is the scale per decade. $\square$

The fit of the model in levels thus tests nothing. What identifies β_P is how a parish's share moves when a coffeehouse arrives and its neighbors' do not, the fundamentals held fixed.

Corollary E.2 (Inversion in β_P). *In terms of $\tilde{\psi}_{nt}$, the shares and access in share form, Theorem E.6 reads*

$$\ln \tilde{\psi}_{nt} = \ln \sigma_{nt} - \beta_P \ln \kappa_{nt}(\sigma_t) + \tilde{c}_t, \quad \tilde{c}_t = (1 - \beta_P) \ln S_t + \frac{\varepsilon_P c_t}{1 + \gamma \varepsilon_P},$$

with $\tilde{c}_t$ common to all parishes.

Proof. By Theorem E.6, with $S_{nt} = \sigma_{nt} S_t$ and $K_{nt} = S_t \kappa_{nt}(\sigma_t)$ from Lemma E.1,

$$\begin{aligned} \ln \tilde{\psi}_{nt} &= \frac{\varepsilon_P}{1 + \gamma \varepsilon_P} \ln \psi_{nt} \\ &= \ln S_{nt} - \frac{\varepsilon_P \eta}{1 + \gamma \varepsilon_P} \ln K_{nt} + \frac{\varepsilon_P c_t}{1 + \gamma \varepsilon_P} \\ &= \ln \sigma_{nt} + \ln S_t - \beta_P (\ln S_t + \ln \kappa_{nt}(\sigma_t)) + \frac{\varepsilon_P c_t}{1 + \gamma \varepsilon_P} \\ &= \ln \sigma_{nt} - \beta_P \ln \kappa_{nt}(\sigma_t) + (1 - \beta_P) \ln S_t + \frac{\varepsilon_P c_t}{1 + \gamma \varepsilon_P} \\ &= \ln \sigma_{nt} - \beta_P \ln \kappa_{nt}(\sigma_t) + \tilde{c}_t. \end{aligned}$$

□

E.7. EQUILIBRIUM IN CHANGES

A counterfactual is a second geography $D' = (D'_{nt})$ with the same fundamentals ψ_{nt} and $\bar{u}$ and the same pool $\bar{L}$. When $\beta_P < 1$ its equilibrium exists and is unique, the shares by Theorem E.5 and the total by Theorem E.4, so we can introduce the following notation: $\hat{x} = x'/x$ is the ratio of a counterfactual value to its observed value.

Definition E.6 (Baseline access shares). Given (ϕ, δ) and an observed equilibrium, we define the baseline access share of parish i in the access of parish n as

$$\omega_{nit} = \frac{e^{-\delta d_{ni}} m_{it} S_{it}}{\sum_j e^{-\delta d_{nj}} m_{jt} S_{jt}}. \quad (\text{E.15})$$

Theorem E.7 (Access in changes). $\hat{K}_{nt} = \hat{m}_{nt} \sum_i \omega_{nit} \hat{m}_{it} \hat{S}_{it}$.

Proof. We have

$$\begin{aligned}
\widehat{K}_{nt} &= \frac{K'_{nt}}{K_{nt}} = \frac{m'_{nt} \sum_i e^{-\delta d_{ni}} m'_{it} S'_{it}}{m_{nt} \sum_j e^{-\delta d_{nj}} m_{jt} S_{jt}} \\
&= \frac{m'_{nt}}{m_{nt}} \sum_i \frac{e^{-\delta d_{ni}} m'_{it} S'_{it}}{\sum_j e^{-\delta d_{nj}} m_{jt} S_{jt}} \\
&= \frac{m'_{nt}}{m_{nt}} \sum_i \frac{e^{-\delta d_{ni}} m_{it} S_{it}}{\sum_j e^{-\delta d_{nj}} m_{jt} S_{jt}} \cdot \frac{m'_{it} S'_{it}}{m_{it} S_{it}} \\
&= \widehat{m}_{nt} \sum_i \omega_{nit} \widehat{m}_{it} \widehat{S}_{it}.
\end{aligned}$$

□

Theorem E.8 (Exact hat and sufficiency). *In a counterfactual,*

$$\begin{aligned}
\widehat{K}_{nt} &= \widehat{m}_{nt} \sum_i \omega_{nit} \widehat{m}_{it} \widehat{S}_{it}, & \widehat{R}_{nt} &= \widehat{K}_{nt}^\eta \widehat{S}_{nt}^{-\gamma}, & \widehat{\sigma}_{nt} &= \frac{\widehat{R}_{nt}^{\varepsilon_P}}{\sum_k \sigma_{kt} \widehat{R}_{kt}^{\varepsilon_P}}, \\
\widehat{S}_t &= \frac{\widehat{V}_t^{\varepsilon_S}}{(1 - \pi_t) + \pi_t \widehat{V}_t^{\varepsilon_S}}, & \widehat{S}_{nt} &= \widehat{\sigma}_{nt} \widehat{S}_t,
\end{aligned}$$

with $\widehat{V}_t^{\varepsilon_P} = \sum_k \sigma_{kt} \widehat{R}_{kt}^{\varepsilon_P}$. No level of ψ or $\bar{u}$ is needed, and $\bar{L}$ enters only through π_t . The fundamental is absent from $\widehat{R}_{nt}$ because it is held fixed and enters the return multiplicatively,

$$\widehat{R}_{nt} = \frac{\psi_{nt} K_{nt}^\eta S_{nt}'^{-\gamma}}{\psi_{nt} K_{nt}^\eta S_{nt}^{-\gamma}} = \widehat{K}_{nt}^\eta \widehat{S}_{nt}^{-\gamma}.$$

Moreover, whatever the value of π_t , the counterfactual shares solve

$$\widehat{\sigma}_{nt} = \frac{\widehat{\kappa}_{nt}^{\beta_P}}{\sum_k \sigma_{kt} \widehat{\kappa}_{kt}^{\beta_P}}, \quad \widehat{\kappa}_{nt} = \widehat{m}_{nt} \sum_i \omega_{nit} \widehat{m}_{it} \widehat{\sigma}_{it},$$

in which $(\eta, \gamma, \varepsilon_P)$ appear only through β_P .

Proof. Access is Theorem E.7, and the return ratio is the display. For the shares, dividing the two Fréchet share equations of Theorem E.2,

$$\begin{aligned}
\widehat{\sigma}_{nt} &= \frac{\sigma'_{nt}}{\sigma_{nt}} = \frac{R_{nt}'^{\varepsilon_P} / \sum_k R_{kt}'^{\varepsilon_P}}{R_{nt}^{\varepsilon_P} / \sum_k R_{kt}^{\varepsilon_P}} \\
&= \frac{\widehat{R}_{nt}^{\varepsilon_P}}{\sum_k R_{kt}'^{\varepsilon_P} / \sum_k R_{kt}^{\varepsilon_P}} \\
&= \frac{\widehat{R}_{nt}^{\varepsilon_P}}{\sum_k (R_{kt}^{\varepsilon_P} / \sum_j R_{jt}^{\varepsilon_P}) \widehat{R}_{kt}^{\varepsilon_P}} = \frac{\widehat{R}_{nt}^{\varepsilon_P}}{\sum_k \sigma_{kt} \widehat{R}_{kt}^{\varepsilon_P}},
\end{aligned}$$

and the denominator is $\widehat{V}_t^{\varepsilon_P} = V_t^{\varepsilon_P} / V_t^{\varepsilon_S} = \sum_k R_{kt}^{\varepsilon_P} / \sum_k R_{kt}^{\varepsilon_S}$. For the total, with $\pi_t = V_t^{\varepsilon_S} / (\bar{u}^{\varepsilon_S} + V_t^{\varepsilon_S})$ from (E.8) and $\bar{u}, \bar{L}$ unchanged,

$$\begin{aligned} \widehat{S}_t &= \frac{S'_t}{S_t} = \frac{\bar{L} V_t^{\varepsilon_S} / (\bar{u}^{\varepsilon_S} + V_t^{\varepsilon_S})}{\bar{L} V_t^{\varepsilon_S} / (\bar{u}^{\varepsilon_S} + V_t^{\varepsilon_S})} \\ &= \widehat{V}_t^{\varepsilon_S} \frac{\bar{u}^{\varepsilon_S} + V_t^{\varepsilon_S}}{\bar{u}^{\varepsilon_S} + V_t^{\varepsilon_S} \widehat{V}_t^{\varepsilon_S}} \\ &= \frac{\widehat{V}_t^{\varepsilon_S}}{(1 - \pi_t) + \pi_t \widehat{V}_t^{\varepsilon_S}}. \end{aligned}$$

For sufficiency, by Lemma E.1 $\widehat{K}_{nt} = \widehat{S}_t \widehat{\kappa}_{nt}$ with $\widehat{\kappa}_{nt} = \kappa_{nt}(\sigma'_t) / \kappa_{nt}(\sigma_t) = \widehat{m}_{nt} \sum_i \omega_{nit} \widehat{m}_{it} \widehat{\sigma}_{it}$ (the computation of Theorem E.7 with σ in place of S), and $\widehat{S}_{nt} = \widehat{\sigma}_{nt} \widehat{S}_t$, so

$$\begin{aligned} \widehat{R}_{nt} &= (\widehat{S}_t \widehat{\kappa}_{nt})^\eta (\widehat{\sigma}_{nt} \widehat{S}_t)^{-\gamma} = \widehat{\kappa}_{nt}^\eta \widehat{\sigma}_{nt}^{-\gamma} \widehat{S}_t^{\eta-\gamma} \\ \Rightarrow \widehat{\sigma}_{nt} &= \frac{\widehat{\kappa}_{nt}^{\eta \varepsilon_P} \widehat{\sigma}_{nt}^{-\gamma \varepsilon_P} \widehat{S}_t^{(\eta-\gamma) \varepsilon_P}}{\sum_k \sigma_{kt} \widehat{\kappa}_{kt}^{\eta \varepsilon_P} \widehat{\sigma}_{kt}^{-\gamma \varepsilon_P} \widehat{S}_t^{(\eta-\gamma) \varepsilon_P}} = \frac{\widehat{\kappa}_{nt}^{\eta \varepsilon_P} \widehat{\sigma}_{nt}^{-\gamma \varepsilon_P}}{\widehat{D}_t}, \quad \widehat{D}_t = \sum_k \sigma_{kt} \widehat{\kappa}_{kt}^{\eta \varepsilon_P} \widehat{\sigma}_{kt}^{-\gamma \varepsilon_P} \\ \Rightarrow \widehat{\sigma}_{nt}^{1+\gamma \varepsilon_P} &= \frac{\widehat{\kappa}_{nt}^{\eta \varepsilon_P}}{\widehat{D}_t} \\ \Rightarrow \widehat{\sigma}_{nt} &= A \widehat{\kappa}_{nt}^{\beta_P}, \quad A = \widehat{D}_t^{-1/(1+\gamma \varepsilon_P)} \quad (\eta \varepsilon_P / (1 + \gamma \varepsilon_P) = \beta_P) \end{aligned}$$

Thus, using that the shares sum to one in both worlds,

$$1 = \sum_k \sigma_{kt} \widehat{\sigma}_{kt} = A \sum_k \sigma_{kt} \widehat{\kappa}_{kt}^{\beta_P} \Rightarrow A = \frac{1}{\sum_k \sigma_{kt} \widehat{\kappa}_{kt}^{\beta_P}}. \quad \square$$

The system of Theorem E.8 is solved by iterating it from $\widehat{\sigma}_t = \mathbf{1}$. Its share equation in logs is $\ln \widehat{\sigma}_{nt} = \beta_P \ln \widehat{\kappa}_{nt} - \ln \sum_k \sigma_{kt} \widehat{\kappa}_{kt}^{\beta_P}$, so substituting $\widehat{\kappa}_{nt} = \widehat{m}_{nt} \sum_i \omega_{nit} \widehat{m}_{it} \widehat{\sigma}_{it}$, a pass in log shares $x_n = \ln \widehat{\sigma}_{nt}$ is

$$x_n \mapsto \beta_P \ln \left(\sum_i a_{ni} e^{x_i} \right) + c, \quad a_{ni} = \widehat{m}_{nt} \omega_{nit} \widehat{m}_{it}, \quad c = -\ln \sum_k \sigma_{kt} \widehat{\kappa}_{kt}^{\beta_P}.$$

The weights $a_{ni} > 0$ are fixed, and c , the normalization, is the same for every parish.

Theorem E.9 (Rate of convergence). *Let $a_{ni} > 0$, let $T(\mathbf{x})_n = \beta_P \ln \sum_i a_{ni} e^{x_i} + c_n$ for $\mathbf{x} \in \mathbb{R}^N$ and constants $c_1, \dots, c_N$, with $\beta_P \geq 0$, and let $\text{sp}(\mathbf{z}) = \max_n z_n - \min_n z_n$ be the span seminorm of $\mathbf{z} \in \mathbb{R}^N$. Then*

$$\text{sp}(T(\mathbf{x}) - T(\mathbf{y})) \leq \beta_P \text{sp}(\mathbf{x} - \mathbf{y}).$$

Proof. Let $\mathbf{v} = \mathbf{x} - \mathbf{y}$, $\bar{v} = \max_i v_i$, and $\underline{v} = \min_i v_i$. Since $e^{x_i} = e^{y_i} e^{v_i}$ and $e^{\underline{v}} \leq e^{v_i} \leq e^{\bar{v}}$,

$$\begin{aligned} \underline{v} + \ln \sum_i a_{ni} e^{y_i} &\leq \ln \sum_i a_{ni} e^{x_i} \leq \bar{v} + \ln \sum_i a_{ni} e^{y_i} \\ \Rightarrow T(\mathbf{x})_n - T(\mathbf{y})_n &= \beta_P \left(\ln \sum_i a_{ni} e^{x_i} - \ln \sum_i a_{ni} e^{y_i} \right) \in [\beta_P \underline{v}, \beta_P \bar{v}] \\ &\Rightarrow \text{sp}(T(\mathbf{x}) - T(\mathbf{y})) \leq \beta_P (\bar{v} - \underline{v}) = \beta_P \text{sp}(\mathbf{x} - \mathbf{y}). \square \end{aligned}$$

Theorem E.10 (The aggregate multiplier). *Take the hat system of Theorem E.8 with $\beta_P < 1$, and let $\bar{L} \rightarrow \infty$, so that $\pi_t \rightarrow 0$. Let $M_t = (\sum_k \sigma_{kt} \widehat{\kappa}_{kt}^{\beta_P})^{1/\beta_P}$ be the access change averaged over London with the observed shares as weights. Then*

$$\ln \widehat{V}_t = \eta \ln(\widehat{S}_t M_t) - \gamma \ln \widehat{S}_t,$$

so the aggregate block in changes reads $\ln \widehat{S}_t = \beta_S \ln \widehat{\bar{K}}_t$ with $\widehat{\bar{K}}_t = \widehat{S}_t M_t$, and

$$\ln \widehat{S}_t = \frac{\beta_S}{1 - \beta_S} \ln M_t,$$

with a positive multiplier.

Proof. Corollary E.1 holds in both equilibria, with the same $\tilde{\psi}_{kt}$ since the fundamentals are fixed. Its last display is $\bar{\kappa}'_t / \bar{\kappa}_t = M_t$, so $\widehat{\bar{K}}_t = \widehat{S}_t M_t$, and its first display in ratios is

$$\ln \widehat{V}_t = \eta \ln \widehat{\bar{K}}_t - \gamma \ln \widehat{S}_t = \eta \ln(\widehat{S}_t M_t) - \gamma \ln \widehat{S}_t,$$

which is the first display. As $\bar{L} \rightarrow \infty$, $\pi_t \rightarrow 0$ and the total of Theorem E.8 is $\ln \widehat{S}_t = \varepsilon_S \ln \widehat{V}_t$, so

$$\begin{aligned} \ln \widehat{S}_t &= \varepsilon_S \eta \ln(\widehat{S}_t M_t) - \varepsilon_S \gamma \ln \widehat{S}_t \\ \Rightarrow (1 + \gamma \varepsilon_S) \ln \widehat{S}_t &= \varepsilon_S \eta \ln \widehat{\bar{K}}_t \\ \Rightarrow \ln \widehat{S}_t &= \beta_S \ln \widehat{\bar{K}}_t = \beta_S (\ln \widehat{S}_t + \ln M_t) \\ \Rightarrow (1 - \beta_S) \ln \widehat{S}_t &= \beta_S \ln M_t \\ \Rightarrow \ln \widehat{S}_t &= \frac{\beta_S}{1 - \beta_S} \ln M_t. \end{aligned}$$

The division by $1 - \beta_S$ is legitimate and the multiplier positive because $\beta_S \leq \beta_P < 1$. $\square$

E.8. RELOCATION AND CREATION

The change in a parish's count from a counterfactual to the observed equilibrium, $S_{nt} - S'_{nt}$, splits, as in the decomposition of Bennet (1920, p. 457), into relocation, the change in its share at the average total, and creation, its average share of the change in the total.

Theorem E.11 (Relocation and creation). Let $\bar{S}_t = \frac{1}{2}(S_t + S'_t)$ and $\bar{\sigma}_{nt} = \frac{1}{2}(\sigma_{nt} + \sigma'_{nt})$. Then

$$S_{nt} - S'_{nt} = \underbrace{\bar{S}_t(\sigma_{nt} - \sigma'_{nt})}_{\text{relocation}} + \underbrace{\bar{\sigma}_{nt}(S_t - S'_t)}_{\text{creation}}, \quad (\text{E.16})$$

the first term summing to zero over n and the second to $S_t - S'_t$.

Proof. Expanding the products,

$$\begin{aligned} \bar{S}_t(\sigma_{nt} - \sigma'_{nt}) + \bar{\sigma}_{nt}(S_t - S'_t) &= \frac{1}{2}(S_t + S'_t)(\sigma_{nt} - \sigma'_{nt}) + \frac{1}{2}(\sigma_{nt} + \sigma'_{nt})(S_t - S'_t) \\ &= \frac{1}{2}(S_t\sigma_{nt} - S_t\sigma'_{nt} + S'_t\sigma_{nt} - S'_t\sigma'_{nt} \\ &\quad + \sigma_{nt}S_t - \sigma_{nt}S'_t + \sigma'_{nt}S_t - \sigma'_{nt}S'_t) \\ &= \sigma_{nt}S_t - \sigma'_{nt}S'_t = S_{nt} - S'_{nt}, \end{aligned}$$

and

$$\begin{aligned} \sum_n \bar{S}_t(\sigma_{nt} - \sigma'_{nt}) &= \bar{S}_t(1 - 1) = 0, \\ \sum_n \bar{\sigma}_{nt}(S_t - S'_t) &= \frac{1}{2}(1 + 1)(S_t - S'_t) = S_t - S'_t. \end{aligned} \quad \square$$

E.9. ESTIMATION, SIMULATION, AND THE AGGREGATE EFFECT

Procedure E.1 uses the 244 mathematical practitioners of [Taylor \(1954\)](#) and the connections among them documented there and in the Six Degrees of Francis Bacon database ([Warren et al., 2016](#)). From Procedure E.2 on, S_{nt} counts those 244 completed with the scientific instrument makers found by later research, the 252 of [Clifton \(1995\)](#) and [Hay et al. \(2025\)](#), 93 of whom are already among the practitioners, so 403 applied scientists in all. A matched person enters once, with the earlier first year and the later last year of the two sources and the addresses of both, and everyone is counted once in a parish and decade. They make 797 scientist-decades in 1650–1690, each scientist counted in every decade they were active, 458 of them in parishes that had a coffeehouse by then.

Procedure E.1 (Access, (ϕ, δ)). By Theorem E.1, in logs,

$$\mathbb{E}[T_{nit}] = \exp(\ln \chi \rho_t + \phi(D_{nt} + D_{it}) - \delta d_{ni} + \ln S_{nt} + \ln S_{it}), \quad (\text{E.17})$$

which is (5) with $\mu_t = \ln \chi \rho_t$. The parameters of this conditional mean are consistently estimated by Poisson pseudo-maximum likelihood. The documented connections are undated, so in estimation the mean is summed over the decades in which both practitioners are active, in the parishes where each works in that decade, and the likelihood is evaluated on person dyads, which puts the decade terms inside that sum, one free term per decade.

Procedure E.2 (The share elasticity, β_P). Indirect inference, in three steps.

1. Run the event study on the observed counts. This gives the post-treatment coefficients $\hat{\tau} = (\hat{\tau}_0, \hat{\tau}_{10}, \dots, \hat{\tau}_{40})$ and their standard errors $\hat{s}_0, \hat{s}_{10}, \dots, \hat{s}_{40}$. The

pre-treatment coefficients are not matched, as they are checks on the design.

2. For a trial β_P , build the panel the model implies at the observed geography and run step 1 on it:

- (a) *Fundamentals.* The fundamentals are taken time-invariant, $\tilde{\psi}_{nt} = \tilde{\psi}_n$. For the trial β_P , recover $\tilde{\psi}_n$ from the pre-coffeehouse decades 1600–1640 by Corollary E.2, $\ln \tilde{\psi}_n = \ln \sigma_n^{\text{pre}} - \beta_P \ln \kappa_n^{\text{pre}} + \tilde{c}$ with $\tilde{c} = 0$, where $\sigma_n^{\text{pre}} = \lambda_n / \sum_k \lambda_k$ is the pre-period share, $\lambda_n = (\sum_{t \leq 1640} S_{nt} + \frac{1}{2})/5$ the count per decade smoothed by one half, which keeps the parishes with nobody recorded before 1650 from being held at a zero share by construction, and κ_n^{pre} the access these shares imply without coffeehouses ($m_{nt} = 1$).⁶³
- (b) *Counts.* For each decade $t = 1600, \dots, 1690$, with the observed indicators D_{nt} , the fixed $\tilde{\psi}_n$, and the trial β_P , solve the share block: with $Q_{nit} = m_{nt} e^{-\delta d_{ni}} m_{it} \tilde{\psi}_i$, that is, (E.13) at $C_t = 1$, iterate $\kappa_{nt} \leftarrow \sum_i Q_{nit} \kappa_{it}^{\beta_P}$ in logarithms from any start until it stops changing (Theorem E.5). The shares are (E.12), and since solving at $C_t = 1$ only rescales κ_t (Theorem E.5), they are

$$\sigma_{nt}(\beta_P) = \frac{\tilde{\psi}_n \kappa_{nt}^{\beta_P}}{\sum_k \tilde{\psi}_k \kappa_{kt}^{\beta_P}}.$$

The model's count is $\sigma_{nt}(\beta_P) S_t$ with S_t the observed London total.

- (c) *Coefficients.* Let $\tau(\beta_P)$ denote the post-treatment coefficients of step 1 on the panel $\sigma_{nt}(\beta_P) S_t$.

3. Estimate

$$\widehat{\beta}_P = \arg \min_{\beta_P \in [0, 0.999]} (\widehat{\tau} - \tau(\beta_P))' W (\widehat{\tau} - \tau(\beta_P)), \quad W = \text{diag}(\widehat{s}_0^{-2}, \widehat{s}_{10}^{-2}, \dots, \widehat{s}_{40}^{-2}), \quad (\text{E.18})$$

by running step 2 once per trial value of β_P . Figure E1 shows the estimated coefficients and the model's at the estimate. The weighting is diagonal so that the estimate fits the five coefficients themselves rather than the contrasts between them that the inverse covariance weights.

β_S is not identified within London. By Theorem E.3 it enters the model only through the total S_t , and S_t enters the share block only through c_t , which is common to all parishes, so no within-London contrast depends on it. It is bounded, $0 < \beta_S \leq \beta_P$. Yet, it can be estimated from the aggregate margin, in two designs, each resting on one of the following lemmas.

Lemma E.9 (The aggregate margin). *Let $\theta = \beta_S / (1 - \beta_S)$, so that $\ln \widehat{S}_t = \theta \ln M_t$ by Theorem E.10.*

- (a) A uniform shock. *If $\widehat{\kappa}_{kt} = e^{2\phi}$ for every parish k , then $\ln \widehat{S}_t = 2\phi\theta$.*

⁶³The weight of one half matters for β_P , which is 0.390, 0.485, 0.560, and 0.670 at weights 0, 1/4, 1/2, and 1, but hardly for the aggregate effect of Procedure E.4, 242 to 246 of the 797 scientist-decades at $\beta_S = 0.39$, with 173 to 189 of them in the parishes that had a coffeehouse by the decade. At weight 0, β_P also falls below the instrument makers' $\beta_S = 0.43$ of Procedure E.3, which $\varepsilon_S \leq \varepsilon_P$ rules out.

(b) The bracket. With $u_k = \ln \widehat{\kappa}_{kt}$, $\sum_k \sigma_{kt} u_k \leq \ln M_t \leq \max_k u_k$.

Proof. (a) Since $\sum_k \sigma_{kt} = 1$,

$$M_t = \left(\sum_k \sigma_{kt} e^{2\phi\beta_P} \right)^{1/\beta_P} = e^{2\phi} \implies \ln \widehat{S}_t = \theta \ln e^{2\phi} = 2\phi\theta.$$

(b) Write $\ln M_t = \frac{1}{\beta_P} \ln \sum_k \sigma_{kt} e^{\beta_P u_k}$. Since the weights σ_{kt} sum to one,

$$e^{\beta_P \sum_k \sigma_{kt} u_k} \leq \sum_k \sigma_{kt} e^{\beta_P u_k} \leq e^{\beta_P \max_k u_k} \sum_k \sigma_{kt} = e^{\beta_P \max_k u_k},$$

where the lower-end inequality follows from Jensen's inequality for the convex function e^z and the upper-end one from $e^{\beta_P u_k} \leq e^{\beta_P \max_j u_j}$ for every k . Taking logarithms and dividing by $\beta_P > 0$ gives the bracket. $\square$

Lemma E.10 (The aggregate margin against a control). *Let S'_t be the total with the coffeehouses of decade t removed, so that $\widehat{S}_t = S'_t/S_t$, and let $a_t = -\ln M_t$ be the access lost. Let S_{Rt} be a control series that moves with the same fundamentals up to a linear trend, $\ln S'_t - \ln S_{Rt} = c_0 + c_1 t + e_t$, with t counted in decades. Then, with $\Delta x_t = x_t - x_{t-1}$,*

$$\Delta \ln \frac{S_t}{S_{Rt}} = \theta \Delta a_t + c_1 + \Delta e_t,$$

and $0 \leq a_t \leq -\sum_k \sigma_{kt} \ln \widehat{\kappa}_{kt}$ when no parish gains access, $\widehat{\kappa}_{kt} \leq 1$ for every k .

Proof. By Lemma E.9, $\ln S'_t - \ln S_t = \theta \ln M_t = -\theta a_t$, so

$$\begin{aligned} \ln S_t - \ln S_{Rt} &= \ln S'_t + \theta a_t - \ln S_{Rt} = c_0 + c_1 t + \theta a_t + e_t, \\ \Delta \ln \frac{S_t}{S_{Rt}} &= \theta \Delta a_t + c_1(t - (t-1)) + \Delta e_t = \theta \Delta a_t + c_1 + \Delta e_t. \end{aligned}$$

For the bounds, with $u_k = \ln \widehat{\kappa}_{kt}$ as in Lemma E.9(b), every $u_k \leq 0$ when no parish gains access. Multiplying the bracket of Lemma E.9(b) by -1 reverses its inequalities, and $-\max_k u_k \geq 0$ because every $u_k \leq 0$, so

$$0 \leq -\max_k u_k \leq a_t \leq -\sum_k \sigma_{kt} u_k. \quad \square$$

Procedure E.3 (The aggregate margin, β_S). Both designs read the aggregate response off the European panel of Section 6, 200 cities of ten thousand or more inhabitants in 1600 over 1600–1690, 37 of them with a coffeehouse before 1700, the outcome the count of applied mathematics titles printed in the city (the Universal Short Title Catalogue) or of scientific instrument makers active there (Webster and Webster, 2007).

1. *Cities.* A city whose applied scientists do not move to other cities is the economy of Theorem E.10. Its first coffeehouse, read as reaching the whole city, is $\widehat{m}_{kt} = e^\phi$

in every parish, so by Theorem E.8 $\widehat{\kappa}_{kt} = e^{2\phi}$ for every k and the shares do not move, and by Lemma E.9(a) it raises the city's log count by $2\phi\theta$. We estimate this rise in Section 6 as $\ln(1 + b/\text{base})$, with b the mean post-entry effect over event times 0 to +30 and base the treated cities' mean count in the decade before entry, so $\theta = \ln(1 + b/\text{base})/(2\phi)$ and $\beta_S = \theta/(1 + \theta)$. Titles give $\beta_S = 0.35$ (s.e. 0.12) and instrument makers $\beta_S = 0.43$ (0.15), with parametric-bootstrap standard errors over b and $\widehat{\phi}$.⁶⁴

2. *London against the other European cities.* Removing London's coffeehouses in decade t is $\widehat{m}_{nt} = e^{-\phi D_{nt}}$. The share system of Theorem E.8 then gives $\widehat{\kappa}_{nt}$, which is below one in every parish and decade, and a_t follows at the observed counts and shares, $(\widehat{\phi}, \widehat{\delta})$ and β_P . With the sum over the other 199 cities as the control S_{Rt} , Lemma E.10 makes θ the coefficient of Δa_t in the least-squares regression of $\Delta \ln(S_{Lt}/S_{Rt})$, with S_{Lt} London's count in the panel, on Δa_t and a constant over 1600–1690, nine differences. Titles: $\theta = 0.43$ (s.e. 1.37), $\beta_S = 0.30$. Instrument makers: $\theta = 0.23$ (0.67), $\beta_S = 0.18$. In a parametric bootstrap over θ , β_S has upper ends of 0.76 and 0.61 and no lower end, since $\theta = -1$, where $\theta/(1 + \theta)$ is unbounded, lies within two standard errors of both estimates.

The remaining parameters are not estimated. γ and ε_P are not identified, since the data speak to β_P alone (Corollary E.2). Both can be set from the quantitative urban literature, which measures them in other places and centuries: modern Berlin and London give the premises share $\gamma = 0.225$ and the commuting dispersion $\varepsilon_P = 5$ (Ahlfeldt et al., 2015; Heblich et al., 2020), so that $\eta = \beta_P(1 + \gamma\varepsilon_P)/\varepsilon_P = 0.238$ by (E.4), and post-war American occupational choice gives ε_S between 1.5 and 2.7, with 2 the base case (Burstein et al., 2015; Galle et al., 2017; Hsieh et al., 2019; Lagakos and Waugh, 2013). The implied $\beta_S = \beta(2) = 0.33$, and 0.27 to 0.40 over that range, is a check on the estimates of Procedure E.3, 0.18 to 0.43, not a measurement of β_S .

Procedure E.4 (The aggregate effect). With (ϕ, δ) , β_P and β_S in hand, the no-coffeehouse counterfactual is computed decade by decade for 1650–1690, in ratios and from the observed decade alone, with the fundamentals held fixed (Theorem E.8). For one decade:

1. *Baseline.* From the observed counts and geography, the shares $\sigma_{nt} = S_{nt}/S_t$, the access shares ω_{nit} of Definition E.6, and the treatment ratios $\widehat{m}_{nt} = e^{\phi(D'_{nt} - D_{nt})}$, here $\widehat{m}_{nt} = e^{-\phi D_{nt}}$ since every coffeehouse is removed ($D'_{nt} = 0$). Parishes with $S_{nt} = 0$ stay at zero.
2. *Shares.* Solve the share system of Theorem E.8 for $\widehat{\sigma}_{nt}$ and $\widehat{\kappa}_{nt}$ by iterating it from $\widehat{\sigma}_t = \mathbf{1}$ until nothing changes. By Theorem E.9 each pass contracts the error by β_P , and the solution is the unique one of Theorem E.5 applied to the kernel $\widehat{m}_{nt}\omega_{nit}\widehat{m}_{it}$.

⁶⁴The standard error is half the distance between the 16th and 84th percentiles of 20,000 draws of b and $\widehat{\phi}$ from normal distributions at their estimates and standard errors, which matches the standard deviation when the draws are normal. The standard deviation itself is not informative here: $\beta_S = \theta/(1 + \theta)$ is unbounded as θ approaches -1 , and the few draws that come close dominate it. The draws also have heavier tails than a normal distribution, with 2.5th and 97.5th percentiles of 0.06 and 0.67 for titles and -0.20 and 0.75 for instrument makers.

3. *Total.* $\ln \widehat{S}_t = \frac{\beta_S}{1-\beta_S} \ln M_t$ by Theorem E.10.
4. *Counts.* The counterfactual count is the observed count times its ratio, $S'_{nt} = \widehat{S}_{nt} S_{nt}$ with $\widehat{S}_{nt} = \widehat{\sigma}_{nt} \widehat{S}_t$ (Theorem E.8), and the decade's aggregate effect is $\sum_n (S_{nt} - S'_{nt})$. Theorem E.11 splits each parish's change into relocation and creation.

Procedure E.1 gives $\phi = 0.554$ and $\delta = 0.277$, Procedure E.2 $\beta_P = 0.560$, and Procedure E.4 turns each value of β_S from Procedure E.3 into the aggregate effect of Table E1, with $\beta_S = 0$ and the bound $\beta_S = \beta_P$ alongside. We prefer the city estimates for their statistical power: they compare 37 treated cities with 163 never treated, whereas London against the other cities is one series of nine differences, which leaves β_S without a lower end. Their average, $\beta_S = 0.39$, is the preferred value. At it the parishes that had a coffeehouse by the decade lose 182 of the 243 scientist-decades and the others a net 61, and coffeehouses account for 46 percent of the rise in the number of applied scientists from the 1640s to the 1690s (40 to 52 at the two city estimates).

E.10. FIGURES

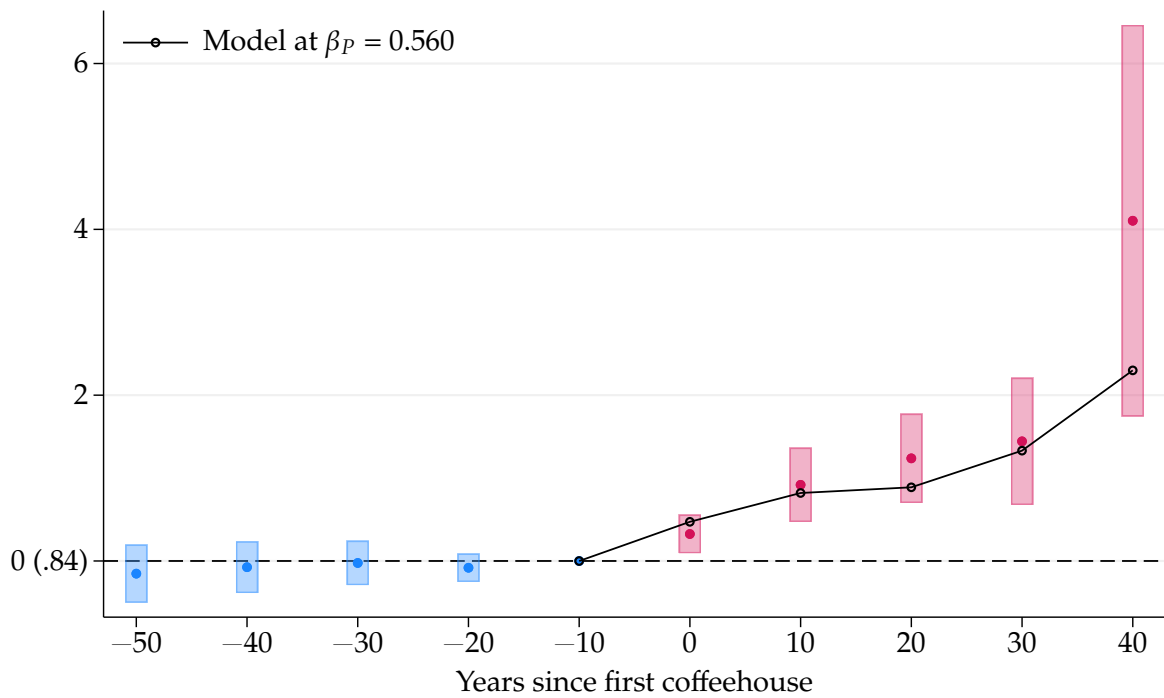


FIGURE E1. THE EVENT STUDY AND THE MODEL AT THE ESTIMATED β_P

Notes: Dots with bars: the coefficients $\hat{\tau}_h$ of the event study on the observed counts of the combined roster, step 1 of Procedure E.2, with 95 percent intervals, the decade before entry normalized to zero, drawn as in Figure 5 (blue before entry, red after, the treated parishes' mean count in the decade before entry beside the zero tick). Line with hollow markers: the post-treatment coefficients $\tau(\hat{\beta}_P)$ of the same event study on the panel the model implies, step 2, at the estimate $\hat{\beta}_P = 0.560$ of (E.18). Only the post-treatment coefficients are matched. [Return to pages 35, 95]

E.11. TABLES

TABLE E1—THE AGGREGATE EFFECT OF REMOVING EVERY COFFEEHOUSE, 1650–1690

	β_S	Aggregate effect		Gain of the treated parishes	
		Count	Percent	Relocation, %	Creation, %
No aggregate response	0.00	0	0	100	0
London against Europe, titles	0.30	172	22	31	69
London against Europe, makers	0.18	96	12	46	54
Cities, titles	0.35	212	27	26	74
Cities, makers	0.43	276	35	21	79
Average of the two city estimates	0.39	243	31	23	77
Upper bound, $\beta_S = \beta_P$	0.56	405	51	14	86

Notes: Procedure E.4, the five decades summed, on the combined roster of applied scientists. Count in scientist-decades, percent of the 797 observed. The last two columns take the gain of the parishes that had a coffeehouse by the decade, $S_{nt} - S'_{nt}$ summed over those parishes and the five decades, and split it into the relocation and creation terms of Theorem E.11, summed the same way, each in percent of the gain. Relocation is the part that the counterfactual places in other London parishes, and creation the part it places outside science. β_S from Procedure E.3: titles are the applied mathematics titles printed in a city and makers its scientific instrument makers. Light shading marks the two city estimates and dark shading their average, the preferred value. [Return to page 98]

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